CAS-Ohm 1.0: A Profile-Specific Experimental Response and Calibration Model

Little Henderson

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CHAPTER 1

Introduction

1. What this monograph is about

This monograph develops, at full technical resolution, a single candidate object: CAS-Ohm 1.0, a profile-specific experimental response and calibration model for steered neural-network activations. The candidate proposes that, within a narrowly declared experimental regime, a declared intervention coordinate — the steering alpha applied along a fixed direction vector at a fixed test point in a fixed model — and an observed activation coordinate — the directional projection of the resulting activation — may stand in a local, approximately proportional relation, and that the proportionality parameter, once fitted on a registered calibration split and carried with uncertainty, may predict held-out response. The candidate prediction is

$$\hat{y} = \frac{a}{r_{\text{eff}}},$$

where a is the alpha dose, $\hat{y}$ is the predicted activation response, and r_{eff} is the effective resistance in model-specific units of alpha per activation.

Stated so briefly, the proposal sounds like a modest exercise in curve fitting, and in one sense it is exactly that. The reason it warrants a monograph is not the arithmetic but the discipline surrounding the arithmetic. The historical source material from which CAS-Ohm 1.0 descends — a quantification guide, an operationalization addendum, and an appendix to a printed-circuit-board-styled engineering text — reached repeatedly for the vocabulary of Ohm's law: voltage, current, resistance, series and parallel composition, gain cascades, hidden resistance inferred from residuals. That vocabulary is seductive precisely because it comes bundled with an enormous inherited apparatus of intuitions, identities, and network theorems, almost none of which have been earned in the setting where the sources deploy them. A preserved source audit, reproduced in full later in this monograph, registers eight distinct conflicts or under-specifications across those three sources: mixed measurement scales, incompatible full-scale calibration methods, a missing dimensional bridge between score-valued and activation-valued quantities, two mathematically distinct definitions of gain, a mnemonic classification axis overloaded with two independent meanings, network composition rules asserted without validation, residual quantities pressed into causal service they cannot bear, and quantization band boundaries left without a precision rule.

CAS-Ohm 1.0 is the constructive response to that audit. It is a typed proposal in which every quantity carries its kind, unit, measurement provenance, calibration binding, and uncertainty; in which the historically observed failure modes are either unrepresentable in the type system or fail closed at validation; in which the Ohm-shaped prediction is admissible only under explicit anti-circularity and split-discipline conditions; and in which every empirical question the analogy raises is registered as a falsifiable hypothesis with a preregistered closure protocol rather than asserted as a fact. The candidate's own status vocabulary is blunt: at the time of writing, every CAS-Ohm claim is OPEN / RING 4 / NOT LICENSED /

HELD. Nothing in this monograph reports a validated empirical result about any live model, and nothing in it should be read as one.

This first chapter fixes the frame for everything that follows. Section 2 states precisely what CAS-Ohm borrows from Ohm's law — the *shape* of the relation and nothing else — and Section 3 states, with equal precision, the four things it refuses. Section 4 sets out the terminology discipline that separates the candidate's quantities from physical electrical quantities. Section 5 explains why dimensional consistency, though enforced everywhere in the candidate, is a necessary condition for meaningfulness and not a sufficient condition for empirical validity. Section 6 states the three contributions of the work. Section 7 records the charter status of this document and the authority boundary that governs every claim in it. Section 12 is a reading guide to the monograph.

2. The borrowed shape

Ohm's law, in its elementary physical form, relates the potential difference across a conductor, the current through it, and a material- and geometry-dependent parameter called resistance. Three structural features of that relation — and only those three — are borrowed by CAS-Ohm.

DEFINITION 1.1 (The borrowed shape). The *borrowed shape* of Ohm's law consists of exactly three components:

(1) a *dose coordinate*: a scalar quantity that the experimenter declares and controls, swept over a registered grid;
(2) a *response coordinate*: a scalar quantity that the experiment observes, measured by a declared method under a declared calibration profile; and
(3) a *fitted proportionality parameter*: a scalar relating the two, estimated from calibration data by a declared fit method, carried with uncertainty, and evaluated on data disjoint from the data used to estimate it.

In the CAS-Ohm instantiation, the dose coordinate is the steering alpha a: the scalar coefficient by which a fixed, unit-norm, sign-declared direction vector is scaled before being added to (or otherwise composed with) the activation at a declared layer, token-position set, and hook type of a declared model. The response coordinate is the activation response y: the directional projection of the observed activation onto a declared effect direction at a declared test point. The fitted proportionality parameter is the effective resistance r_{eff}, whose units are alpha per activation, so that the candidate prediction $\hat{y} = a/r_{\text{eff}}$ is dimensionally coherent by construction.

The borrowed shape is deliberately austere. It does not include the physics of conduction, the microscopic theory that explains why some materials are ohmic over wide ranges, the thermodynamic account of dissipation, Kirchhoff's laws, the network theorems that follow from them, or the metrological infrastructure — standard resistors, Josephson voltage standards, quantum Hall resistance standards — that gives physical electrical measurement its extraordinary reproducibility. Every one of those omissions is intentional, and several of them are enforced by the type system rather than left to authorial restraint.

The intellectual tradition the borrowed shape actually belongs to is not circuit physics but dose-response modeling: the empirical study, ubiquitous in pharmacology, toxicology, and agronomy, of how a controlled stimulus maps to an observed response, with characteristic regions — a sub-threshold region where response is indistinguishable from baseline, an approximately linear middle region, and a saturation region where further dose produces diminishing response [**38**]. CAS-Ohm's dose-response region classification (which includes cutoff,

active, saturation, breakdown, non-ohmic, and unstable regions, together with an unassessed default) descends from that tradition, not from electrical engineering. The Ohm vocabulary survives only as the name of the candidate and as the mnemonic for the fitted-slope-through-the-origin form of the active-region model; the analytical machinery is dose-response analysis under metrological discipline.

A second point deserves emphasis before any formalism appears: borrowing a shape licenses nothing. The relation $\hat{y} = a/r_{\text{eff}}$ is a hypothesis schema, not a law. Whether any (model, layer, direction, test point, corpus, alpha grid) configuration actually exhibits a local linear response region; whether a fitted r_{eff} predicts held-out response better than trivial baselines; whether independent manipulation of proposed constraint mechanisms moves response with the predicted inverse slope — these are the registered hypotheses of the falsification program, every one of which may fail. The monograph's stance is that the analogy is valuable exactly insofar as it is testable and rejectable, and worthless the moment it is treated as inherited truth.

Example 1.2 (Schematic reading of the borrowed shape). To fix intuitions without asserting any result, consider a purely schematic run. An experimenter declares a calibration profile: an exact model identity (weights, configuration, and tokenizer digests), a layer, a token-position set, a unit-norm direction with declared sign and provenance, an alpha grid, a full-scale method, fit and uncertainty methods, and controls. Sweeping a over the grid on the calibration split, the experimenter records typed observations of y. Suppose — schematically — the region classifier finds a contiguous sub-grid classified ACTIVE. A fit on the calibration split produces r_{eff} with a standard uncertainty and interval, in role CALIBRATED HELD-OUT bound to the split reference. Only now is $\hat{y} = a/r_{\text{eff}}$ computable on the disjoint held-out split, and only its held-out performance — against zero-intervention, random-direction, wrong-layer, and norm-matched controls — would bear on the local-response hypothesis. Every step of this schematic is specification; whether any real configuration completes it with favorable numbers is precisely what no part of this monograph claims.

3. The four refusals

The candidate is defined as much by what it refuses as by what it proposes. Four refusals are load-bearing; each is stated here as an explicit claim boundary and enforced later by the ontology, the audit dispositions, or the capability gates.

Claim Boundary 1.3 (No physical law). CAS-Ohm 1.0 asserts no physical law. No CAS-Ohm quantity is a physical voltage, current, resistance, charge, flux, power, or any other SI electrical quantity. No CAS-Ohm record is physical telemetry. The unit ontology contains no volt, ampere, or ohm; those labels are forbidden aliases for model quantities, and a hypothetical physical profile would require separate explicit typing, separate instrumentation, and separate validation that do not exist in the current candidate.

The first refusal is not a stylistic disclaimer but a structural fact about the type system. Chapter 3 presents a unit ontology whose members are `ALPHA`, `ACTIVATION`, `ALPHA_PER_ACTIVATION`, `DIMENSIONLESS`, and `FRACTION`, together with the `Q10` and `Q100` score scales. There is no representable way, within the candidate, to record a value in volts. The generic CAS PCBA vocabulary elsewhere in the wider project contains a member `Unit.VOLT` for entirely different purposes; the candidate's position, developed in Chapter 3, is that the existence of that vocabulary member licenses nothing — neither physical CAS telemetry nor a CAS-Ohm unit.

Claim Boundary 1.4 (No electrical implementation). CAS-Ohm 1.0 describes no electrical circuit, device, board, or component, and no electrical behavior of any computing substrate. The series, parallel, and gain-composition operations defined in the candidate are structural research calculations over typed records, capability-gated and unauthenticated; they are not claims that any network of model components behaves like a resistor network.

The second refusal addresses the most tempting inheritance of all: network theorems. Series addition of resistances, reciprocal aggregation of parallel resistances, and multiplication of cascaded gains are, in circuit theory, consequences of conservation laws that hold with extraordinary generality in physical circuits. Nothing analogous has been established for steered activations. The source audit found the network rules presented in the Guide as working machinery, applied in Appendix I with partial warnings, and correctly demoted in the addendum to preregistered predictions. The candidate adopts the strictest reading: composition operations exist only as gated structural calculations whose empirical content is exactly the registered composition and redundancy hypotheses, and no more.

Claim Boundary 1.5 (No attention measurement). CAS-Ohm 1.0 does not measure attention. No CAS-Ohm quantity is a measurement of a model's attention weights, attention allocation, or any cognitive construct named "attention," and no CAS-Ohm record supports the product wording "measures attention."

The third refusal severs a specific historical thread. The source Guide computed a quantity it called raw attention as a ratio of score-valued quantities, inviting the reading that the framework measures how much attention a system pays to something. The candidate's response coordinate is nothing of the sort: it is a directional projection of an activation vector — a geometric quantity relative to a declared direction, at a declared layer and token position, under a declared calibration profile. Whether such projections relate in any lawful way to attention mechanisms in the architectural sense, or to attention in any psychological sense, is outside the candidate's claim surface entirely.

Claim Boundary 1.6 (No causal law). CAS-Ohm 1.0 asserts no causal law. A fitted response relation is an observational-and-interventional summary within one calibration profile; it does not identify mechanisms, does not license counterfactuals beyond its registered regime, and does not support causal attributions from residual signs. Causal interpretation requires an explicit causal model and identifiable interventions [27], neither of which the candidate supplies.

The fourth refusal has teeth in a specific place: residual diagnostics. The sources repeatedly used the sign of a prediction residual to suggest hidden opposition, extra amplification, or hidden resistance — in one case treating a positive residual on a safety-relevant configuration as evidence of a bypass. The candidate's residual discipline, developed later in the monograph, permits residuals to rank follow-up probes and nothing else: a residual never proves a hidden policy, a bypass, an opposition, a cause, or a safety failure, and at zero or zero-compatible predicted response the diagnostic ratios are censored rather than reported. The registered residual-localization hypothesis states the exact condition under which any stronger language could ever be earned — blinded implanted faults localized above shuffled and matched controls — and commits to withdrawing causal language if that test fails.

These four refusals interlock. A reader who grants the candidate its dose-response framing but silently retains an electrical mental model will over-trust the network operations; a reader who accepts the projection-based response coordinate but hears "attention" will over-read the semantics; a reader who accepts the fitted slope but forgets the causal boundary will

convert residuals into accusations. The monograph therefore repeats the refusals wherever the corresponding machinery appears, at the cost of some redundancy, because the failure mode being defended against is precisely the quiet reactivation of inherited intuitions.

4. Terminology discipline

The historical sources use V, I, and R throughout, by analogy. This monograph and the candidate do not. The terminological rule is simple and total.

RULE 1.7 (Terminology discipline). In every CAS-Ohm context, the dose coordinate is named the *alpha dose* and written a; the response coordinate is named the *activation response* and written y; the fitted proportionality parameter is named the *effective resistance* and written r_{eff}, with units of alpha per activation. The words *volt*, *ampere*, and *ohm*, the symbols V, I, and R in their electrical senses, and every derived electrical unit are reserved for physical profiles, which do not exist in the current ontology. Source quotations retain source notation, always inside an explicit attribution to the source.

The rule does more work than it may appear to. First, it prevents dimensional camouflage. When a score-valued quantity on a 0–10 scale is called R and divided into a quantity called V, the notation itself suggests that the quotient is a current-like quantity, and the sources fell into exactly this trap: the audit's third conflict records that the predicted-response formula retained the form V/R_{total} even after V had been reinterpreted as a steering alpha and R remained score-derived, leaving the prediction dimensionally unmoored. Under the discipline, the same error is impossible to write down without noticing: a constraint score in `Q10` or `Q100` units simply is not an r_{eff} in `ALPHA_PER_ACTIVATION` units, and the type system rejects the division at the unit-validation layer.

Second, the rule keeps the analogy honest about its direction of travel. Physical Ohm's law earned its network theorems from conservation laws; the candidate's composition rules are hypotheses. Writing r_{eff} rather than R is a persistent, low-cost reminder that the parameter is an effective, profile-relative, fitted quantity — closer in spirit to an effective dose-response slope than to a material property.

Third, the rule fixes the reading of the one place where electrical vocabulary legitimately appears: the forbidden-alias list. The unit ontology names VOLT, AMPERE, and OHM precisely in order to forbid them. This is not decoration. An ontology that merely omitted the electrical units would be vulnerable to a later "helpful" extension adding them; an ontology that names them as forbidden aliases makes the refusal itself part of the contract, testable by the golden rejection fixtures that accompany the implementation.

A related discipline governs the word *resistance* itself. The candidate uses *effective resistance* only for the fitted `ALPHA_PER_ACTIVATION` parameter with a declared role, and *constraint score* for the score-valued quantities the sources sometimes called resistance. The two are distinct quantity kinds with disjoint allowed units, and the audit's binding disposition — constraint scores never masquerade as effective resistance — is enforced at validation. Where the monograph must discuss the sources' conflated usage, it does so in quotation and with the conflict explicitly cited.

5. Dimensional consistency is necessary, not sufficient

Everything in the candidate is dimensionally disciplined: every quantity kind has a closed set of allowed units, every uncertainty carries the unit of its measurement, and the prediction

$\hat{y} = a/r_{\text{eff}}$ type-checks only because r_{eff} carries alpha-per-activation units. It is therefore worth stating plainly what this discipline does and does not buy.

Dimensional analysis, in the tradition formalized by Buckingham [34], establishes that physically meaningful relations must be expressible in terms of dimensionless combinations of the governing quantities, and that a dimensionally inconsistent equation cannot be a correct description of anything. The contrapositive is the useful direction here: dimensional consistency is a *necessary* condition for a relation to be a candidate description at all. An equation that adds a score to a projection, or divides an alpha by a 0–100 constraint score and calls the result an activation, is not wrong about the world; it is not yet about the world. The audit's mixed-scale conflict — the worked example in which a saturation threshold of 8.5 on a 0–10 scale was compared against region fractions on a 0–1 scale, with the corrected component value being 0.85 — is exactly such a pre-empirical failure, and the candidate's separation of `Q10`, `Q100`, and `FRACTION` units exists to make that class of failure unrepresentable.

But necessity is not sufficiency, and the distinction is the hinge of the entire claim discipline. A dimensionally impeccable relation can still be empirically false, vacuous, or circular. Three specific insufficiencies matter for CAS-Ohm.

First, *fit does not imply prediction.* Given any paired sample (a, y) with $y \neq 0$, the same-sample identity $r = a/y$ followed by $\hat{y} = a/r$ reproduces y exactly, with perfect dimensional hygiene and zero empirical content. The candidate rejects this circularity by construction: an effective resistance is admissible for prediction only in the role CALIBRATED HELD-OUT (or a separately validated network-equivalent role), which requires a registered calibration split disjoint from the evaluation split, a declared fit method, and carried uncertainty. The apparent-same-sample role exists in the ontology precisely so that the circular quantity can be recorded, labeled as what it is, and barred from prediction.

Second, *units do not certify measurement.* A value in `ACTIVATION` units is not thereby a trustworthy observation; it might be a design declaration, a model-derived computation, or a score. The measurement-kind axis of the ontology — design value, instrumented, behavioral proxy, model derived, human score, LLM score — is orthogonal to the unit axis, and the non-fungibility rules (a derived value never becomes instrumented; a score never becomes telemetry) prevent dimensional correctness from laundering provenance. This is elementary metrological discipline [32, 33], adopted without any claim that CAS-Ohm's instruments are themselves validated.

Third, *dimensional coherence does not select the model.* Within the same unit system one can write a proportional law, a saturating law, a threshold law, or an arbitrary monotone curve; dimensional analysis is silent among them. Which, if any, describes a given configuration is an empirical question for the falsification program, to be answered under calibration discipline that represents parameter uncertainty and model discrepancy rather than anointing a best fit as truth [37], and under the separation of software verification from operational validation [23].

The monograph therefore treats dimensional consistency as a floor: a precondition enforced everywhere, celebrated nowhere. Any prose in later chapters that appears to move from "the units work out" toward "the relation holds" should be read against this section and rejected; no such move is licensed.

6. The three contributions

The monograph makes three contributions. None is an empirical finding about any live model; all are artifacts of specification, preservation, and program design.

6.1. An ontology in which the observed failure modes are unrepresentable. The first contribution is the quantity and evidence ontology of Chapter 3: a typed system in which every numeric record carries its quantity kind (one of sixteen), its unit (from a closed unit ontology with forbidden electrical aliases), its measurement kind (one of six, non-fungible), its method identity, its calibration-profile digest, its estimate-or-observation status, its uncertainty and confidence interval, its sample size, its raw-artifact and control references, its derivation parents, and — where applicable — its missing-or-censored reason. No bare binary float is admitted anywhere in the evidence layer.

The design criterion for the ontology was neither generality nor elegance but a specific negative property: each failure mode surfaced by the source audit should be either unrepresentable or fail-closed. The mixed-scale comparison cannot be written because `Q10`, `Q100`, and `FRACTION` are distinct units with disjoint validation ranges and no silent conversion. The dimensionally unmoored prediction cannot be written because effective resistance is the only quantity licensed to bridge alpha and activation, and it demands its own units, role, and split binding. The score-as-telemetry slide cannot occur because measurement kinds are non-fungible and instrumented records demand instrument identity, calibration binding, raw evidence, and sample information. The residual-as-cause leap is blocked because residual emission requires an admitted active-region prediction with propagated uncertainty and is censored at zero-compatible response. The ontology is, in this sense, the audit's negative image.

6.2. A preserved audit at evidence-record granularity. The second contribution is the preserved source audit itself. Eight conflicts or under-specifications across the three sources are registered in a machine-readable evidence record, each with disclosed locator systems (rendered PDF pages for the Guide; OOXML paragraph and table-row indices for the addendum, for which no page number is asserted because both available page-rendering routes failed; UTF-8 line numbers for Appendix I; PDF and book pages for the PCBA gain definitions), the conflicting facts side by side, and a binding disposition for the candidate.

Preservation, rather than harmonization, is the point. It would have been easy — and it would have been malpractice — to smooth the sources into a single retrospectively coherent story. The preserved audit makes that impossible: downstream code and prose cannot claim one coherent inherited scale, one full-scale method, or one gain law where the sources demonstrably provide several, and every disposition in the candidate is traceable to the specific conflict it resolves or quarantines. The audit is thus an engineering result in its own right: it converts historical incoherence from a latent hazard into an explicit, cited, machine-checkable boundary condition.

6.3. A fail-closed candidate with a falsification program. The third contribution is the candidate itself: a specified, implemented, and test-covered system — calibration-profile contract; independent dose-response-region and semantic-outcome classification axes; exact decimal quantization with a boundary-preservation guarantee; capability-gated structural network operations; residual censoring; and a strict split between structural research-packet completeness and authenticated Evidence-Packet eligibility, the latter constant false in this candidate — together with a falsification program of five registered hypotheses (local response, effective-resistance prediction, gain composition, redundancy, residual localization) and a preregistered closure protocol fixing controls, falsifiers, negative-outcome retention, and permitted claim language before any run.

The phrase *fail-closed* carries specific weight. Wherever the candidate encounters a record that is score-only, zero-instrumentation, missing-provenance, mixed-scale, unknown-profile,

or tampered, it rejects rather than degrades; wherever a capability gate's conditions are unmet, the operation is unavailable rather than approximate; wherever authentication conditions are unmet — which, in this candidate, is always — eligibility is false rather than provisional. The bounded software evidence that exists (a passing test suite exercising rejection paths, boundary cases, and gate behavior at a pinned receipt) establishes tested software behavior only, and the monograph will say so every time it is mentioned.

7. Charter status and the authority boundary

7.1. Provisional technical report, not an eighth charter paper. The Project Coherence charter, which governs the wider research program this work sits inside, lists seven planned papers. No standalone CAS-Ohm paper is among them. Until and unless a charter amendment exists, this monograph and the research paper it extends are *provisional technical reports*: internal working artifacts, drafted to fix the candidate's specification and claim discipline at full resolution, and explicitly not representable as an eighth charter paper, as a charter-approved publication, or as any other charter-level deliverable.

This is not modesty theater; it is a governance fact with operational consequences. Charter papers pass through the program's admission machinery — internal cross-review, independent external review, claim-ledger reconciliation — before any claim in them changes status. This monograph has passed through none of that: drafting it promotes no claim, and both internal cross-review and independent external review remain open. A reader who cites this monograph should cite it as what it is: a draft technical report describing a candidate whose every substantive claim is registered as open.

7.2. The authority boundary.

Claim Boundary 1.8 (Authority boundary). This monograph establishes no live execution, no scientific conclusion, no physical measurement, no causal effect, no safety or clinical conclusion, no product capability, no license, no canon admission, and no disclosure. All CAS-Ohm claims referenced herein hold status OPEN / RING 4 / NOT LICENSED / HELD. Authenticated Evidence-Packet eligibility is constant false in this candidate. Structural consistency, canonical digests, replayed attestations, and passing test suites are integrity and replay evidence only; none is reviewer, legal, product, or scientific authority.

The boundary deserves a paragraph of unpacking, because its final sentence names the subtlest hazard. The candidate is unusually rich in integrity machinery: canonical byte encodings, SHA-256 digests over every typed record, optional Ed25519 fixture-key attestations that are verified and replayed, exhaustive lineage. It is natural — and wrong — to let that machinery accrete into an impression of authority. A digest proves that bytes have not changed; it does not prove the bytes were ever true. A replayed attestation proves a fixture key once signed a payload; it does not conjure an independent reviewer. Provenance structure is not truth or authority [22]. The candidate encodes this as an architectural split — completeness and eligibility are independent axes, and no accumulation of structural evidence can move a packet from complete to authoritative — and the monograph adopts the same split in its prose: descriptive thoroughness in the chapters that follow is never to be read as an authority claim.

A final component of the boundary concerns forbidden product wording. The claim ledger records, as a NOT LICENSED product claim, any blanket representation of design scores, behavioral proxies, or model-derived values as physical telemetry; and the specification forbids, in any product context, the phrases "physical telemetry," "measures attention," "electrical

law," "causal by construction," "cross-model comparable," "proves hidden resistance," and any safety or verdict claim. The monograph mentions these phrases only to forbid them.

8. Relation to the sources and to the wider program

8.1. The three sources. Three documents constitute the audited lineage. The *Quantification Guide* is the earliest and most ambitious: it introduces the electrical vocabulary, defines score-valued drive and constraint quantities on a 0–10 scale (constraint on 0.1–10.0), computes a raw attention quantity as their ratio, presents series, parallel, and gain-cascade rules as working machinery, reads residual signs diagnostically, and displays a four-band quantization table with two-decimal band edges. The *operationalization addendum* is the pivotal reinterpretation: it maps the dose coordinate to a steering alpha and the response coordinate to a directional activation projection — the move that makes the proposal empirically testable at all — while retaining, in places, score-derived resistance components inside the predictive formula, and it is also the source that correctly demotes the network rules to preregistered predictions and separates dose-response signatures from effect-axis outcomes. *Appendix I* of the PCBA-styled engineering text is the most self-critical source: it rescales constraint to 0–100, moves saturation and cutoff to 0–1 fractions, identifies the Guide's mixed-scale worked example as an error, defines a per-path rail-and-floor full-scale procedure and warns it does not transfer, frames the predictive relation as testable design arithmetic rather than physics, and warns that co-present guardrails are not parallel routes — while nonetheless itself treating a positive residual on a safety board as a bypass.

The candidate's relation to these sources is neither adoption nor repudiation but audit-mediated inheritance. Where the sources agree and the content survives typing, the candidate adopts it. Where they conflict, the conflict is preserved as a first-class record and the candidate binds a disposition that is deliberately the most conservative reading — distinct quantity types over silent conversion, separately named calibration methods over a preferred one, preregistered predictions over inherited theorems, censored residuals over diagnostic ones. Where a source under-specifies, the candidate specifies exactly and marks the specification as its own choice rather than as recovered intent. Chapter 3 and the audit chapter make each of these moves explicit; nothing is harmonized silently.

8.2. Position inside the program. Within the wider research program, CAS-Ohm 1.0 occupies a deliberately peripheral position. It inherits the program's canonical-encoding contract (the evidence bounds of Chapter 3 are an inheritance, not an invention), its claim-ledger discipline, its ring-based admission model, and its separation of structural consistency from authorization. It contributes back, so far, only working material: a typed candidate, a preserved audit, and registered hypotheses. It depends on no result from the program's other tracks, and no other track depends on it; a total failure of the CAS-Ohm falsification program would falsify the Ohm-shaped proposal and nothing else. This isolation is intentional. A candidate whose sources required an eight-conflict audit is exactly the kind of object that should be quarantined until its own program closes, and the monograph's authority boundary (Section 7) is the textual form of that quarantine.

8.3. How to disagree with this monograph. A specification-first document invites two distinct kinds of disagreement, and the claim discipline benefits from keeping them separate. The first kind targets the specification: a reader may hold that a binding disposition resolves a source conflict the wrong way, that a field of the observed-value record is redundant or missing, that a capability gate is too strict or too lax, or that a registered hypothesis is

malformed. Such disagreements are welcome and actionable — they are exactly what the open internal cross-review is for — and because every disposition cites its conflict and every gate cites its rationale, a critic can attack the precise joint rather than the gestalt. The second kind targets the empirical question: a reader may expect the local-response hypothesis to fail, or the composition rules to be artifacts. Such expectations are not disagreements with this monograph, which asserts neither the hypotheses nor their negations; they are predictions about the falsification program's outcome, and the preregistered protocol is designed so that either outcome, once obtained under controls, will be recorded without erasure. The one form of engagement the monograph forecloses is treating its descriptive machinery as if it carried empirical or product authority — in either direction.

9. Status vocabulary and claim discipline

Because every substantive statement in this monograph is qualified by a status, the status vocabulary itself must be defined before it is used further. The vocabulary is inherited from the wider program's claim-ledger discipline and applied here without modification.

9.1. The status line. Each CAS-Ohm claim carries a four-part status. OPEN states the research status: the claim is registered, unresolved, and awaiting evidence; nothing about it has been accepted or rejected. RING 4 states the admission ring: the outermost ring of the program's admission model, in which artifacts are working material — specifications, candidates, drafts — that have crossed none of the review gates separating working material from admitted results. NOT LICENSED states the license status: no claim may be used to support a product statement, a marketing statement, a safety statement, or any external representation. HELD states the disclosure status: the material is not released, and drafting or circulating it internally effects no disclosure.

The four parts are independent axes, not degrees of a single scale, and the candidate's implementation embeds all four constants into the canonical payload of every measurement record, so that the status travels with the bytes: any future consumer of a CAS-Ohm record will find `OPEN`, admission ring `4`, `NOT LICENSED`, and `HELD` inside the digested content itself, not in detachable metadata. A record cannot be quoted out of its status without altering its digest.

9.2. What drafting does and does not do. A recurring hazard in long-form technical writing is that the act of careful exposition comes to feel like validation: a mechanism described in detail across many pages acquires, in the reader's mind, the solidity of a mechanism that works. The claim discipline addresses this directly. Drafting this monograph promotes no claim. Describing the anti-circularity rule does not establish that any non-circular fit succeeds; describing the capability gates does not establish that any capability's conditions have ever been met on live data; describing the falsification program does not establish that any hypothesis will survive it. The monograph's descriptive completeness is an input to future review, not a substitute for it.

Symmetrically, the discipline constrains the author as much as the reader. No sentence in this monograph may assert, of any live model, that a linear response region exists, that a fitted effective resistance predicted anything, that a network composition rule held, or that a residual localized a fault. Where the temptation to write such a sentence arises — and in expository writing about a response model it arises constantly — the licensed forms are the conditional ("if the local-response hypothesis survives its controls, then…"), the registered ("the redundancy hypothesis states…"), and the negative ("no such result exists at the time

of writing"). The reader is invited to audit the monograph against this rule; any violation found is a defect of the text, not a leak of hidden evidence.

9.3. Bounded software evidence. One category of evidence does exist and is cited where relevant: bounded software evidence. At the pinned receipt of the current gate, the candidate's test suite passes — sixty-five test methods and fifty-eight parameterized subtests in the combined focused receipt — exercising quantity and unit rejection, canonical-evidence limits, calibration identity, split separation, exact quantization boundaries, structural network constraints, residual censoring, uncertainty propagation, lineage, and fail-closed capability behavior, including synthetic known-fault boards (linear, cutoff, saturation, nonlinear, parallel, confounded, bypass) constructed to verify that the software detects what it claims to detect.

The scope of this evidence is exactly its name: it establishes tested software behavior, at a pinned revision, on synthetic inputs. It is verification, not validation [23]. A suite that proves the classifier assigns band boundaries by exact decimal comparison proves nothing about whether any real configuration occupies a linear region; a synthetic bypass board that the residual machinery flags proves nothing about bypasses in any live model. The monograph cites the software evidence often, because fail-closed behavior is a genuine engineering property worth establishing; it converts that evidence into empirical support for the response model nowhere, because no such conversion is valid.

10. Methodological stance: specification before experiment

It is reasonable to ask why a candidate with no admitted empirical results warrants a specification of this length. The answer is the monograph's central methodological commitment: for a proposal whose historical sources demonstrably conflated scales, units, methods, and claim types, *specification must precede experiment*, because an experiment run against an ambiguous specification produces evidence about nothing in particular.

Consider what an alpha-sweep experiment inherits from the specification layer. Which full-scale method normalizes the response — a board-declared constant, a per-path rail-and-floor measurement, or a corpus percentile — changes every normalized number downstream; the audit found all three in the sources, incompatible and unmarked. Whether a saturation threshold lives on a 0–10 scale or a 0–1 scale decides whether a worked example is in-region or seven times out of range; the audit found exactly that discrepancy. Whether gain means a Fisher–Rao displacement ratio or a local Jacobian projection derivative decides what a cascade experiment even measures; the audit found both, defined in different sources, with no bridge. An experiment run before these questions are pinned does not test the response model; it tests the experimenter's implicit choices, and its results cannot be compared, replicated, or even clearly stated.

The candidate therefore front-loads disambiguation. Every choice the sources left open is either fixed by a binding disposition (with the conflict preserved so the fix is visible as a fix), parameterized into the calibration profile (so that incompatible choices become distinct profiles with distinct digests rather than silent variants), or excluded by the type system. Only after that layer is stable does the falsification program define runs — and by then, each registered hypothesis can state exactly which profile, grid, controls, and fit methods it binds, which is what makes its eventual outcome interpretable and its preregistration meaningful [37].

This stance has a cost, which the monograph accepts explicitly: it defers every interesting empirical question. The reader who wants to know whether steered activations are locally

ohmic will not learn it here, and the author does not know either. What the reader will obtain is the precise, typed, falsifiable form of the question — which, for a research lineage that previously could not state the question without dimensional error, is the necessary first result.

11. Notation and conventions

The following conventions hold throughout the monograph.

Quantities. The alpha dose is a, in `ALPHA` units. The activation response is y, in `ACTIVATION` units; a predicted response is $\hat{y}$. The effective resistance is r_{eff}, in `ALPHA_PER_ACTIVATION` units, strictly positive, and always accompanied — in any admissible use — by a role, a fit method, a split binding, and an uncertainty. Residuals are $\delta_y = y_{\text{obs}} - y_{\text{pred}}$. Score-valued quantities on the 0–10 and 0–100 scales are written with their units (`Q10`, `Q100`) rather than with suggestive symbols.

Enumerations. Implementation enumerations are reproduced in `typewriter` face with their exact member spellings, including underscores, as in `CALIBRATED_HELD_OUT` or `CORPUS_PERCENTILE_PER_TES` Small-caps forms such as INSTRUMENTED are prose references to the same members; where the two could diverge, the typewriter form is authoritative.

Exactness. All boundary-sensitive arithmetic in the candidate is exact decimal arithmetic, never binary floating point; the monograph's worked numbers are to be read as exact decimals. Digests are SHA-256 over canonical bytes; equality of records means equality of digests.

Profiles. Every quantity is relative to a calibration profile, identified by its canonical digest. Where the profile is clear from context it is suppressed from the notation; nothing transfers across profiles without a stated comparison rule, and no suppressed profile subscript should be read as cross-profile generality.

Citations. Numbered citations refer to the external literature. Source documents of the audit (the Guide, the addendum, Appendix I, and the PCBA text) are referred to by those names, with locators, and are deliberately not treated as citable scientific literature: they are audited inputs, and their conflicting content is evidence about the lineage, not authority about the world.

12. Reading guide

The monograph is organized so that a reader can always locate the epistemic status of what is being read: background, specification, preserved conflict, or registered hypothesis.

Chapter 2 is a tutorial survey of the five literatures the candidate stands on: dose-response modeling, with its response regions, linear ranges, and saturation phenomena [38]; metrology, with its vocabulary of quantities, units, measurement kinds, uncertainty, and traceability [32, 33]; the statistical calibration of computer models, including parameter uncertainty, model discrepancy, and train-versus-held-out discipline [37]; verification and validation as distinct activities [23]; the limits of causal inference from observational fits [27]; activation steering and representation engineering as the empirical setting, at primer depth [39]; and provenance modeling [22]. Everything in that chapter is background: none of it is a CAS-Ohm novelty, and none of it smuggles an empirical claim about CAS-Ohm.

Chapter 3 presents the quantity and evidence ontology in full: the observed-value record field by field with the rationale for each field; all sixteen quantity kinds; all six measurement kinds with examples and the non-fungibility rules; the unit ontology with its forbidden aliases; and the canonical evidence bounds — exact-type tagging, cycle awareness, unsafe-Unicode

rejection, and the declared depth, node, container, text, output, and integer limits, with host-exception normalization.

Subsequent chapters, outside the scope of this volume's opening movement, develop the preserved source audit conflict by conflict; the calibration-profile contract and the response model with its anti-circularity rule; the two independent classification axes and the exact quantization scheme with its boundary-preservation proposition; the capability-gated structural network operations; residual censoring; the completeness-versus-eligibility split; the five registered hypotheses and the preregistered closure protocol; the memristive extension boundary; and the candidate's limitations, stated without softening.

Three reading conventions apply throughout. First, environments carry status: a *definition* or *ruleenv* states candidate specification; a *conflict* preserves a source disagreement without resolving the underlying question; a *hypothesis* is registered and open; a *boundary* bounds claims and is normative for every reading of the surrounding text. Second, enumerated implementation values are typeset in `typewriter` face and reproduce the implementation exactly; where the specification prose and the implementation enumerate slightly differently (as with the dose-response regions, where the implementation retains a `BREAKDOWN` member alongside the audit's five-region summary), both are recorded and the discrepancy is stated rather than smoothed. Third, the status line OPEN / RING 4 / NOT LICENSED / HELD is not boilerplate to be skimmed; it is the single most important sentence in the monograph, and every other sentence is subordinate to it.

With the frame fixed, the monograph now turns to background: the established literatures whose disciplines the candidate borrows, and against whose standards its falsification program will eventually be judged.

CHAPTER 2

Background: Dose-Response Modeling, Metrology, and Activation Steering

1. Purpose, scope, and status of this chapter

This chapter is a tutorial survey of the established literatures on which CAS-Ohm 1.0 stands. It covers, in order: dose-response modeling, the empirical tradition from which the candidate takes its analytical frame (Section 2); metrology, the science of measurement, from which the candidate takes its quantity-and-uncertainty discipline (Section 3); the statistical calibration of computer models, from which the candidate takes its treatment of fitted parameters, model discrepancy, and split discipline (Section 4); verification and validation of models, from which the candidate takes the distinction between tested software and validated science (Section 5); the limits of causal inference from observational fits, which bound what any fitted response relation can ever mean (Section 6); activation steering and representation engineering in neural networks, which constitute the empirical setting in which the candidate's quantities would be measured (Section 7); and provenance modeling, from which the candidate takes its lineage vocabulary (Section 8). A closing synthesis (Section 9) states what this background licenses for the rest of the monograph and, just as importantly, what it does not.

Two status rules govern every sentence that follows, and the reader is asked to hold the chapter to them.

Rule 2.1 (Background is not novelty). Nothing in this chapter is a contribution of the CAS-Ohm project. Every concept, method, distinction, and caution surveyed here belongs to its own literature and is presented as such. Where later chapters adopt one of these concepts, the adoption is an application of standard discipline, never a claim of invention.

Rule 2.2 (Background smuggles no empirical claims). Nothing in this chapter asserts, implies, or presupposes any empirical result about CAS-Ohm 1.0. Descriptions of steering experiments, dose-response regions, calibration successes, or provenance systems are descriptions of the external literature and of generic methodology. In particular, the survey of activation steering in Section 7 describes what that literature does and reports at primer depth; it does not state or suggest that any CAS-Ohm quantity has been measured, that any CAS-Ohm hypothesis has support, or that the phenomena the literature reports transfer to the candidate's setting. All CAS-Ohm claims remain OPEN / RING 4 / NOT LICENSED / HELD.

The chapter is deliberately elementary in places. The candidate's historical sources went wrong not on advanced questions but on elementary ones — what a unit is, what a fitted parameter can claim, what a residual can prove — and the corrective is a survey that states the elementary things carefully, with citations to the authoritative treatments, so that later chapters can invoke them by reference rather than re-arguing them.

2. Dose-response modeling

2.1. The setting. Dose-response analysis is the branch of applied statistics concerned with how a measured response varies as a function of a controlled stimulus, conventionally called the dose. The tradition is oldest and most developed in the biological sciences — pharmacology, toxicology, weed science, environmental risk assessment — where the dose is a chemical concentration or exposure and the response is growth, mortality, binding, or some other measured endpoint, but its formal apparatus is domain-general: a design variable swept over a grid, a response observed with error, and a parametric or semi-parametric curve fitted to summarize the relation [38].

Three features of the tradition matter for this monograph. First, dose-response analysis is unapologetically *empirical*: the fitted curve is a summary of observed behavior under a protocol, not a derivation from first principles, and its parameters — slopes, midpoints, asymptotes — are estimates with standard errors, not constants of nature. Second, the tradition is organized around *characteristic regions* of the dose axis, described next, which give qualitative structure to an otherwise arbitrary curve. Third, the tradition has accumulated hard-won cautions about extrapolation, model selection, and design — cautions that transfer directly to any new domain that borrows the frame.

2.2. Model families. The workhorse models of dose-response analysis are sigmoidal. The log-logistic family, in its four-parameter form, expresses the mean response at dose x as

$$f(x) = c + \frac{d - c}{1 + \exp\big(b\,(\log x - \log e)\big)},$$

with lower asymptote c, upper asymptote d, steepness parameter b, and midpoint (the dose of half-maximal effect) e; specializations fix $c = 0$ or $d = 1$, and relatives such as the Weibull and log-normal families trade tail behavior against the same qualitative shape [38]. What all of these share is the sigmoid's three-act structure: a low-dose plateau, a rising middle, and a high-dose plateau. Threshold variants add an explicit dose below which the mean response is exactly at baseline; hormetic variants permit non-monotone behavior at low dose. The family chosen constrains what the analyst can express: a model without an upper asymptote cannot represent saturation, and a model without a threshold parameter will interpolate smoothly through what may in truth be a hard onset.

Selection among families is itself a statistical act with consequences. Information criteria, lack-of-fit tests against richer alternatives, and comparison across nested specializations are standard practice, and the literature is insistent that the fitted family is a modeling choice to be reported and defended, not a fact discovered in the data [38]. An analyst who fits only a straight line and reports a slope has not shown the relation is linear; an analyst who fits a sigmoid and reports a midpoint has not shown saturation exists. The curve constrains the summary; the data constrain the curve only within the family offered.

2.3. Response regions. The qualitative anatomy of a sigmoidal dose-response relation is conventionally described in regions, and this anatomy — not any electrical analogy — is the ancestral home of the region vocabulary used later in this monograph.

Sub-threshold region. At sufficiently low dose, the response is statistically indistinguishable from baseline: whatever mechanism couples dose to response has not been engaged above the noise floor. Operationally, the boundary of this region is a detection question as much as a mechanism question — it depends on assay noise, replication, and the baseline's own

variability — and the literature distinguishes carefully between "no effect" and "no detected effect."

Approximately linear region. In the rising middle of a sigmoid, the curve is locally well-approximated by a straight line, and for many designs the analyst's primary quantity is exactly the local slope in this region: the marginal response per unit dose. The linear region is where proportional summaries are honest, where relative potencies of two treatments can be compared by slope ratios, and where the classical machinery of linear calibration applies. Its boundaries are model-dependent estimates, not observables, and reporting a slope without reporting the region over which linearity was assessed is a recognized malpractice.

Saturation region. At high dose the response approaches its upper asymptote: receptors are occupied, transport is maximal, the instrument clips, or the underlying process has no further room to respond. Marginal response per unit dose falls toward zero, and two consequences follow that the literature emphasizes. First, doses in the saturation region are nearly uninformative about the slope parameter — the design's information about proportionality lives in the middle. Second, summaries computed in saturation are treacherous: a ratio of response to dose declines mechanically with dose even when the underlying relation is perfectly regular, so a declining ratio is not evidence of a changing mechanism.

Beyond saturation. Some assays exhibit qualitative breakdown at extreme dose — toxicity supervening on the measured endpoint, instrument overload, loss of the preparation — where the response is no longer a measurement of the original quantity at all. The honest treatment is exclusion with a recorded reason, not inclusion as data.

Two further phenomena complete the anatomy. *Non-monotone* relations (hormesis in the biological literature) violate the sigmoid's ordering assumption outright and demand richer families. And *instability* — irreproducibility of the response at fixed dose across replicates beyond assay noise — signals that the preparation, not the curve, is the problem; no fitted parameter from an unstable preparation deserves trust.

2.4. Estimation and inference. Fitting a dose-response model is, computationally, nonlinear regression: the family's parameters are estimated by minimizing a loss — ordinarily least squares under an assumption of independent errors, or a transformation- or weight-adjusted variant when the response variance changes with its mean, as it commonly does near asymptotes [38]. Several inferential practices from this setting deserve explicit record, because each has a disciplinary echo later in the monograph.

First, *parameter uncertainty is part of the estimate.* The fitted slope, midpoint, and asymptotes come with standard errors derived from the curvature of the loss surface (or from resampling, when the asymptotic approximations are doubtful — bootstrap intervals are routine in the applied literature), and downstream quantities computed from the parameters inherit uncertainty through the delta method or through resampling. A reported midpoint without an interval is regarded as an incomplete result, not a bold one.

Second, *variance structure is a modeling decision.* Dose-response data are frequently heteroscedastic — noisier in the rising region than at the plateaus, or vice versa — and ignoring this does not merely widen intervals; it can bias which region dominates the fit. Weighted fitting, variance modeling, and transform-both-sides approaches are the standard responses, and the choice made is reported with the model [38].

Third, *replicates and lack-of-fit are the honesty check.* With genuine replicates at each dose, the pure-error variance is estimable independently of any curve, and a lack-of-fit comparison asks whether the chosen family's residual variation exceeds what replicates alone

explain. A family that fails lack-of-fit is not rescued by the elegance of its parameters; it is replaced or the failure is reported.

Fourth, *designs are evaluated by the information they carry about the target parameter.* If the analysis's purpose is a slope in the linear region, doses concentrated at the plateaus are wasted; if the purpose is an asymptote, the reverse. Optimal-design considerations — placing doses where the parameter of interest is most informed — are part of the tradition's standard toolkit, and their qualitative lesson (the grid is chosen for the question, and a grid chosen for another question may be uninformative for this one) transfers to any borrowed setting.

2.5. Derived quantities: effective doses and relative potency. The tradition's headline summaries are typically *derived* quantities: the effective dose ED_{50} (the dose producing half-maximal response, coinciding with the midpoint parameter in the standard families), its generalizations ED_p for other fractions, and the *relative potency* of two treatments, defined — under an assumption of similar curve shape — as the ratio of doses producing equal response [38]. Two cautions attach. Derived quantities inherit and often amplify parameter uncertainty: an ED_p far from the sampled region can have an interval spanning orders of magnitude, and reporting the point value without the interval misrepresents the experiment. And comparability constructs like relative potency are *conditional on shape similarity*: when two curves differ in slope or asymptote, a single potency ratio does not exist, and forcing one manufactures a number without a referent. The general lesson — a derived summary exists only under the assumptions of its derivation, and those assumptions are checkable and reportable — is applied repeatedly in later chapters, where every derived CAS-Ohm quantity carries its derivation parents and method identity precisely so that the conditioning assumptions remain visible.

2.6. Design and the flat-response ambiguity. A methodological staple of the tradition is that the dose grid is part of the inference. A grid confined to low doses cannot distinguish a sub-threshold relation from an insensitive assay; a grid confined to high doses cannot distinguish saturation from a ceiling artifact; a two-point grid cannot detect curvature at all. Good designs bracket the anticipated midpoint, replicate at each dose, and include controls at zero dose.

Of particular relevance to later chapters is what may be called the *flat-response ambiguity*: an observed response that does not vary with dose is compatible with at least four distinct states of affairs — the dose is sub-threshold; the response is saturated; the assay is insensitive or mis-targeted; or the measurement failed. Distinguishing them requires information beyond the flat curve itself: positive controls, grid extension, assay validation. The dose-response literature treats this as elementary; it is recorded here because the corresponding conflation (reading every flat response as one particular failure) is precisely the mnemonic overload that the source audit found in the candidate's lineage, and the disposition there — classify the dose-response signature and the semantic outcome independently — is an application of this standard point, not an innovation.

2.7. Calibration curves and working ranges. A sibling of the dose-response curve, from analytical measurement, is the *calibration curve*: the fitted relation between an instrument's indication and the known values of reference materials, used thereafter to convert indications into reported values. The two curves are formally similar — a controlled abscissa, a measured ordinate, a fitted relation — but their inferential directions differ: dose-response analysis *characterizes* the relation as the object of interest, while calibration-curve practice

inverts it, reading unknown inputs off measured outputs. The inversion imports every property of the fitted curve into every subsequent measurement, which is why analytical practice surrounds calibration curves with explicit range discipline.

Three range concepts from that practice complete the vocabulary this monograph borrows. The *limit of detection* is the smallest input distinguishable from blank with stated confidence — below it, indications are reported as "not detected," never as small numbers. The *limit of quantification* is the smallest input measurable with stated relative uncertainty — between detection and quantification, presence may be asserted but magnitude may not. The *working range* (or linear range, when linearity is the fitted form) is the interval over which the calibration relation has been established with acceptable lack-of-fit — and reported values outside it are extrapolations of the calibration, flagged as such or refused. Together these define a regime discipline: an instrument's conversion relation is trusted only inside a characterized envelope, with distinct sub-thresholds for "say nothing," "say present," and "say how much."

The relevance to a response-model candidate is structural. Any proposal that fits a relation between an intervention coordinate and a response coordinate, and then uses the fitted parameter to predict, is running calibration-curve logic; it thereby inherits the obligations of that logic — established range, declared limits below which no quantitative claim is made, refusal to report outside the envelope — independently of any question about mechanism. The candidate's cutoff and saturation gating and its refusal to emit diagnostics from out-of-region operating points, described in later chapters, are these obligations transcribed; the obligations themselves are as old as analytical chemistry.

2.8. What the tradition offers this monograph. From dose-response analysis, the candidate inherits: the framing of a controlled scalar dose and measured scalar response; the region anatomy (sub-threshold, approximately linear, saturating, broken-down, non-monotone, unstable) as a classification of *signatures* rather than mechanisms; the insistence that linearity is a locally assessed, design-dependent, reported property rather than an assumption; the caution that ratios computed outside the linear region mislead; and the flat-response ambiguity. What the tradition does not offer — and what none of this chapter should be read as offering — is any evidence that steered neural activations exhibit these anatomies. That is an open empirical question, registered as such.

3. Metrology: quantities, units, uncertainty, traceability

3.1. Why metrology, and which metrology. Metrology is the science of measurement and its applications. Its two foundational reference documents are the International Vocabulary of Metrology, which fixes the concepts and terms — quantity, unit, measurand, calibration, traceability — [32], and the Guide to the Expression of Uncertainty in Measurement, which fixes the treatment of measurement uncertainty [33]. The candidate borrows this conceptual discipline wholesale while claiming none of its institutional substance: CAS-Ohm has no accredited laboratories, no reference standards, no interlaboratory comparisons, and no validated instruments, and the monograph will not pretend otherwise. What it takes from metrology is the *grammar* of careful measurement talk, because the historical sources failed at exactly that grammar.

3.2. Quantity, kind of quantity, and unit. In the Vocabulary's usage, a *quantity* is a property of a phenomenon, body, or substance that has a magnitude expressible as a number and a reference [32]. Quantities partition into *kinds*: length and wavelength are of the same

kind; length and mass are not; and — the point with teeth — quantities of different kinds are not comparable, addable, or convertible merely because both are numbers. A *unit* is a conventionally chosen reference quantity of the same kind, in terms of which magnitudes are expressed; a *quantity value* is the pairing of a number with a unit.

Three consequences of this grammar recur throughout the monograph. First, a number without its unit and kind is not a quantity value at all; it is a numeral. Second, scales that happen to share a numeric range are not thereby the same kind: a score on 0–10 and a fraction on 0–1 differ in kind even where their numbers overlap, and a conversion between them is a substantive claim requiring a defined relation, not a rescaling one may perform silently. Third, ratios and products of quantities have derived kinds determined by their arguments; dividing a quantity of one kind by a quantity of another yields a third kind, which either has a place in the analyst's ontology or signals that the operation was meaningless. The Vocabulary also distinguishes *ordinal* quantities — those, like many scores, for which magnitude comparison is meaningful but differences and ratios are not — and warns that arithmetic on ordinal quantities requires explicit justification. Score-valued quantities in any framework should be presumed ordinal until shown otherwise.

3.3. Measurand, measurement, and calibration. The *measurand* is the quantity intended to be measured — and the Vocabulary is emphatic that specifying it is a substantive act: the "same" informal quantity under different conditions, definitions, or procedures is a different measurand [32]. A *measurement* is the process of experimentally obtaining quantity values attributable to the measurand, and it presupposes a measurement procedure, a measuring system, and a description sufficient for another practitioner to repeat it. *Calibration*, in the Vocabulary's sense, is the operation that establishes a relation between values indicated by a measuring system and values provided by standards, with their uncertainties, and then permits corrected measurement results; it is not a synonym for adjustment, tuning, or fitting in general.

The distinction between what a system *indicates* and what one is licensed to *report* as a measurement result is the deepest single lesson metrology offers a new field. An indication becomes a measurement result only through a chain: a defined measurand, a documented procedure, a calibration relation with uncertainty, and a stated coverage. Skipping the chain and reporting indications as results is the generic form of several failures the source audit catalogued.

3.4. Uncertainty. The Guide's framework treats every measurement result as an estimate accompanied by a *measurement uncertainty*: a non-negative parameter characterizing the dispersion of values attributable to the measurand [33]. Uncertainty evaluations are classified by method: *Type A* evaluations use statistical analysis of repeated observations (standard deviations of means, regression residuals); *Type B* evaluations use other knowledge — manufacturer specifications, prior data, physical bounds, expert judgment — encoded as probability distributions. Both types yield *standard uncertainties*; components combine through the law of propagation of uncertainty (in first order, via sensitivity coefficients and covariances) into a *combined standard uncertainty*, which may be scaled by a coverage factor into an *expanded uncertainty* defining an interval with stated coverage probability.

Two disciplines from this framework are load-bearing later. First, *uncertainty travels with the value*: a derived quantity's uncertainty is computed from its parents' uncertainties and covariances, not invented downstream, and a value whose uncertainty was never evaluated should be marked as such rather than silently treated as exact. Second, *covariance is*

not optional: when combined quantities share error sources, ignoring their covariance misstates the combined uncertainty, sometimes drastically; where covariances are unavailable, the honest course is to declare the approximation being made rather than let a zero-covariance assumption pass as fact.

3.5. Error, repeatability, and reproducibility. The Vocabulary draws a distinction, easily blurred in casual usage, between measurement *error* — the difference between a measured value and a reference value, a signed quantity that is in general unknowable when the true value is unknowable — and measurement *uncertainty* — the analyst's quantified doubt, a non-negative dispersion parameter that is always evaluable [32, 33]. The older error-analysis tradition spoke of random and systematic errors; the uncertainty framework re-expresses the same physics without requiring knowledge of true values: systematic effects, once recognized, are corrected for, and the imperfect knowledge of the correction contributes a (Type B) uncertainty component, while random effects contribute (Type A) components estimated from dispersion. The practical rule is that recognized systematic effects are never left inside a reported value as silent bias; they are either corrected with an uncertainty on the correction or declared as a limitation.

Conditions of measurement are stratified with equal care. *Repeatability* conditions hold everything fixed — same procedure, same operators, same system, same conditions, same location, short time span — and characterize the floor of dispersion attainable by the setup. *Reproducibility* conditions vary what practice actually varies — different laboratories, operators, instruments, occasions — and characterize the dispersion a claim must survive to be portable. The gap between the two is itself diagnostic: results that repeat but do not reproduce indicate uncontrolled condition-dependence, and the metrological response is to promote the offending condition into the measurand's specification (a different condition then defines a different measurand) rather than to average over it silently. The transferable form of this rule appears throughout the candidate's design: configuration details that change results — model revision, layer, direction provenance, corpus, seeds — belong inside the profile identity, so that variation across them is variation across declared measurands, never noise smuggled into one.

3.6. The calibration hierarchy. Institutional metrology arranges instrument calibrations in a *hierarchy*: national standards realize units; reference standards are calibrated against them; working standards against those; and field instruments against working standards — each step documented, each contributing uncertainty, so that the uncertainty of a field measurement provably includes every link above it [32]. Two structural features of the hierarchy are worth extracting from their institutional setting. First, uncertainty is *monotone down the chain*: no link can report less uncertainty than its parent chain supplies, and any claimed shortcut (a field result more certain than its reference) is a red flag, not an achievement. Second, the hierarchy makes *staleness* a first-class concept: calibrations expire, drift is monitored between calibrations, and a result traceable only through an expired or drifted link loses its status until recalibration. Both features transfer to any lineage-disciplined computational system as design requirements — propagated uncertainty that can only grow through derivation, and expiry-and-drift metadata on every calibration-like binding — and both appear, in exactly that role, in the candidate's calibration-profile contract discussed in later chapters.

3.7. Traceability. *Metrological traceability* is the property of a measurement result whereby it can be related to a stated reference through a documented, unbroken chain of

calibrations, each contributing to the stated uncertainty [32]. The definition repays close reading: traceability is a property of *results*, not of instruments or laboratories; the chain must be *documented* and *unbroken*; and every link must carry uncertainty. In institutional metrology the chain terminates in realizations of the SI units; the concept, however, is general — a result is traceable to whatever reference the chain actually reaches, and no further.

The transferable content for a computational setting is the chain discipline itself: every reported value should be connected, by explicit and inspectable links, to the raw artifacts and procedures that produced it, with each link's method and uncertainty recorded. What does not transfer — and must not be claimed — is the institutional authority of the SI chain. A digest-linked lineage is documentation, not accreditation; a complete chain to a raw artifact establishes where a number came from, not that the number measures what its label says. That further step is validation, treated in Section 5.

3.8. Missing values, censoring, and the discipline of absence. A final piece of measurement discipline concerns values that are *not there*. Statistical practice distinguishes several kinds of absence, and the distinctions carry inferential weight. A value may be *missing* — the measurement was not made, or was made and lost — in which case the salient question is whether the missingness is related to the value that would have been observed; analyses that ignore informative missingness are biased in ways no downstream care repairs. A value may be *censored* — the measurement was made but yields only partial information, as when a response falls below a detection limit (left-censoring) or an instrument clips at full scale (right-censoring); censored observations are not zeros, are not the limit value, and are not discardable: each of those common substitutions injects a specific, well-documented bias, and the correct treatments (likelihood contributions from the censoring interval, or explicit interval-valued records) are standard. And a value may be *undefined* — the quantity's defining ratio or operation has no value at the observed operating point, as when a ratio's denominator is zero; reporting a number anyway manufactures data.

The transferable discipline is representational: an evidence system must be able to say, of any field, "absent, for this stated reason" as a first-class recorded fact, distinct from zero, distinct from any sentinel numeral, and preserved through derivation — so that a downstream consumer can distinguish a measured zero from a censored observation from an undefined quantity from a value never sought. This background point, elementary in biostatistics and survival analysis, is the rationale for the missing-or-censored-reason field of the candidate's observed-value record and for its censoring of residual diagnostics at zero-compatible response, both described in later chapters.

4. Calibration of computer models

4.1. The setting and the vocabulary collision. A second, distinct sense of "calibration" governs the statistical literature on computer models. Here the object is a *simulator*: a deterministic or stochastic code mapping inputs and unknown parameters to outputs, standing in for a physical or empirical system. *Calibration* in this sense is the use of observed field data to learn about the unknown parameters — and, crucially, about the simulator's inadequacy [37]. The collision with the metrological sense of calibration (Section 3.3) is unfortunate but entrenched; this monograph disambiguates by context and, where necessary, by the phrases *instrument calibration* and *model calibration*.

4.2. Parameter uncertainty and model discrepancy. The canonical formulation writes the observed data as simulator output at the true-but-unknown parameter value, plus

a *model discrepancy* term, plus observation error:

$$z_i = \rho\,\eta(x_i, \theta) + \delta(x_i) + \varepsilon_i,$$

where η is the simulator, θ the calibration parameters, $\delta(\cdot)$ an unknown function representing the systematic inadequacy of the simulator even at the best parameter value, and ε_i observation error [37]. Two commitments distinguish this framework from naive fitting. First, θ is treated as uncertain *after* calibration: the analysis yields a posterior distribution, not a point estimate anointed as truth, and predictions propagate that posterior. Second — the framework's signature — the discrepancy term is carried explicitly: the model is assumed inadequate, the inadequacy is estimated alongside the parameters, and predictive uncertainty includes it. Omitting δ does not make a model adequate; it makes the analysis overconfident and biases the parameter estimates, which absorb the inadequacy in uninterpretable ways.

The framework also names an identifiability problem that any borrower must respect: with a flexible discrepancy, the data cannot always separate "different parameter value" from "different inadequacy," so the physical interpretability of calibrated parameters is a claim requiring argument, not a free consequence of good fit. A calibrated parameter is, in the first instance, the parameter value that makes the simulator-plus-discrepancy predict well — which is a weaker and more honest thing than the true value of a physical constant.

4.3. Train/held-out discipline and the circularity failure. Whatever the inferential machinery, the elementary discipline of predictive assessment is the separation of the data that estimate a model from the data that evaluate it. Assessing a fit on its own training data measures the model's capacity to absorb those data — which increases with flexibility — rather than its capacity to predict; the degenerate limit is the pure circularity in which a parameter is defined from a sample and then "verified" by reproducing that same sample exactly. Held-out evaluation, cross-validation, and prospective validation on later data are the standard remedies, each with known caveats (leakage between splits, repeated peeking, distribution shift between calibration and use).

For a proportionality model the degenerate limit is worth writing out because it is so easily disguised. Given one paired observation (a, y) with $y \neq 0$, define $r := a/y$; then predict $\hat{y} := a/r$ and observe $\hat{y} = y$ exactly. The agreement is an algebraic identity, carries no information about any other dose, and survives every dimensional check. Its only reliable signature is procedural: the same sample appears on both sides. Hence the discipline — registered splits, fit on one, evaluate on the other, report uncertainty from the fit — is not methodological ornament but the minimal condition under which a fitted slope means anything at all [37]. Later chapters encode this discipline structurally; the point here is that the discipline itself is standard, old, and non-negotiable.

4.4. Emulation and computational practice. When the simulator is expensive, the calibration framework interposes an *emulator*: a statistical surrogate — in the canonical treatment, a Gaussian process — trained on a designed set of simulator runs and used in place of the simulator inside the calibration computation, with the emulator's own approximation uncertainty propagated alongside parameter uncertainty and discrepancy [37]. The details of emulation matter less for this monograph than the epistemic pattern it exemplifies: every layer interposed between data and conclusion — surrogate, interpolation, preprocessing — is itself a model with its own inadequacy, and a disciplined analysis carries that inadequacy explicitly rather than treating the layer as transparent. The pattern generalizes to computational measurement pipelines of any kind: a normalization step, a projection, a quantization

each transform the quantity, and each transformation either has a declared method with an assessable error or constitutes an undeclared modeling assumption.

A second point of computational practice deserves record: *the design of calibration runs is part of the inference.* Which parameter settings are simulated, over what ranges, with what space-filling properties, determines what the calibration can learn — exactly parallel to the dose-grid point of Section 2.6. Sparse or lopsided designs yield posteriors that reflect the design as much as the system, and the standard practice is to report the design alongside the result.

4.5. Validation of calibrated models and extrapolation. Calibration and validation interact in a way the literature is careful about: data used to calibrate cannot also validate, since the calibrated model has been conformed to them. Validation of a calibrated model therefore requires either held-out field data, prospective data gathered after freezing the model, or structured cross-validation with the calibration repeated inside each fold [37]. Moreover, a calibrated-and-validated model is validated *over the domain of the validation data*; the discrepancy term, being an empirical estimate of inadequacy where data exist, provides no warrant where they do not, and extrapolation beyond the calibration-and-validation envelope reintroduces unquantified inadequacy. The framework thus arrives, by its own route, at the same regime-boundedness that causal inference imposes (Section 6): a fitted, uncertainty-carrying, discrepancy-aware model earns claims inside its envelope and speculation outside it. Later chapters implement this as an expiry-and-envelope discipline on calibration profiles; the principle recorded here is standard.

5. Verification and validation

5.1. Two questions that must not be merged. The simulation literature distinguishes two questions about any model-bearing artifact. *Verification* asks: was the artifact built correctly — does the implementation faithfully realize its specification, free of the defects that testing can expose? *Validation* asks: was the correct artifact built — does the model represent the target system accurately enough for its intended use [23]? The questions have different evidence types, different audiences, and different failure modes, and a mature practice keeps them separate: a model can be perfectly verified and wholly invalid (a flawless implementation of a wrong idea), or roughly right in conception yet unusable because its implementation is untrustworthy.

Verification evidence is the familiar apparatus of software quality: unit and property tests, boundary-case fixtures, golden rejection tests (inputs that *must* fail, verifying that guards guard), replay of pinned computations, static analysis. Validation evidence is comparison against the world: agreement with independent data not used in construction, behavior under conditions spanning the intended operating envelope, expert scrutiny of assumptions, and — in strong practices — validation performed or reviewed by parties independent of the developers [23]. The literature further ties validity to *intended use*: a model is not valid simpliciter but valid for a stated purpose over a stated domain, and reuse outside that domain restarts the validation question.

5.2. A catalogue of validation techniques. The simulation literature has accumulated a standard catalogue of validation techniques, and rehearsing a representative subset clarifies how varied — and how demanding — validation evidence is compared with verification evidence [23]. *Comparison to other models*: outputs are checked against independently

developed models of the same system, valued precisely because shared errors between independent efforts are less likely. *Degenerate tests*: parameters are pushed to values with known degenerate behavior (zero arrival rate, infinite capacity) and the model is required to reproduce the known limit. *Extreme-condition tests*: the model must remain plausible at the boundaries of its input space, where errors concentrate. *Event and face validity*: occurrence patterns and input-output behavior are reviewed by people knowledgeable about the system for gross implausibility — weak evidence alone, but cheap and early. *Historical data validation*: part of the historical record drives the model, the remainder tests it — the split discipline of Section 4.3 in its simulation form. *Parameter variability and sensitivity analysis*: inputs are varied and the direction and magnitude of output changes checked against system knowledge; parameters found sensitive demand accurate estimation. *Predictive validation*: the model forecasts, and the system is then observed. *Internal validity*: replicated stochastic runs are checked for a stability of results without which no single run means anything. *Traces*: individual entities are followed through the model logic to confirm the implemented behavior is the intended one — a technique straddling the verification-validation border.

Two features of the catalogue merit emphasis. First, no single technique validates; practices accumulate evidence across techniques, matched to the model's intended use and the cost of error. Second, several techniques presuppose access to the target system — historical data, predictive trials, expert knowledge of system behavior — and are simply unavailable when no comparable system data exist; a practice honest about this reports which techniques were unavailable rather than substituting the available ones and claiming completeness.

5.3. Documentation, independence, and accreditation. The literature further separates the *performance* of verification and validation from its *evaluation*. Documentation — of assumptions, techniques applied, results, and known limitations — is treated as constitutive: undocumented validation is unevaluable and therefore, for consumers of the model, indistinguishable from no validation. Independence enters in degrees: developer self-assessment is the weakest arrangement; independent verification and validation by parties without a stake in the model's acceptance is materially stronger; and formal *accreditation* — an authority's determination that a model is acceptable for a specific purpose — is an institutional act distinct from both, performed against declared criteria [23]. The mapping onto this monograph's status vocabulary is direct and is stated once here: everything the candidate's own project produces, including this monograph, is developer self-assessment; the program's internal cross-review would constitute a first independence step; external independent review a second; and nothing corresponding to accreditation exists or is claimed. The status line OPEN / RING 4 / NOT LICENSED / HELD is, among other things, a compact encoding of where in this hierarchy the candidate currently sits: at the bottom.

5.4. Operational consequence for this monograph. The consequence for reading everything that follows is mechanical. Test-suite results — however extensive — are verification: they establish that specified behavior, including specified rejection behavior, occurs at a pinned revision on constructed inputs. They cannot establish that the specification captures anything true about steered neural activations; that is validation, requires the falsification program's controlled runs against held-out data, and has not occurred. When later chapters cite passing tests over synthetic known-fault fixtures, the citation licenses exactly the sentence "the software detects the synthetic fault it was built to detect" and no sentence about live models. This is the standard reading [23], stated once here so that later citations can be brief.

6. Limits of causal inference from observational fits

6.1. The ladder. Modern causal inference, in the structural tradition, organizes claims into three strictly ordered levels: *association* (what the joint distribution of observed variables looks like), *intervention* (what would happen to one variable if another were set by external action), and *counterfactuals* (what would have happened in a particular case had circumstances differed) [27]. The central formal lesson is that these levels do not reduce to one another: no amount of associational data, by itself, answers interventional questions, because distinct causal structures can generate identical joint distributions. Interventional content enters only through interventional assumptions (encoded, e.g., as a structural causal model or graph) or interventional data (actual experiments), and counterfactual content requires still more.

The apparatus that makes this precise includes the do-operator, distinguishing $P(y \mid \mathrm{do}(a))$ — the distribution of y under external setting of a — from the observational $P(y \mid a)$; graphical criteria for when interventional quantities are *identifiable* from observational ones given an assumed graph; and the systematic study of *confounding*, the presence of common causes that induce association without causation [27].

6.2. What a fitted slope is and is not. Where the dose is genuinely randomized and externally set by the experimenter, a dose-response fit does estimate an interventional quantity — for the manipulated variable, in the sampled population, over the sampled range. This is the classical warrant of the randomized experiment, and it is why the falsification program's registered designs randomize the alpha dose and include controls. But the warrant is exactly as wide as the randomization and no wider, and three restrictions matter even in the best case.

First, *regime-boundedness*: an interventional relation estimated over a grid supports claims about doses within (approximately) that grid, under that protocol, on that population; extrapolation beyond the grid or transport to a different population are additional assumptions, not consequences [27]. Second, *mechanism-silence*: even a well-estimated interventional slope says nothing about *how* the effect is mediated; competing mechanistic stories — direct paths, mediated paths, multiple parallel paths — can share the same low-dimensional interventional summary, and distinguishing them requires interventions on the mediators themselves. Third, and most relevant to the candidate's lineage, *residual non-identification*: the deviation of an observation from a fitted prediction is a composite of model misspecification, measurement error, unmodeled heterogeneity, and — possibly — unmodeled causes. Attributing a residual's sign to a specific hidden cause is a causal claim at the interventional or counterfactual level, and it is not identified by the fit; making it honestly requires an explicit causal model in which the hidden cause appears, plus interventions (e.g., deliberately implanted faults with controls) that could distinguish it from the alternatives [27]. The audit's residual conflict, and the candidate's censoring discipline, are downstream of exactly this point.

6.3. Adjustment and its perils. Between the poles of pure observation and randomized intervention lies the everyday practice of *adjustment*: regressing an outcome on a treatment while controlling for covariates, in the hope that conditioning removes confounding. The structural literature gives this practice both its warrant and its boundary [27]. The warrant is the identification result: if a covariate set satisfies the appropriate graphical criterion relative to an assumed causal graph — blocking the spurious (back-door) paths between

treatment and outcome without opening others — then the adjusted association equals the interventional quantity, and observational data suffice. The boundary is everything the warrant presupposes: the graph is an assumption, not an observation; omitting a genuine confounder leaves bias in place; and, less intuitively, conditioning can *create* bias — adjusting for a variable that is a common effect of treatment and outcome (a collider), or a consequence of treatment on the causal path, induces spurious association or blocks part of the effect being estimated. "Control for everything measurable" is therefore not a conservative strategy but an incoherent one; which covariates to adjust for is determined by causal structure, and without a defended structure, adjusted estimates have no defended interpretation.

The lesson is recorded here because steering-style analyses invite exactly this practice: comparing fitted parameters across layers, directions, prompts, or checkpoints while "controlling" for whatever covariates are convenient. The background licenses such comparisons as descriptive summaries only; elevating any of them to a causal contrast would require the graph-level argument this section describes, which no part of the candidate supplies or claims.

6.4. Observational overlays. Finally, not every quantity in a steering-style experiment is randomized even when the dose is: the choice of layer, direction, corpus, and test point is typically made adaptively, on the basis of prior observation, and comparisons *across* those choices are observational comparisons subject to selection and confounding in full force. A fitted parameter that predicts well within its profile carries no license to explain *why* one profile differs from another. The general rule — interventional claims extend exactly as far as the interventions — covers both this and the preceding restrictions, and it is the rule later chapters encode.

7. Activation steering and representation engineering

7.1. Status of this section. This section surveys, at primer depth, the empirical literature within which a CAS-Ohm experiment would be situated: activation steering and, more broadly, representation engineering in neural networks. The status rules of Section 1 apply with maximum force here. What follows describes what published work in this area does and the kinds of phenomena it reports [39]; it does not assert that those phenomena are settled, that they transfer to any particular model or configuration, or — above all — that any CAS-Ohm measurement has been performed or any CAS-Ohm hypothesis supported. The candidate treats every quantitative question about steering in its own setting as open.

7.2. Activations, directions, and readouts. A transformer-style network computes, at each layer and token position, a high-dimensional activation vector; the sequence of these vectors along the depth of the network (often described via the residual stream, the running sum to which each block adds its contribution) constitutes the network's internal state for a given input. Interpretability research has long explored the hypothesis that behaviorally meaningful features are represented, at least in part, as *directions* in these activation spaces: that movement of the activation along a particular vector corresponds to a change in some property of the computation or the output.

Two operational primitives follow from this picture and recur throughout the literature. The first is the *directional readout*: given a unit-norm direction v at a declared layer and position, the scalar projection $\langle h, v\rangle$ of the activation h onto v is recorded as a measurement of "how much" of the putative feature is present. The second is the *activation intervention*: the activation is modified in place during the forward pass — most simply by adding a scaled direction, $h \mapsto h + \alpha v$ — and the effect on downstream computation or generated text is

observed. The scalar α is the steering coefficient: its sign selects the direction's orientation, and its magnitude the strength of the intervention.

7.3. Steering vectors. The activation-addition literature constructs steering vectors in several ways, of which three families suffice for a primer. *Difference-based* constructions take the difference of activations induced by contrasting inputs — for instance, a pair of prompts embodying opposite values of the property of interest — optionally averaged over many pairs, on the reasoning that the difference cancels shared content and retains the contrast [39]. *Probe-based* constructions train a linear classifier (a probe) to predict the property from activations and take the probe's weight vector as the direction. *Optimization-based* constructions search directly for a vector whose addition produces a target behavior. Each family carries characteristic caveats discussed in the literature: difference vectors inherit whatever else differs between the contrasting inputs; probe directions reflect what is *decodable*, which need not be what is *used* by the network; optimized vectors can exploit idiosyncrasies of the objective. Across families, reported practice varies in normalization (unit norm or not), in the layer and token positions targeted, and in whether the same vector is applied at one position or broadcast across many — all of which are consequential experimental parameters rather than nuisance details.

7.4. Interventions at layers, and coefficient sweeps. The intervention's locus — which layer, which positions, which hook point within the block — is empirically consequential in the reported literature: the same nominal direction applied at different depths can produce effects of different sign, size, or character, and part of experimental practice is sweeping or selecting the layer [39]. Equally standard is the *coefficient sweep*: applying the intervention over a grid of α values and recording, at each, either a downstream behavioral measure or an internal readout. Published sweeps report a range of qualitative shapes — regions where effects grow with coefficient, regions where output degrades into disfluency at large magnitude, asymmetries between positive and negative coefficients — and the literature is candid that evaluation is difficult: behavioral effects are measured by classifiers, human raters, or model-based judges, each with their own error, and internal readouts measure the projection, not the behavior. The primer-level summary is that the coefficient sweep is exactly a dose-design (Section 2.6) applied to activation space, with all of that tradition's questions — where is the informative range, what does flatness mean, when does the preparation break — reappearing in new clothes. Whether any given configuration exhibits a clean region anatomy is, in each case, an empirical finding of that experiment, not a property of the method.

7.5. Controls in the steering setting. Because a steering intervention changes the activation by construction, the inferentially hard question is never whether *something* changed but whether the change is attributable to the direction's semantic content rather than to generic perturbation. The literature's response is a family of controls, which this survey records as standard practice: *zero-intervention* runs establishing the baseline; *random-direction* controls of matched norm, testing whether an arbitrary vector of the same size produces comparable effects; *wrong-layer* and *wrong-position* applications, testing locality; *shuffled or mismatched* pairings between inputs and vectors; and norm-matching generally, since effect size scales with perturbation magnitude and unmatched comparisons confound content with size. The methodological point — a difference measured against no control, or against an unmatched control, does not support a content-specific claim — is standard experimental logic; the steering literature's contribution is the specific inventory of controls appropriate to activation space [39].

7.6. Evaluation difficulties and reproducibility cautions. The steering literature is, at its best, candid about the difficulty of evaluating its own interventions, and a primer should transmit the cautions along with the techniques [39]. The central difficulty is that the quantity of ultimate interest — a change in model *behavior* along a semantic dimension — is not directly readable from any scalar. Behavioral evaluation routes through instruments that are themselves models or judgments: trained classifiers over outputs (with their own error rates and blind spots), human raters (with inter-rater disagreement and prompt-order effects), or judge models (whose failure modes may correlate with the very interventions under study). Internal evaluation routes through readouts like directional projections, which are cheap and precise but measure the representation, not the behavior; the coupling between a projection shift and a behavioral shift is itself an empirical hypothesis, not an identity. A result established under one evaluation instrument does not automatically hold under another, and the literature's stronger studies report multiple instruments and their disagreements.

Reproducibility cautions compound the evaluation difficulty. Reported steering effects are, in the published record, sensitive to experimental parameters that casual summaries omit: the model checkpoint (effects need not survive fine-tuning or even re-runs of training), the tokenizer's segmentation of the probe prompts, the layer and position conventions (off-by-one conventions differ across codebases), the norm convention for the direction, and the sampling temperature of generation-based evaluations. Sign conventions deserve particular mention: a direction and its negation are equally valid outputs of a difference or probe construction, and cross-study comparison of "positive" steering presupposes a shared orientation convention that must be checked, not assumed. None of these cautions is a criticism of the literature; they are the ordinary condition of a young experimental field, recorded here because they translate directly into requirements on any measurement ontology that would operate in this setting — checkpoint digests, tokenizer digests, explicit hook and position identities, declared norms and signs — requirements whose implementation in the candidate is described in Chapter 3.

7.7. Representation engineering more broadly. Activation addition is one member of a broader family sometimes collected under the name representation engineering: techniques that read or write network-internal representations to characterize or influence behavior. On the reading side, the family includes linear probing at scale, concept-activation analyses, and unsupervised decompositions of activation spaces; on the writing side, ablations that remove a direction (projecting activations onto its orthogonal complement), targeted patches that transplant activations between runs, and learned interventions trained against behavioral objectives. The family matters to this survey for one structural reason: it supplies *alternative operationalizations* of what is informally the same manipulandum. A "feature" can be operationalized as a probe direction, a difference direction, a decomposition component, or an ablation target, and these operationalizations need not agree — a probe may decode what an ablation shows to be causally inert, and a steering vector may act through content no probe was trained to see. The metrological translation (Section 3.3) is exact: each operationalization defines a different measurand, and results attach to the measurand actually used, not to the informal feature name. The candidate's insistence that direction *provenance* — probe-derived, difference-derived, or otherwise — be part of every test-point identity is the mechanical form of this point.

7.8. What this setting supplies and what it leaves open. For this monograph, the steering literature supplies the empirical coordinates a response-model candidate needs: a controllable scalar dose (the steering coefficient), a measurable scalar response (a directional projection at a declared test point), an inventory of consequential experimental parameters

(model identity, layer, positions, hook, direction provenance, norm, sign, corpus), and a control repertoire. It leaves open — explicitly, as reported open questions in that literature and untouched ones in this project — whether projection responses to coefficient sweeps are lawful in any reusable sense, whether fitted summaries transfer across prompts, layers, or checkpoints, and whether any composition rules hold. The candidate's registered hypotheses are bets on some of these questions; this chapter, being background, places no bets.

8. Provenance

8.1. The PROV model. The W3C provenance recommendations define a domain-neutral model for recording how things came to be. Its core ontology, PROV-O, is built from three node classes and a small set of relations [22]. *Entities* are things — documents, datasets, records, values. *Activities* are occurrents that use and generate entities over time. *Agents* are things bearing responsibility — persons, organizations, software. The load-bearing relations include: an entity *was generated by* an activity; an activity *used* entities; an entity *was derived from* another entity; an entity or activity *was attributed to* or *associated with* an agent; and an agent *acted on behalf of* another. From these primitives, arbitrarily deep lineage graphs can be assembled: every value can point to the activities that produced it, the entities those consumed, and the agents responsible, recursively to the raw sources.

Provenance modeling of this kind is what makes lineage *inspectable*: given a questioned result, one can walk the graph to its inputs, identify every transformation, and ask of each link whether it was appropriate. It underwrites reproducibility (the graph documents what to redo), attribution (the graph names who and what is responsible), and impact analysis (the graph shows what downstream results a defective input contaminates).

8.2. Qualified relations, bundles, and plans. Beyond its core triangle, the PROV model supplies refinements that recur in serious lineage systems [22]. *Qualified relations* expand a bare edge into a described one: instead of merely asserting that an entity was derived from another, a qualified derivation can name the activity that effected it, the specific usage and generation events involved, and the time; instead of merely associating an activity with an agent, a qualified association can name the *plan* — the procedure or protocol — the agent was following. *Roles* type the manner of participation: the same activity may use one entity as input data and another as configuration, and the role distinction preserves which was which. *Bundles* make provenance itself first-class: a bundle is a named set of provenance assertions whose own provenance — who asserted this history, when, following what — can then be described, enabling meta-lineage and the auditing of audits. Collections and their memberships allow lineage over aggregates without losing the identities of members.

These refinements matter here because the naive reading of provenance — a bare ancestry graph — is too weak for measurement discipline. A measurement's lineage must record not just *that* a value descends from raw artifacts but *by what method*, *under which configuration*, *in what role* each parent participated, and *in which order* — distinctions that map onto qualified derivations, plans, and roles. The candidate's typed parent references (role-tagged, position-ordered, digest-identified) and method identities are, in PROV terms, qualified derivations under declared plans; the correspondence is noted in Chapter 3, and recorded here as an application of the standard model rather than a new idea.

8.3. Structure is not truth. The recommendation itself is careful about what provenance establishes, and the caution deserves independent statement because it bounds a hazard this monograph has already flagged (Chapter 1, authority boundary). A provenance record

is an *assertion* about history, made by some agent; its existence establishes that the assertion was made, not that it is true, and a well-formed graph can describe a fabricated history as fluently as a real one [22]. Trust in provenance therefore requires machinery beyond structure: integrity protection (so records cannot be silently altered), authentication (so asserting agents are who they claim), and review (so the asserted history is checked against independent evidence). Cryptographic digests and signatures supply the first and part of the second; nothing in the provenance layer supplies the third. The transferable rule — *provenance structure is documentation, never authority* — is the background form of the candidate's split between structural completeness and authenticated eligibility, and it is why that split cannot be closed by accumulating more structure.

9. Synthesis: what the background licenses

Assembled, the seven literatures surveyed here license a specific, limited methodological package, which the remainder of the monograph draws on by reference.

From dose-response analysis: the dose/response framing, the region anatomy as signature classification, locality of linear summaries, and the flat-response ambiguity [38]. From metrology: the quantity/kind/unit grammar, the measurand's procedure-dependence, uncertainty as an obligatory companion of every value with propagation and declared covariance treatment, and chain-style traceability as documentation [32, 33]. From model calibration: parameter uncertainty after fitting, explicit model discrepancy, the identifiability caution on interpreting calibrated parameters, and the train/held-out discipline whose violation is circularity [37]. From verification and validation: the strict separation of built-correctly from built-the-correct-thing, and the confinement of test-suite evidence to the former [23]. From causal inference: the association/intervention/counterfactual hierarchy, regime-bounded warrants for randomized doses, mechanism-silence of interventional summaries, and the non-identification of causes from residuals [27]. From the steering literature: the empirical coordinates — coefficients, directions, projections, layers — their consequential experimental parameters, and the control repertoire [39]. From provenance: the entity/activity/agent lineage model and the rule that structure is not truth [22].

Table 1 summarizes the mapping from each background discipline to the concepts this monograph draws from it and the locus of their later application. The table is a navigation aid, not an argument; each row's content is exactly what the corresponding section stated, no more.

What the background does not license is equally definite. It does not license any claim that steered activations obey a proportional law — the dose-response frame classifies signatures; it does not guarantee their existence. It does not license treating scores as measurements, indications as results, or lineage as authority. It does not license causal readings of fitted slopes beyond their randomized, regime-bounded warrant, nor any causal reading of residuals. It does not license transporting steering phenomena reported elsewhere into claims about the candidate's own setting. Every empirical sentence about CAS-Ohm 1.0 therefore remains exactly where Chapter 1 left it: registered, controlled for, and open. The next chapter turns from the literatures to the candidate's first contribution — the typed ontology that makes this discipline mechanical.

Discipline	Concepts drawn	Later locus
Dose-response [38]	regions; local linearity; flat ambiguity	region axis; active-region gates
Metrology [32, 33]	kind/unit grammar; uncertainty; traceability; censoring discipline	quantity kinds; unit ontology; lineage; censored fields
Model calibration [37]	parameter uncertainty; discrepancy; split discipline	profile contract; fit roles; anti-circularity rule
V&V [23]	verification/validation split; independence hierarchy	bounded software evidence; status vocabulary
Causal inference [27]	intervention hierarchy; adjustment limits; residual non-identification	residual censoring; claim boundaries
Steering [39]	doses, directions, projections; controls; parameter sensitivity	test-point identity; controls in registered hypotheses
Provenance [22]	lineage model; qualified relations; structure is not truth	parent refs; method identities; eligibility split

TABLE 1. Background disciplines, the concepts this monograph draws from each, and where the candidate applies them. Every row is background applied, not novelty claimed.

CHAPTER 3

The Quantity and Evidence Ontology

1. The no-bare-numbers principle

Every failure catalogued in the source audit shares a common enabling condition: somewhere, a number traveled without its identity. A threshold of 8.5 moved between a 0–10 scale and a 0–1 scale because nothing traveled with it to say which scale it lived on. A score-valued constraint divided an alpha because nothing marked the quotient as dimensionally undefined. A residual's sign became a bypass accusation because nothing recorded that the prediction it deviated from was uncertain, same-sample, or saturated. The ontology developed in this chapter is the candidate's structural answer: it makes the identity of a value — its kind, unit, provenance, uncertainty, and status — part of the value itself, inseparable from it by construction.

Rule 3.1 (No bare numbers). No bare binary float is admitted anywhere in the CAS-Ohm evidence layer. Every numeric record is a typed observed value carrying, at minimum: quantity kind; exact decimal value; unit identifier; measurement kind; method identifier; calibration-profile digest; estimate-or-observation status; uncertainty and confidence interval; sample size; raw-artifact and control references; derivation parents; and, where applicable, a missing-or-censored reason.

The rule has two halves, and both do work. The first half — no *bare* numbers — means that a numeral without its record is not evidence: it cannot be stored, compared, aggregated, or cited within the candidate. The second half — no bare *binary floats* — addresses precision: values are exact decimals, not IEEE floating-point approximations, because the candidate's quantization and boundary rules (developed in a later chapter) turn on exact comparison against decimal cut points, and a binary float that is "close to" 2.5 is, for those purposes, simply not 2.5. Exactness at the representation layer is what makes boundary behavior provable rather than platform-dependent.

This chapter proceeds as follows. Section 2 presents the observed-value record field by field, with the rationale for each field — what failure it prevents or what discipline it encodes. Section 3 describes how the implementation realizes the record as a validated, frozen, canonically hashable structure. Section 5 defines all sixteen quantity kinds. Section 6 defines all six measurement kinds, with examples, and states the non-fungibility rules. Section 7 presents the unit ontology, including the forbidden electrical aliases and the disposition of the generic vocabulary member `Unit.VOLT`. Section 8 tabulates the quantity-unit admissibility relation and the range and role constraints. Section 9 specifies the canonical evidence bounds. Section 11 collects the chapter's invariants.

Throughout, the epistemic status of Chapter 1 governs: this chapter describes a specification and its implementation. Bounded software evidence exists that the validators behave as specified on constructed inputs; no claim is made that any instrument populating these records is validated, and all CAS-Ohm claims remain OPEN / RING 4 / NOT LICENSED / HELD.

2. The observed-value record, field by field

Definition 3.2 (Observed value). An *observed value* is the atomic evidence record of the candidate. Its fields are: (1) quantity kind; (2) exact decimal value; (3) unit identifier; (4) measurement kind; (5) method identifier; (6) calibration-profile SHA-256 digest; (7) estimate-or-observation status; (8) uncertainty; (9) confidence interval; (10) sample size; (11) raw-artifact references; (12) control references; (13) derivation parents; and (14) missing-or-censored reason. Fields 1–5 are unconditionally required; the remainder are conditionally required by measurement kind, quantity kind, and role, under rules stated in this chapter, and are never silently defaulted.

The following subsections take the fields in order. For each, the format is the same: what the field is, and why the record cannot do without it.

2.1. Quantity kind. The quantity kind names *what kind of thing* the value is a magnitude of, drawn from the closed sixteen-member enumeration of Section 5. It is the record's first field because it governs everything downstream: which units are admissible, which range constraints apply, which roles are required, and which operations the value may participate in.

Rationale. The metrological grammar of Chapter 2 holds that quantities of different kinds are not comparable or combinable merely because both are numbers [**32**]. The audit's mixed-scale conflict is the lineage's own demonstration: a saturation threshold and a region fraction, numerically comparable, were compared — and the comparison was meaningless. A closed enumeration (rather than a free-text label) makes kind-checking mechanical: an operation either has a rule for the kinds it receives or fails closed. The enumeration is deliberately fine-grained — sixteen kinds where a physicist might expect four — because the candidate's hazard is precisely the collapse of distinct score-like and measurement-like quantities into one another; distinctions that matter are promoted to kinds so that the type system, not authorial vigilance, enforces them.

2.2. Exact decimal value. The value is an exact decimal number — implemented as an arbitrary-precision decimal, never a binary float.

Rationale. Three reasons compound. First, boundary correctness: quantization bands and validation ranges are defined by decimal cut points, and exact decimal comparison is total and unambiguous where binary rounding is not. Second, digest stability: canonical bytes are computed over the exact value, so two records agree in digest exactly when they agree in value; a float's platform- and path-dependent representation would make digest equality an accident. Third, honesty about precision: a decimal carries exactly the digits that were asserted, no more; it cannot silently manufacture fifteen significant figures from an instrument that reported three.

2.3. Unit identifier. The unit identifier names the unit, drawn from the closed unit ontology of Section 7, and must be admissible for the quantity kind under the relation of Section 8.

Rationale. A number and a kind still underdetermine a magnitude: a constraint score of 65 on the 0–100 scale and one of 6.5 on the 0–10 scale are the same assertion in different units, and only the unit field distinguishes them. Making the unit a validated field — with a closed vocabulary and a per-kind admissibility set — converts the audit's silent rescaling hazard into a hard error: a `Q10` value cannot be stored in a `Q100` slot, and no conversion

exists unless a conversion method is invoked explicitly, producing a new record with its own lineage.

2.4. Measurement kind. The measurement kind names *how* the value came to be known, drawn from the six-member enumeration of Section 6: declared by design, produced by an instrument chain, inferred from behavior, computed from other records, or scored by a human or a language model.

Rationale. Provenance of method is orthogonal to kind and unit: the same quantity kind in the same unit may be a design declaration or an instrumented observation, and the difference is the whole difference between a specification and a fact. The audit's score-as-telemetry hazard — design scores and model-derived values drifting into presentation as measurements — is blocked at exactly this field, together with the non-fungibility rules that prevent the field from being upgraded by association.

2.5. Method identifier. The method identifier names the specific procedure — fit method, projection method, scoring rubric, normalization rule — by which the value was produced, as a stable identifier resolvable within the project's method registry.

Rationale. The Vocabulary's lesson that a measurand is procedure-dependent (Chapter 2) means two values produced by different procedures are answers to different questions even when kind and unit agree. The audit's full-scale conflict is the canonical instance: board-declared, rail-and-floor, and corpus-percentile full-scale procedures yield incomparable normalizations, and the candidate's disposition — separately named methods, never interchangeable — is enforceable only because every record names its method. The method identifier is also what makes replication possible in principle: a record whose method cannot be named cannot be redone.

2.6. Calibration-profile digest. The calibration-profile field carries the SHA-256 digest of the calibration profile under which the value was produced — the frozen bundle of model identity, test point, direction, corpus, grid, methods, tolerances, and review state described in a later chapter.

Rationale. Every CAS-Ohm quantity is profile-relative; nothing transfers across profiles without a stated comparison rule. Binding the profile by digest (rather than by name) means the binding is to the profile's exact content: any change to any profile field produces a different digest, and records produced under the changed profile are visibly foreign to the old one. This is the mechanism that turns the reproducibility cautions of the steering literature — checkpoint sensitivity, convention sensitivity — into bookkeeping rather than confusion: a record from a different checkpoint is not a conflicting measurement of the same thing; it is a measurement of a different thing, and its digest says so.

2.7. Estimate-or-observation status. This field marks whether the value is an *observation* (a recorded outcome of a measurement act) or an *estimate* (a value inferred, fitted, or predicted from other records).

Rationale. The distinction is the record-level shadow of the anti-circularity rule: a fitted effective resistance and a predicted response are estimates, and the discipline that estimates be evaluated against observations they were not fitted to requires that the two be distinguishable at the record level. It also protects summary statistics: an average of observations and a model's fitted mean may be numerically close, but only one of them is data.

2.8. Uncertainty. The uncertainty field carries the value's standard uncertainty, in the same unit as the value, together with the method identifier of the uncertainty evaluation.

Rationale. The Guide to uncertainty's central discipline — every reported value is an estimate with quantified doubt [33] — is here made structural: uncertainty is a field of the record, not a footnote, and derived records compute their uncertainty from their parents' rather than inventing it downstream. Where uncertainty has genuinely not been evaluated, the honest representation is not a zero (which asserts perfect knowledge) but an explicit absence with a stated reason, borne by the missing-or-censored field. The unit-agreement requirement (uncertainty in the unit of the value) prevents a subtle scale error the audit's lineage invites: an uncertainty computed on one scale silently attached to a value on another.

2.9. Confidence interval. The confidence-interval field carries a lower bound, an upper bound, and a confidence level strictly between zero and one; the value must lie within the interval, and both bounds must themselves satisfy the record's kind and unit constraints.

Rationale. A standard uncertainty alone presumes a distributional convention to be interpretable as coverage; the interval field makes coverage explicit. Requiring the bounds to satisfy the same validation as the value closes a loophole: an interval that strays outside the quantity's admissible range (a fraction with an upper bound above one, a strictly positive resistance with a lower bound at zero) is asserting probability mass on impossible values, and the validator rejects it rather than letting impossible coverage stand.

2.10. Sample size. The sample-size field records the number of independent underlying observations, as a strictly positive integer, wherever the value summarizes repeated measurement.

Rationale. Dispersion-based uncertainty is meaningless without its n; a mean of three sweeps and a mean of three thousand carry the same units and possibly the same value, and everything about their evidential weight differs. Recording n also enables downstream aggregation to be done correctly — combined uncertainties depend on the sizes — and enables reviewers to detect the specific pathology of grand claims resting on tiny samples.

2.11. Raw-artifact references. Raw-artifact references point, by identifier, URI, SHA-256 digest, and media type, to the raw evidence underlying the value: sweep logs, generation transcripts, projection dumps, rating forms.

Rationale. This field is the traceability chain's first link (Chapter 2): a value that cannot point to raw bytes is testimony, not evidence. Digest-carrying references make the pointing tamper-evident — the referenced artifact either hashes to the recorded digest or the chain is visibly broken — and the per-measurement-kind requirements of Section 6 make raw references mandatory exactly where the value's authority depends on them: instrumented values and all three score-like kinds must carry them; a design declaration need not, because its authority is the declaring document, referenced as a source.

2.12. Control references. Control references point to the control conditions relevant to the value — zero-intervention baselines, random-direction runs, wrong-layer applications, norm-matched comparisons — and, in the implementation, each control is additionally bound to the test point it controls for and to the method by which its relevance is asserted.

Rationale. The steering literature's inventory of controls (Chapter 2) is only worth having if control linkage survives into the record: an effect value whose controls are "available somewhere" is an effect value without controls. The relevance binding matters because a control is only a control *for something*: a zero-intervention baseline at the wrong layer controls

for nothing, and the binding forces the claim of relevance to be explicit and reviewable rather than implied by proximity.

2.13. Derivation parents. Derivation parents identify, in order and by role, the records from which a derived value was computed: each parent carries a role name, a position, an object type, and the parent's SHA-256 digest.

Rationale. This is the qualified-derivation discipline of the provenance model [22] in typed form. Ordered, role-tagged parents make derivations auditable (which record served as numerator, which as denominator), replayable (the derivation can be recomputed from the identified parents), and contamination-traceable (a defective parent's descendants are mechanically enumerable). The ordering and role constraints are validated — positions contiguous from zero, roles distinct, parent identities distinct — so that a derivation cannot silently duplicate a parent or leave its operand structure ambiguous.

2.14. Missing-or-censored reason. Where a field or a value is absent, censored, or undefined, this field records why, as declared text: below detection, saturated instrument, undefined ratio at zero response, not evaluated, withheld.

Rationale. The censoring discipline of Chapter 2 requires that absence be a stated fact, not a silent hole: a censored observation treated as zero, or an undefined ratio reported as a number, each inject known biases. The field is also where the candidate's residual discipline surfaces at the record level: at zero-compatible predicted response, the residual ratios are not small numbers — they are censored, and this field carries the reason forward so that downstream consumers cannot mistake a censored record for a measured null.

2.15. The record as a whole. Assembled, the fourteen fields implement a single design intention: *every question a skeptical reviewer would ask of a number has a slot whose emptiness is visible.* What kind of thing is this? (kind.) On what scale? (unit.) How was it obtained? (measurement kind, method.) Under what fixed conditions? (profile digest.) Is it data or inference? (estimate-or-observation.) How sure are you? (uncertainty, interval, sample size.) Where are the raw bytes? (raw references.) Against what was it controlled? (control references.) From what was it computed? (parents.) And if any answer is missing — why? (missing-or-censored reason.) A record that cannot answer fails validation; a record that answers falsely is exposed by its digests. What no record can do is decline to be asked.

3. Implementation realization

The specification's observed-value record is realized in the implementation as a frozen, slotted measurement structure together with a provenance sub-structure, and the mapping deserves explicit statement because later chapters cite implementation behavior.

The measurement structure — `MeasurementValue` in the implementation — carries the field name, quantity kind, exact decimal value, unit, measurement kind, provenance, optional uncertainty, optional sample size, optional resistance role, and optional missing-or-censored reason. The provenance sub-structure — `FieldProvenance` — carries the source references, method identifier, optional calibration-profile digest, optional instrument identity, optional test-point identity, raw-artifact references, control references and control bindings, optional dataset-split reference, derivation parents, and optional operation binding. The estimate-or-observation status of the specification is realized jointly by the measurement kind (scores and derivations are never observations of the measurand), the resistance role where applicable, and the dataset-split binding that distinguishes calibration-fitted estimates from held-out evaluations.

Validation is constructor-time and total: a structure that violates any rule of this chapter cannot be instantiated, so no invalid record ever exists to be filtered later. Exact-type checks reject subclasses and look-alikes (a provenance must be exactly a `FieldProvenance`; a reference exactly an `ArtifactRef`), closing the class of bypasses in which a permissive subtype smuggles altered behavior under a valid type name. Records are immutable after construction, and each exposes canonical bytes and a SHA-256 digest over a canonical payload that embeds — alongside the fields — the candidate's four status constants, so that research status `OPEN`, admission ring `4`, license status `NOT LICENSED`, and disclosure status `HELD` are inside the digested content of every measurement.

Two further realization details recur later. First, certain roles are *factory-only*: a network-equivalent resistance cannot be constructed by ordinary client code at all — its constructor demands a private factory token held by the network-operation module — so the only path to such a record passes through the capability gates. Second, uncertainty is validated *as a quantity*: its unit must be the measurement's unit, its interval must contain the value, and its bounds must independently satisfy the quantity's own range constraints, realizing the interval discipline of Section 2.9.

4. Supporting identity structures

The observed-value record does not stand alone: several of its fields refer to typed identity structures that are themselves validated, frozen, and canonically digestible. This section describes each, because the record's guarantees are only as strong as the identities it binds.

4.1. Artifact references. An artifact reference — `ArtifactRef` in the implementation — is the atomic pointer of the evidence layer: an artifact identifier, a URI, a SHA-256 digest, and a media type, all mandatory. The digest is the reference's substance; the URI is a courtesy. A consumer resolving the reference is obligated to verify that the resolved bytes hash to the recorded digest, and a resolution that does not is a broken chain, not a soft mismatch. The media type is recorded so that raw evidence is interpreted as what it was — a UTF-8 log, a JSON sweep record, a PDF page render — rather than sniffed; content sniffing is a classic route to interpretation drift between producers and reviewers.

Distinctness discipline applies wherever references are aggregated: within a record's source references, raw-artifact references, or control references, duplicate digests are rejected. Duplication in evidence lists is either an error or padding, and the validator treats both as failure rather than tolerating inflated evidence counts.

4.2. Control bindings. A control binding — `ControlBinding` — joins an artifact reference to two further mandatory facts: the digest of the test point the control is relevant to, and the method identifier under which its relevance is asserted. The record-level rule (Section 2.12) that a control is only a control *for something* is realized here: an unbound control artifact is inert, and the implementation additionally requires that a record's flat control-reference list agree exactly with the artifacts of its control bindings — sorted by digest, with no extras and no omissions — so that the reviewable list and the binding structure cannot drift apart. A control cited without a relevance method is unconstructible; a control bound to a foreign test point is visible as such by digest comparison.

4.3. Model identity. A model identity — `ModelIdentity` — names the subject system exactly: a model identifier, a revision string, and three digests — weights, configuration, and tokenizer. All five fields are mandatory, and the three-digest design is deliberate: the steering literature's reproducibility cautions (Chapter 2) include effects that fail to survive checkpoint

changes, configuration differences that alter hook semantics, and tokenizer differences that re-segment the very prompts a direction was derived from. A model named only by its marketing string is not an experimental subject; a model named by weights, configuration, and tokenizer digests is. Two records disagreeing about any of the three digests concern different subjects, whatever their identifiers say.

4.4. Instrument identity. An instrument identity — `InstrumentIdentity` — names the measuring apparatus: an instrument identifier, a kind, a version, and the digest of the implementing code. In this candidate the instruments are software — projection hooks, sweep harnesses, logging shims — and the implementation digest is the field that makes "the same instrument" a checkable claim rather than a folk assertion. The metrological rule that an indication becomes a result only through an identified measuring system (Chapter 2) is realized by making instrumented records unconstructible without this identity (Section 6.1).

4.5. Test-point identity. A test-point identity — `TestPointIdentity` — is the most information-dense identity in the ontology, because it fixes *where and how* the model was probed. It binds: the full model identity; the layer; a non-empty tuple of distinct, non-negative token positions; the direction identifier and the digest of the direction vector's bytes; the direction's provenance, from a closed three-member enumeration (`PROBE_DERIVED`, `JACOBIAN_DERIVED`, `DIFF_DERIVED`); the direction norm, required to equal exactly the decimal one; the direction sign, from the closed enumeration (`POSITIVE`, `NEGATIVE`); and the hook type.

Each requirement answers a specific hazard recorded in the background survey. The positions are a validated tuple because position conventions differ across codebases and off-by-one ambiguity is a real replication killer; distinctness is enforced because a duplicated position double-counts its contribution silently. The direction is bound by digest because "the same direction" must mean the same bytes, not the same name. Provenance is mandatory because probe-derived, Jacobian-derived, and difference-derived directions are different measurands (Chapter 2); a record that omitted provenance would invite exactly the operationalization conflation the survey warns against. The unit-norm requirement — the norm must equal exactly 1 — removes perturbation magnitude from the direction and locates it entirely in the alpha dose, so that dose values are comparable within a profile; a direction of norm 2 would silently double every dose. And the sign is explicit because a direction and its negation are equally natural outputs of every derivation method, and cross-record comparison presupposes an orientation convention that must be recorded, not assumed.

4.6. Uncertainty structure. The uncertainty structure — `Uncertainty` — realizes the uncertainty and confidence-interval fields jointly: a unit (which must equal the measurement's unit); a non-negative standard uncertainty; interval bounds with lower not exceeding upper; a confidence level strictly between zero and one; and the method identifier of the uncertainty evaluation. The containing record additionally requires that its value lie within the interval and that both bounds independently satisfy the quantity's own validation (Section 2.9). The mandatory method identifier deserves emphasis: an interval without its evaluation method — Type A dispersion, bootstrap, declared Type B bound — cannot be weighed, and the field forces the question "uncertain according to what?" to have a recorded answer [**33**].

4.7. Parent references and operation bindings. A parent reference — `ParentRef` — carries a role, a position, an object type, and the parent's digest; the containing provenance validates that positions run contiguously from zero, that roles are distinct, and that no parent identity repeats (Section 2.13).

An operation binding — `OperationBinding` — is the heavyweight lineage structure attached to records produced by gated operations. It binds: the authority-manifest digest; a canonically ordered, duplicate-free tuple of attestation digests (which may be empty); the capability under which the operation ran, from the closed capability enumeration; the trust mode and authorization state, from closed enumerations with a validated correspondence — structural-research mode may claim only unauthenticated-structural status, and integrity-configuration mode only unauthenticated-integrity status, so that *no combination of fields can express an authenticated state*; the caller-supplied replay time and optional clock-receipt digest; the runtime identifier; references to the dependency lockfile and the executing code, both digest-carrying; and the covariance limitation.

The covariance limitation is a fixed declared constant — stating that cross-input covariance is unavailable and a zero-covariance approximation is in use — and the validator rejects any binding that omits or alters it. This is the candidate's implementation of the background rule that covariance treatment must be declared rather than assumed (Chapter 2): the approximation is not hidden in a footnote but hashed into every gated result, so no aggregate can later claim its uncertainty accounted for dependence it never modeled.

5. The sixteen quantity kinds

The quantity-kind enumeration has exactly sixteen members. Each is defined below with its admissible units (anticipating Section 8) and its role in the candidate. The member spellings are the implementation's, verbatim.

Definition 3.3 (`DRIVE_INTERVENTION`). The declared intervention magnitude: the steering alpha applied along the profile's direction at the profile's test point. Unit: `ALPHA`. This is the dose coordinate a of the response model — the quantity the experimenter sets, not measures. Its records are typically design values (the grid point intended) or instrumented confirmations (the coefficient actually applied, as logged by the intervention harness).

Definition 3.4 (`DRIVE_SCORE`). A score-valued rating of drive on the ten-point scale. Unit: `Q10`. This kind exists to give the lineage's score-valued drive quantities a home that is *not* the dose coordinate: a rated drive of 7.5 is a score, participates in no dimensional arithmetic with activations, and cannot be substituted for an `ALPHA`-valued intervention.

Definition 3.5 (`SIGNED_ACTIVATION`). A directional activation projection retaining its sign: the inner product of an activation with the declared unit direction. Unit: `ACTIVATION`. The sign carries orientation relative to the profile's declared direction sign; a negative value is a projection opposite the direction, not an error.

Definition 3.6 (`FLOW_MAGNITUDE`). The magnitude of a directional activation response — the response coordinate y in its unsigned form. Unit: `ACTIVATION`; constrained non-negative. Flow magnitude is the quantity the response model predicts; it is related to signed activation by an explicit magnitude derivation, recorded with parents, never by silent absolute-value-taking.

Definition 3.7 (`NORMALIZED_FLOW`). A flow magnitude normalized by the profile's full-scale response and rescaled to the ten-point display scale. Unit: `Q10`. Normalized flow is profile-relative twice over — through the measurement and through the full-scale method — and its records must bind the profile digest for exactly that reason. It exists for display and quantization; the exact pre-clipping value and the overload ratio are separate kinds, preventing the normalization from destroying information.

Definition 3.8 (`CONSTRAINT_SCORE`). A score-valued rating of constraint or damping intensity. Units: `Q10` or `Q100` — the only kind with two admissible units, because the lineage's two constraint scales both required a typed home. The two units are not interconvertible without an explicit conversion method. A constraint score is never an effective resistance: the audit's dimensional-bridge disposition — constraint scores never masquerade as resistance — is enforced by this kind's unit set, which excludes `ALPHA_PER_ACTIVATION`.

Definition 3.9 (`EFFECTIVE_RESISTANCE`). The fitted proportionality parameter r_{eff} of the response model. Unit: `ALPHA_PER_ACTIVATION`; constrained strictly positive; requires an explicit resistance role (calibrated held-out, apparent same-sample, network equivalent, or design proposal) with role-specific validation described in Section 8. This is the only kind licensed to bridge the alpha and activation dimensions.

Definition 3.10 (`RESISTANCE_DELTA`). A difference of two effective resistances under one profile — the paired-delta quantity by which the lineage's addendum proposed to infer contextual resistance changes. Unit: `ALPHA_PER_ACTIVATION`. As a difference it may be negative; it carries its two parent resistances as ordered derivation parents, and it inherits their split and role discipline through them.

Definition 3.11 (`CAPABILITY_SCORE`). A score-valued rating of a capability or performance dimension on the ten-point scale. Unit: `Q10`. Like drive score, it exists to keep the lineage's rated quantities typed and quarantined from measurement arithmetic.

Definition 3.12 (`GAIN_MAGNITUDE`). A local gain: the magnitude of a derivative or ratio relating an effect coordinate to a cause coordinate between declared test points. Unit: `DIMENSIONLESS`; constrained non-negative. The audit's gain conflict — Fisher–Rao displacement gain versus local Jacobian projection gain, distinct until a bridge exists — lives at the method layer: both profiles of gain occupy this one kind, distinguished mandatorily by method identifier, so that any aggregation across the two methods is visible and rejectable.

Definition 3.13 (`EFFECT_PROJECTION`). The projection of a downstream effect onto a declared effect direction at a declared downstream test point — the effect-side analogue of signed activation, kept as its own kind so that cause-side and effect-side projections cannot be conflated in gain derivations. Unit: `ACTIVATION`.

Definition 3.14 (`SATURATION_FRACTION`). The fraction of a sweep, grid, or operating envelope classified as saturated. Unit: `FRACTION`, on $[0, 1]$. This kind is the typed repair of the audit's mixed-scale conflict: the lineage's sat = 8.5 was a fraction-kind assertion written on a score scale, and its corrected value 0.85 is representable only here.

Definition 3.15 (`CUTOFF_FRACTION`). The fraction of a sweep, grid, or operating envelope classified as cutoff — below response threshold. Unit: `FRACTION`, on $[0, 1]$. Kept distinct from saturation fraction because the flat-response ambiguity (Chapter 2) makes the two ends of the curve evidentially different despite their shared flatness.

Definition 3.16 (`OVERLOAD_RATIO`). The ratio of a raw response to the profile's full-scale response, retained exactly and without clipping. Unit: `DIMENSIONLESS`; constrained non-negative; values above one are the point of the kind. Overload ratio is what preserves information the display normalization discards: a normalized flow clipped at ten records *that* the scale was exceeded, the overload ratio records *by how much*.

Definition 3.17 (`FLOW_RESIDUAL`). The signed difference between an observed and a predicted flow, $\delta_y = y_{\text{obs}} - y_{\text{pred}}$, under one profile with a registered prediction. Unit:

ACTIVATION. A flow residual requires calibrated resistance and propagated uncertainty in its derivation, is censored at zero-compatible predicted response, and — per the audit's residual disposition — ranks follow-up probes without ever identifying a cause.

Definition 3.18 (RELATIVE_ERROR). The magnitude of a residual relative to the magnitude of the prediction or observation it deviates from, under a declared convention named by the method identifier. Unit: DIMENSIONLESS; constrained non-negative; undefined — and therefore censored, not computed — where its denominator is zero or zero-compatible.

Three observations about the enumeration as a whole. First, it is *closed*: there is no other-kind escape hatch, and a quantity that fits no kind cannot be recorded until the ontology is revised — deliberately, with a new digest for every profile that uses it. Second, it is *partitioned by discipline*: four kinds are score-valued (DRIVE_SCORE, CONSTRAINT_SCORE, CAPABILITY_SCORE, and the display-scale NORMALIZED_FLOW), five are activation-dimensioned measurements (SIGNED_ACTIVATION, FLOW_MAGNITUDE, EFFECT_PROJECTION, FLOW_RESIDUAL, and the alpha-dimensioned DRIVE_INTERVENTION), two are the resistance bridge (EFFECTIVE_RESISTANCE, RESISTANCE_DELTA), and the remainder are dimensionless diagnostics (GAIN_MAGNITUDE, OVERLOAD_RATIO, RELATIVE_ERROR) and fractions (SATURATION_FRACTION, CUTOFF_FRACTION); no arithmetic crosses these families without an explicitly typed bridging kind. Third, it is *audit-derived*: nearly every kind exists because some conflict in the lineage required the distinction it draws, and the enumeration should be read as the audit's dispositions compiled into types.

6. The six measurement kinds and non-fungibility

6.1. The six kinds. The measurement-kind enumeration has exactly six members, spelled as in the implementation.

Definition 3.19 (DESIGN_VALUE). A value declared by a board, profile, or specification — intended, not observed. *Example:* the alpha grid point $a = 2.0$ that a sweep protocol declares it will apply; a board's declared full-scale response under the declared-design full-scale method. A design value's authority is the declaring document; it requires source references but not raw measurement artifacts, and no accumulation of design values ever constitutes an observation.

Definition 3.20 (INSTRUMENTED). A value produced by an identified instrument chain with calibration binding. *Example:* a signed activation projection logged by the declared projection hook at the declared layer and positions, under a declared calibration profile, with the sweep log retained as a raw artifact. Instrumented records carry the heaviest obligations: instrument identity, test-point identity, calibration-profile digest, and raw-artifact references are all mandatory, and their absence is a constructor-time failure, not a warning.

Definition 3.21 (BEHAVIORAL_PROXY). A value inferred from external behavior rather than internal measurement. *Example:* a classifier's rate of target-property outputs over generated text at a given alpha, used as a proxy for the intervention's behavioral effect. Behavioral proxies require raw-artifact references (the transcripts and classifier outputs) and their proxy status is permanent: however well a proxy tracks an internal quantity, its records remain proxies.

Definition 3.22 (MODEL_DERIVED). A value computed from other records. *Example:* an effective resistance fitted from calibration-split sweep records; a residual computed from an observation and a registered prediction; a normalized flow computed from a flow magnitude and a full-scale value. Model-derived records must carry derivation parents — a derived

value with no parents is unconstructible — and their evidential weight is a function of their parents', never greater.

DEFINITION 3.23 (`HUMAN_SCORE`). A value rated by a person. *Example:* a rater's 0–10 judgment of how strongly a generated passage expresses the steered property, with the rating form retained as a raw artifact. Human scores require raw-artifact references and are permanently score-kind: no aggregation, agreement statistic, or averaging converts rated judgment into telemetry.

DEFINITION 3.24 (`LLM_SCORE`). A value rated by a language model. *Example:* a judge model's 0–10 rating of the same passage, with the judging prompt and response retained as raw artifacts. LLM scores carry the same obligations and the same permanent score status as human scores, plus a caution recorded in Chapter 2: judge-model failure modes may correlate with the interventions under study, so LLM-scored evidence is never sole support for an effect claim.

6.2. Non-fungibility.

RULE 3.25 (Kind non-fungibility). Measurement kinds do not convert, upgrade, or launder. In particular: (i) a derived value does not become instrumented because one of its parents — or all of them — was instrumented; (ii) a score does not become telemetry because it is stored, aggregated, or displayed alongside measurements; (iii) a behavioral proxy does not become an internal measurement because it correlates with one; (iv) a design value does not become an observation because the design was executed. Derivation preserves parent lineage; it does not inherit parent status.

The rule's rationale is that measurement kind encodes the *worst-case question a reviewer must ask*, and that question never improves through composition. A resistance fitted from instrumented sweeps is exactly as good as the fit — so it is `MODEL_DERIVED`, and its parents say what it was fitted from. An average of ten instrumented projections is a derived summary of instrumented data — reviewable as such, but not itself the output of an instrument, and the distinction bites precisely when the averaging method is wrong. In the other direction, the rule forbids the audit's score-as-telemetry slide at the level where it historically happened: not in any single dramatic act, but by drift — scores stored in measurement tables, displayed on measurement dashboards, and eventually cited as measurements. Under the ontology, the kind travels inside the digested record, and no storage location, display context, or citation practice can alter it.

The implementation enforces the rule both by validation (each kind's mandatory fields, per Section 3) and by construction discipline for the sensitive cases: the resistance roles that carry predictive authority are constructible only as `MODEL_DERIVED`, so an "instrumented resistance" — a category error, since resistance is a fitted parameter — is unrepresentable outright.

6.3. Obligations by kind, in summary. Table 1 summarizes the mandatory provenance obligations attached to each measurement kind by the constructor-time validation, as implemented. The pattern is deliberate: obligations scale with the authority the kind could be mistaken for carrying. Instrumented records — the only kind whose name suggests telemetry — carry the heaviest load; design values — whose authority is the declaring document — the lightest.

Measurement kind	Mandatory beyond baseline	Rationale in brief
`DESIGN_VALUE`	— (source refs, method)	authority is the declaration
`INSTRUMENTED`	instrument, test point, profile digest, raw artifacts	indication-to-result chain must be complete
`BEHAVIORAL_PROXY`	raw artifacts	proxy must expose its basis
`MODEL_DERIVED`	derivation parents	no computation without inputs
`HUMAN_SCORE`	raw artifacts	rating must expose its record
`LLM_SCORE`	raw artifacts	judging must expose prompt and response

TABLE 1. Mandatory provenance obligations by measurement kind. Baseline obligations (non-empty source references and a method identifier) apply to every record.

6.4. Design rationale: granularity and closure. Two recurring design questions deserve answers on the record, because reviewers of earlier drafts asked both.

Why sixteen quantity kinds and not four? A minimal dimensional analysis would need only alpha-valued, activation-valued, ratio-valued, and dimensionless quantities. The finer grain exists because the candidate's threat model is not dimensional error alone but *semantic* conflation within a dimension: signed activation, flow magnitude, effect projection, and flow residual all live in `ACTIVATION` units, yet substituting one for another in a derivation is always an error — a residual is not an observation, an effect-side projection is not a cause-side one, a magnitude has discarded a sign that a gain derivation may need. Each such substitution the audit's lineage either committed or invited is blocked by a kind distinction. The cost of granularity is verbosity in producers; the benefit is that the reviewer's question "which quantity, exactly?" is answered by the record rather than by archaeology. The candidate pays the verbosity.

Why closed enumerations rather than extensible registries? Extensibility is the conventional virtue, and the candidate refuses it at every layer — kinds, units, measurement kinds, roles, provenances, regions, outcomes are all closed. The reason is the digest discipline: profiles, records, and gates are compared by canonical bytes, and an open vocabulary makes two deployments' vocabularies silently diverge while their digests continue to compare. A closed enumeration changes only by revising the ontology itself, which changes the digests of everything downstream — making evolution *loud*, which is the property the candidate wants. Extension points do exist, but they are quarantined: the implementation provides a non-claim extension structure whose namespace must begin with `x-nonclaim-`, allowing auxiliary payloads to travel with records while being, by their very namespace, incapable of stating a claim. Anything that would state a claim must enter the closed vocabularies through revision and review.

7. The unit ontology

7.1. The seven units. The unit enumeration has exactly seven members, spelled as in the implementation: `ALPHA`, the unit of steering dose; `ACTIVATION`, the unit of directional projection magnitude; `ALPHA_PER_ACTIVATION`, the unit of effective resistance and resistance deltas; `DIMENSIONLESS`, for ratios and gains whose dimensions cancel; `FRACTION`, for proportions constrained to $[0, 1]$; and the two score scales `Q10` (range $[0, 10]$) and `Q100` (range $[0, 100]$).

Several deliberate features of this small ontology deserve statement. *Alpha and activation are primitive and mutually irreducible:* neither is defined in terms of the other, and no dimensionless rescaling connects them; the only bridge is the ratio unit `ALPHA_PER_ACTIVATION`, reserved to the resistance kinds. *Dimensionless and fraction are distinct:* both are pure numbers, but `FRACTION` carries the $[0, 1]$ constraint as part of its meaning (a fraction of 1.2 is a validation error), while `DIMENSIONLESS` does not (an overload ratio of 1.2 is the point); collapsing them would force one constraint to be wrong. *The score scales are units, not ranges:* `Q10` and `Q100` are treated as genuinely different units of score-kind quantities, so the lineage's two constraint scales coexist without ambiguity and without silent conversion — a `Q100` record rescaled to `Q10` is a new derived record with a conversion method and a parent, or it is nothing.

7.2. Forbidden aliases.

Rule 3.26 (Forbidden electrical aliases). The physical unit labels VOLT, AMPERE, and OHM — and by extension every SI electrical unit — are forbidden aliases for CAS-Ohm model quantities. No CAS-Ohm record may carry them; no display, export, or documentation layer may attach them to CAS-Ohm values; and their absence from the unit enumeration is a contract clause, verified by golden rejection fixtures, not an oversight to be repaired.

The rule's form is unusual — ontologies do not usually legislate about members they lack — and Chapter 1 gave the reason: an ontology that merely omitted the electrical units would invite a later extension to add them, whereas naming them as forbidden makes the refusal itself part of the specification, with tests that must be deliberately deleted before the refusal can erode. The forbidden-alias rule is the unit-layer enforcement of the monograph's first refusal (no physical law): it guarantees that the candidate cannot even *state* a physical electrical claim in its own evidence layer.

A hypothetical future physical profile — an actual instrumented electrical measurement context, should the wider project ever have one — would require separate explicit typing: its own unit members, its own quantity kinds, its own instrument and calibration machinery, admitted through the project's ordinary review discipline. Nothing in the current candidate anticipates, licenses, or partially implements such a profile.

7.3. The `Unit.VOLT` disposition.

One vocabulary fact requires explicit quarantine. The wider project's generic PCBA vocabulary — the board-description layer from which the candidate's lineage borrows its engineering idiom — contains a member `Unit.VOLT`, present there as part of a general-purpose unit vocabulary for describing printed-circuit-board artifacts.

Rule 3.27 (`Unit.VOLT` licenses nothing). The existence of `Unit.VOLT` in the generic PCBA vocabulary licenses nothing for CAS-Ohm: it does not license physical CAS telemetry, does not create a CAS-Ohm unit, does not soften the forbidden-alias rule, and does not constitute partial implementation of a physical profile. The member belongs to a different vocabulary with a different subject matter; its import into any CAS-Ohm context is a validation error.

The disposition is recorded at this length because vocabulary adjacency is exactly the mechanism by which the historical sources drifted: an electrical idiom available nearby became an electrical semantics assumed everywhere. The candidate's defense is to name the adjacency and sever it explicitly.

8. Admissibility, ranges, and roles

The quantity-unit admissibility relation is total and closed: every quantity kind has a declared set of admissible units, every (kind, unit) pair outside the relation is rejected at construction, and the relation is reproduced in Table 2 exactly as implemented.

Quantity kind	Admissible unit(s)	Range constraint
DRIVE_INTERVENTION	ALPHA	—
DRIVE_SCORE	Q10	$[0, 10]$
SIGNED_ACTIVATION	ACTIVATION	—
FLOW_MAGNITUDE	ACTIVATION	≥ 0
NORMALIZED_FLOW	Q10	$[0, 10]$
CONSTRAINT_SCORE	Q10, Q100	$[0, 10]$ / $[0, 100]$
EFFECTIVE_RESISTANCE	ALPHA_PER_ACTIVATION	> 0; role required
RESISTANCE_DELTA	ALPHA_PER_ACTIVATION	—
CAPABILITY_SCORE	Q10	$[0, 10]$
GAIN_MAGNITUDE	DIMENSIONLESS	≥ 0
EFFECT_PROJECTION	ACTIVATION	—
SATURATION_FRACTION	FRACTION	$[0, 1]$
CUTOFF_FRACTION	FRACTION	$[0, 1]$
OVERLOAD_RATIO	DIMENSIONLESS	≥ 0
FLOW_RESIDUAL	ACTIVATION	—
RELATIVE_ERROR	DIMENSIONLESS	≥ 0

TABLE 2. The quantity-unit admissibility relation and range constraints, as implemented. Scale ranges attach to the unit (Q10, Q100, FRACTION); sign constraints attach to the kind; EFFECTIVE_RESISTANCE additionally requires an explicit resistance role.

The role discipline attached to effective resistance completes the admissibility layer. Every EFFECTIVE_RESISTANCE record must declare exactly one role: CALIBRATED_HELD_OUT, the only role admissible for prediction, which additionally requires MODEL_DERIVED measurement kind and a bound dataset-split reference identifying the held-out split; APPARENT_SAME_SAMPLE, the typed home of the circular quantity $r = a/y$, recordable so that same-sample fits can be documented and studied but barred from prediction; NETWORK_EQUIVALENT, constructible only through the capability-gated network-operation factory (Section 3); and DESIGN_PROPOSAL, a declared design-layer value with no predictive standing. A resistance role on any other quantity kind is a validation error, as is an effective resistance without one. This four-way role split is the record-level realization of the anti-circularity rule: the circular fit is not forbidden from existing — forbidding it would merely drive it underground — it is forbidden from *mattering*, by a role that every consumer can see and every gate checks.

9. Canonical evidence bounds

Every structure in the ontology exposes canonical bytes and a SHA-256 digest, and every downstream comparison in the candidate — record identity, profile binding, parent linkage, tamper detection, attestation replay — reduces to digest equality. The canonical encoding is therefore itself part of the trusted computing base, and it is governed by an inherited contract — the wider project's canonical-encoding rules — whose CAS-Ohm-relevant clauses are specified here.

Invariant 3.28 (Exact-type tagging). Canonical encoding is exact-type driven: every encoded node carries its exact type, subclasses and duck-typed look-alikes are rejected rather than encoded as their nearest ancestor, and two structures of different exact types never share canonical bytes. Type confusion — the classic route by which a permissive encoder lets a crafted object impersonate a valid one — is thereby excluded at the byte layer.

Invariant 3.29 (Cycle awareness). The encoder detects reference cycles and rejects cyclic structures rather than recursing to exhaustion. Evidence records are finite trees by specification; a cycle is not an exotic input but a malformed or adversarial one, and the defined behavior is a validation error, not a stack fault.

Invariant 3.30 (Unsafe-Unicode rejection). Text admitted into canonical evidence is screened, and unsafe Unicode is rejected: control and format characters, direction-override and other visually deceptive code points, unpaired surrogates, and related hazard classes cannot enter a canonical record. The screen exists because digested text is displayed, compared, and reviewed by humans, and a code point that renders one way while hashing another is an attack on exactly the equality the digests are trusted to witness.

Invariant 3.31 (Declared resource bounds). Canonical encoding operates under declared, validated limits on: nesting depth; total node count; per-container size; per-text length; total output size; and integer magnitude, the last bounded at 2,048 bits. Inputs exceeding any limit are rejected with the candidate's validation error. The bounds make encoding cost predictable and deny the class of resource-exhaustion inputs — pathological nesting, enormous containers, unbounded numerals — that unbounded encoders admit by default.

Invariant 3.32 (Host-exception normalization). Failures raised by the host runtime during encoding — in particular, integer-conversion failures on magnitudes the platform cannot process — are caught and normalized to the candidate's own validation error. No platform exception leaks through the canonical layer: the encoder's error surface is its own, closed, and identical across host environments.

Two remarks situate these bounds. First, they are *inherited*, not invented: the candidate adopts the project's canonical-encoding contract, and the invariants above are that contract's clauses restated for this evidence layer; consistency with the wider project is itself part of the point, since cross-layer digest comparison presupposes one encoding. Second, they are *load-bearing for everything else in this chapter*: the no-bare-numbers rule stores exact decimals whose digests must be stable (exact-type tagging); the provenance chains are trees whose finiteness must be enforced (cycle awareness); method identifiers and reasons are text that reviewers will read (Unicode screening); and adversarial or defective producers must be met with defined rejection, not undefined behavior (bounds and normalization). A record system with softer bytes would leave every rule stated earlier enforceable only in principle.

10. Worked constructions and rejections

The ontology's behavior is best fixed by examples: one construction that succeeds, and a series that fail. All values below are schematic — they illustrate validation, and assert nothing empirical.

Example 3.33 (A valid instrumented observation). Consider recording a single directional projection observed during a sweep. The record asserts: quantity kind `SIGNED_ACTIVATION`;

exact decimal value `-0.4375`; unit `ACTIVATION`; measurement kind `INSTRUMENTED`. Its provenance carries: a source reference to the sweep protocol document; the projection method identifier; the calibration-profile digest; an instrument identity naming the projection hook implementation by digest; a test-point identity binding the model (by weights, configuration, and tokenizer digests), the layer, the position tuple, the direction digest with `DIFF_DERIVED` provenance, exact unit norm, and `POSITIVE` sign; a raw-artifact reference to the sweep log whose bytes hash to the recorded digest; and two control bindings — a zero-intervention baseline and a norm-matched random direction — each tied to the same test-point digest with a declared relevance method, with the flat control list agreeing exactly with the bindings. An uncertainty structure in `ACTIVATION` units carries the standard uncertainty from repeated passes, an interval containing the value, a confidence level of `0.95`, and the dispersion method identifier; the sample size records the number of passes. Construction succeeds, the record freezes, and its digest — which internally binds `OPEN`, ring `4`, `NOT LICENSED`, `HELD` — becomes its permanent identity.

Example 3.34 (Rejected: mixed scale). A producer attempts to record a saturation fraction of `8.5` — the lineage's own error. As `SATURATION_FRACTION` with unit `FRACTION`, the value violates the $[0, 1]$ range and is rejected. Retyping it as `DRIVE_SCORE` in `Q10` would validate numerically — but then it is a score, unusable where a fraction is demanded, and the admissibility relation blocks every downstream slot that expects `FRACTION`. There is no representation in which `8.5` functions as a region fraction; the corrected value `0.85` must be produced, as a new record with its own method and lineage.

Example 3.35 (Rejected: dimensional bridge violation). A producer attempts to register an effective resistance whose value was obtained from a constraint score: quantity kind `EFFECTIVE_RESISTANCE`, unit `Q100`. The admissibility relation rejects the pair outright — resistance admits only `ALPHA_PER_ACTIVATION`. Re-labeling the unit without a genuine calibrated bridge fails differently: the record now demands a resistance role; role `CALIBRATED_HELD_OUT` demands `MODEL_DERIVED` kind, derivation parents, and a held-out dataset-split reference, none of which a rescaled score possesses. The audit's disposition — constraint scores never masquerade as resistance — is thus not one check but a gauntlet, and every path through it fails closed.

Example 3.36 (Rejected: laundered provenance). A producer computes a mean of instrumented projections and records it with measurement kind `INSTRUMENTED`, reasoning that all parents were instrumented. The record fails: its provenance carries derivation parents but no instrument identity for the mean itself, and no instrument exists that measured the mean. Recorded correctly as `MODEL_DERIVED`, it must carry its parents — and the constructor rejects a model-derived record with an empty parent tuple, so the derivation cannot hide its inputs either. Non-fungibility (Rule 3.25) is enforced from both directions.

Example 3.37 (Rejected: circular resistance in a predictive slot). A producer fits $r = a/y$ on a single sample and attempts to register it for prediction. As `APPARENT_SAME_SAMPLE`, the record constructs — the circular quantity is documentable — but every predictive gate requires `CALIBRATED_HELD_OUT`, and the role is inside the digest, unforgeable after the fact. Claiming `CALIBRATED_HELD_OUT` directly fails for want of a dataset-split reference; supplying a split reference whose artifact does not resolve to registered disjoint splits fails at review; and `NETWORK_EQUIVALENT` is unreachable entirely, its constructor demanding the factory token that only the gated network module holds. The circularity rejection of Chapter 1 is, at this layer, a property of reachable states.

Example 3.38 (Rejected at the byte layer). Finally, three sub-record rejections. A method identifier containing a direction-override code point is rejected by the Unicode screen before any semantic validation runs. A derivation graph in which a record transitively references itself is rejected by cycle detection. An integer of 2,500 bits — however it arose — is rejected by the integer bound, and if the host runtime itself fails first on the conversion, that failure surfaces as the candidate's validation error, not as a platform exception (Invariant 3.32). The evidence layer's rejections are uniform all the way down.

These examples are the ontology's specification exercised end to end, and each corresponds to test methods in the pinned suite. Per the standing discipline (Chapter 2): that correspondence establishes tested software behavior on constructed inputs — verification — and nothing else.

11. Summary invariants

The chapter closes by collecting its content into the invariants that later chapters cite.

Invariant 3.39 (Typed evidence). Every CAS-Ohm numeric record is an observed value per Definition 3.2: kind-typed from the closed sixteen-member enumeration, unit-typed from the closed seven-member ontology under the admissibility relation of Table 2, measurement-kinded from the closed six-member enumeration with kind-specific mandatory provenance, exact-decimal valued, profile-bound where its kind requires it, uncertainty-carrying or explicitly censored, lineage-complete, and canonically digestible under the bounds of Section 9. Anything less does not exist in the evidence layer.

Invariant 3.40 (Non-fungibility). Measurement kinds never convert: derived never becomes instrumented, scores never become telemetry, proxies never become internal measurements, designs never become observations (Rule 3.25).

Invariant 3.41 (Dimensional closure). The only bridge between the `ALPHA` and `ACTIVATION` dimensions is the resistance family in `ALPHA_PER_ACTIVATION` units, with mandatory roles; score units bridge nothing; physical electrical units exist nowhere and are forbidden as aliases (Rules 3.26 and 3.27).

Invariant 3.42 (Fail-closed validation). All validation is constructor-time and rejecting: invalid records are never created, never flagged-but-stored, never repaired silently. The evidence layer's error surface is the candidate's own validation error, everywhere.

These invariants are specification, verified as software behavior at the pinned test receipt and nowhere validated as science. On that foundation — typed quantities that cannot lose their identities, provenance that cannot be laundered, and bytes that cannot be confused — the following chapters build the preserved source audit and the calibration and response machinery, without ever asking the reader to trust a number that cannot answer for itself.

CHAPTER 4

The Preserved Source Audit

The single most consequential decision in the CAS-Ohm 1.0 candidate is a negative one: the decision *not* to harmonize its sources. Three primary documents—the Quantification Guide, the operationalization addendum, and PCBA Appendix I—together with the PCBA engineering book's gain definitions, describe what is nominally one framework. Read side by side, they do not agree. They disagree about numeric scales, about how a full-scale reference is obtained, about what the symbols V, I, and R denote, about what the gain β is, about whether a mnemonic is a classification, about whether network arithmetic is established or conjectural, about what a residual can identify, and about where quantization band boundaries fall. A conventional specification effort would have resolved these disagreements editorially: pick the reading that seems most sensible, silently rewrite the others to match, and present the result as the framework's single inherited meaning. The CAS-Ohm audit refuses that move. Every disagreement found during source inspection is preserved as a first-class, machine-readable evidence record in `CAS_OHM_SOURCE_CONFLICT_EVIDENCE_v0.1.json` (schema `project_coherence.cas_ohm_source_conflict_evidence.v0.1`, status DRAFT / OPEN / NOT FROZEN, as of 2026-08-04), and every downstream artifact—the typed ontology of Chapter 3 where present, the calibration profile of Chapter 5, the response model of Chapter 6—is designed so that the conflict cannot be papered over by construction.

This chapter reproduces that audit at evidence-record granularity. For each of the eight registered conflicts we give the evidence records verbatim-faithfully (source identifier, locator, and recorded fact), name the conflict kind under which the record classifies the disagreement, analyze at length why the disagreement matters to any downstream quantitative use, and state the binding disposition and candidate rule that the CAS-Ohm 1.0 type system enforces in response. We close with the record's own summary fields: the conflict count, the gate effect, and the claim boundary—and with an argument that the preservation itself, independent of any future empirical outcome, is an engineering result.

A standing caution governs everything below. The audit record preserves *conflicts about texts*. It does not adjudicate which text is right, because on the evidence available none of them is established: no source has an admitted empirical validation, and the candidate as a whole is RING 4 / NOT LICENSED / HELD. The dispositions quoted in this chapter are candidate rules—typed constraints a successor implementation enforces so that the ambiguity is representable and inert rather than invisible and active. They are not verdicts on the sources.

1. The four sources and their locator systems

The evidence record opens with a source manifest. Four sources are registered, each with a content digest, a declared locator system, and a declared inspection method. We reproduce the manifest faithfully before discussing why its structure matters.

src-cas-ohm-guide-01: (the Quantification Guide). SHA-256 `89AE13C0A5D6208E328E9EDC55F816609B`
Locator system: *PDF page number*. Inspection: relevant cited pages were rendered and visually inspected.

src-cas-ohm-addendum-01: (the operationalization addendum). SHA-256 `0B62BBC66828A273E5458C`
Locator system: ***`python-docx`*** *paragraph and table-row index*. Inspection: the OOXML content was inspected structurally. LibreOffice failed with a PNG writer error and Word export did not complete, so *no page-number claim is made for this source*.

src-cas-ohm-pcba-app-i: (PCBA Appendix I). SHA-256 `9FFFAC3E16B43C6D809ECE913FD8C09E40C65F`
Locator system: *UTF-8 Markdown line number*. Inspection: the cited line ranges were inspected directly from the source bytes.

src-pcba-engineer-20260726: (the PCBA engineering book, cited for its gain definitions). SHA-256 `294320B39FA521EAC086E99E2D9DB913E2B6CFF0754FF82DB1EF966424566274`.
Locator system: *PDF page number and printed book page*. Inspection: the cited gain-definition pages were rendered and visually inspected.

1.1. Why disclosed locator systems matter. A citation is a claim of the form "at this position in that artifact, this fact appears." The claim has three parts—artifact identity, position, and fact—and each part can fail independently. The source manifest hardens all three. Artifact identity is fixed by a SHA-256 content digest, so "the Guide" is not a title that could silently drift across revisions but a specific byte sequence. The fact is recorded verbatim in the evidence record, as the following sections reproduce. The position, however, is the part most often left informal in technical writing, and it is the part the manifest is most careful about, because *position is only meaningful relative to a rendering procedure*, and different artifact formats admit different renderings.

Consider each system in turn.

PDF pages, rendered and visually inspected. A PDF fixes its own pagination, so "PDF page 6" is well defined given the artifact bytes—but only if someone actually rendered page 6 and looked at it. The inspection note records that this was done. The distinction matters: a page number harvested from a table of contents, a search index, or a prior citation is a transcription, not an observation. The manifest asserts observation.

`python-docx` *paragraph and table-row indices, with no page number asserted.* A DOCX file has no intrinsic pagination at all. Its page breaks are computed at render time by a layout engine, and different engines (or the same engine with different fonts, margins, or versions) paginate differently. The honest locator for a DOCX artifact inspected without a successful rendering is therefore a *structural* coordinate: the index of a paragraph or a table row in the OOXML element sequence, which is a property of the bytes themselves. The manifest goes further and records *why* no page number is asserted: both available page-rendering routes failed—LibreOffice with a PNG writer error, and Word export did not complete. This is a small but exemplary act of negative disclosure. The auditors could have guessed a page number, or cited one from memory, and no reader would likely have caught it. Instead the record states the failure and confines its claims to coordinates it can defend. Every addendum citation in this chapter therefore reads "paragraph n" or "table row k," never "page p."

UTF-8 line numbers from exact bytes. Appendix I is a Markdown text file. Line numbers in a text file are exact byte-level coordinates: given the artifact digest, "lines 20–31" names a unique substring, checkable by anyone holding the bytes, with no rendering step in between. This is the strongest locator system of the four, and the record uses it wherever the source format permits.

PDF and printed book pages together. The PCBA engineering book exists both as a PDF and as a printed volume whose pagination differs (front matter, trim, and imposition shift the count). Citing both—"PDF page 56 (book page 40)"—lets a reader verify against either physical form without a conversion table.

Remark 4.1. The reason to belabor locator systems is that the entire audit inherits its credibility from them. A conflict record asserts that two sources say incompatible things. That assertion is only as strong as the reader's ability to open each source at the cited position and confirm the recorded fact. If locators were informal—"somewhere in the Guide"—the conflicts would be paraphrases, contestable and unfalsifiable. Because every locator is exact relative to a disclosed rendering procedure (or, for the addendum, exact relative to the bytes with the rendering honestly declared failed), each conflict record is an independently checkable claim. The audit practices on itself the evidential discipline the CAS-Ohm ontology imposes on measurements: no bare assertions, provenance always attached, and missing capabilities (here, DOCX pagination) recorded as missing rather than interpolated.

1.2. Anatomy of a conflict record. Each of the eight entries in the evidence record has the same shape, which this chapter mirrors section by section.

Definition 4.2 (Conflict record). A conflict record consists of: (i) a stable identifier (for example `OHM-CONFLICT-SCALE-AND-SATURATION`); (ii) a *kind*, drawn from a small vocabulary describing the failure mode (`DIRECT_NUMERIC_CONFLICT`, `INCOMPATIBLE_CALIBRATION_METHODS`, `UNIT_UNDERSPECIFICATION`, `DISTINCT_MATHEMATICAL_PROFILES`, `MNEMONIC_OVERLOAD`, `VALIDATION_STATUS_CON` `CAUSAL_OVERINTERPRETATION_RISK`, `PRECISION_UNDERSPECIFICATION`); (iii) an evidence list, each element binding a source identifier from the manifest, a locator in that source's declared locator system, and the recorded fact; (iv) a *disposition*, naming the strategy the candidate adopts (for example `PRESERVE_AS_PROFILE_CONFLICT`); and (v) a *candidate rule*, the typed constraint a successor implementation enforces.

The kinds are worth a moment of attention because they are a taxonomy of how multi-document specifications fail. A `DIRECT_NUMERIC_CONFLICT` is the crudest failure: two sources assert incompatible numbers for the same named thing. `UNIT_UNDERSPECIFICATION` and `PRECISION_UNDERSPECIFICATION` are subtler: no single assertion is wrong, but jointly the sources fail to determine a quantity's dimension or a boundary's behavior. `INCOMPATIBLE_CALIBRATION_METHODS` and `DISTINCT_MATHEMATICAL_PROFILES` mark places where one name covers several genuinely different procedures or definitions. `MNEMONIC_OVERLOAD` marks a teaching device promoted into a classification. `VALIDATION_STATUS_CONFLICT` marks disagreement not about content but about epistemic standing—is a rule established or conjectural? And `CAUSAL_OVERINTERPRETATION_RISK` marks language that invites causal conclusions its own source elsewhere disclaims. Most large specification corpora contain instances of most of these kinds; what is unusual here is only that they are indexed.

Table 1 summarizes the eight records for orientation; the sections that follow reproduce and analyze each in full.

1.3. Verbatim fidelity. One further discipline governs the presentation in this chapter. The "fact" field of each evidence record is a fixed string in the JSON artifact, and this chapter reproduces those strings faithfully rather than paraphrasing them. The distinction is not pedantry. A paraphrase of a conflict is an interpretation, and interpretations of conflicting texts tend, by small increments, toward whichever resolution the paraphraser privately favors—which is exactly the harmonization pressure the audit exists to resist. If the record

Conflict (short name)	Kind	Disposition
Scale and saturation	DIRECT_NUMERIC_CONFLICT	PRESERVE_AS_PROFILE_CONFLICT
Full-scale profile	INCOMPATIBLE_CALIBRATION_METHODS	SEPARATE_NAMED_PROFILES
Dimensional bridge	UNIT_UNDERSPECIFICATION	REQUIRE_CALIBRATED_UNIT_BRIDGE
Gain definition	DISTINCT_MATHEMATICAL_PROFILES	NO_UNPROVED_IDENTIFICATION
ASOM axes	MNEMONIC_OVERLOAD	SPLIT_RESPONSE_AND_SEMANTIC_AXES
Network rules	VALIDATION_STATUS_CONFLICT	CAPABILITY_GATE_EACH_OPERATION
Residual identification	CAUSAL_OVERINTERPRETATION_RISK	NON_IDENTIFYING_DIAGNOSTIC_ONLY
Q4 boundaries	PRECISION_UNDERSPECIFICATION	DEFINE_EXACT_HALF_OPEN_BINS

TABLE 1. The eight preserved conflicts of CAS_OHM_SOURCE_CONFLICT_EVIDENCE_v0.1.json.

says Appendix I identifies sat = 8.5 "as a mixed-scale error whose component value is 0.85," then a restatement such as "Appendix I corrects the Guide's threshold" would already have taken a side (that Appendix I is the corrector rather than a second conflicting voice), and a restatement such as "the sources disagree slightly about thresholds" would have shrunk a category error to a rounding quibble. Analysis in this chapter is therefore kept structurally separate from evidence: the *conflict* environments carry the recorded facts with their locators, and the analysis subsections argue about consequences without ever restating the facts in altered form. Readers auditing this monograph against the JSON record should find every fact string, every locator, every kind, every disposition, and every candidate rule matching the artifact—and should treat any divergence they find as an error in this text, not a permissible gloss.

2. Conflict 1: Scale and Saturation

PRESERVED CONFLICT 4.3 (OHM-CONFLICT-SCALE-AND-SATURATION; kind DIRECT_NUMERIC_CONFLICT). The evidence records are:

- SRC-CAS-OHM-GUIDE-01, PDF page 6: "Drive is 0.0–10.0, constraint R is 0.1–10.0, and raw attention is computed as V/R."
- SRC-CAS-OHM-GUIDE-01, PDF page 12: "The worked CJT uses sat = 8.5 as a component threshold."
- SRC-CAS-OHM-PCBA-APP-I, lines 20–31 and 51–59: "Constraint R is 0–100, saturation/cutoff/threshold are 0–1, and sat = 8.5 is identified as a mixed-scale error whose component value is 0.85."

Disposition: PRESERVE_AS_PROFILE_CONFLICT.

2.1. Analysis. This is the audit's cleanest conflict and its most instructive, because the disagreement is numeric and total. The Guide's page 6 places the constraint quantity R on a 0.1–10.0 scale. Appendix I's lines 20–31 place the same-named quantity on a 0–100 scale. These ranges are not related by a stated conversion; they simply coexist under one symbol. Worse, the Guide's worked example on page 12 uses sat = 8.5 as a component threshold, while Appendix I—in the very lines that fix saturation, cutoff, and threshold quantities to the interval 0–1—identifies that same 8.5 as a *mixed-scale error* whose intended component value is 0.85. One source's worked pedagogy is another source's diagnosed bug.

Why does this matter beyond tidiness? Because every quantitative operation downstream of these definitions is scale-sensitive. Consider the innocent formula on Guide page 6: raw attention is V/R. If R is drawn from 0.1–10.0, the quotient for a given V differs by up to a

factor of ten from the quotient computed with R drawn from 0–100—and at the low end of the Appendix I scale, $R = 0$ is admissible, making the quotient undefined where the Guide's floor of 0.1 guaranteed it finite. A threshold comparison such as "is saturation above sat?" flips its truth value depending on whether the comparand lives in $[0, 1]$ or $[0, 10]$: a fraction of 0.9 is above a threshold of 0.85 and far below a threshold of 8.5. A system that ingests both sources without typing the scales will, with certainty rather than mere risk, compare quantities across scales and produce numbers that are wrong by an order of magnitude while looking perfectly well formed. Mixed-scale failure is silent: nothing about the floating-point value 8.5 announces which scale it came from.

The sat = 8.5 case deserves emphasis as a documented instance of the failure actually occurring *inside the source corpus itself*. This is not a hypothetical about what a careless integrator might do; the Guide's own worked example, per Appendix I's reading, already committed the mixed-scale error that the framework's downstream users would be warned against. When a specification's teaching material contains the bug its appendix diagnoses, the specification cannot be said to carry one coherent scale system, and any implementation claiming to inherit "the" scale system is choosing among contradictory parents without saying so.

It is worth being precise about what the third evidence record does and does not assert. Appendix I *identifies* sat = 8.5 as a mixed-scale error and states that its component value is 0.85. That is a reading by one source of another source's text—a diagnosis, with a proposed intended value—not a repair the audit endorses. The audit's posture is deliberately one level up: it records that the Guide asserts 8.5, that Appendix I calls 8.5 an error with intended value 0.85, and that these positions conflict. A harmonizing editor would have adopted Appendix I's diagnosis, rewritten the Guide's example, and moved on; the audit instead preserves both, because adopting the diagnosis would presuppose exactly the scale-priority question (which source's scale system governs?) that the conflict exists to keep open. Unit-confusion failures are a storied genre in engineering practice generally, and their common anatomy is visible here in miniature: two communities of practice, each internally consistent, each producing artifacts that interoperate numerically while diverging semantically, with no type system standing between them. The lesson the broader engineering literature draws—encode the unit in the type, not in the documentation—is precisely the candidate rule's content.

2.2. Disposition and candidate rule.

Rule 4.4 (Candidate rule for Conflict 1). "Q10 score, Q100 constraint score, normalized fraction, alpha dose, activation projection, and effective resistance are distinct quantity types; no silent conversion is allowed."

The disposition `PRESERVE_AS_PROFILE_CONFLICT` means the candidate does not decide whether R "really" lives on 0.1–10.0 or 0–100. Instead the type system makes the question unaskable in ambiguous form: a value carries `Q10` or `Q100` or `FRACTION` as its unit kind, the quantity kind constrains which unit kinds are admissible, range checks are enforced per unit ($[0, 10]$ for `Q10`, $[0, 100]$ for `Q100`, $[0, 1]$ for `FRACTION`), and no arithmetic path converts between them without an explicit, provenance-carrying operation. Under this rule the historical sat = 8.5 is not silently repaired to 0.85; it is *unrepresentable* as a saturation fraction, because a fraction-typed field rejects 8.5 at validation. The conflict is preserved in the record and neutralized in the types.

3. Conflict 2: The Full-Scale Profile

PRESERVED CONFLICT 4.5 (`OHM-CONFLICT-FULL-SCALE-PROFILE`; kind `INCOMPATIBLE_CALIBRATION_METHO`
The evidence records are:

- SRC-CAS-OHM-GUIDE-01, PDF page 6: "I_{FS} is a selected board-specific full-scale attention value."
- SRC-CAS-OHM-ADDENDUM-01, paragraph 30 and operationalization table row 13: "I_{FS} is the per-test-point 95th percentile of absolute directional projection over a fixed hashed corpus."
- SRC-CAS-OHM-PCBA-APP-I, lines 40–49 and 263–267: "I_{FS} is measured once per path at drive rail and constraint floor and does not transfer."

Disposition: `SEPARATE_NAMED_PROFILES`.

3.1. Analysis. I_{FS} is the full-scale reference against which raw response is normalized; every Q10 display value and every Q4 band assignment (Conflict 4.17) is a ratio with I_{FS} in the denominator. The three sources give three procedures for obtaining it, and the procedures differ in every property that matters for a denominator.

The Guide's I_{FS} is *declared*: a board-specific full-scale value that someone selected. It is a design decision, not a measurement; it has no sampling distribution, no uncertainty, and no dependence on any corpus. The addendum's I_{FS} is *estimated*: the 95th percentile of the absolute directional projection, computed per test point, over a fixed hashed corpus. It is a statistic; it inherits sampling variability from the corpus, changes if the corpus changes, and is pinned down only because the corpus is content-addressed ("fixed hashed"). The Appendix I I_{FS} is *probed at an extreme operating point*: measured once per path with drive at its rail and constraint at its floor, and—the source says so explicitly—it does not transfer.

These are not three descriptions of one number. In general, a declared design ceiling, an empirical 95th percentile over ordinary traffic, and a maximal excursion under rail/floor forcing will be three different magnitudes, and nothing in any source relates them. The consequences propagate through every normalized quantity. The same raw response yields different Q10 values under the three denominators, hence potentially different Q4 bands, hence different region classifications where thresholds are compared, hence different downstream diagnostics. Two reports both stating "$I_Q = 7.8$" agree on nothing unless they also agree on which full-scale procedure produced the denominator—and under the corpus-percentile method, on which corpus, at which percentile, and at which test point; and under the rail/floor method, on which path, since that source warns the value does not transfer even between paths.

The transferability warning is the sharpest edge. A declared full-scale might plausibly be reused across a family of boards; a rail/floor probe is explicitly local. Averaging the three, or substituting one when another is unavailable, would produce a denominator with no defined meaning at all.

The three methods also differ in their failure modes, which is the deeper reason they cannot be interchanged even "approximately." A declared design full-scale fails by becoming stale: the board changes, the declaration does not, and nothing in the number signals the divergence—only a drift-check discipline external to the value can catch it. A corpus percentile fails by corpus shift and by tail sensitivity: the 95th percentile is an order statistic of the projection distribution's upper tail, exactly the part of a distribution most affected by a handful of outlying documents, so an apparently minor corpus revision can move the

denominator while every individual document looks innocuous. A rail/floor probe fails by extrapolation: it characterizes the system at an extreme operating point that ordinary operation never visits, so its relevance to mid-range normalization is an assumption, not an observation. Three methods, three distinct threat models, three different maintenance obligations. A framework that let one number stand for all three would not merely blur magnitudes; it would blur which failure its operators needed to be watching for. Keeping the methods separately named keeps their maintenance disciplines separately owned.

3.2. Disposition and candidate rule.

RULE 4.6 (Candidate rule for Conflict 2). "Board-selected, corpus-percentile, and rail/floor full-scale procedures remain different calibration methods with different profile digests."

The disposition `SEPARATE_NAMED_PROFILES` is implemented as the `IFSMethod` enumeration in the candidate code (`RAIL_AND_FLOOR_PER_PATH`, `CORPUS_PERCENTILE_PER_TEST_POINT`, `DECLARED_DESIGN_FULL_SCALE`), examined field by field in Chapter 5. Because the method is a field of the calibration profile and the profile's canonical digest covers it, two normalizations performed under different full-scale methods are performed under different profile digests, and the profile-relativity principle (nothing transfers across profiles without a stated comparison rule) blocks their naive comparison mechanically. No method is anointed correct; all three are representable; none is substitutable for another.

4. Conflict 3: The Dimensional Bridge

PRESERVED CONFLICT 4.7 (`OHM-CONFLICT-DIMENSIONAL-BRIDGE`; kind `UNIT_UNDERSPECIFICATION`). The evidence records are:

- SRC-CAS-OHM-GUIDE-01, PDF pages 6, 10, and 18: "V, R, and I are design-scale quantities related by $I_{\text{pred}} = V/R_{\text{total}}$."
- SRC-CAS-OHM-ADDENDUM-01, operationalization table rows 1, 2, 5, and 12: "V is steering alpha, I is directional activation projection, contextual R is inferred from a paired delta, and I_{pred} still uses V/R_{total}."
- SRC-CAS-OHM-PCBA-APP-I, lines 90–108 and 326–339: "The relation is presented as testable design/calibration arithmetic, not as physical voltage or a measured live-model law."

Disposition: `REQUIRE_CALIBRATED_UNIT_BRIDGE`.

4.1. Analysis. Between the Guide and the addendum, the symbols keep their algebra and change their referents. On the Guide's pages, V, R, and I are design-scale scores—abstract magnitudes on the score scales of Conflict 4.3—and $I_{\text{pred}} = V/R_{\text{total}}$ is arithmetic among scores. In the addendum's operationalization rows, V has become a steering alpha (an intervention magnitude applied to a model), I has become a directional activation projection (an observed quantity in a model's representation space), and contextual R is inferred from a paired delta. Yet the prediction equation is carried over unchanged: I_{pred} still uses V/R_{total}.

Dimensional analysis makes the gap precise. For $\hat{I} = V/R_{\text{total}}$ to be dimensionally coherent with V in alpha units and I in activation units, R_{total} must carry units of alpha per activation [34]. But the R components the sources supply are score-derived—dimensionless design-scale numbers from Conflict 4.3's (mutually conflicting) ranges. Dividing an alpha by a dimensionless score yields alpha, not activation: the formula, as inherited, predicts a quantity in the wrong dimension. Nothing in any source supplies the missing conversion—the

calibrated bridge that would turn a score-derived resistance into a quantity with alpha-per-activation units. This is why the record's kind is `UNIT_UNDERSPECIFICATION` rather than a direct conflict: no single source asserts a falsehood; jointly they fail to define the one constant that would make the central equation mean anything.

Appendix I's contribution is a guard on interpretation: the relation is testable design/calibration arithmetic, not physical voltage and not a measured live-model law. This matters in both directions. It blocks the physical reading (nothing here is electrically a volt, an ampere, or an ohm; the candidate's unit ontology makes those forbidden aliases), and it blocks the empirical reading (writing the equation does not make it a law of any live model; whether any (a, y) pair from a real system satisfies a linear relation is exactly what the registered hypotheses of Chapter 12 where present leave to controlled experiments).

The double negation in Appendix I's framing—not physical, not yet empirical—leaves the equation in a precisely intermediate status that the candidate's vocabulary makes explicit: it is *design arithmetic*, a relation among declared and calibrated quantities that is testable in principle once its units are fixed, and nothing more until tested. The physical exclusion is enforced in the unit ontology as a forbidden-alias rule: VOLT, AMPERE, and OHM are not merely absent from the unit enumeration but affirmatively barred as labels for model quantities, so that the analogical vocabulary of the sources (V, I, R) can never quietly reify into a claim of electrical measurement. The empirical exclusion is enforced by the response model's admissibility conditions (Chapter 6): the equation may be evaluated only with a calibrated, uncertainty-carrying, held-out-validated effective resistance, which as of this writing does not exist for any live system.

The practical stakes are high because $\hat{I} = V/R_{\text{total}}$ is the framework's load-bearing equation: it is the prediction whose residuals drive the diagnostics of Conflict 4.15 and whose network compositions drive Conflict 4.13. An equation with an undefined dimension in its denominator contaminates everything built on it. Any residual computed from such a prediction is a difference between an observed activation and a quantity of indeterminate dimension; any series sum of such resistances aggregates numbers whose common unit was never established.

4.2. Disposition and candidate rule.

RULE 4.8 (Candidate rule for Conflict 3). "Effective resistance may be used only with an explicit alpha-unit per activation-unit calibration; constraint scores never masquerade as that quantity."

Under `REQUIRE_CALIBRATED_UNIT_BRIDGE`, the candidate splits the overloaded R into types that cannot be confused: `CONSTRAINT_SCORE` (unit `Q10` or `Q100`) for the score-scale quantity, and `EFFECTIVE_RESISTANCE` (unit `ALPHA_PER_ACTIVATION`, strictly positive) for the quantity admissible in the prediction $\hat{y} = a/r_{\text{eff}}$. There is no constructor from one to the other. An effective resistance comes into existence only through a declared fit on a calibration split under a full calibration profile, carrying uncertainty and a role tag (Chapter 6); a constraint score can never be passed where an effective resistance is required, because the unit kinds differ and unit checking is mandatory. The Guide's equation shape is thereby preserved as a *hypothesis about a calibrated quantity* while the dimensional hole that made its inherited form unmeaning is closed by type-level refusal rather than by silent reinterpretation.

5. Conflict 4: The Gain Definition

PRESERVED CONFLICT 4.9 (`OHM-CONFLICT-GAIN-DEFINITION`; kind `DISTINCT_MATHEMATICAL_PROFILES`). The evidence records are:

- SRC-PCBA-ENGINEER-20260726, PDF page 56 (book page 40) and PDF page 153 (book page 137): "CJT beta is defined by Fisher–Rao small- or large-signal displacement ratios on the open simplex."
- SRC-CAS-OHM-ADDENDUM-01, operationalization table row 6: "Beta is operationalized as a local Jacobian/projection derivative between cause and effect test points."
- SRC-CAS-OHM-GUIDE-01, PDF page 13: "Stage beta values multiply in cascades while attention flow is local at each stage."

Disposition: `NO_UNPROVED_IDENTIFICATION`.

5.1. Analysis. The symbol β names a gain—an amplification factor between a cause stage and an effect stage. The PCBA engineering book defines it geometrically: a ratio of Fisher–Rao displacements on the open probability simplex, in small-signal or large-signal form. The addendum operationalizes it analytically: a local Jacobian/projection derivative between cause and effect test points—a directional sensitivity of one projection with respect to another in a model's representation space. The Guide, meanwhile, uses β compositionally: stage values multiply along cascades, while attention flow remains local at each stage.

The Fisher–Rao definition and the Jacobian/projection definition are different mathematical objects in almost every respect that could matter. They live on different spaces: the open simplex with its information geometry, versus activation space with a Euclidean projection structure. They have different invariance properties: Fisher–Rao displacement is invariant under the simplex's natural reparameterizations, while a projection derivative depends on the chosen direction vectors at both test points. They respond differently to signal size: the book itself distinguishes small- and large-signal forms, while a local derivative is by construction a small-signal object. There is no theorem in any of the sources—and the record is explicit that none is currently supplied—identifying the two, nor even a coordinate map under which such an identification could be stated.

The Guide's cascade rule then quietly raises the stakes. Multiplying stage gains along a cascade is exactly the operation that is *most* sensitive to which definition is in force, because composition of local derivatives follows a chain rule only when adjacent stages' coordinates match, and composition of Fisher–Rao displacement ratios has no chain rule at all without further structure. Under the multiplication rule, an error of identification does not stay local: it compounds geometrically with cascade depth. Three stages of a modest 30% definitional discrepancy are a factor of 2.2; the deeper the cascade, the worse. (Those numbers are illustrative arithmetic about error propagation, not measurements.)

The general lesson is about symbol overloading across document genres. A definitional text (the book), an operationalization (the addendum), and a usage idiom (the Guide's cascades) each pulled β toward their own needs. Each is internally reasonable. Jointly they leave "the gain" denoting two unbridged mathematical profiles plus a composition rule that presupposes the bridge.

5.2. Disposition and candidate rule.

RULE 4.10 (Candidate rule for Conflict 4). "Fisher–Rao gain and local Jacobian/projection gain are separately named profiles until a theorem or experiment supplies a coordinate bridge."

NO_UNPROVED_IDENTIFICATION is the mathematical analogue of Conflict 4.5's disposition: one name, several referents, therefore several names. The candidate types gain magnitudes as dimensionless quantities bound to a declared method identity, so a Fisher–Rao displacement gain and a Jacobian/projection gain carry different method identifiers and cannot be pooled, averaged, or multiplied together by accident. The gain composition operation (Chapter 9 where present) is capability-gated and requires adjacent coordinate agreement precisely so that the Guide's cascade multiplication can be *tested* as the registered hypothesis CLM-CAS-OHM-GAIN-COMPOSITION rather than assumed. If a future theorem or a controlled experiment supplies the bridge, the identification becomes a recorded, evidenced event; until then the two profiles remain separate by name and by type.

6. Conflict 5: ASOM Axis Overload

PRESERVED CONFLICT 4.11 (OHM-CONFLICT-ASOM-AXES; kind MNEMONIC_OVERLOAD). The evidence records are:

- SRC-CAS-OHM-GUIDE-01, PDF page 11: "ASOM is explicitly session-derived and overlays Starved/Aligned/Oversaturated/Misaligned onto cutoff/active/saturation/breakdown."
- SRC-CAS-OHM-ADDENDUM-01, paragraphs 14–15 and ASOM table rows 1–4: "The addendum uses dose-response signatures plus effect-axis outcomes and warns that flat response has multiple causes."

Disposition: SPLIT_RESPONSE_AND_SEMANTIC_AXES.

6.1. Analysis. ASOM—Starved, Aligned, Oversaturated, Misaligned—is, per the Guide's own page 11, session-derived: a mnemonic overlay mapping four evocative words onto the four dose-response regions cutoff, active, saturation, and breakdown. The overlay is pedagogically appealing because it compresses two questions into one word. But the two questions are different in kind. *Where on the dose-response curve is the system operating?* is a question about response shape: is response flat at low dose (cutoff), rising (active), flat at high dose (saturation), anomalous (breakdown)? *Is the effect the intended one?* is a question about semantic outcome: did the intervention move the targeted quantity in the targeted way, the opposite way, or off-axis? The mnemonic fuses them: "Aligned" answers the second question with the vocabulary slot of the first, and "Misaligned" imports a semantic verdict into what is structurally a region label.

The addendum resists the fusion. Its paragraphs 14–15 and ASOM table rows 1–4 keep dose-response signatures and effect-axis outcomes as separate columns, and issue the warning that unravels the overlay: flat response has multiple causes. A flat dose-response curve is compatible with cutoff (dose below threshold), with saturation (dose beyond the linear region), with a wrong direction vector (the probe measures an axis the intervention does not move), with genuine insensitivity, and with instrument failure. A mnemonic that reads "flat" as "Starved" has silently chosen one cause out of five. Symmetrically, an active region says nothing about intent: a system can respond briskly and in exactly the wrong direction, which the fused vocabulary can only render as a paradox ("actively misaligned") rather than as the ordinary factored state (region ACTIVE, outcome OPPOSITE).

There is also an honest-ignorance problem the mnemonic cannot express. Four words partition every observation into four verdicts; there is no ASOM cell for "the response region is known but the semantic outcome has not been assessed," although that is the *normal* state of a fresh measurement: region classification needs only the dose-response sweep, while outcome classification needs an independent determination of what the intervention was supposed to do and whether it did it. A vocabulary without an unassessed value forces premature verdicts

by construction—every observation must land somewhere, so it lands in whichever verdict its shape most resembles. The two-axis scheme, by contrast, admits `UNASSESSED` on each axis independently, so a system can honestly report "region ACTIVE, outcome UNASSESSED" and downstream consumers can distinguish "verified as intended" from "not yet examined"—a distinction on which, for a framework aimed at auditing, essentially everything depends.

The failure kind, `MNEMONIC_OVERLOAD`, names a pattern endemic to maturing frameworks: a teaching aid, introduced honestly as such ("session-derived"), hardens through repetition into an analytical category, and downstream text begins drawing inferences from the mnemonic's connotations rather than from the measurements the mnemonic was compressing. The cost is not hypothetical. A classifier emitting "Misaligned" as a single-axis verdict makes a semantic claim on the strength of a response shape; on a system under audit, that is precisely the kind of unearned conclusion the whole evidence discipline exists to prevent.

6.2. Disposition and candidate rule.

RULE 4.12 (Candidate rule for Conflict 5). "Dose-response region and semantic outcome are classified independently; active does not imply intended and non-ohmic does not imply misaligned."

Under `SPLIT_RESPONSE_AND_SEMANTIC_AXES`, the candidate carries two enumerations: dose-response region (`UNASSESSED`, `CUTOFF`, `ACTIVE`, `SATURATION`, `BREAKDOWN`, `NON_OHMIC`, `UNSTABLE`—the source's breakdown region retained) and semantic outcome (`UNASSESSED`, `INTENDED`, `OPPOSITE`, `OFF_AXIS`). Every classification is a pair; no region value mechanically determines an outcome value. ASOM survives—but demoted to what the Guide said it was: a derived display label computed from the pair after the fact, never an input, never an axis, and undefined whenever the pair does not fit its four cells. The mnemonic keeps its pedagogy and loses its authority.

7. Conflict 6: Network Rules

PRESERVED CONFLICT 4.13 (`OHM-CONFLICT-NETWORK-RULES`; kind `VALIDATION_STATUS_CONFLICT`). The evidence records are:

- SRC-CAS-OHM-GUIDE-01, PDF pages 13–14 and 18: "Series addition, reciprocal parallel aggregation, and beta multiplication are presented as working network rules."
- SRC-CAS-OHM-PCBA-APP-I, lines 206–254: "The same arithmetic is applied to paths, with a separate warning that co-present guardrails are not parallel routes and flow remains local."
- SRC-CAS-OHM-ADDENDUM-01, paragraphs 16–19 and falsification table rows P2–P3: "Gain composition and parallel conductance are preregistered predictions whose rules must be dropped if the tests fail."

Disposition: `CAPABILITY_GATE_EACH_OPERATION`.

7.1. Analysis.

Here the three sources agree on the arithmetic and disagree on its standing. The rules in question are the circuit-theory triad: series resistances add; parallel resistances aggregate by reciprocals; stage gains multiply. The Guide presents them as *working rules*—usable machinery. Appendix I applies the same arithmetic to paths but attaches a structural warning: co-present guardrails are not parallel routes, and flow remains local. The addendum reclassifies the rules entirely: gain composition and parallel conductance are *preregistered predictions*, indexed as falsification rows P2 and P3, whose rules must be dropped if the tests fail.

A `VALIDATION_STATUS_CONFLICT` is in some ways more dangerous than a numeric conflict, because the disputed content circulates freely. A reader of the Guide alone will compute series equivalents and cascade gains as routinely as an electrical engineer computes them for resistor networks—for resistors, the rules are consequences of established physics. Here they are borrowed shape. Whether ablating one of two "parallel" safety mechanisms changes system response according to reciprocal aggregation is an empirical question about a particular system, and the addendum has the epistemically correct posture: state the prediction, register the test, and commit in advance to dropping the rule on failure.

Appendix I's warning identifies the specific modeling error most likely to produce false parallel structure. Two guardrails that are co-present—both active on the same trajectory—are not thereby alternative routes between the same endpoints. Parallel aggregation presupposes genuine alternatives: distinct branches with common endpoints such that flow divides among them. Co-presence is a weaker, easier-to-observe relation, and conflating it with parallelism imports an aggregation formula into a topology that does not support it. The further warning that "flow remains local" blocks the complementary error at the series end: chaining stage arithmetic into a claim about a global conserved flow through the chain, when nothing establishes a conservation law for attention flow across stages.

The record therefore preserves a three-way disagreement about *how much is known*: working rule (Guide), conditionally applicable arithmetic with topological caveats (Appendix I), falsifiable conjecture (addendum). An implementation must choose a posture, and the choice is consequential: treat conjecture as machinery and every downstream aggregate silently inherits an unvalidated assumption; treat machinery as forbidden and legitimate structural exploration is blocked.

The capability-gate disposition is best understood as a third posture that dominates both extremes. Against the permissive posture, it ensures that no aggregate can exist without a machine-checked record of the preconditions it satisfied and—equally important—of the epistemic label it carries: every network result in the candidate is stamped as an unauthenticated structural research calculation, bound to its trust mode, its authority manifest identity, and its declared covariance limitation, so the result cannot later be quoted shorn of its status. Against the prohibitive posture, it preserves the ability to *compute* the structural quantities the registered hypotheses will need: testing whether parallel conductance predicts ablation response (hypothesis row P3) requires computing the reciprocal aggregate to have a prediction to test, and a framework that forbade the arithmetic outright could never run its own falsification program. The gate thus operationalizes the addendum's preregistration posture in code: the arithmetic runs, its outputs are labeled conjectural, and the label can only be removed by the registered experiments—never by repetition or lapse of memory.

7.2. Disposition and candidate rule.

Rule 4.14 (Candidate rule for Conflict 6). "No series, parallel, or gain composition is enabled without coordinate, unit, topology, active-region, and evidence compatibility under the exact profile."

`CAPABILITY_GATE_EACH_OPERATION` threads the needle. The operations exist in the candidate—but only as unauthenticated structural research calculations, each behind a gate that checks precisely the preconditions the sources' caveats identify: compatible coordinates (against the false-parallelism error), a single alpha-per-activation unit (against Conflict 4.7), genuine acyclic topology with distinct endpoints (against co-presence masquerading as parallelism), active-region components with adequate linearity (against composing outside the region where

a linear rule could even be a candidate), and complete evidence under one exact profile. Every aggregate binds its child order, topology, and the declared covariance limitation, and duplicate or cyclic paths fail closed. The addendum's falsification posture is preserved intact: rows P2–P3 correspond to registered hypotheses, and the gates ensure that structural results produced meanwhile are labeled as exactly that—structure, not validated network law.

8. Conflict 7: Residual Identification

Preserved Conflict 4.15 (`OHM-CONFLICT-RESIDUAL-IDENTIFICATION`; kind `CAUSAL_OVERINTERPRETATIO`
The evidence records are:

- SRC-CAS-OHM-GUIDE-01, PDF pages 10 and 16: "Residual sign is used to suggest hidden opposition, extra amplification, or hidden resistance, while the page also says residual alone is insufficient."
- SRC-CAS-OHM-PCBA-APP-I, lines 171–202 and 260–277: "R_{hidden} is called a residual rather than a component, but positive residual on a safety board is treated as a bypass and saturated-stage residuals are meaningless."
- SRC-CAS-OHM-ADDENDUM-01, paragraph 19 and falsification table row P4: "Hidden content is last in the diagnostic walk and residual validity requires independently identified causes above a shuffled-control baseline."

Disposition: `NON_IDENTIFYING_DIAGNOSTIC_ONLY`.

8.1. Analysis. A residual is a subtraction: observed response minus predicted response. The temptation the record documents is the promotion of that subtraction into an identification—reading the sign of δ as the presence of a named hidden mechanism. The Guide exhibits the temptation and its own antidote on the same pages: residual sign "suggests" hidden opposition, extra amplification, or hidden resistance, while the same page concedes that residual alone is insufficient. Appendix I shows the slide completed in two steps: first terminological (R_{hidden} is "called a residual rather than a component"—the name of a model-mismatch symptom is dressed as a circuit element), then inferential (a positive residual *on a safety board* is treated as a bypass). It simultaneously notes a condition under which residuals carry no information at all: at a saturated stage, where predicted response is flat, residuals are meaningless.

Why is the identification invalid? Because the residual is computed against a fitted model, and every deficiency of that model lands in the residual indistinguishably. A nonzero δ is compatible with: a genuine unmodeled mechanism; misfit of r_{eff} (wrong region, drifted calibration); a wrong or degraded direction vector; ordinary noise at the recorded uncertainty; a stale profile; or an error in the observation chain. Sign does not separate these. Inferring "bypass" from "positive residual on a safety board" is an inference to the most alarming member of an equivalence class the data cannot resolve—and its cost profile is asymmetric in the worst way, since a bypass claim about a safety mechanism triggers exactly the kind of downstream action that should require the *most* evidence, on the strength of a statistic that here carries the least. The general point is Pearl's: causal identification requires an explicit causal model and identifiable interventions, not signs of differences from a curve fit [27].

The addendum again supplies the disciplined posture, in two parts. Hidden content is *last* in the diagnostic walk: before entertaining a hidden mechanism one must exhaust the identifiable causes—region, calibration, direction, noise, staleness. And residual validity is itself an empirical claim with a registered test (falsification row P4): residual diagnostics must localize independently identified causes above a shuffled-control baseline. If shuffled

controls localize as well as real residuals, the diagnostic is noise and its causal vocabulary must be withdrawn.

The "diagnostic walk" ordering deserves a further word, because it encodes a cost asymmetry that pure statistics does not capture. The identifiable causes ahead of hidden content in the walk—wrong region, stale calibration, degraded direction, noise—are all *checkable* at bounded cost: each has an associated record in the profile or the observation chain that can be examined directly. "Hidden mechanism," by contrast, is unbounded: it names the complement of everything modeled, so no finite check can confirm it and almost any anomaly can be retrofitted to it. Placing it last is therefore not a mere tidiness preference; it is the only ordering under which the diagnostic process terminates against checkable facts before reaching the unfalsifiable residue. Inverting the walk—reaching for the hidden mechanism first, as the "bypass" reading does—converts every calibration bug and every noisy run into apparent evidence of concealed behavior, which for a safety-relevant audit is the most corrosive possible failure: it manufactures alarming findings at exactly the rate of the auditor's own instrumentation defects.

8.2. Disposition and candidate rule.

Rule 4.16 (Candidate rule for Conflict 7). "Residuals retain propagated uncertainty and may rank follow-up probes; they never prove a hidden policy, bypass, or cause. Zero or zero-compatible observations are censored."

`NON_IDENTIFYING_DIAGNOSTIC_ONLY` assigns residuals their honest job. In the candidate, a residual exists only for an admitted active-region prediction, with instrumented held-out observations, matching test points and profile, calibrated resistance, and propagated uncertainty; at zero or zero-compatible predicted response—the saturated-stage case Appendix I flags—no residual quantity is emitted at all (the observation is censored with a recorded reason rather than divided into a meaningless ratio). What a residual may do is *rank follow-up probes*: allocate investigative attention. What it may never do is appear as the proof of a hidden policy, bypass, opposition, or cause. The registered hypothesis CLM-CAS-OHM-RESIDUAL-LOCALIZATION keeps even the ranking use on probation: blinded implanted faults, matched controls, and withdrawal of causal language on failure.

9. Conflict 8: Q4 Boundaries

Preserved Conflict 4.17 (`OHM-CONFLICT-Q4-BOUNDARIES`; kind `PRECISION_UNDERSPECIFICATION`). The evidence record is:

- SRC-CAS-OHM-GUIDE-01, PDF page 7: "Q4 is displayed with two-decimal bands 0.00–2.49, 2.50–4.99, 5.00–7.49, and 7.50–10.00 without a rule for higher-precision boundary values."

Disposition: `DEFINE_EXACT_HALF_OPEN_BINS`.

9.1. Analysis. This is the audit's smallest conflict and the easiest to underestimate. The Guide's Q4 quantizer displays four bands with two-decimal endpoints: 0.00–2.49, 2.50–4.99, 5.00–7.49, 7.50–10.00. As a display convention for two-decimal values this is complete. As a specification of a function on finer-grained inputs it is not, because the bands as written do not cover the line: the value 2.495 falls between 2.49 and 2.50, in no band. What happens to it depends on an unstated rounding rule—and the two natural resolutions disagree. Round-then-bin sends 2.495 to 2.50 (round-half-up) and thence to the second band; bin-by-truncation keeps it under 2.50 in the first. The specification gap is only two decimal places wide, but a

classification function undefined on a set of inputs, with plausible readings that disagree on those inputs, is not one function; it is a family of functions wearing one name.

The failure has an aggravating property the other conflicts lack: it sits exactly where decisions concentrate. Band boundaries are where a quantizer's output changes, so any downstream logic keyed to bands—reporting tiers, threshold alarms, comparisons across runs—changes behavior precisely at the values the specification leaves undefined. Systems accumulate values near their thresholds (thresholds are typically set where the interesting mass is), so the undefined set is visited in practice, not just in principle. Worse, an implementation that rounds before binning couples the classification to display precision: increase the displayed decimals and boundary values silently change bands, so the same measurement classifies differently depending on a formatting choice. A quantization band should be a fact about a value, not about its printout.

9.2. Disposition and candidate rule.

Rule 4.18 (Candidate rule for Conflict 8). "The successor uses exact unrounded comparison against 2.5, 5, and 7.5; normalization may not round a value across a band boundary."

`DEFINE_EXACT_HALF_OPEN_BINS` completes the specification in the only way that composes cleanly: half-open intervals $[0, 2.5)$, $[2.5, 5)$, $[5, 7.5)$, $[7.5, \infty)$ evaluated on the exact pre-rounding value, with boundary points assigned to the upper band by the half-open convention. The candidate implements the comparison over exact rational arithmetic on the retained pre-clipping ratio, so no representable input is ambiguous and display rounding, which happens afterward and never re-enters classification, cannot move a value across a boundary (Chapter 8 where present states this as a proposition). The Guide's two-decimal table remains exactly what it was—a correct display of the bands at two decimals—while the function it under-specified is now total.

10. Interactions Among the Conflicts

The eight records are registered independently, but they are not independent in effect. Reading them as a system reveals a dependency structure that explains why the candidate's dispositions had to be typed constraints rather than local patches: several conflicts compound, and a resolution of one that ignored its neighbors would leak.

Conflict 1 feeds Conflict 3. The dimensional bridge problem is aggravated by the scale problem beneath it. Even if one resolved Conflict 4.7 by fiat—declaring that R_{total} shall carry alpha-per-activation units obtained from some conversion of constraint scores—the conversion would have to know which score scale it was converting *from*, and Conflict 4.3 establishes that the sources disagree about exactly that. A bridge from an ambiguous abutment is ambiguous end to end: the same nominal constraint value of, say, 5 would bridge to resistances differing by an order of magnitude depending on whether it was read on the 0.1–10.0 or the 0–100 scale. The candidate rule for Conflict 1 (distinct quantity types, no silent conversion) is therefore a load-bearing prerequisite for the candidate rule of Conflict 3 (explicit calibrated bridge only): the bridge requirement is meaningful because the scales it would bridge from are typed.

Conflict 2 feeds Conflict 8. Q4 band assignment is a comparison of a normalized ratio against fixed cut points, and the ratio's denominator is I_{FS}. Conflict 4.5 means the denominator is method-relative; Conflict 4.17 means the comparison was boundary-underspecified. The two compound multiplicatively: a value near a band boundary under one full-scale method may sit far from any boundary under another, so the boundary rule's importance

is itself method-dependent, and no statement of the form "this response is Q3" is complete without both the exact bin convention (Conflict 8's disposition) and the full-scale method with its parameters (Conflict 2's disposition) attached. This is why the candidate binds the `IFSMethod` into the profile digest and computes bands from exact pre-rounding ratios: each disposition closes one half of a two-part ambiguity in every quantized report.

Conflict 3 underlies Conflicts 6 and 7. Series addition and reciprocal parallel aggregation are operations *on resistances*; the residual is a difference between an observation and a prediction *made with a resistance*. Until the dimensional bridge exists, the operands of Conflict 4.13's arithmetic and the predictions behind Conflict 4.15's diagnostics are quantities of undefined dimension. The capability gates on network operations therefore check unit kind before topology—an aggregate of score-typed pseudo-resistances is rejected at the unit check, before the question of whether its branches are genuinely parallel even arises. Likewise the residual pipeline requires a calibrated resistance with the correct unit as a precondition for emitting any residual at all. Had the audit "resolved" Conflict 3 by tolerating score-derived resistances, the gates of Conflicts 6 and 7 would be guarding operations whose operands were already meaningless.

Conflict 4 arms Conflict 6. The Guide's cascade rule multiplies stage gains; Conflict 4.9 shows "gain" has two unbridged definitions. The composition gate's requirement of adjacent coordinate agreement and a single declared gain method is what prevents a cascade from mixing Fisher–Rao displacement ratios with Jacobian derivatives—a mixture that would be numerically computable (both are dimensionless) and semantically void. This is the general pattern: dimensionless quantities are the most dangerous operands in the system, because unit checking alone cannot distinguish them, so method identity must travel with the value.

Conflict 5 disciplines Conflict 7's vocabulary. The residual temptation ("positive residual on a safety board is a bypass") and the mnemonic temptation ("flat response is Starved") are the same move at different altitudes: collapsing a measurement-shaped observation into a semantic verdict without the evidence that would distinguish the verdict from its alternatives. The split-axes disposition gives the system a place to put honest ignorance—semantic outcome `UNASSESSED`—and the residual disposition uses the same discipline: a residual may flag *that* something is unexplained, never *what* explains it. Both dispositions route the missing determination to the same destination: independent evidence under registered protocols, not connotation.

Adverse Finding 4.19. The eight conflicts form a dependency web in which the scale and bridge conflicts (1 and 3) are upstream of nearly everything quantitative, the method-identity conflicts (2 and 4) are upstream of all cross-context comparison and composition, and the interpretation conflicts (5 and 7) govern what may be said about the results. No single conflict could have been resolved in isolation without either presupposing a resolution of its upstream neighbors or silently exporting its ambiguity downstream. This is the structural justification for the audit's uniform strategy: preserve every conflict, and encode every disposition at the type level, where the dependencies are enforced in composition order automatically.

11. The Record as a Whole

The evidence record closes with summary fields that are as much a part of the audit as the conflicts themselves, and we reproduce them faithfully.

Conflict count: 8. The count is a completeness assertion with a precise scope: these eight records are what the inspection of the cited sources, at the cited locators, under the declared

inspection methods, registered as conflicts or under-specifications. It is not an assertion that the sources contain exactly eight problems; it is an assertion that the audit found and preserved these eight, and that downstream artifacts claiming to rest on the audit inherit exactly this list—no silent additions, no silent deletions.

Gate effect: `KEEP_OPEN`. The audit feeds a gated development process, and a source audit that discovered eight unresolved conflicts could not honestly close anything. `KEEP_OPEN` records that the audit's effect on its gate is to hold it open: the conflicts are preserved, not resolved; the registered hypotheses that would resolve the resolvable ones (network rules, residual validity, gain composition) remain untested; and no downstream claim may cite the audit as if it had settled what it explicitly left unsettled.

The claim boundary. The record's own boundary statement reads: "This record preserves source conflicts and under-specifications. It does not validate CAS-Ohm empirically, identify physical quantities, license a product claim, or authorize model intervention." The four exclusions are worth reading as the four ways an audit is most easily misused. An audit can be waved as empirical validation ("the sources were audited, so the model works"); the boundary denies it. It can be read as physical identification ("V and R were audited, so they measure something physical"); denied. It can be marketed ("audited" as a product adjective); denied. And it can be treated as operational authorization ("the audit approves intervening on a model"); denied. The record also carries its status fields—admission ring 4, license status NOT LICENSED, disclosure status HELD—so that the boundary travels with the bytes.

The status fields merit unpacking, since they recur on every artifact in the candidate's evidence chain and their meanings are load-bearing. Admission ring 4 places the record at the outermost ring of the project's admission scheme: material that may be worked on and cited internally but that has earned no promotion toward canonical status. NOT LICENSED states that no use of the record grounds a product, marketing, or capability claim—the record can be true and complete and still license nothing, because licensing is a separate act by a separate authority that has not occurred. HELD states that the record is not disclosed: its existence and content are internal, and nothing in this monograph's reproduction of it changes its disclosure posture, since the monograph inherits the same status labels (see the front matter). Finally, the record's document status—DRAFT / OPEN / NOT FROZEN, as of 2026-08-04—means the record itself may still be amended by its owners; this chapter documents the record as of that date, and a reader holding a later revision should trust the artifact's own version fields over this text.

CLAIM BOUNDARY 4.20. Everything this chapter asserts is bounded by the record's own claim boundary. The chapter documents that eight conflicts were found, how they were evidenced, and what typed rules the candidate adopts in response. It establishes nothing about whether any CAS-Ohm prediction is empirically accurate, whether any audited quantity is physical, or whether any use of the framework is licensed. All of that remains OPEN / RING 4 / NOT LICENSED / HELD.

12. Why Preserving Conflicts Is an Engineering Result

It may seem deflationary to end a chapter of conflicts with no resolutions. The claim of this section is the opposite: for a framework in CAS-Ohm's epistemic position, the preserved audit is worth more than any resolution currently available, for four reasons.

First, harmonization without evidence is fabrication. Every conflict above could have been "resolved" editorially: declare R to be 0–100, declare I_{FS} to be the corpus percentile,

declare β to be the Jacobian derivative, declare the network rules valid. Each declaration would have manufactured exactly the coherence the sources demonstrably lack, and every downstream artifact would then cite "the framework" for choices that were in fact one editor's unevidenced preference. The audit's refusal to harmonize is a refusal to launder editorial choice into inherited authority. Where a choice genuinely must be made to proceed (Q4 bins, the split classification axes), the record makes it as a *named candidate rule* attached to a conflict—visible, attributable, and reversible—rather than as a silent rewrite of the sources.

Second, preserved conflicts are executable. Because each disposition is implemented as a type-level constraint—distinct quantity kinds with no silent conversion, an `IFSMethod` enumeration inside the profile digest, separate unit kinds for scores and resistances, separately named gain profiles, paired classification axes, capability gates on network arithmetic, censoring rules for residuals, exact half-open bins—the audit is not a document that future engineers must remember to consult. It is enforced at validation time. The historically observed failure modes (the mixed-scale sat = 8.5, the score-as-resistance division, the residual-as-bypass leap) are not merely warned against; they are unrepresentable or fail closed in the candidate's data model. An audit that compiles is a different artifact from an audit that admonishes.

Third, the record converts ambiguity into falsifiable structure. Several conflicts are not decidable by inspection because they are empirical questions in disguise: whether series and parallel rules hold, whether gains compose, whether residuals localize faults. The audit routes each to a registered hypothesis with a preregistered failure consequence (drop the rule, withdraw the causal language). This is the correct fate for conjectural machinery—neither adopted nor discarded, but staged for controlled testing with the decision criteria fixed in advance.

Fourth, the audit is itself evidence about process. Independent of CAS-Ohm's eventual fortunes, the record demonstrates a reusable method for inheriting a multi-document specification honestly: fix artifact identities by digest; disclose locator systems and inspection procedures, including failed ones; record disagreements verbatim with exact positions; classify the failure kinds; attach dispositions as typed, named rules; and bind the whole under an explicit claim boundary. Any successor framework, and any auditor of this one, can check the work. The alternative process—silent harmonization—leaves no residue to check.

Fifth, preservation is what mature measurement practice does with unresolved definition. Metrological convention distinguishes sharply between a quantity, the procedure that realizes it, and the uncertainty of its values [**32**, **33**]; when two procedures realize different quantities under one name, the metrological response is not to average them but to name them separately and record the difference until a comparison experiment relates them. The audit's dispositions—separate named profiles for I_{FS} methods, separately named gain profiles, distinct quantity types for scales—are this convention applied to an inherited specification corpus. Nothing about the practice is novel; what would have been novel, and wrong, is exempting a machine-learning-adjacent framework from it on the grounds that its quantities are "only scores." The audit treats score scales with the same definitional seriousness a standards laboratory would apply to a physical unit, precisely because the candidate's entire claim to discipline rests on refusing the score/measurement blur documented in the sources themselves.

The chapters that follow build directly on the dispositions established here. Chapter 5 develops the calibration profile whose fields exist largely because Conflicts 1, 2, and 3 demand them; Chapter 6 develops the response model and the anti-circularity rule that Conflict 3's calibrated bridge makes possible and that Conflict 7's discipline makes necessary. Throughout, the eight records of this chapter function as the monograph's negative space: the places

where the sources do not speak with one voice, held open on purpose, so that nothing built above them can quietly pretend that they do.

CHAPTER 5

The Calibration Profile

Every quantity in CAS-Ohm 1.0 is relative to a calibration profile. This chapter develops that object field by field: what each field binds, why the binding is mandatory rather than advisory, what failure the field's absence would permit, and how the fields jointly support the two principles that organize the whole candidate—*profile relativity* (no quantity means anything except under its profile, and nothing transfers across profiles without a stated comparison rule) and *anti-circularity* (an effective resistance is fit on calibration data and earns its keep only by predicting held-out data, developed fully in Chapter 6).

The motivating observation is simple and easy to underestimate. A directional activation projection is not a property of "the model" in the abstract. It is a property of a specific weights file, read by specific code, under a specific runtime and numeric precision, at a specific layer and token position, projected onto a specific direction vector of specific provenance, normalized against a full-scale value obtained by a specific method from a specific corpus, with specific seeds. Change any element and the number changes—sometimes negligibly, sometimes qualitatively, and *which* of those it will be is exactly what nobody knows in advance for systems of this kind. Metrological practice long ago internalized the consequence: a measurement result is inseparable from the procedure and conditions that produced it, and a reported value without its conditions is not a result but a rumor [**32**, **33**]. The calibration profile is the candidate's mechanization of that doctrine. It is a single, frozen, content-addressed record binding every condition that could plausibly move a number, so that "the same measurement" has an exact meaning: same profile digest, nothing less.

The chapter's claim discipline matches the rest of the monograph. The profile is an implemented data contract with validation rules exercised by bounded software tests; no live profile for any production model has been admitted, no field's adequacy has been empirically validated, and the whole apparatus is RING 4 / NOT LICENSED / HELD. What is defended here is the *design argument*: why each field exists and what its omission would break.

1. The Profile as the Unit of Meaning

DEFINITION 5.1 (Calibration profile). A calibration profile is an immutable record binding: a profile identifier; the exact model identity (model, configuration, tokenizer, weights); the code revision and dependency lock; the runtime identity and dtype; temporal validity (creation, validity start, expiry) and drift state with its checking evidence; the registered test points; the direction vectors with identity, norm, sign, and provenance; the corpus manifest and its rights record; the calibration and validation split references; the random seed and randomization method; the alpha grid with its endpoints; the full-scale (I_{FS}) method, its per-test-point full-scale values, and, where applicable, its percentile parameter; the fit method and its receipt; the uncertainty method; the response-region method; the tolerance profile; the control references and per-test-point control bindings; the source references; and a review state. The profile's canonical digest is the SHA-256 of its canonical encoding, and every measurement made under the profile binds that digest.

INVARIANT 5.2 (Profile relativity). No CAS-Ohm quantity is interpretable except relative to the calibration profile whose digest it binds. No quantity, threshold, region boundary, full-scale value, fitted resistance, or classification transfers to another profile—including a profile differing in a single field—without an explicitly stated and separately evidenced comparison rule.

The invariant sounds austere; it is in fact merely honest. Two profiles that differ in, say, dtype may produce projections that differ in the third decimal or in the first, and absent a comparison experiment no one knows which. Declaring the numbers comparable by default would be an empirical claim made without evidence. The candidate's default is therefore incomparability, with comparability available as a first-class, evidence-bearing assertion when someone does the work to establish it. This inverts the usual failure mode of measurement systems, in which comparability is assumed until an incident proves otherwise; here incomparability is assumed until a comparison rule is admitted. Chapter 4 showed why the inversion is forced: the sources themselves demonstrate (Conflict 4.5) that even the framework's own full-scale denominators were never one quantity.

A second consequence of Definition 5.1 deserves emphasis before the field-by-field walk: the profile is *immutable*. There is no "update a profile" operation. A changed condition—new weights, a revised corpus, a different tolerance—produces a new profile with a new digest, and history accumulates as a set of distinct profiles rather than as mutations of one. Immutability is what makes the digest meaningful: a digest that could survive edits would identify nothing.

2. Identity of the Measured System

The profile's model identity binds five values: a model identifier, a revision string, and three content digests—weights, configuration, and tokenizer. The rationale for the identifier and revision is ordinary bookkeeping; the rationale for the three digests is that names and revisions are assertions while digests are checkable facts. Model repositories reuse names across incompatible checkpoints; configurations change activation functions and attention variants without renaming the model; tokenizers change vocabulary and merge rules in ways that silently re-index every token position a test point names. Each of the three artifacts can vary independently of the other two, and each variation invalidates measurements in a different way: changed weights change every activation; a changed configuration can change the meaning of "layer 12"; a changed tokenizer changes which token sits at position 7 of a probe prompt, which is enough to re-aim every token-position-bound test point at different content.

The profile also requires that its model artifact references *resolve*: the set of digests among the artifact references must equal the set of digests named in the model identity. This closes a subtle gap between naming and having. A profile that named a weights digest without a resolvable artifact reference would be unfalsifiable—nobody could re-derive its measurements, because nobody could obtain its weights. The resolution rule makes the profile self-supporting: everything it names, it references, and the evidence registry can demand the bytes.

3. Code Revision, Dependency Lock, Runtime, and Dtype

Measurements are produced by code, and code has the same identity problems as models. The profile binds a code reference (the exact revision of the measuring implementation) and a dependency lock reference (the closed set of third-party versions beneath it). The dependency lock matters because the measuring code's behavior is not a function of its

own source alone: numerical libraries change reduction orders, kernel selections, and default tolerances between versions, and any of these can move low-order digits of an activation projection. A measurement that binds its code revision but not its dependency closure has fixed the recipe while leaving the ingredients loose. The lock reference freezes the closure; re-execution under a different closure is, by Invariant 5.2, a different profile.

The runtime identity and dtype fields record the execution substrate and the numeric precision of the computation. Dtype is the more consequential of the two and the easier to lose track of, since frameworks freely and sometimes implicitly cast between precisions. Half-precision, bfloat16, and single-precision evaluations of the same network on the same input are three different computations with three different rounding profiles; whether their differences matter for a given direction projection is an empirical question that cannot even be *asked* unless the precision is recorded. The profile makes the recording mandatory, so that the question is always askable and never silently begged.

4. Instruments

The profile binds a non-empty tuple of instrument identities—each carrying an instrument identifier, a kind, a version, and an implementation content digest—together with implementation references that must resolve exactly the set of implementation digests the instruments name. An instrument, in the candidate's vocabulary, is the identified software chain that produces an instrumented measurement: the hooking and projection code that reads an activation, applies the direction, and emits the raw value. The term is borrowed deliberately from laboratory practice, and so is the discipline: a laboratory result is attributed to a specific instrument with a calibration binding, not to "the lab" [**32**].

Two rationales drive the field. First, the measurement-kind rules of the ontology make `INSTRUMENTED` the most privileged kind—the only kind that can anchor structural research-packet completeness—and privilege must be checkable: an instrumented measurement must name a registered instrument, and the profile must hold the bytes of that instrument's implementation, or the claim "this number came from an identified instrument chain" is unverifiable decoration. Second, instruments version independently of the main code revision. The measuring harness may be stable across many profile revisions while a projection kernel changes, or vice versa; giving instruments their own identities and digests means the question "did the instrument change between these two profiles?" has a mechanical answer, field by field, rather than requiring a code archaeology expedition.

The uniqueness rule (instrument identifiers unique within a profile) and the resolution rule (implementation references exactly matching declared digests) mirror the model-artifact discipline of Section 2, and for the same reason: naming without having is the failure mode, and exact set equality between names and holdings is the check.

5. Temporal Validity, Drift, and Expiry

A profile carries three timestamps—created at, valid from, expires at—ordered by the validation rule (creation not after validity start, validity start strictly before expiry), plus a drift state and a drift-check evidence reference. Together they implement a position the candidate takes deliberately: *calibration decays*. Nothing about a frozen weights file decays, of course; what decays is the environment of assumptions around it—upstream serving stacks change, corpora age relative to usage, directions estimated from one data distribution meet another. A calibration regime without expiry drifts into exactly the "stale profile" condition

that Chapter 4's residual discipline identified as a standing alternative explanation for anomalies. The profile therefore expires by construction, and expiry is checked mechanically: the currency assertion fails outside the validity window.

The drift state is a four-valued enumeration—`CURRENT`, `STALE`, `FAILED`, `REVOKED`—bound to evidence by the drift-check reference. `CURRENT` asserts a passing recent check; `STALE` that the check is overdue; `FAILED` that a check ran and did not pass; `REVOKED` that the profile was withdrawn by decision. The distinction between `STALE` and `FAILED` preserves an important asymmetry of ignorance: an overdue check means *we do not know* whether calibration holds, while a failed check means *we know it does not*. Both block operation (only `CURRENT` passes the currency assertion), but they license different remediations and different retrospective doubts about recent results.

Operationally, the profile's currency assertion requires the conjunction: review state `REVIEWED`, drift state `CURRENT`, and the replay time inside the validity window. All admitted operations evaluate it against a caller-supplied replay time—which, as Chapter 4's eligibility discussion notes, is a historical replay coordinate, never an authenticated assertion about the actual present.

6. Test Points

A test point is the profile's answer to "measured *where*?" Each registered test point binds: the model identity (which must equal the profile's model—a test point cannot smuggle in a different system); the layer; the token positions (a non-empty tuple of distinct non-negative indices); the direction vector's identifier, content digest, provenance, norm, and sign; and the hook type (where in the layer's computation the activation is read). Test points must be unique within a profile, and every per-test-point structure elsewhere in the profile—full-scale values, control bindings—must cover the registered test points exactly.

The rationale is locality. CAS-Ohm's response model is explicitly a *local* hypothesis: a dose-response relation at a particular place in a particular computation, not a global property of a model. All of the audit's warnings about transfer (full-scales that "do not transfer," flow that "remains local") are warnings that locality is real and unforgiving. The test point is locality made into an identity: two measurements are at the same place if and only if their test-point digests are equal, which requires agreement on layer, positions, direction, sign, and hook type simultaneously. Anything less—"layer 12, roughly the same place"—is not an identity, and the type system does not accept it as one.

The hook type deserves a word, since it is the field most often left implicit in informal practice. A transformer layer is not a point; it is a pipeline of sublayer computations, and "the activation at layer 12" could mean the residual stream before the block, after attention, after the MLP, or after the final layer norm—numerically different vectors with different relationships to the direction vector. Recording the hook type turns an ambiguous phrase into an exact coordinate.

7. Direction Vectors: Identity, Norm, Sign, and Provenance

The direction vector is the most epistemically delicate object in the profile. Every response magnitude in the framework is a projection onto some direction; the direction, in effect, *is* the instrument's aperture. The profile binds four aspects.

Identity. Each direction carries an identifier and a content digest, and the profile's direction references must resolve exactly the digests named by its test points—the same resolution

discipline as for model artifacts, for the same reason: a direction that cannot be re-obtained byte-for-byte cannot anchor a reproducible measurement.

Norm. The candidate fixes direction norm to exactly one (validated as `Decimal("1")`, an exact equality, not a tolerance). Projections scale linearly with the direction's norm, so unnormalized directions would fold an arbitrary multiplicative constant into every response value and every fitted resistance. Fixing the norm removes a gauge freedom that would otherwise make numbers incomparable even within one profile.

Sign. A direction and its negation define the same line; a projection onto one is the negative of the projection onto the other. Sign conventions are thus pure convention—but an *unrecorded* convention is a latent sign error, and a sign error in a response model flips the apparent semantic outcome (`INTENDED` versus `OPPOSITE`) wholesale. The profile records the sign explicitly so that the convention is an auditable fact rather than tribal knowledge.

Provenance. The candidate enumerates three direction provenance kinds, and the enumeration is closed:

`PROBE_DERIVED`.: The direction is the weight vector of a probe—typically a linear classifier—trained to separate labeled activation examples. Its epistemic character: the direction is an *estimate from data*, inheriting the probe corpus, its labels, its regularization, and its sampling noise. Probe directions answer "along which axis do these labeled classes separate?"—which is related to, but not the same as, "along which axis does the model *use* this concept?"

`JACOBIAN_DERIVED`.: The direction is obtained from differentiation—a gradient or Jacobian row of some downstream quantity with respect to the activation. Its character: the direction is a *local analytic object*, exact for the computation at the linearization point but valid only there; it answers "what perturbation most moves this output, here?"

`DIFF_DERIVED`.: The direction is a difference of activation statistics between two conditions (for example, means under contrasting prompts). Its character: a *contrast estimate*, simple and robust, but credited with meaning only to the extent the two conditions differ in one thing.

Three kinds, three different failure surfaces: probe directions fail with their training distribution, Jacobian directions fail away from their linearization point, and diff directions fail when the contrast was confounded. The profile does not rank them—each is legitimate—but it refuses to let them blur, for the by-now familiar reason: a number whose production method is unrecorded cannot be doubted *specifically*, and undoubtable numbers are the enemy. The provenance field also feeds the registered reproducibility obligations (cross-split direction reproducibility appears among the closure experiments), which are stated per-provenance because the reproduction procedure differs by kind.

A corollary worth stating explicitly: directions of different provenance are not interchangeable even when they point in nearly the same place. A probe direction and a diff direction for "the same concept" at the same test point are estimates of related but distinct estimands, with distinct error structures; a resistance fitted against projections on one is not a resistance for the other, and the profile machinery enforces this automatically because the direction digest participates in the test-point identity, which participates in everything downstream. If two provenances' directions are claimed equivalent for some purpose, that claim is a comparison rule in the sense of Invariant 5.2—an assertable, evidence-bearing statement—not a default. The candidate's posture is uniform: geometric proximity is measurable, semantic equivalence is a claim, and claims need evidence.

8. Corpus Manifest and Rights

Two references bind the data environment: the corpus manifest and the corpus rights record. The manifest is a content-addressed inventory of the corpus from which calibration stimuli, probe data, and (under the percentile method) full-scale statistics are drawn. Content addressing does for the corpus what digests do for weights: "the corpus" stops being a mutable directory and becomes an exact set of bytes, so that corpus-derived statistics—above all the 95th-percentile full-scale of Conflict 4.5—are anchored to something that cannot silently gain or lose documents.

The rights reference is the field most likely to strike a technical reader as out of place, and its inclusion is deliberate. A calibration built on data whose use rights are unestablished is a calibration that may not be lawfully re-executed, shared with a reviewer, or retained—which makes every downstream measurement irreproducible in the practical sense that matters, whatever its technical merits. Binding a rights record into the profile digest makes rights status part of measurement identity: a profile cannot be complete without it, and a rights change (a revoked license, a newly discovered encumbrance) is visible as exactly what it is, a change to the conditions under which the numbers exist.

9. Splits, Seeds, and Randomization

The profile binds a calibration split reference and a validation split reference, and its validation rules require the two digests to differ. This pair of fields is the structural footprint of the anti-circularity rule: the data on which the effective resistance is fit (calibration split) and the data on which its predictions are scored (validation split) are distinct artifacts, distinguished at the type level, before any statistics happen. Chapter 6 develops the statistical argument; here we note only the design point that the split separation is a *profile* fact, not a habit of the analyst. An analyst cannot forget the held-out discipline without constructing a profile that fails validation.

The random seed (a non-negative integer) and the randomization method (an identifier for how randomness was consumed—assignment of items to splits, ordering of sweeps, any stochastic element of the fit) close the remaining degrees of freedom. Randomization without a recorded seed is irreproducibility by construction; a seed without a recorded method is a seed for an unspecified generator consumed in an unspecified order, which reproduces nothing. Together they make the stochastic parts of calibration replayable, and—as important—they make *selective* reruns visible: a report whose seed differs from its profile's seed is advertising that something was rerun.

10. The Alpha Grid

The alpha grid is the ordered tuple of intervention doses at which the dose-response relation is probed. The validation rules are specific: at least three doses; exact decimal values; strictly increasing and duplicate free; the declared minimum and maximum equal to the grid's endpoints; and zero included.

Each clause has a job. *Three doses minimum*: two points determine a line exactly, leaving zero residual degrees of freedom—no way to detect curvature, hence no way to distinguish a genuinely linear response from any monotone response whatsoever; three is the least count at which "linearity" is even falsifiable, and the region classifier's own minimum (interior points with neighbors on both sides) matches it. *Exactness and strict ordering*: the grid is part of measurement identity, and a grid with representation-dependent values or ambiguous ordering would leak platform noise into the profile digest. *Endpoints declared*: the declared

range is the region over which the profile claims to have probed; predictions outside it are extrapolations, and the admissibility machinery can only know that if the range is a recorded fact. *Zero included*: the zero dose is the null intervention—the control column built into the grid itself. Without it, the fit has no anchor at the origin and the zero-intervention control comparisons required by the closure experiments have no in-grid reference.

The grid also encodes a modeling humility: it is a *grid*, not a distribution. The candidate does not assume responses at unprobed doses; the response region machinery classifies behavior at and between probed doses, and the tolerance profile (Section 13) governs how much interpolation credence the fit may claim.

11. The Full-Scale Method

Conflict 4.5 established that the sources supply three incompatible procedures for the full-scale reference I_{FS}. The profile's response is the `IFSMethod` enumeration, with the audit's disposition implemented literally: one method per profile, declared up front, inside the digest.

`RAIL_AND_FLOOR_PER_PATH`**.:** The Appendix I procedure: I_{FS} measured once per path at drive rail and constraint floor. An extreme-operating-point probe; local by construction, with the source's own warning that it does not transfer.

`CORPUS_PERCENTILE_PER_TEST_POINT`**.:** The addendum procedure: the per-test-point 95th percentile of absolute directional projection over a fixed hashed corpus. A statistic, requiring the percentile parameter; the profile's validation rules enforce that the percentile is present exactly when this method is declared and lies strictly between 0 and 100.

`DECLARED_DESIGN_FULL_SCALE`**.:** The Guide procedure: a board-selected design value. A declaration rather than a measurement, typed accordingly (its natural measurement kind is a design value, not an instrumented one).

Rule 5.3 (Non-interchangeability of full-scale methods). The three `IFSMethod` values name different calibration procedures producing different quantities. No value obtained under one method may be substituted for, averaged with, or compared against a value obtained under another except through an explicitly admitted comparison rule. Because the method is bound into the profile digest, any cross-method comparison is a cross-profile comparison and is blocked by default under Invariant 5.2.

Whatever the method, the profile must carry a full-scale *value record* for every registered test point, exactly one each. The value record is itself typed evidence: a strictly positive value in activation units, an uncertainty whose interval is positive and contains the value, a sample size, and non-empty source references. Note the asymmetry with the method enumeration: the method says how the denominator was obtained; the value record proves the denominator exists, with uncertainty, for each place it will be used. Normalization (Chapter 8 where present) divides by these values and propagates their uncertainty into every Q10 score; a full-scale without uncertainty would launder the denominator's error out of every displayed value.

12. Fit Method, Uncertainty Method, and the Fit Receipt

Three fields govern how the effective resistance is estimated: the fit method identifier, the uncertainty method identifier, and the fit receipt reference. The method identifiers are deliberately identifiers rather than enumerations: the candidate does not legislate a unique estimator (least squares on the active region, robust variants, and others are all conceivable),

but it insists the choice be named, because estimator choice moves results and an unnamed estimator is an unfalsifiable one. The uncertainty method is bound separately because uncertainty computation is a procedure in its own right (analytic propagation, bootstrap, and interval methods differ), and the audit's discipline requires every emitted uncertainty to say how it was obtained [**33**].

The fit receipt reference binds the artifact recording the actual fit run: inputs consumed, estimates produced, diagnostics observed. Its presence in the profile closes the loop between declaration and execution—the profile does not merely promise a fit method, it references the receipt of the fit that was actually performed. In the Kennedy–O'Hagan framing of computer model calibration, the receipt is where the calibration data, the parameter posterior (or point estimate with its uncertainty), and the discrepancy evidence live [**37**]; Chapter 6 returns to what the receipt must show before a resistance earns the `CALIBRATED_HELD_OUT` role.

13. Tolerances

The tolerance profile is a five-field record of exact decimals: an absolute tolerance, a relative tolerance, a residual floor, a minimum linearity coefficient of determination, and a maximum off-target fraction. Validation requires the absolute tolerance and residual floor strictly positive and the relative tolerance, linearity minimum, and off-target maximum in the unit interval.

The rationale is that every downstream judgment of the form "close enough," "linear enough," or "clean enough" must be a comparison against a declared number rather than an analyst's mood. The absolute and relative tolerances govern agreement checks between predicted and observed response. The residual floor sets the magnitude below which a residual is treated as indistinguishable from zero—the operational meaning of "zero-compatible" in the residual-censoring rule of Conflict 4.15. The linearity minimum is the floor an active-region fit must clear before any operation gated on "adequate linearity" (notably the network compositions of Conflict 4.13) may proceed. The off-target maximum bounds the fraction of response appearing on axes other than the target direction before the measurement is disqualified as off-axis-contaminated.

Putting the five numbers in the profile—rather than in code defaults or per-analysis choices—has two effects. It makes tolerance-shopping visible: loosening a tolerance to admit a marginal result requires minting a new profile with a new digest, an auditable event. And it makes strictness comparable: two profiles' tolerance records can be diffed field by field, so "profile B is laxer than profile A" is a checkable statement.

14. Response-Region Rules

The response-region method identifier names the procedure by which sweep observations are classified into the dose-response regions (`CUTOFF`, `ACTIVE`, `SATURATION`, `BREAKDOWN`, `NON_OHMIC`, `UNSTABLE`, with `UNASSESSED` as the honest default). The field exists in the profile for the same reason as the fit method: region assignment is a procedure with alternatives, region labels gate consequential operations (predictions are admissible only in the active region; network compositions require active-region components; saturated-stage residuals are censored), and a gate whose criterion is not frozen is not a gate. Binding the method into the digest means a reclassification standard cannot drift under a fixed profile: the same sweep, replayed, classifies the same way.

15. Controls and Control Bindings

The profile binds a set of control references and a set of per-test-point control bindings, with validation rules enforcing exact correspondence: the bound control artifacts must exactly equal the declared control references, every binding must name a registered test point, and the bindings must cover every test point. Controls here are the comparison conditions the closure experiments require—zero-intervention runs, random same-norm directions, wrong-layer and wrong-token probes, label shuffles, norm-matched perturbations—represented as artifacts with relevance methods binding them to the specific test points they control for.

The design point is that controls are part of the *calibration*, not an optional garnish on the analysis. A measurement under the profile carries the profile's control set; the validation machinery rejects non-design measurements whose control references do not equal the profile's. This forecloses the quiet failure in which controls exist somewhere in a repository but the measurement that mattered was taken without them. The coverage rule (every test point controlled) similarly forecloses the partial version, where the well-behaved test points get controls and the interesting one does not.

16. Review State

The profile carries a review state: `UNASSESSED`, `IN_REVIEW`, or `REVIEWED`; the implementation additionally provides `FLAG_FOR_HUMAN_REVIEW` for records requiring escalation. Only `REVIEWED` passes the currency assertion—an unreviewed profile can be constructed, stored, and studied, but no operation that requires a current profile will run under it.

The field encodes a separation the candidate is careful about everywhere: *structural validity is not endorsement.* A profile that passes every validation rule is well formed; whether its choices are *sensible*—is this corpus appropriate, are these tolerances defensible, is this direction provenance suited to this question—is a judgment no type system can make. The review state is the socket where human judgment plugs into the machinery, and its default is the honest `UNASSESSED`. Note what the field does not claim: `REVIEWED` records that review occurred within whatever process governs the record; it is not an independent-authority attestation, which belongs to the separate authority machinery of Chapter 11 where present, and which the candidate implements with constant-false authenticated eligibility.

17. The Canonical Digest

Every field above participates in the profile's canonical payload, which is serialized by the candidate's canonical encoding (exact-type tagged, cycle-aware, bounded) and hashed to produce the profile's SHA-256 digest. The digest is the profile's identity for every purpose: measurements bind it, operation contexts snapshot it and re-verify it on every subsequent operation (the context fails closed if the profile, its authority manifest, its attestations, or the replay time change after admission), and cross-profile comparisons are detected by digest inequality.

Three design properties make the digest trustworthy. *Totality*: the payload includes every semantically relevant field—including the status constants (admission ring, license status, disclosure status) and the schema label—so no meaningful variation escapes the hash. *Canonical form*: field ordering and collection ordering are normalized (references sorted by digest, bindings by their own hashes), so equality of content implies equality of bytes implies equality of digest, with no representation-dependent false negatives. *Exactness*: numeric

fields are exact decimals, not binary floats, so the digest cannot be perturbed by platform-dependent rounding in the serialization path. The result is that "same profile" is decidable by string comparison, which is the property every downstream fail-closed check relies on.

18. The Operation Context: Admission and Revalidation

The profile does not act alone; it acts inside an *operation context*, the runtime object through which every measurement validation, normalization, and structural operation flows. A context binds the profile, the evidence registry holding the referenced bytes, the authority configuration with its trust mode, any capability attestations, the caller-supplied replay time, and optionally a clock-receipt record. At admission, the context performs the full battery: it revalidates the evidence registry, resolves the authority manifest's bytes against its digest, asserts the profile current at the replay time (review state `REVIEWED`, drift `CURRENT`, replay time inside the validity window), and demands that every artifact the profile references—model artifacts, instrument implementations, directions, splits, corpus, rights, controls, sources, full-scale sources, the fit receipt—be present and resolvable. Then it snapshots: the profile digest, the manifest digest, the attestation digests, the replay time, and the clock-receipt digest are recorded, and every subsequent operation re-verifies the snapshot before doing anything else. Any drift—a profile object swapped, an attestation added, a replay time edited—fails the operation closed.

The design rationale is time-of-check versus time-of-use. A validation performed once at construction protects nothing if the validated objects can change before use; the snapshot-and-revalidate pattern makes the context's guarantees continuous rather than momentary. For the audit posture of this monograph, the significant property is that the context is where profile discipline becomes *unavoidable*: there is no API path to a normalization or a structural operation that does not pass through a context, hence through the currency assertion, hence through the profile's review, drift, and validity gates. The context also fixes the epistemic status of time: the replay time is caller-supplied history, useful for replaying past states, and the context's authenticated-currency answer is constant false—no admitted external time authority exists in the candidate, and the machinery says so rather than pretending.

19. What Each Field Defends Against

A compact way to review the chapter is by threat: for each family of fields, the failure it renders unrepresentable or fail-closed. Table 1 gives the mapping.

Threat 5.4. The table's coverage claim should itself be read with the monograph's discipline: these are the failures the fields are *designed* to foreclose, at the data-contract level, as exercised by bounded software tests. Whether the fields suffice against adversarial construction of profiles, against collusion between a profile author and a reviewer, or against failure modes not yet enumerated is unassessed; the authenticated authority layer that would address the adversarial cases is exactly the part the candidate implements as constant-false eligibility (Chapter 11 where present).

20. An Illustrative Construction

Construction 5.5 (Hypothetical profile walk-through; all values illustrative). The following is a fabricated example to make the field interactions concrete. Every value is invented for exposition; nothing here describes a real model, corpus, or measurement.

An analyst wishes to calibrate a response model at one test point of a hypothetical checkpoint. She constructs: a model identity naming the checkpoint with its three digests, and

Field family	Failure defended against
Model identity + artifact resolution	silent checkpoint/config/tokenizer substitution; unobtainable weights behind a name
Code ref + dependency lock	behavior drift from unpinned libraries; "same code" with different numerics
Runtime + dtype	precision-dependent results with no record of precision
Instruments + implementation refs	privileged `INSTRUMENTED` claims with unverifiable production chains
Validity window + drift state + check ref	stale calibration operating as current; unknown-vs-known-bad conflation
Test points + hook type	"roughly the same place" passing as identity
Direction identity/norm/sign/provenance	gauge freedom in magnitudes; sign flips in outcomes; method-blind pooling of estimates
Corpus manifest + rights	mutable corpora under fixed statistics; legally unreexecutable calibrations
Distinct splits + seed + method	circular fitting; unreplayable randomness; silent selective reruns
Alpha grid rules	unfalsifiable linearity; undeclared extrapolation; missing null dose
`IFSMethod` + full-scale records + percentile rule	denominator-method conflation (Conflict 4.5); uncertainty-free denominators
Fit/uncertainty/region method ids + receipt	unnamed estimators; bare point estimates; drifting region criteria; promised-but-unperformed fits
Tolerance profile	analyst-relative "close enough"; invisible tolerance-shopping
Controls + bindings + coverage	control-free measurements; selectively controlled test points
Review state	structural validity mistaken for endorsement
Canonical digest	identity by name rather than content; representation noise in identity

TABLE 1. Profile fields and the failures they foreclose.

three artifact references resolving them; a code reference at her harness revision and a dependency lock from her environment export; runtime identifier and dtype (say, `"runtime-x"` and `"float32"`); a validity window of ninety days from a drift check whose passing record she references, drift state `CURRENT`; one instrument identity for her projection hook with its implementation digest resolved; one test point at layer `"blocks.12.hook_resid_post"`, token positions $(5, 9)$, with a `DIFF_DERIVED` direction of unit norm, positive sign, and resolved digest; corpus and rights references for her fixed prompt set; two split references partitioning it (digests distinct, as required); seed 17 with a named shuffling method; an alpha grid $(-1.0, -0.5, 0, 0.5, 1.0)$ with endpoints declared and zero present; `IFSMethod.CORPUS_PERCENTILE_PER_TEST_POI` with percentile 95 and one full-scale record for her test point (value, uncertainty interval, sample size, sources); fit method `"ols-active-region-v1"` with a receipt reference; uncertainty method `"bootstrap-1000-v1"`; region method `"piecewise-slope-v1"`; tolerances (absolute 0.05, relative 0.10, residual floor 0.02, linearity minimum 0.95, off-target maximum 0.20); control references and bindings covering the test point (zero-dose, random-direction, wrong-layer artifacts); source references to her design notes; review state `UNASSESSED`.

The profile validates structurally—every rule of this chapter passes—and receives its digest. Note what she can and cannot now do. She cannot run a normalization: the currency assertion fails on `UNASSESSED`. After review, with state `REVIEWED` (a new profile, hence a new digest—immutability), operations under an admitted context become possible. If she later regenerates her direction vector, even "equivalently," its digest changes, so the test point changes, so the profile changes; her earlier full-scale record no longer covers the new test point, and validation forces her to remeasure it. The contract has done its job: an easy-to-overlook invalidation (new direction, old denominator) was surfaced as a hard error at construction time rather than as a subtly wrong number at analysis time.

21. Profile Relativity in Practice

It is worth spelling out what Invariant 5.2 concretely forbids and what it permits, because the invariant is the profile's entire point.

Forbidden by default: comparing Q10 scores across profiles (different denominators, per Rule 5.3); pooling sweep observations across profiles into one fit; carrying a fitted r_{eff} from one model revision to the next ("the model barely changed" is an empirical claim requiring evidence, not a default); quoting a region classification under a profile whose drift state is not current; and every flavor of cross-model league table—"model A has lower resistance than model B"—absent an admitted comparison rule. The last of these is the politically hardest to resist, since ranking is what audiences want; it is also the least defensible, compounding every incomparability at once (different weights, tokenizers, geometries, direction provenances, and full-scale denominators).

Permitted: anything within one digest, subject to the other gates; and cross-profile statements made *through* an explicitly admitted comparison rule—a registered procedure with its own evidence establishing that a specific quantity is comparable across a specific pair of profiles under specific conditions. The registered checkpoint-replication hypothesis is an instance: replicate on a held-out checkpoint *under a prespecified comparison rule*. The invariant does not make cross-profile science impossible; it makes it a deliberate, evidenced act instead of an ambient assumption.

Remark 5.6. Profile relativity generalizes the familiar rule that quantities in different units may not be added without conversion [**34**]. A profile is, in effect, a very large compound unit: the "unit" in which a CAS-Ohm number is expressed includes the weights digest, the direction, the denominator method, the dtype, and the rest. Two numbers in different compound units need a conversion—the comparison rule—and the burden of producing it lies with whoever asserts comparability. The type system merely refuses to pretend the conversion is the identity.

22. The Profile Against the Audit

The profile can also be read as the point-by-point answer to Chapter 4. Each preserved conflict deposits at least one field.

Conflict 4.3 (scale and saturation) appears as the insistence on exact decimal values everywhere and on typed quantities bound to unit kinds; the profile contains no field where a number floats free of its scale. Conflict 4.5 (full-scale profile) deposits the `IFSMethod` enumeration, the percentile-presence rule, and the per-test-point full-scale records with mandatory uncertainty—the audit's "separate named profiles" realized as literally separate digests. Conflict 4.7 (dimensional bridge) deposits the entire fit apparatus: the split references, fit method, fit receipt, and uncertainty method exist so that an alpha-per-activation quantity

can come into existence by calibration rather than by relabeling a score. Conflict 4.9 (gain definition) deposits the method-identity discipline—dimensionless quantities carry method identifiers because unit checking cannot distinguish them. Conflict 4.11 (ASOM axes) deposits the response-region method field, which fixes the region axis's procedure while leaving the semantic axis to independent assessment. Conflict 4.13 (network rules) deposits the tolerance profile's linearity minimum and the region method, which are the preconditions the capability gates consult. Conflict 4.15 (residual identification) deposits the residual floor—the operational meaning of "zero-compatible"—and the control bindings that make shuffled and matched controls a structural requirement rather than an aspiration. Conflict 4.17 (Q4 boundaries) deposits the exactness discipline itself: band assignment is computable from exact pre-rounding values only because the profile's numeric fields are exact from end to end.

This mapping is the concrete sense in which the audit "compiles" into the candidate: the conflicts of Chapter 4 are not commemorated in a document appendix but disassembled into validation rules that run on every profile ever constructed.

23. Anatomy of a Comparison Rule

Invariant 5.2 repeatedly defers to "a stated comparison rule." Since the deferral carries so much weight, the candidate owes a statement of what such a rule must contain, even though none has yet been admitted. A comparison rule, to be admissible, must at minimum declare: (i) the two profile digests it relates, exactly—not families, not patterns; (ii) the specific quantity kinds it renders comparable, with all others remaining incomparable; (iii) the transformation, if any, applied in the comparison (identity, rescaling, requantization), with its own uncertainty contribution; (iv) the evidence establishing the comparability—which, for anything beyond definitional identity, means a measurement campaign under both profiles with agreement scored against declared tolerances; (v) its own validity window and drift obligations, since comparability decays for the same reasons calibration does; and (vi) its own review state. In other words, a comparison rule is itself a profile-like object: typed, evidenced, digested, expiring, reviewed. The symmetry is intentional. If crossing between profiles were cheaper than staying inside one, the system would leak at the crossings; making the crossing artifact carry the same discipline as the endpoints closes the arbitrage.

Remark 5.7. The prespecified comparison rule required by the registered checkpoint-replication hypothesis is the intended first instance: a rule relating a profile on one checkpoint to a profile on a held-out checkpoint, admitted *before* the replication runs, so that the replication cannot quietly tune the comparison to the result. As of this writing no such rule exists; the slot for it does.

24. Estimating Effective Resistance Under a Profile

The profile exists so that one quantity can be estimated honestly: the effective resistance r_{eff}, in alpha-per-activation units, for a given test point. The estimation requirements, stated here and developed statistically in Chapter 6, are:

(1) *Fit on the calibration split only.* The estimator consumes dose-response observations drawn from the calibration split referenced by the profile, at doses from the profile's alpha grid, under the profile's seed and randomization method.
(2) *Declared method, receipted run.* The estimator is the profile's named fit method; the run is recorded in the fit receipt with its diagnostics.

(3) *Uncertainty mandatory.* The estimate carries an uncertainty produced by the profile's named uncertainty method; a bare point estimate is not an admissible resistance.
(4) *Region-conditioned.* The fit claims validity only in the active region as classified by the profile's response-region method; observations in cutoff, saturation, breakdown, or unstable regions do not certify it.
(5) *Predict held-out data.* The estimate's admissible role, `CALIBRATED_HELD_OUT`, is earned by prediction on the validation split—data the fit never saw—scored against the profile's tolerances. Defining $r = a/y$ on a sample and "verifying" $\hat{y} = a/r$ on the same sample is the circular procedure the candidate rejects outright.
(6) *Discrepancy acknowledged.* Following the computer-model calibration literature, the estimation must represent not only parameter uncertainty but model discrepancy—the structured difference between the best-fitting linear response and the system's actual response—rather than treating the best fit as truth [37]. A calibration that hides discrepancy inside its parameter estimate will predict confidently and wrongly; one that models discrepancy knows where its linear hypothesis stops.

Every requirement lands on a profile field: (1) on the split references and randomization fields; (2) on the fit method and receipt; (3) on the uncertainty method; (4) on the response-region method; (5) on the distinct validation split and the tolerance profile; (6) on the receipt's required diagnostics.

Two interactions between estimation and the rest of the profile deserve explicit note. First, estimation interacts with *drift*: a resistance fitted under a profile inherits the profile's validity window, and the drift-check discipline is what stands between "was calibrated" and "is calibrated." A fitted r_{eff} under a profile whose drift state has left `CURRENT` is not merely older evidence; it is evidence whose preconditions are no longer certified, and the currency assertion removes it from service automatically rather than trusting consumers to check dates. Second, estimation interacts with *region classification* bidirectionally: the fit claims validity only in the active region, but the active region is itself identified from sweep data by the profile's region method. The candidate handles the circularity the only defensible way—both procedures are frozen in the profile before either runs, so neither can be adjusted to flatter the other, and the region method's verdicts are replayable facts under the digest. This is the sense in which the profile is not bureaucracy around the science but the science's own preconditions, itemized. The chapter's boundary, however, is unchanged from the monograph's: these are requirements an estimate *would* have to meet. No estimate meeting them—no admitted live calibration of any production system—exists in the current evidence, and nothing in this chapter asserts otherwise.

25. What the Profile Is Not

Because the profile is the most infrastructure-like object in the candidate, it attracts misreadings that the chapter should close by naming.

The profile is not a certification. Passing profile validation means the record is well formed and self-consistent: references resolve, splits differ, grids are ordered, coverage rules hold. It does not mean the calibration is good, the direction meaningful, the corpus adequate, or the model safe. The review state carries human judgment about sensibleness; the authenticated-eligibility machinery (constant false in this candidate) would carry independent authority; neither is implied by structural validity, and nothing in the digest should be read as a seal of quality.

The profile is not a claim that its fields are sufficient. The field set was derived from the audit's conflicts and from general reproducibility practice; it forecloses the enumerated failures of Table 1. Whether further conditions move numbers in ways the profile does not record—hardware nondeterminism below the dtype level, undocumented runtime kernel dispatch, environmental dependencies outside the lock—is an open empirical question. The honest statement is that the profile records every condition the candidate's authors currently know to matter, and that discovering another such condition would mean extending the schema, versioning it, and treating older profiles' comparability to newer ones as—again—a comparison-rule question rather than an assumption.

The profile is not portable authority. A profile is bytes; anyone can construct one. Its digest proves integrity (these exact fields, unaltered), never provenance in the social sense (a trustworthy party constructed this). The trust architecture that would bind profiles to accountable parties—protected key lifecycles, independent authority manifests, trusted time—is deliberately unimplemented, and the candidate's operation contexts say so on every binding by carrying unauthenticated authorization states. Treating a structurally valid profile from an unknown source as trustworthy input would be a category error the type system permits but the process rules forbid.

The profile is not the science. It is the container the science requires. The registered hypotheses of Chapter 12 where present—local response above matched controls, held-out predictive resistance, gain composition, redundancy, residual localization—are the science, and every one of them presupposes a profile precisely so that a failed prediction is informative: with all conditions frozen and recorded, a miss cannot be blamed on an unrecorded condition, which is what makes the hypotheses falsifiable in the first place. A framework that let conditions float would be unfalsifiable in the weakest sense—its failures would always have an alibi. The profile removes the alibis, for successes and failures alike.

Claim Boundary 5.8. This chapter specifies a data contract and its rationale. It establishes no empirical adequacy of any field choice, no validated calibration of any live system, no comparability of any pair of profiles, and no authority beyond RING 4 / NOT LICENSED / HELD. Where the chapter argues that a field prevents a failure, the claim is architectural—the failure becomes unrepresentable or fail-closed in the candidate's types—not a claim that deployed systems exhibiting the failure have been observed or measured.

CHAPTER 6

The Response Model and Anti-Circularity

The center of CAS-Ohm 1.0 is one small equation and one large refusal. The equation is the candidate prediction

$$\hat{y} \;=\; \frac{a}{r_{\text{eff}}},$$

a proposed local linear relation between an intervention dose and an observed response under a fixed calibration profile. The refusal is the anti-circularity rule: the candidate rejects, categorically and at the type level, the procedure that computes $r = a/y$ on a sample, computes $\hat{y} = a/r$ on the same sample, observes $\hat{y} = y$, and announces a validated law. This chapter develops both—the equation with its admissibility conditions, and the refusal with its statistical justification—then works through hypothetical numeric illustrations, distinguishes verification from validation in Sargent's sense [**23**], and closes by delimiting exactly what a validated response model would and would not license.

The claim posture is stated once and holds throughout: everything here is RING 4 / NOT LICENSED / HELD. The response model is a registered hypothesis with an implemented guard apparatus, not an established law; no live calibration exists; every number in the worked examples is fabricated for exposition and is labeled as such where it appears.

1. The Candidate Prediction

DEFINITION 6.1 (Candidate response prediction). Fix a calibration profile P (Chapter 5) and a registered test point t of P. Let a be an alpha dose (quantity kind `DRIVE_INTERVENTION`, unit `ALPHA`) within P's declared alpha range, and let r_{eff} be an effective resistance at t (quantity kind `EFFECTIVE_RESISTANCE`, unit `ALPHA_PER_ACTIVATION`, strictly positive). The candidate prediction for the directional activation response is $\hat{y} = a/r_{\text{eff}}$, a quantity in `ACTIVATION` units, asserted—as hypothesis, not fact—to approximate the observed response y within P's declared tolerances, for doses in P's active region at t.

Three features of the definition do quiet work. First, the *units close*: alpha divided by alpha-per-activation yields activation, so the prediction is dimensionally coherent by construction [**34**]—which, after Conflict 4.7, is not a triviality but a repair. The sources' inherited $I_{\text{pred}} = V/R_{\text{total}}$ divided an intervention by a score and produced dimensional nonsense; the candidate's version is well-formed precisely because r_{eff} is required to be the calibrated bridge quantity and nothing else. A constraint score in the denominator is not a smaller error; it is a type error, rejected before arithmetic.

Second, the *strict positivity* of r_{eff} is a validation rule, not a hope. A zero resistance would make the prediction undefined; a negative one would smuggle a sign inversion into a quantity whose sign conventions live elsewhere (in the direction's recorded sign). The candidate's quantity validation rejects non-positive effective resistance outright, so the prediction's denominator is well-defined whenever the prediction is constructible at all.

Third, the prediction is *local three times over*: to the profile (no transfer across digests without a comparison rule), to the test point (no transfer across layers, positions, directions,

or hooks), and to the active region (no claim at doses classified cutoff, saturated, breakdown, non-ohmic, or unstable). The equation's resemblance to a global physical law is the analogy the framework borrows; the typed reality is a patch of claimed linearity, fenced on all sides.

2. Admissibility Conditions

The prediction may be *evaluated* whenever its operands exist; it may be *admitted*—used as the reference against which residuals are computed, regions confirmed, or hypotheses scored—only under conditions that the implementation enforces as guards. Stated as a rule:

RULE 6.2 (Admissibility of the response prediction). A prediction $\hat{y} = a/r_{\text{eff}}$ is admissible only if:

(1) r_{eff} is strictly positive and carries an uncertainty record produced by the profile's declared uncertainty method;
(2) r_{eff} binds the profile's calibration and validation split references—the data it was fit on and the data it was scored on—as resolvable artifacts;
(3) r_{eff} was produced by the profile's declared fit method, with the fit receipt resolvable;
(4) r_{eff} holds the role `CALIBRATED_HELD_OUT`, or the role `NETWORK_EQUIVALENT` with the separate validation that role requires;
(5) the dose a lies within the profile's declared alpha range, and the operating classification at the test point is `ACTIVE` on the region axis;
(6) the operation context is admitted and current: profile `REVIEWED`, drift `CURRENT`, replay time within validity.

The role vocabulary in condition (4) is the load-bearing novelty. The candidate types every effective resistance with a *resistance role*, an enumeration of four values, and the roles are not interchangeable:

`CALIBRATED_HELD_OUT`.: Fit on the calibration split, scored on the disjoint validation split against the profile's tolerances. The only role, besides a validated network equivalent, that admits prediction. The role's constructor requirements enforce a held-out split reference: a resistance cannot even claim the role without binding the data that scored it.

`APPARENT_SAME_SAMPLE`.: Computed as a/y from the very observations it would "predict." The candidate allows this object to exist—it is a legitimate descriptive statistic, the observed dose-response ratio—but the role bars it from admission as a predictor. Its name is its warning label.

`NETWORK_EQUIVALENT`.: Produced by a capability-gated structural composition (series or parallel aggregation, Chapter 9 where present). Factory-constructed only: the implementation refuses hand-built instances, so a network equivalent carries its full derivation or does not exist. Admissible for prediction only under the separate validation its gating demands.

`DESIGN_PROPOSAL`.: A declared value—someone's proposal for what the resistance ought to be, typed as design information. Never admissible for prediction; useful as a registered target that later calibration may confirm or refute.

The enumeration is the anti-circularity rule made structural. In an untyped system, the same floating-point number could serve as descriptive ratio, fitted parameter, composed aggregate, and design guess, with its history carried in comments if at all. Here the four histories are four types, the guards read the type, and the tautological one is stopped at the door by its name.

3. The Anti-Circularity Rule

RULE 6.3 (Anti-circularity). The following procedure is rejected regardless of its numerical outcome: compute $r_{\text{same}} = a/y$ on a sample of observations; compute $\hat{y} = a/r_{\text{same}}$ on that same sample; report the agreement of $\hat{y}$ with y as validation of the response model. A defensible test fits r_{eff} on a registered calibration split and evaluates its predictions on a disjoint held-out split, with uncertainty and discrepancy represented [37].

The rejection is not a stylistic preference; the rejected procedure is mathematically incapable of failing, and a test that cannot fail confirms nothing.

PROPOSITION 6.4 (Same-sample identity is tautological). *Let (a, y) be any observation with $a \neq 0$ and $y > 0$. Define $r_{\text{same}} = a/y$ and $\hat{y} = a/r_{\text{same}}$. Then $\hat{y} = y$ identically, for every such observation, regardless of whether any linear (or any other) relation holds between dose and response.*

PROOF. By substitution: $\hat{y} = a/r_{\text{same}} = a/(a/y) = a \cdot (y/a) = y$. The computation uses only the field axioms; no property of the data enters. Hence the "agreement" $\hat{y} = y$ is an algebraic identity, true for observations drawn from a linear system, a saturating system, a random number generator, or a constant. Observing it carries zero bits of information about the dose-response relation. □

REMARK 6.5. The multi-point version is only superficially better. Given n observations (a_i, y_i), computing per-point ratios $r_i = a_i/y_i$ and "predicting" each y_i from its own r_i commits the identity n times. Fitting a single r to all n points and scoring it on the same n points breaks the exact identity but not the epistemic problem: the score now measures in-sample goodness of fit, which improves mechanically with every degree of freedom granted to the fit and says nothing about behavior on data the fit has not seen. The rejected procedure and its multi-point relatives sit on a continuum of self-scoring; the rule draws the admissibility line where the scoring data and the fitting data are disjoint by construction.

3.1. The statistical framing. Why is disjointness the line? The statistical answer has three layers.

Layer one: selection optimism. A parameter chosen to minimize error on a sample is, by that very selection, tuned to the sample's noise as well as its signal. Its error on the selection sample is therefore biased low—optimistically—relative to its error on fresh data, and the bias grows with the flexibility of the fit relative to the data. This is the elementary overfitting phenomenon, and the one-parameter case at issue here is its minimal instance: even a single fitted slope captures some noise, and a same-sample score inherits that optimism. Held-out evaluation removes the bias by breaking the dependence between the noise the fit absorbed and the noise the score encounters.

Layer two: the hypothesis is the shape, not the number. What the response model actually asserts is not "there exists a number r such that $y \approx a/r$ on these points"—for any single point that is trivially true, and for n points it is true whenever the data are roughly proportional—but "the dose-response relation at this test point, in this region, is stably linear, such that a slope estimated yesterday predicts tomorrow." Stability under transfer from fitting data to fresh data is the entire content of the claim. A same-sample test does not weakly test this claim; it does not touch it. The train/test separation is thus not a methodological garnish but the operational definition of the thing being asserted.

Layer three: discrepancy is not noise. Even a held-out test can mislead if the model's error structure is misrepresented. The Kennedy–O'Hagan calibration framework makes a

distinction the candidate adopts wholesale: parameter uncertainty (our estimate of r_{eff} is imprecise) is not model discrepancy (the true dose-response curve is not exactly linear, so even the best possible r_{eff} leaves a structured, dose-dependent residual) [**37**]. A calibration that forces all misfit into parameter uncertainty will report a confidently wrong slope with honest-looking error bars; one that models discrepancy separately learns where the linear hypothesis degrades—which, for a framework whose region classifier must detect the edges of linearity, is not a nuisance term but the signal that draws the fence. The admissibility conditions therefore require the fit receipt to represent both, and the tolerance profile's linearity minimum is the declared threshold at which discrepancy evidence disqualifies the active-region claim itself.

Remark 6.6 (Relation to the audit). Anti-circularity is the response model's local instance of the audit-wide pattern of Chapter 4: a numerically computable operation (divide, refit, compare) whose availability tempts users into an epistemic claim its computation cannot support. The residual conflict (Conflict 4.15) documented the same pattern one step downstream—residuals promoted to causes—and the network conflict (Conflict 4.13) one step up—compositions promoted to laws. In all three cases the candidate's answer is identical: keep the computation, type its output honestly, and gate the claim on evidence the computation itself cannot manufacture.

4. Worked Hypothetical Illustrations

The following examples use fabricated numbers chosen for arithmetic clarity. *They are illustrations of logic, not measurements: no model was probed, no activation observed, and no value below describes any real system.* Units are the candidate's model units (alpha, activation, alpha-per-activation) and are as hypothetical as the numbers.

Example 6.7 (The circular procedure cannot fail). *(Hypothetical.)* Suppose a single sweep observation at dose $a = 2.0$ alpha yields response $y = 0.8$ activation. The circular analyst computes $r_{\text{same}} = 2.0/0.8 = 2.5$ alpha-per-activation, then "predicts" $\hat{y} = 2.0/2.5 = 0.8$ activation, matching observation exactly, and declares the relation validated.

Now corrupt the data deliberately: replace the response with a value drawn from a table of random digits, say $y = 0.137$. The same procedure gives $r_{\text{same}} = 2.0/0.137 \approx 14.599$ and $\hat{y} = 2.0/14.599 \approx 0.137$: perfect agreement again. The procedure certifies random digits as obeying the response law with the same confidence as any real signal—which is to say, its certification is worthless (Proposition 6.4). In the candidate's types, both computed ratios are constructible, but each is an `EFFECTIVE_RESISTANCE` with role `APPARENT_SAME_SAMPLE`, and Rule 6.2 refuses them admission as predictors. The refusal costs nothing real: the descriptive ratio remains available as description.

Example 6.8 (A held-out test that fails informatively). *(Hypothetical.)* Suppose a calibration split contains sweep observations at doses $(0.5, 1.0, 1.5)$ alpha with responses $(0.25, 0.50, 0.75)$ activation—exactly proportional, slope 0.5 activation per alpha, so the declared fit method estimates $r_{\text{eff}} = 2.0$ alpha-per-activation with (say) a bootstrap interval $[1.9, 2.1]$. In-sample agreement is perfect. The validation split probes the withheld doses $(2.0, 2.5)$ and observes responses $(0.82, 0.84)$ activation, whereas the model predicts $\hat{y} = 2.0/2.0 = 1.00$ and $2.5/2.0 = 1.25$. Against a hypothetical absolute tolerance of 0.05 activation, both predictions miss badly (errors 0.18 and 0.41).

The failure is informative in exactly the way the circular test could never be: the response flattens near 0.8–0.85 activation while predictions keep climbing, the signature of entry into a saturation region. The correct disposition under the candidate's machinery is not "the model

is refuted everywhere" but "the active region does not extend to these doses": the region classifier should bound the active region below 2.0 alpha, the resistance's claim contracts to the bounded region, and the saturated observations are censored from residual diagnostics per Conflict 4.15's rule. The held-out design converted disagreement into a map of where the linear hypothesis lives—precisely the discrepancy structure that Kennedy–O'Hagan-style calibration is built to expose [37].

EXAMPLE 6.9 (A held-out test that passes, and what it would license). *(Hypothetical.)* Suppose instead the validation observations at doses $(0.75, 1.25)$ alpha—interior to the calibrated range—are $(0.36, 0.64)$ activation against predictions $(0.375, 0.625)$: errors of 0.015 within the hypothetical absolute tolerance 0.05, with the linearity minimum met and off-target contamination below its ceiling. Then, and only then, may the fitted quantity hold role `CALIBRATED_HELD_OUT`, and predictions become admissible—under this profile, at this test point, for doses inside the calibrated range, in the active region, while the profile remains current.

Note how little even the pass licenses. It does not extend one dose beyond the grid's endpoints (extrapolation was never scored). It does not transfer to the neighboring layer, a re-derived direction, or next month's checkpoint (different test point or profile). It does not assert that the intervention *causes* the response in the interventionist sense without the causal-model apparatus that claim requires [27]. And it is two hypothetical numbers agreeing with two other hypothetical numbers: in the real program, the pass threshold is a registered hypothesis with controls (zero-dose, random-direction, wrong-layer, label-shuffle, norm-matched) whose defeat is also required—agreement alone, without superiority over matched controls, is not evidence of a direction-specific response at all.

5. Verification Versus Validation

Sargent's classical decomposition distinguishes the questions a simulation or model effort must keep separate: conceptual model validation (is the model's theory a defensible account of the problem entity?), computerized model verification (does the implementation correctly implement the specification?), and operational validation (do the implemented model's outputs agree with the real system's behavior in the domain of intended use?) [23]. The candidate maps onto the decomposition cleanly, and the mapping is the cleanest way to state what has and has not been done.

Conceptual model validation: partially addressed, negatively. The source audit (Chapter 4) establishes that the inherited conceptual model was not even internally univocal—eight conflicts—and the candidate's typed reconstruction restores internal coherence. Internal coherence is the precondition of conceptual validity, not its establishment; whether a local linear dose-response theory is a *defensible account* of steered activations is exactly what the registered hypotheses will assess.

Computerized model verification: this is what the bounded software evidence actually is. The test suite at the Gate 4 receipt (65 passing test methods, 58 parameterized subtests) exercises the guards, the types, the canonical encoding, the split-distinctness rules, the exact quantization boundaries, the fail-closed behaviors—the implementation against its specification. Synthetic known-fault boards (linear, cutoff, saturating, nonlinear, parallel, confounded, bypass) verify that the classifier and guards respond to designed inputs as specified. All of this is verification in Sargent's sense: it establishes that the software does what the design says, on inputs the designers constructed. It establishes nothing about any real model's behavior.

Operational validation: not performed. No live system has been calibrated; no held-out prediction against a real activation has been scored; the registered hypotheses are open. The candidate's entire status vocabulary—RING 4 / NOT LICENSED / HELD, constant-false authenticated eligibility, roles that no real quantity yet holds—exists to keep this cell of the matrix visibly empty until the preregistered closure protocol fills it.

REMARK 6.10. The distinction dissolves a misreading to which test counts invite themselves. "65 tests pass" is evidence about the correspondence between code and specification; it is categorically incapable of being evidence about the correspondence between specification and world. A perfectly verified implementation of a false theory passes all its tests. Conversely, operational validation without verification is uninterpretable—an agreement between a buggy implementation and reality confirms nothing about the theory. The program needs both, in order, and it currently has the first only.

6. What a Validated Response Model Would and Would Not License

Suppose—counterfactually as of this writing—that the closure protocol ran to completion for some profile: preregistered claim, frozen environment, randomized held-out sweeps, all controls defeated, independent recomputation, replication on a held-out checkpoint under a prespecified comparison rule, independent review. What then?

It would license: interpolative prediction of directional activation response at the calibrated test points, within the calibrated dose range, in the active region, under the current profile, with stated uncertainty; the use of such predictions as reference curves for residual *ranking* (never identification) per Conflict 4.15; region classification with the confidence the validated fit supports; and—if the corresponding composition hypotheses also closed—the network equivalences within their own validated scope. It would, in short, license the framework's *instrumental* uses: prediction, screening, prioritization, all profile-local.

It would not license: causal claims ("the dose causes the response via mechanism M")—interventionist causal language requires an explicit causal model and identifiable interventions, which a fitted slope is not [27]; physical claims of any kind—the units remain model units, and the forbidden-alias rule keeps volt/ampere/ohm out of reach permanently, validation or no; transfer—to other profiles, models, layers, directions, corpora, or dtypes—without separately admitted comparison rules; aggregate or league-table claims across systems; "measures attention" or any construct-identification claim, which would require its own validity argument connecting the projection to the construct; safety verdicts, product capabilities, or marketing predicates, which are licensing acts reserved to authorities the candidate does not contain; and permanence—a validated profile drifts, expires, and returns to `UNASSESSED` obligations like any other.

The asymmetry of the two lists is the chapter's summary. Validation, even at its best, buys a bounded instrument: a locally trustworthy slope with error bars, inside a fence. Everything outside the fence—cause, physics, transfer, safety, authority—requires evidence of a different kind, which no amount of dose-response agreement accumulates. The candidate's types are built so that the fence travels with every number; the purpose of this chapter has been to show that the fence is not caution appended to the science but the exact shape of what the science, done honestly, would have shown.

CLAIM BOUNDARY 6.11. This chapter defines a hypothesis and its admission conditions. The response model is unvalidated: no admissible `CALIBRATED_HELD_OUT` resistance exists for any live system, all worked numbers above are fabricated illustrations, and nothing here establishes empirical accuracy, causal interpretation, physical identification, or licensed use.

The anti-circularity rule and admissibility guards are implemented and verification-tested; their scientific worth is exactly the worth of the experiments they are waiting for.

CHAPTER 7

Independent Classification Axes

1. Why two axes, and why they are independent

Every classification decision in CAS-Ohm 1.0 descends from a single retained source conflict: the Quantification Guide's ASOM mnemonic (Starved/Aligned/Oversaturated/Misaligned) overlays semantic judgments onto dose-response regions, while the operationalization addendum states plainly that dose-response signatures and effect-axis outcomes are distinct and that a flat response has multiple causes. The candidate resolves this not by choosing one source over the other but by refusing the overlay as a matter of type discipline: the implementation classifies two independent axes and derives any mnemonic label afterward, as a display-only partial function that is permitted to return nothing.

The first axis answers a question about the *shape* of the local dose-response relation at a registered test point: as the declared alpha dose is varied over adjacent points of a sweep, does the observed flow magnitude respond linearly at the expected slope, not at all, only up to a ceiling, or in some other way? This is the question dose-response analysis has always asked, and the candidate deliberately inherits the vocabulary of that tradition rather than the vocabulary of circuit diagnosis: a region label is a statement about a fitted local curve under a declared design, in the sense of nonlinear dose-response regression [38], not a statement about what the underlying system *is* or *wants*.

The second axis answers a question about the *direction* of whatever response exists: did the activation move along the registered direction vector, against it, or somewhere else in activation space? This question is answerable, in principle, even when the magnitude question is degenerate, and unanswerable in practice in many situations where the magnitude question is crisp. The two questions have different evidence requirements, different failure modes, and different silence conditions. Collapsing them into one label is precisely the ASOM overload the source audit preserved as a conflict, and the whole architecture of this chapter is the refusal to perform that collapse.

Throughout this chapter the enumerations are reproduced exactly as implemented in the candidate's typed ontology (`src/cas_ohm_v1/model.py`). The dose-response region enumeration has seven members; the semantic outcome enumeration has four. Nothing in this chapter asserts that any classification has been produced against a live model: the artifact status of the entire candidate remains OPEN / RING 4 / NOT LICENSED / HELD, and every numeric example below is an illustrative construction over synthetic or hypothetical sweep data, not an empirical result.

CLAIM BOUNDARY 7.1 (Ring 4 status of every classification). A region label or outcome label emitted by the candidate is a statement about recorded sweep observations under one calibration profile, produced by tested software. It is not a scientific conclusion, a safety finding, a measurement of attention, or a causal diagnosis. No classification in this chapter is licensed for any representation beyond structural research use.

2. The dose-response region axis as implemented

Definition 7.2 (Dose-response region enumeration). The implemented enumeration `DoseResponseRegion` has exactly seven members, in declaration order: `UNASSESSED`, `CUTOFF`, `ACTIVE`, `SATURATION`, `BREAKDOWN`, `NON_OHMIC`, `UNSTABLE`. No other region label exists in the type system, and the classification container (`OperatingClassification`) rejects any value that is not an exact member of this enumeration.

A structural fact about the implementation deserves emphasis before the region-by-region treatment, because it governs how several members must be read. The candidate distinguishes two classification modes. In `NUMERIC_DERIVED` mode, a factory function derives the region from at least three sweep observations bracketing an interior index, using only the numeric neighbors, the profile's expected resistance, and the profile's tolerance thresholds; the factory then binds the exact observation tuple, its canonical digest, and the classification into one assessment object that replays deterministically. In `EXTERNAL_UNVERIFIED` mode, a region label supplied from outside is carried as an explicitly unverified annotation and is forbidden from embedding any derived sweep claim. The numeric deriver, as implemented, emits only four of the seven members: `CUTOFF`, `SATURATION`, `NON_OHMIC`, and `ACTIVE`. The members `UNASSESSED`, `BREAKDOWN`, and `UNSTABLE` are reachable only as typed carriers of external or absent judgment. This asymmetry is not an oversight; it is the candidate's honesty about what its three-point local classifier can and cannot distinguish, and it is recorded here because a reader of the enumeration alone would not see it.

Invariant 7.3 (Numeric emission subset). In `NUMERIC_DERIVED` mode the implemented classifier emits a region in {`CUTOFF`, `SATURATION`, `NON_OHMIC`, `ACTIVE`} and never emits `UNASSESSED`, `BREAKDOWN`, or `UNSTABLE`. The latter three members are admissible only in `EXTERNAL_UNVERIFIED` mode, where the assessment may not carry observations, an index, or an expected resistance.

The numeric derivation itself, for the record, proceeds as follows. Given strictly increasing drives at the predecessor, current, and successor observations, the classifier computes the backward slope, the forward slope, the expected slope $1/r_{\text{eff}}$ from the supplied expected resistance, a slope floor equal to the expected slope times the profile's relative tolerance, and a relative slope error equal to the larger of the two slope deviations divided by the expected slope. Classification is then a short guarded cascade: flow at or below the absolute tolerance yields `CUTOFF`; a flat forward slope with a live backward slope yields `SATURATION`; a relative slope error above the relative tolerance yields `NON_OHMIC`; otherwise `ACTIVE`. Every threshold in this cascade is a profile-bound tolerance, not a universal constant, and the same sweep classified under a different tolerance profile may land in a different region. Region labels are therefore profile-relative twice over: through the calibration identity that produced the observations and through the tolerance identity that partitioned them.

2.1. `UNASSESSED`. *Signature.* There is none. `UNASSESSED` is the typed absence of a region judgment: no sweep was classified, or an external annotation declined to commit. It is the region axis analogue of a missing value that has been made unignorable by being given a name.

Causes. A profile may have been constructed and reviewed without any sweep yet run; an external record may carry semantic information but no dose-response evidence; a pipeline may deliberately defer region classification until controls are in place. In all such cases the honest label is `UNASSESSED`, and the type system carries it forward rather than defaulting to anything.

What it does not imply. `UNASSESSED` does not imply `ACTIVE`-by-default, does not imply that the response is small, and does not imply that assessment failed. It implies only that no admissible classification event has occurred. Critically, an `UNASSESSED` region participates in the no-diagnosis conditions of Section 8: since residual diagnosis requires the `ACTIVE` region and `UNASSESSED` is not `ACTIVE`, nothing downstream fires.

2.2. `CUTOFF`**.** *Signature.* The observed flow magnitude at the classified point is at or below the profile's absolute tolerance (`delta_abs`): within measurement resolution, nothing is flowing. In the implemented cascade this test fires first, so a point that is simultaneously flat and tiny is `CUTOFF`, not `SATURATION`.

Causes. The candidate deliberately does not enumerate causes inside the label, because the causes are many and the label identifies none of them. A dose below an activation threshold; a direction vector that fails to intersect the mechanism it was intended to probe; an intervention applied at a layer or token position where the target computation has not yet formed or has already been consumed; a hard upstream gate; an instrumentation fault that zeroes the projection; a full-scale or units error that shrinks the recorded value below tolerance. All produce the same signature.

What it does not imply. `CUTOFF` does not imply that the system is *starved* of the driven content, that larger doses would produce response, that the mechanism is absent, or that the intervention is safe or ineffective in any behavioral sense. It is a statement that this sweep, at this point, under this profile, recorded no resolvable flow. The mnemonic reading (STARVED) is available only through the overlay of Section 6 and only when the semantic axis independently reads `INTENDED`, which is itself a subtle condition: a semantic judgment about a response that is below resolution inherits all the fragility of dividing by small numbers, which is why downstream ratio diagnostics are censored rather than computed (Chapter 10 of this monograph).

2.3. `ACTIVE`**.** *Signature.* Flow is resolvable, both local slopes exist, and the relative slope error against the profile's expected resistance is within the relative tolerance. Informally: the local dose-response relation looks like the calibrated line, at the calibrated slope, within declared error.

Causes. An `ACTIVE` label is evidence-compatible with the candidate's central hypothesis — a local linear response region — but it is equally compatible with a locally linear stretch of a globally nonlinear curve, with a coincidental slope match at one interior point, or with compensating errors in the full-scale and direction conventions. Three points cannot distinguish these; the label does not pretend to.

What it does not imply. This is the region most exposed to overreading, so the negative space is listed at length. `ACTIVE` does not imply `INTENDED`: the magnitude of the projection can track the dose perfectly while the induced activation moves off the registered axis or against it (Section 5). `ACTIVE` does not imply causal efficacy of the driven content on behavior; it is a statement about a projection coordinate, not about outputs. `ACTIVE` does not imply that the linearity extends beyond the bracketing points, that the region would survive a different seed or corpus, or that the fitted resistance transfers to any other test point, model revision, or profile. And `ACTIVE` at one interior index does not imply `ACTIVE` anywhere else on the same sweep: region labels are pointwise.

2.4. `SATURATION`**.** *Signature.* The backward slope is alive (above the slope floor) while the forward slope is flat (at or below it): response grew on the way in and has stopped growing

on the way out. The implemented test is deliberately one-sided and local; it detects a ceiling entered between the predecessor and successor, not the global supremum of the response.

Causes. Genuine ceilings in the projected activation; normalization ceilings (a full-scale chosen low, so that the instrument clips before the system does); representational ceilings such as norm growth limits at the hooked site; or a sweep whose upper drives simply left the calibrated alpha range's informative interior. The label does not distinguish a system that saturates from an instrument that clips, and the candidate's separation of the overload ratio from the clipped display value (Chapter 8) exists exactly so that this ambiguity is carried in the record rather than erased by it.

What it does not imply. `SATURATION` does not imply *oversaturation* in any semantic or safety sense; does not imply that the response at the ceiling is aligned with the intended direction; does not imply that dose reduction would restore linear response (hysteresis and state dependence are explicitly out of scope of the memoryless candidate); and does not imply that the ceiling is a property of the model rather than of the measurement chain. Saturated-stage residuals are, per the source audit itself, meaningless; the no-diagnosis machinery enforces this by refusing residual computation outside `ACTIVE`.

2.5. `BREAKDOWN`**.** *Signature.* None is implemented numerically. `BREAKDOWN` is retained in the enumeration from the source material — the Guide's fifth region, in which response ceases to be governed by the modeled relation at all, by analogy with dielectric breakdown — but the three-point classifier has no test that emits it, and the audit's five-region summary and the implementation's seven-member enumeration are both recorded in the seed paper for exactness. As implemented, a `BREAKDOWN` label can enter the system only as an `EXTERNAL_UNVERIFIED` annotation.

Causes. Hypothetically: interventions large enough to destroy the local structure being measured — degenerate outputs, representation collapse, drives far beyond the calibrated grid. The candidate takes no position on when this occurs, because it has no admitted evidence that it occurs, and the calibrated alpha range gate refuses predictions outside the grid in any case.

What it does not imply. An external `BREAKDOWN` label carries no numeric evidence by construction (the external mode forbids embedded sweeps), so it implies nothing quantitative whatsoever. It is a typed way of recording that some outside process asserted catastrophic non-response, pending verification that the candidate itself cannot supply. It does not imply damage, harm, jailbreak, or any behavioral state; the electrical connotation of the borrowed word licenses nothing.

2.6. `NON_OHMIC`**.** *Signature.* Flow is resolvable and the point is not saturated, but the relative slope error against the expected resistance exceeds the relative tolerance: the response is doing something, and that something is not the calibrated line. Convex, concave, kinked, and merely miscalibrated local shapes all land here.

Causes. Genuine nonlinearity of the projection in the swept range; a calibrated resistance that is simply wrong for this neighborhood (fit elsewhere, drifted, or fit on contaminated data); interference between the steering intervention and other active mechanisms; or tolerance misconfiguration. The label is a residual category in the taxonomic sense: it collects everything resolvable that neither matches the line nor exhibits the specific saturation signature.

What it does not imply. `NON_OHMIC` does not imply `MISALIGNED`; this is one of the two implications the seed paper singles out for explicit denial. A response can be strongly nonlinear in magnitude while moving exactly along the intended axis, and the mnemonic overlay

accordingly returns no label at all for an intended non-ohmic point. It also does not imply that the Ohmic model has been falsified in any global sense — one point exceeding one tolerance under one profile is not a test of the registered hypotheses, whose closure protocol is preregistered and has not been run.

2.7. `UNSTABLE`. *Signature.* None is implemented numerically; like `BREAKDOWN`, `UNSTABLE` is reachable only as an external annotation. Its intended meaning, inherited from the sources, is non-reproducibility: the same dose at the same point under the same profile fails to yield a stable response distribution — variance rather than shape is the pathology.

Causes. Seed sensitivity, sampling noise dominating the projection, context-dependent activation at the hooked site, or drift within a session. The three-point classifier, which sees a single sweep, is structurally incapable of assessing reproducibility, which is why it never emits this label; assessing it would require repeated sweeps under controlled randomization, which is exactly item eight of the closure experiment list (bootstrap uncertainty and cross-split reproducibility), an experiment that has not been admitted.

What it does not imply. `UNSTABLE` does not imply that the underlying system is stochastic in any interesting sense, that the direction is wrong, or that the profile is invalid; it implies only that a stable pointwise response summary was not obtainable by whatever external process asserted the label. It is also a no-diagnosis condition: unstable points emit nothing downstream.

3. A worked derivation, and what an assessment binds

The region taxonomy is easiest to hold onto with the classifier's arithmetic run once in the open. All numbers below are illustrative constructions over a synthetic sweep; nothing is a measurement.

Example 7.4 (Illustrative three-point classification). Let the profile declare expected resistance $r_{\text{eff}} = 2.0$ alpha-per-activation, absolute tolerance `delta_abs` $= 0.05$, relative tolerance `delta_relative` $= 0.2$, off-target maximum 0.3, and residual floor 0.01. Consider the ordered sweep observations (drive, flow, off-axis): (1.0, 0.50, 0.05), (2.0, 1.00, 0.08), (3.0, 1.48, 0.10), classified at the interior index. The expected slope is $1/2.0 = 0.5$. The backward slope is $(1.00 - 0.50)/(2.0 - 1.0) = 0.50$; the forward slope is $(1.48 - 1.00)/(3.0 - 2.0) = 0.48$; the relative slope error is $\max(|0.50 - 0.5|, |0.48 - 0.5|)/0.5 = 0.04$, well inside 0.2. The current flow 1.00 exceeds `delta_abs`, so not `CUTOFF`; the forward slope 0.48 exceeds the slope floor $0.5 \times 0.2 = 0.1$, so not `SATURATION`; the relative slope error is inside tolerance, so not `NON_OHMIC`: region `ACTIVE`. The semantic ratio is $0.08/\max(1.00, 0.01) = 0.08 \leq 0.3$: outcome `INTENDED`. Overlay: `ALIGNED`.

Now perturb the same sweep three ways. (a) Flows (0.98, 1.00, 1.01): the backward slope collapses to 0.02 and the forward to 0.01, both under the slope floor — but the saturation test requires a *live* backward slope with a flat forward slope, and here both are flat while the flow is large, so the cascade falls through to the relative-slope-error test, which reads $\max(|0.02 - 0.5|, |0.01 - 0.5|)/0.5 = 0.98 > 0.2$: region `NON_OHMIC` — a uniformly flat plateau at high flow, entered before the observed window, is locally indistinguishable from a wrong slope, and the classifier says the modest thing rather than the narrative thing. (b) Flows (0.50, 1.00, 1.02): backward slope 0.50 (live), forward slope $0.02 \leq 0.1$ (flat): region `SATURATION` — the ceiling was entered *inside* the window, which is what the one-sided signature detects. (c) Off-axis column (0.30, 0.40, 0.45) with the original flows: the semantic ratio becomes 0.40, exceeding 0.3: outcome `OFF_AXIS`, region still `ACTIVE`, overlay `MISALIGNED` — the independence principle in one line, since nothing about the magnitude cascade changed.

The example also shows the sense in which tolerances are *classification parameters*, not implementation detail. Halving `delta_relative` to 0.1 reclassifies perturbation (a) no differently but tightens the `ACTIVE` verdict of the base case (0.04 still passes); raising it to 1.0 would absorb perturbation (a) into `ACTIVE`. Every region label is therefore implicitly subscripted by the tolerance profile that produced it, and the candidate makes the subscript explicit: the tolerance block is part of the calibration profile, the profile digest is bound into every assessment, and two assessments with different tolerance digests are formally incomparable even over identical observations.

What the assessment object binds, beyond the two axis labels, completes the account. A numeric assessment is factory-only (direct construction is rejected by token) and carries: the classification pair; the mode (`NUMERIC_DERIVED`); the profile's response-region method identity; the calibration-profile digest; the registered test-point digest; an evidence reference whose digest must equal the canonical digest of the embedded observation tuple (at least three points, with an interior index); the index and expected resistance; the profile's control bindings; and a full operation binding (Chapter 9's Definition on aggregate binding enumerates the binding fields). At consumption time the assessment is not trusted but *replayed*: the validator re-resolves the evidence artifact, requires byte equality with the embedded observations, re-derives the classification from the observations, index, expected resistance, and current profile tolerances, and requires the re-derived pair to equal the stored pair. A classification that cannot be reproduced from its own bound evidence does not validate — which is the concrete meaning of the earlier statement that region labels fail loudly.

By contrast, an `EXTERNAL_UNVERIFIED` assessment must carry *no* observations, no index, and no expected resistance — the type rejects an external label that smuggles derived-looking numbers — and consumers are required to treat it as an unverified annotation: the diagnostic layer, in particular, replays numeric assessments and refuses external ones outright. The two modes thus split exactly along the loud/silent failure line: numeric assessments carry their own disproof machinery; external ones carry a label and an explicit admission that no machinery stands behind it.

4. The semantic outcome axis as implemented

Definition 7.5 (Semantic outcome enumeration). The implemented enumeration `SemanticOutcome` has exactly four members: `UNASSESSED`, `INTENDED`, `OPPOSITE`, `OFF_AXIS`.

The numeric deriver assesses the semantic axis from a single ratio: the off-axis activation magnitude at the current point divided by the flow magnitude (floored by the profile's residual floor to avoid division by values compatible with zero). If that ratio exceeds the profile's off-target maximum, the outcome is `OFF_AXIS`; otherwise `INTENDED`. Two consequences follow and are recorded candidly. First, the numeric path never emits `OPPOSITE`: distinguishing anti-directional movement from off-axis movement requires signed projection evidence that the magnitude-only sweep observation does not carry, so `OPPOSITE` is, in the current implementation, an externally assessable outcome only. Second, the numeric path never emits `UNASSESSED`, because it refuses to run at all without the observations that make the ratio computable; `UNASSESSED` is what an assessment carries when the semantic question was not put to any evidence.

4.1. `UNASSESSED`**.** The typed absence of a directional judgment. A profile can classify region crisply and still have no admissible answer to "did it move the intended way," most obviously when no behavioral or activation-space comparison against the registered direction

has been performed. `UNASSESSED` on the semantic axis is a hard silence condition: the residual diagnostic explicitly rejects assessments whose semantic outcome is `UNASSESSED`, on the principle that a residual against an intention nobody assessed is an invitation to invent the intention afterward.

4.2. `INTENDED`**.** The observed activation moved along the registered direction, with off-axis contamination below the profile's tolerance. `INTENDED` does not imply that the magnitude was appropriate (it combines with every region, including `CUTOFF` and `SATURATION`), does not imply that the behavioral consequence of the activation movement matches anyone's intention at the output level, and does not imply endorsement of the intervention. The word "intended" binds to the registered direction vector and to nothing else: change the registration and the same physical event reads differently. This is deliberate; intention lives in the profile, where it is frozen, digested, and auditable, not in the interpreter's retrospect.

4.3. `OPPOSITE`**.** The observed activation moved against the registered direction: a signed reversal, not merely leakage. As noted, the current numeric path cannot produce this label; it exists in the enumeration because the source material treats reversal as a first-class outcome (the inverted-response reading of a steering intervention) and because a future signed-evidence classifier must have somewhere typed to land. An `OPPOSITE` label, where externally asserted, does not imply psychological reactance, adversarial optimization, safety-training interference, or any mechanism at all; mechanism attribution from outcome labels is exactly the inferential step the independence principle below forbids.

4.4. `OFF_AXIS`**.** The observed activation moved, but a supra-tolerance fraction of the movement lies outside the registered direction. This is the label the numeric path emits when the off-axis ratio exceeds the profile's off-target maximum. `OFF_AXIS` does not imply that the off-axis component is harmful, meaningful, or even stable; a direction vector that captures its target imperfectly will manufacture off-axis readings from perfectly on-target mechanisms, and a poorly conditioned projection basis will do the same. The label records a geometric fact about one comparison under one registration; the temptation to read it as "side effects" is a semantic overlay the candidate declines.

5. The independence principle

Rule 7.6 (Axis independence). No dose-response region mechanically determines a semantic outcome, and no semantic outcome mechanically determines a region. Every pair in the product $\{\text{region}\} \times \{\text{outcome}\}$ is admissible to the type system, and the classification container stores the two axes as separate fields with no consistency coupling beyond enumeration membership.

The principle is worth developing at length, because it is the load-bearing wall between this chapter and the mnemonic it replaces.

Active does not mean aligned. The region axis sees the magnitude of a projection; the outcome axis sees its geometry. A dose that recruits a strong, beautifully linear response in the projection coordinate can be recruiting it off the intended axis — the projection tracks whatever component of the induced activation lies along the measurement direction, and a diagonal movement has such a component. Conversely, an intervention can move activation precisely along the intended axis while the magnitude response is grossly nonlinear. In the crossed cases, the mnemonic has no word, and that lexical gap in ASOM is not an accident of vocabulary: it marks a real, observable class of events that a four-word overlay cannot

name. An ACTIVE/OFF_AXIS classification and an ACTIVE/OPPOSITE classification are the two canonical counterexamples to region-implies-outcome, and both are fully admissible records.

Flat response is multi-causal. The converse direction is subtler and more dangerous, because flatness invites narrative. A flat or negligible response at the classified point is evidence-compatible with at least five generating situations, which the candidate insists on keeping distinct: (i) *cutoff* — the dose is below whatever threshold governs resolvable response at this point; (ii) *saturation* — the response exhausted its range before the observed doses, so the local curve is flat at ceiling rather than at floor; (iii) *wrong direction* — the intervention moves activation vigorously, but orthogonally to the measurement direction, so the projection sees nothing; (iv) *insensitivity* — the system genuinely does not respond to this intervention at this site, at any dose; and (v) *measurement failure* — hooks misplaced, units mismatched, full-scale misdeclared, instrument silent. Of these five, the three-point classifier can discriminate only the first two, and only by their local slope signatures; the last three all present as CUTOFF or as low-magnitude ACTIVE noise, and distinguishing them requires controls — random same-norm directions, wrong-layer placements, zero-intervention baselines — that live in the preregistered closure protocol, not in the classifier. A flat point is therefore never, by itself, evidence of starvation, and the candidate's refusal to emit STARVED except through the explicitly decorative overlay is the operational form of that refusal.

The independence principle also has an evidential asymmetry worth recording. Region classification is internally checkable: it replays from the observation tuple and the tolerance profile, and the factory binds the digest of the observations so that replay is exact. Outcome classification, by contrast, is only as good as the direction registration and the off-axis instrumentation, both of which are profile inputs that the classifier consumes on trust. The two axes therefore fail differently — region labels fail loudly, by replay divergence, while outcome labels fail silently, by mis-registration — and treating them as one axis would let the silent failure mode contaminate the loud one. Keeping them separate keeps their error budgets separate, in the same spirit in which model verification and model validation are kept separate activities [23].

6. ASOM as mnemonic overlay

Definition 7.7 (ASOM overlay). The implemented enumeration AsomLabel has four members: STARVED, ALIGNED, OVERSATURATED, MISALIGNED. The overlay is the partial function asom_label() on classifications, defined exactly as follows: if the semantic outcome is OPPOSITE or OFF_AXIS, return MISALIGNED; otherwise, if the semantic outcome is not INTENDED, return nothing; otherwise return STARVED for region CUTOFF, ALIGNED for ACTIVE, OVERSATURATED for SATURATION, and nothing for every other region.

Three properties of this definition carry the argument. First, the overlay is *derived*: it is computed from the two stored axes and stored nowhere, so no ASOM word can drift out of agreement with the axes that justify it, and no process can write an ASOM word directly. Second, the overlay is *partial*: an UNASSESSED outcome yields no label at all, and an INTENDED outcome over BREAKDOWN, NON_OHMIC, UNSTABLE, or UNASSESSED regions likewise yields nothing — the mnemonic simply has no word for most of the product space, and the implementation returns None rather than coining one. Third, the overlay is *lossy in exactly one direction*: MISALIGNED merges OPPOSITE with OFF_AXIS and discards the region entirely, so recovering the axes from the label is impossible. A four-word summary of a twenty-eight-cell product space cannot be information-preserving, and the candidate's position is that it

should not try: the words exist for human pattern recognition in displays and prose, and every load-bearing consumer reads the axes.

RULE 7.8 (No identity theorem). No inference, gate, capability, diagnostic, or claim in the candidate consumes an `AsomLabel`. The overlay never appears in canonical payloads of measurement values, never conditions residual admission, and never participates in packet completeness. Any external document that reasons from STARVED, ALIGNED, OVERSATURATED, or MISALIGNED as if these were classification primitives is reasoning from a display artifact.

7. The full region-by-outcome cross table

Table 1 enumerates the product space. For compactness the outcome columns are abbreviated U (`UNASSESSED`), I (`INTENDED`), Op (`OPPOSITE`), Off (`OFF_AXIS`); each cell records the derived ASOM overlay, if any, with "—" for the overlay's silence.

Region	U	I	Op	Off
UNASSESSED	—	—	MISALIGNED	MISALIGNED
CUTOFF	—	STARVED	MISALIGNED	MISALIGNED
ACTIVE	—	ALIGNED	MISALIGNED	MISALIGNED
SATURATION	—	OVERSATURATED	MISALIGNED	MISALIGNED
BREAKDOWN	—	—	MISALIGNED	MISALIGNED
NON_OHMIC	—	—	MISALIGNED	MISALIGNED
UNSTABLE	—	—	MISALIGNED	MISALIGNED

TABLE 1. The 7×4 classification product and its derived ASOM overlay. The overlay names only six of the twenty-eight cells distinctly; fourteen cells share `MISALIGNED`, and eight cells have no name.

The remainder of this section walks the table row by row, reading each combination for what it can and cannot mean. Throughout, "possible readings" are enumerations of evidence-compatible situations, not diagnoses; selecting among them is exactly what classification alone cannot do.

`UNASSESSED` region. With outcome U, this is the null assessment: a typed placeholder, common for freshly registered test points. With outcome I, an external process asserted directional success without any region evidence — admissible as a record, useless as a premise; the overlay pointedly has no word for it. With Op or Off, someone asserted directional failure with no dose-response context; the overlay says `MISALIGNED`, and the correct reading of that word here is "an unverified external annotation exists," nothing stronger.

`CUTOFF` region. With U: no resolvable flow and no directional assessment — the modal honest state for a dead point, since assessing direction of a sub-tolerance response is numerically fraught. With I: the overlay's `STARVED` cell; the response is below resolution and whatever trace exists is on-axis. Possible readings include genuine sub-threshold dosing, but equally a dead instrument whose noise floor happens to project on-axis; the cell does not separate them. With Op: below-resolution flow asserted to be reversed — a nearly self-undermining combination that should prompt scrutiny of the asserting process rather than of the model. With Off: sub-tolerance magnitude, off-axis geometry; often the signature of measuring the wrong thing in the right place, and a natural trigger for the wrong-direction hypothesis of Section 5, which only controls can confirm.

ACTIVE region. With U: a calibrated-looking response whose direction nobody has assessed; the diagnostic machinery refuses to run on it, precisely because an unassessed intention invites retrofitting. With I: the overlay's `ALIGNED` cell and the candidate's best case — magnitude tracks dose at the calibrated slope, geometry tracks the registration; still a structural statement under one profile, not an endorsement, not transferable, not causal. With Op: linear-in-magnitude, reversed-in-direction; if externally asserted and later verified, this combination would be the single most alarming cell for a steering application, and the candidate's contribution is to make it representable rather than lexically impossible (ASOM has no word for a linear reversal other than the undifferentiated `MISALIGNED`). With Off: strong, lawful magnitude response leaking off-axis — the canonical "active-but-off-axis" counterexample; possible readings span imperfect direction registration, genuinely diffuse mechanisms, and superposed responses, and the cell selects none of them.

SATURATION region. With U: ceiling behavior, direction unassessed; residuals refuse to run, and per the sources saturated-stage residuals would be meaningless anyway. With I: the overlay's `OVERSATURATED` cell — response climbed on-axis and hit a ceiling; readings include a genuine representational ceiling and a clipping instrument, which the overload ratio (kept separately from the clipped display, Chapter 8) exists to help distinguish. With Op: ceiling-flat magnitude with asserted reversal — coherent only if the reversal occurred below the ceiling or in a signed coordinate the sweep does not carry; treat as external assertion pending signed evidence. With Off: the ceiling is real but the content at ceiling is off-target; a reading consistent with saturating the wrong mechanism entirely.

BREAKDOWN region. All four cells are external-only, since the numeric path never emits the region. With U, the cell is a bare catastrophic-response assertion; with I, an internally tense one (broken response, yet on-axis?); with Op or Off, the overlay's undifferentiated `MISALIGNED` again. In every case the assessment carries no observations by construction, and the only defensible consumption of the row is as a flag for human review of the asserting process.

NON_OHMIC region. With U: resolvable, non-linear magnitude response, direction unassessed — an honest and common state for exploratory sweeps. With I: nonlinear but on-axis; the overlay has no word, and rightly so — "misaligned" would be false and "aligned" would launder the nonlinearity. Readings include genuine supra-linear recruitment, miscalibrated expected resistance, and interference; the falsification program's gain-composition and inverse-slope experiments are the registered instruments for narrowing this set. With Op or Off: nonlinear magnitude and adverse geometry; the least constrained cells in the table, compatible with nearly any mechanism story and therefore probative of none.

UNSTABLE region. All cells are external-only. With U: an assertion of irreproducibility, unquantified. With I: "unstable but on-average intended" — a reading that presupposes distributional evidence the record cannot carry, hence flagged, not consumed. With Op or Off: the overlay says `MISALIGNED`, but the honest reading is that neither axis is trustworthy at an unstable point, which is why instability is a no-diagnosis condition rather than a diagnosable state.

Remark 7.9. The table's practical function is negative. It exists so that no combination can be declared impossible by vocabulary, and so that every overlay word can be traced to the cells it covers and the cells it merges. Twenty-two of twenty-eight cells receive either no ASOM word or a shared one; any analysis pipeline that operates at ASOM granularity therefore discards most of the classification structure by construction.

8. No-diagnosis conditions

The candidate's diagnostic layer (residual computation, and everything downstream of it) is guarded by a conjunction of admission conditions, each of which independently silences diagnosis. They are enumerated here as implemented, with the enforcing mechanism named for each.

RULE 7.10 (Silence conditions). No diagnostic is emitted when any of the following holds:

(1) *Region is not `ACTIVE`.* The residual entry point rejects any assessment whose region differs from `ACTIVE`; this covers `CUTOFF`, `SATURATION`, `BREAKDOWN`, `NON_OHMIC`, `UNSTABLE`, and `UNASSESSED` uniformly. Cutoff points have nothing to diagnose; saturated points have meaningless residuals; unstable points have untrustworthy ones.
(2) *Semantic outcome is `UNASSESSED`.* A residual against an unassessed intention is refused outright.
(3) *Profile is uncalibrated or not reviewed.* The operation context admits a calibration profile only in review state `REVIEWED`; any other review state fails the context, so no operation — classification included — can even begin.
(4) *Profile drift state is not current.* A profile whose drift state differs from current is rejected at context admission.
(5) *Replay time is stale.* The caller-supplied `replay_as_of` must satisfy `valid_from` $\leq$ `replay_as_of` $<$ `expires_at`; outside that window the profile is "stale or not yet valid" and the context fails.
(6) *Wrong model or wrong profile.* Every assessment binds the calibration profile digest and a registered test-point digest; an assessment presented against a context whose profile digest differs, or whose test point is unregistered, is rejected. Since the profile binds the exact model, configuration, tokenizer, and weights identities, a wrong-model presentation is a wrong-profile presentation and fails the same check.

Two design commitments underlie the list. The first is that silence is the default and diagnosis is the exception: every condition is checked fail-closed, and a missing precondition produces a typed rejection, never a degraded diagnostic. The second is that the conditions compose by conjunction with no override path — there is no force flag, no administrative bypass, and no trust mode in which a stale profile or an unassessed outcome becomes diagnosable. The alternative — emitting diagnostics with warnings attached — was foreclosed by the source audit's observation that warned-against readings (the residual-as-bypass reading in particular) were made on the same page as the warnings.

9. Replay-time semantics: `replay_as_of`

Every operation context carries a field `replay_as_of`: a caller-supplied timestamp, parsed and canonicalized at admission, snapshotted, and immutable thereafter (any change after admission is a validation error). Profile currency (Rule 7.10, condition 5) is evaluated against this timestamp, and every operation binding embeds it, so that each result records the historical instant *as of which* the profile was asserted current.

RULE 7.11 (Replay time is not authenticated freshness). `replay_as_of` is a replay coordinate, not a clock. It asserts that the computation is being evaluated as of a stated historical time; it does not assert that the stated time is now, that the caller's clock is honest, or that the profile is current in wall-clock reality. The context exposes this directly: its `authenticated_current` property is constantly false, and the optional clock-receipt field,

when present, is itself only a recorded artifact, not a trusted time source. A caller can therefore always choose a `replay_as_of` inside a profile's validity window and obtain classifications against that historical frame; nothing in the candidate prevents this, and nothing in the candidate represents the result as fresh.

The design is deliberate and mirrors the completeness-versus-eligibility split that governs the whole candidate: structural checks (the window inequality, the snapshot immutability, the binding of the timestamp into every result) are implemented and tested; authority checks (a trusted current-time receipt from an admitted external authority) are named, required for authenticated Evidence-Packet eligibility, and absent. The honest reading of any classification record is therefore always conditional: *under profile P, replayed as of t, this sweep classified as such.* Removing the conditionality would require exactly the authenticated infrastructure the candidate declares it does not have.

Remark 7.12 (What this chapter establishes). The classification layer contributes no empirical claim. What it contributes is a shape for future claims: a seven-member region axis and a four-member outcome axis, independent by rule, with a decorative overlay that cannot be consumed, a product space in which no combination is unspeakable, and a silence discipline under which most states of the world produce no diagnosis at all. If the registered hypotheses eventually close against this scaffolding, their content will be expressible without renaming anything; if they fail, the failure will be recordable in the same vocabulary. That neutrality between success and failure is the property the chapter was built to secure.

CHAPTER 8

Exact Normalization and Quantization

1. The problem a quantizer inherits

Of the eight conflicts preserved in the source audit, the quantization conflict is the smallest in surface area and the most instructive in kind. The Quantification Guide displays four score bands with two-decimal endpoints — 0.00–2.49, 2.50–4.99, 5.00–7.49, 7.50–10.00 — and provides no rule for values of higher precision. The conflict is not between two sources; it is between the display and the mathematics. Two-decimal endpoint pairs like 2.49 and 2.50 presuppose that scores arrive already rounded to two decimals, and are silent about every value in the open gap between them. A value of 2.495, a value of 2.4999, a value produced by dividing two exact decimals whose quotient does not terminate at two places — all fall outside the displayed partition. A classification system whose bands are defined on displayed values inherits the rounding rule of its display layer as an invisible classification parameter, and different rounding conventions then classify the same underlying value differently.

The candidate's resolution is structural: classification is computed from the exact pre-rounding value, over half-open bins whose cut points are the exact rationals 5/2, 5, and 15/2; the displayed value is derived afterward and never re-enters classification. This chapter states the exact formulas as implemented, proves the boundary-preservation property they were designed for, works the boundary cases numerically, and explains precisely why the Guide's two-decimal bands underdetermine classification. It closes with the trend-tag discipline, which extends the same exactness-first posture from single values to comparisons across values.

As everywhere in this monograph, the status frame governs: the quantizer is tested software operating on typed records (OPEN / RING 4 / NOT LICENSED / HELD); no normalized score reported here derives from an admitted live-model measurement, and all numeric values in the worked examples are illustrative constructions.

2. Exact normalization as implemented

DEFINITION 8.1 (Normalization formulas). Let raw ≥ 0 be an instrumented flow magnitude in ACTIVATION units at a registered test point, and let $I_{\mathrm{FS}} > 0$ be the calibration profile's full-scale value for that test point, produced under the profile's declared full-scale method. The implemented normalization computes three quantities:

$$q_{10}^{\mathrm{exact}} \;=\; 10 \cdot \frac{\mathrm{raw}}{I_{\mathrm{FS}}}, \qquad \mathrm{overload} \;=\; \frac{\mathrm{raw}}{I_{\mathrm{FS}}}, \qquad q_{10}^{\mathrm{disp}} \;=\; \min\bigl(q_{10}^{\mathrm{exact}},\, 10\bigr).$$

The displayed score q_{10}^{disp} is clipped at 10; the overload ratio is never clipped; and the exact value q_{10}^{exact} is retained as an exact ratio object — numerator raw, denominator I_{FS}, scale 10 — whose value is evaluated in exact rational arithmetic, not in binary floating point.

Three implementation details make the definition load-bearing rather than decorative.

First, the three quantities are constructed together, inside a factory, and sealed into a single result object (`NormalizedFlow`) that cannot be built by any other path: the constructor validates a factory token and rejects direct construction. The result object re-verifies, at admission, the arithmetic identity between its parts — that the stored displayed value equals $\min(10, 10 \cdot \text{raw}/I_{\text{FS}})$ and the stored overload ratio equals raw/I_{FS}, both recomputed from the exact ratio — so a normalized record whose display has drifted from its exact value is unrepresentable, not merely unlikely.

Second, the inputs are typed evidence, not numbers. The raw flow must be a `FLOW_MAGNITUDE` in ACTIVATION units, of measurement kind `INSTRUMENTED`, with a recorded sample size, bound to the same test point for which the profile's full-scale value is registered; the full-scale value itself is resolved from the profile and re-verified against the evidence registry. A quantity of the wrong kind, unit, provenance, or test point is rejected before any arithmetic occurs. The mixed-scale error preserved in the source audit — the Guide's sat = 8.5 read against a 0–1 fraction convention — is in this sense not merely avoided but unrepresentable: a Q10 score, a Q100 constraint score, a fraction, and an activation magnitude are four different unit kinds, and the unit-admissibility table refuses the cross-assignments.

Third, the exact value is carried as a ratio of decimals evaluated through exact rational arithmetic (numerator and denominator lifted to arbitrary-precision rationals), so that the band comparisons of the next section are performed on a representation in which equality and order at the cut points are exact questions with exact answers. No binary float participates in classification.

REMARK 8.2 (Why the overload ratio is a separate quantity). Clipping the display at 10 is a presentation decision inherited from the Q10 score convention; it deliberately destroys information above full scale. The overload ratio preserves exactly the destroyed information: a reading of overload = 1.2 records that the raw flow exceeded the declared full scale by twenty percent, which the clipped display $q_{10}^{\text{disp}} = 10$ cannot distinguish from a reading exactly at full scale. Because the full-scale value is itself method-relative — the profile's declared `IFSMethod` is one of three non-interchangeable procedures, per the preserved full-scale conflict — an overload above 1 is ambiguous between a genuinely extreme response and a full-scale value selected too low for the point; the ratio does not resolve the ambiguity, it preserves it in a typed, dimensionless quantity (`OVERLOAD_RATIO`) that downstream analysis can interrogate. Overload is also the natural bridge to the saturation discussions of Chapter 7: a saturated region label with overload near 1 reads differently from the same label with overload far above 1, and neither reading is licensed automatically.

3. Representation: decimals in, rationals underneath

The exactness claims of this chapter rest on a representation choice that the candidate makes globally and the quantizer inherits: every numeric value in the evidence layer is an exact decimal — a scaled integer, not a binary float — and every ratio that classification consumes is carried as an explicit numerator-denominator pair evaluated in arbitrary-precision rational arithmetic. The choice forecloses three familiar failure classes at once.

First, *representation error at the boundary.* The cut point 2.5 is exactly representable in decimal and in rational arithmetic; a value like 2.4999 is likewise exact. In binary floating point, neither the inputs nor the cut points of a decimal band scheme are generally representable, and a comparison at the boundary inherits up to half an ulp of silent displacement — small, but structurally identical to the rendering gap of Section 7, and platform-honest rather

than convention-honest. The rational route makes the comparison a cross-multiplication of integers, which is exact on every platform.

Second, *division before comparison.* A naive implementation would compute $10 \cdot \mathrm{raw}/I_{\mathrm{FS}}$ as a quotient first and compare afterward, forcing a precision decision (how many digits of quotient?) before the comparison that needed the answer. The implemented route stores the ratio unevaluated — numerator, denominator, scale — and compares by cross-multiplication, so no quotient precision is ever chosen for classification purposes. The displayed quotient, by contrast, is computed in the candidate's declared decimal arithmetic context, where a precision *is* chosen; the asymmetry is the whole design: presentation gets a precision, classification gets none because it needs none.

Third, *drift between stored value and stored band.* Because the band is a function, not a field — it is recomputed from the exact ratio on demand, and the sealed result object verifies its internal identities at construction — there is no stored band to fall out of sync with a stored value. Nothing in the evidence model can hold a record asserting exact value 2.4 and band $Q2$; the assertion is not storable, and this unstorability is the quantizer's contribution to the candidate's general no-bare-numbers discipline.

4. Half-open bins

Definition 8.3 (Q4 bands). Band assignment is a total function of the exact value q_{10}^{exact}, through half-open intervals closed on the left:

$$[0, 2.5) \mapsto Q1, \qquad [2.5, 5) \mapsto Q2, \qquad [5, 7.5) \mapsto Q3, \qquad [7.5, \infty) \mapsto Q4.$$

Equivalently, as implemented: the value is compared in exact rational arithmetic against $5/2$, then 5, then $15/2$, returning the band at the first strict-inequality success and $Q4$ otherwise.

Three observations. First, the bins partition the entire non-negative ray: every admissible exact value, including values above 10 produced by overload, receives exactly one band. The implementation's top band is $[7.5, \infty)$, not $[7.5, 10]$: an overloaded value of 12 is a $Q4$ value with an overload ratio recorded alongside, rather than an error or an out-of-range special case. (The audit's prose summary states the top bin as $[7.5, 10]$; the implemented comparison function has no upper test, and the two agree on all clipped displays while differing on the exact values above 10 that only the overload path produces. This monograph records the implemented behavior.) Second, each cut point belongs to the upper band: 2.5 is $Q2$, 5 is $Q3$, 7.5 is $Q4$. The choice of upward assignment is a convention, but having *some* exact convention is not: it is what makes band assignment a function. Third, the cut points are exact rationals, so membership questions are decidable without tolerance: there is no epsilon, no rounding mode, and no platform dependence in the band function.

5. The boundary-preservation proposition

Proposition 8.4 (Boundary preservation). *Band assignment is computed from* q_{10}^{exact} *by exact comparison against* 2.5, 5, *and* 7.5. *Display derivation — clipping at* 10 *and any subsequent rounding for presentation — cannot move a value across a band boundary, because the displayed value never re-enters classification.*

Proof. The claim decomposes into three parts.

(i) *The band function reads only the exact value.* By construction, the implemented band function accepts only the sealed normalized-flow result and evaluates only its exact-ratio component; the displayed value and the overload ratio are not arguments to any comparison. The band of a record is therefore a function of $(\mathrm{raw}, I_{\mathrm{FS}})$ alone, through the exact quotient.

(ii) *Exact comparison is total and unambiguous at the cut points.* The exact value is a non-negative rational with an explicit numerator and denominator; the cut points $5/2$, 5, and $15/2$ are rationals. For any two rationals, exactly one of $<$, $=$, $>$ holds, and the comparison is computed exactly in integer arithmetic on cross-multiplied numerators and denominators. Hence for every representable input the guarded cascade ($< 5/2$, else < 5, else $< 15/2$, else $Q4$) selects exactly one band. In particular, an input exactly equal to a cut point fails the strict inequality at that cut point and is assigned to the upper band; equality is not a boundary anomaly but an ordinary, deterministically handled case.

(iii) *Display is derived afterward and is inert.* The displayed value $q_{10}^{\mathrm{disp}} = \min(q_{10}^{\mathrm{exact}}, 10)$, and any two-decimal (or other) rounding applied for presentation, are computed from the exact value after classification and are consumed only by presentation surfaces. Since no classification path accepts a displayed value as input — the band function's type signature admits only the sealed result, whose exact component is verified at admission against its own display — there is no composition of implemented functions in which a rounded value determines a band. A display rounding that crosses a cut point (say, $2.4999 \to 2.50$) therefore changes what a human reads, and nothing else.

Combining (i)–(iii): the composite map from $(\mathrm{raw}, I_{\mathrm{FS}})$ to (band, display) computes the band from the exact quotient by total exact comparison and computes the display independently downstream; no display operation participates in, or can alter, band assignment. □

Two small consequences are worth separating from the proposition, because they answer questions a careful reader will ask next.

Lemma 8.5 (Display is monotone and non-injective). *The display map $q_{10}^{\mathrm{exact}} \mapsto q_{10}^{\mathrm{disp}} = \min(q_{10}^{\mathrm{exact}}, 10)$ is order-preserving (weakly) and not injective: it is the identity on $[0, 10]$ and constant on $[10, \infty)$. Consequently the pair (display, overload) determines the exact value — the overload ratio is injective on the whole ray — while the display alone determines it only below full scale.*

Proof. Monotonicity and the two-piece description are immediate from the definition of min. Injectivity of the overload ratio holds because overload $= q_{10}^{\mathrm{exact}}/10$ identically — overload is the same exact quotient at unit scale — so the exact value is recoverable from it by multiplication by 10; the display coincides with the exact value on $[0, 10]$ and discards it above. □

Corollary 8.6 (Band recoverability). *The band of a record is recoverable from its overload ratio alone (compare against $1/4$, $1/2$, $3/4$), and from its display alone if and only if the display is strictly below 10 or the band is $Q4$. In particular, a consumer holding only clipped displays can still classify correctly, because every clipped record displays 10 and $10 \geq 7.5$ lies in $Q4$; what such a consumer cannot do is anything finer than banding in the overload regime.*

Table 1 collects the band structure with its boundary behavior in one place.

Remark 8.7. The proposition is a software-structural statement, verified by the candidate's test suite at the level of golden boundary fixtures; it is not a metrological claim. It says that the quantizer cannot *add* a boundary error to whatever measurement error the raw value already carries; it does not say the raw value is right. Uncertainty attached to the raw flow propagates through normalization as interval and standard uncertainty on the ratio (Section 8); a raw value whose uncertainty interval straddles a cut point yields a band assignment that is exact for the point value and undecided across the interval, and honest reporting of that situation is the trend-tag discipline's concern (Section 10).

Band	Exact interval	Cut point at left	Guide display band	Gap under two decimals
$Q1$	$[0, 2.5)$	0 (closed)	0.00–2.49	$(2.49, 2.50)$ unruled
$Q2$	$[2.5, 5)$	$2.5 \in Q2$	2.50–4.99	$(4.99, 5.00)$ unruled
$Q3$	$[5, 7.5)$	$5 \in Q3$	5.00–7.49	$(7.49, 7.50)$ unruled
$Q4$	$[7.5, \infty)$	$7.5 \in Q4$	7.50–10.00	overload collapsed at 10.00

TABLE 1. Exact half-open bands versus the Guide's printed two-decimal display bands. Every printed endpoint pair leaves an unruled open gap; the implemented cut points rule every value; the top band differs additionally in its treatment of overload.

6. Worked examples

All values below are illustrative constructions, labeled as such; none is a measurement.

EXAMPLE 8.8 (Overload: raw/$I_{\mathrm{FS}} = 1.2$). Take raw $= 6$ activation units at a test point whose declared full scale is $I_{\mathrm{FS}} = 5$. Then

$$q_{10}^{\mathrm{exact}} = 10 \cdot \tfrac{6}{5} = 12, \qquad \text{overload} = \tfrac{6}{5} = 1.2, \qquad q_{10}^{\mathrm{disp}} = \min(12, 10) = 10.$$

Band assignment reads the exact value 12: since $12 \geq 15/2$, the band is $Q4$. The record thus carries four coordinated facts — exact 12, display 10, overload 1.2, band $Q4$ — and the sealed result object verifies their mutual consistency at construction. Note what the display alone would have lost: a reader of "10.00" cannot distinguish this record from a reading exactly at full scale, but the record itself can, and any consumer that reads the typed object rather than its rendering retains the distinction.

EXAMPLE 8.9 (Exact cut point: $q_{10}^{\mathrm{exact}} = 2.5$). Take raw $= 1$ and $I_{\mathrm{FS}} = 4$, so $q_{10}^{\mathrm{exact}} = 10/4 = 5/2$ exactly. The first guard ($< 5/2$) fails on equality; the second guard (< 5) succeeds; the band is $Q2$. The half-open convention assigns every cut point upward, so 2.5 is $Q2$, 5 is $Q3$, and 7.5 is $Q4$, deterministically. Under the Guide's displayed bands the value 2.50 also reads as the second band (2.50–4.99), so at exactly representable two-decimal values the implemented convention and the Guide's display agree; the divergence appears strictly between the displayed endpoints, as the next example shows.

EXAMPLE 8.10 (Near-boundary value: $q_{10}^{\mathrm{exact}} = 2.4999$). Take raw $= 2.4999$ and $I_{\mathrm{FS}} = 10$, so $q_{10}^{\mathrm{exact}} = 2.4999$ exactly. Since $2.4999 < 5/2$, the band is $Q1$. A two-decimal display that rounds half up renders this value as 2.50 — the printed lower endpoint of the Guide's *second* band. A reader classifying from the display would place the record in $Q2$; the implemented classifier, reading the exact value, places it in $Q1$; and by Proposition 8.4 the rendered "2.50" has no way to leak back into classification. The example generalizes: every value in $[2.495, 2.5)$ rounds to 2.50 at two decimals while lying exactly in $Q1$, so the display-classification gap is an interval of positive width at every cut point, not an isolated pathological value.

EXAMPLE 8.11 (Gap value under the Guide's bands: 2.495). The value 2.495 lies in the open gap between the Guide's printed endpoints 2.49 and 2.50: it belongs to neither 0.00–2.49 nor 2.50–4.99 as literally written. The Guide provides no rule for it. The implemented convention answers instantly: $2.495 < 5/2$, band $Q1$. The point of the example is not that the Guide's authors intended otherwise — plausibly they intended truncation, or rounding, or never contemplated the gap — but that "plausibly intended" is not a classification rule,

and three incompatible completions (truncate at two decimals, round half up, round half even) each classify parts of the gap differently.

EXAMPLE 8.12 (Remaining cut points and degenerate inputs). All values illustrative. (i) raw = 2, $I_{\mathrm{FS}} = 4$: exact value 5 exactly; the first two guards fail (at the second, on equality); band $Q3$. (ii) raw = 7.4999, $I_{\mathrm{FS}} = 10$: exact value $7.4999 < 15/2$; band $Q3$, although a two-decimal display renders 7.50, the printed floor of the Guide's top band — the $Q4$-boundary instance of the rendering gap. (iii) raw = 0: exact value 0; band $Q1$ by the closed left end of the first bin; overload 0; and note that a zero *observed* flow is banded without difficulty even though ratio diagnostics elsewhere in the pipeline would censor at a zero-compatible observation — banding divides by the full scale, which is strictly positive by profile construction, never by the observation. (iv) raw = 1.874975, $I_{\mathrm{FS}} = 7.4999$: exact value $10 \cdot 1.874975/7.4999 = 2.5$ exactly (the numerator is 0.25×7.4999); band $Q2$ by upward assignment — an example chosen to show that cut-point equality arises from unremarkable-looking inputs, not only from round numbers.

EXAMPLE 8.13 (Uncertainty straddling a cut point). All values illustrative. A raw flow of 1.24 with standard uncertainty 0.03 against $I_{\mathrm{FS}} = 5$ (with its own small uncertainty) yields exact value 2.48 and, after propagation, an interval on the Q10 coordinate of roughly $[2.34, 2.62]$. The point value is $Q1$; the interval spans the 2.5 cut. The record reports all three facts. What the record does not do is report "$Q1$–$Q2$" or a probability over bands: banding stays exact on the point value, and the interval stays available to any consumer whose question is distributional. A trend or comparison built on this record must use the interval under the declared uncertainty rule of Section 10; a display built on it will render 2.48; and neither operation can move the recorded band.

7. Why two-decimal display bands underdetermine classification

The preceding examples can now be assembled into the general argument, which is the content of the preserved quantization conflict.

A band scheme defined on two-decimal displays is a partition of the *rendered* values $\{0.00, 0.01, \ldots\}$, not of the underlying value space. To classify an underlying value, the scheme must be composed with a rendering rule, and the composition — not the printed bands — is the actual classifier. The Guide prints the bands and omits the rendering rule; therefore the Guide's scheme is not one classifier but a family of classifiers indexed by the omitted rule, and members of the family disagree on every interval of the form $[c - 0.005, c)$ for each cut point $c \in \{2.5, 5.0, 7.5\}$ (under round-half-up rendering; the disagreeing intervals shift with the rendering convention but never vanish). The underdetermination is thus structural, not editorial: no choice of printed two-decimal endpoints can close it, because the gap lives between the printed values.

The conflict is sharpest at the top band — the Q4-boundaries conflict of the source audit — for two compounding reasons. First, the Guide's top band is printed as 7.50–10.00, closed above, which presupposes that no value exceeds 10.00; but the normalization that produces the score admits raw values above full scale, and the clipped display maps all of them to exactly 10.00. Under a display-defined scheme, the entire overload regime collapses into the top endpoint, and the scheme cannot even express the question "by how much was full scale exceeded." The implemented scheme keeps the top band half-open to infinity on the exact value and records overload separately, so the collapse happens only in presentation. Second, the boundary 7.50 itself inherits the rendering gap like every other cut point: a value of 7.4999 renders as 7.50 and would read as top-band under the display while lying exactly in

$Q3$. A two-decimal scheme therefore misclassifies (relative to the exact convention) in both directions at the Q4 boundary — upward across the cut by rounding, and degenerately at the top by clipping — and these are precisely the two failure modes the exact-value-plus-overload design eliminates.

Adverse Finding 8.14 (Display bands are not a classifier). The Guide's two-decimal bands, as printed, do not determine a classification function on exact values; they determine one only relative to an unstated rendering rule, and distinct reasonable rules yield distinct classifications on intervals of positive width at every band boundary. The implemented candidate closes the gap by fiat — exact values, half-open bins, upward cut-point assignment, display derived afterward — and Proposition 8.4 certifies that the closure is complete: no rendering choice can re-open it.

It bears stating why this matters at monograph length rather than as a footnote. Quantized bands are the layer at which numeric evidence becomes categorical language — "$Q1$" reads as *low* and "$Q4$" as *high* — and categorical language is what travels into summaries, dashboards, and decisions. A classification gap at the band boundary is therefore not a cosmetic defect: it is a mechanism by which two honest readers of the same underlying value can carry incompatible categorical claims forward, each traceable to the same source, with the disagreement invisible unless the rendering rule is recovered. The dose-response literature's insistence that analysis be conducted on the modeled continuous response, with categorization applied only at the reporting layer if at all [**38**], is the same discipline in statistical form; the candidate implements it as a type-level separation between an exact ratio that classifies and a display that does not.

8. Uncertainty through normalization

Normalization does not launder uncertainty. The implemented path requires uncertainty on the raw flow and on the full-scale value, and propagates both through the ratio: interval bounds by the monotone quotient rule (lower display bound from lower numerator over upper denominator, upper from upper over lower), standard uncertainty by the first-order combination of the two relative contributions, and clipping applied to the Q10 display bounds at 10 while the overload ratio's bounds remain unclipped. The confidence level attached to the normalized quantities is the minimum of the input confidence levels — a deliberately conservative convention. Treating the normalized best value as if it were the truth, with the uncertainty discarded at the seam, is the calibration sin the wider candidate architecture is built to refuse [**37**]; the quantizer participates by never emitting a normalized value object without its propagated uncertainty attached.

The interaction between propagated uncertainty and exact banding deserves one explicit paragraph, because the two disciplines answer different questions and are easily conflated. Banding is exact on the point value: a record whose exact value is 2.4999 is a $Q1$ record, full stop. But if that record's propagated interval spans $[2.3, 2.7]$, then the statement "the underlying quantity is in $Q1$" is supported only at the point estimate, not across the interval, and the record carries everything a consumer needs to see this: exact value, band, and interval, jointly. The candidate's position is that the band function must not blur itself to accommodate uncertainty (a fuzzy band is a second, worse rendering rule); instead, uncertainty-aware statements are pushed to the comparison layer, where they are made explicit — which is the trend-tag rule.

9. The quantizer and the mixed-scale conflict

The quantization conflict is the audit's eighth; its first — the mixed-scale conflict — is enforced partly here, and the connection deserves explicit statement because the two conflicts compound. The Guide's worked example uses sat = 8.5 as a component threshold on a page whose conventions make saturation a 0–1 fraction; Appendix I identifies the value as a mixed-scale error whose intended magnitude is 0.85. On a Q10 coordinate, 8.5 and 0.85 fall in different bands ($Q4$ and $Q1$); on a fraction coordinate, 8.5 is not even admissible. A quantizer that accepted bare numbers would classify the error fluently — 8.5 is a perfectly bandable float — and the misclassification would propagate wearing a band label of unimpeachable exactness.

The implemented defense is the unit ontology upstream of the band function. The quantizer's input is a typed measurement whose quantity kind (`FLOW_MAGNITUDE`) and unit (ACTIVATION) are checked before normalization; its output Q10 score is a `NORMALIZED_FLOW` in unit `Q10`, range-checked to $[0, 10]$; fractions are a distinct unit kind range-checked to $[0, 1]$; Q100 constraint scores are a third, range-checked to $[0, 100]$; and the unit-admissibility table permits each quantity kind only its declared units, with no conversion function anywhere in the layer. A saturation fraction of 8.5 is a construction-time validation error; a Q10 score used where a fraction is required is a type error; and the Guide's sat = 8.5 is therefore not a value the candidate handles *incorrectly* but a value it cannot admit at all. Exact banding sits behind these gates: exactness is only worth proving about values whose kind and scale are already beyond dispute, which is why this chapter's guarantees are stated for typed inputs and why the mixed-scale defense, though implemented elsewhere, is properly part of the quantization story.

10. Trend tags

A single normalized value invites a band word; a pair of them invites a trend word — *rising*, *falling*, *stable* — and trend words smuggle in even more unstated structure than band words do. A trend is a claim about two or more values *of the same quantity, on the same axis, over a stated interval*, and it is empty or misleading if any of those qualifiers is left implicit: values from different test points do not form a trend; values under different profiles do not form a trend; a difference smaller than the propagated uncertainty does not evidence a trend; and a difference smaller than a declared minimum effect is not a trend worth naming even if statistically resolvable.

RULE 8.15 (Trend-tag admission). A trend tag is admissible only when it declares, explicitly: (i) the axis — which typed quantity, at which test point, under which profile digest, the profile identity fixing model, method, and full-scale convention; (ii) the interval — the ordered replay frame over which the compared values were produced, with the caller-supplied replay semantics of Chapter 7 understood; (iii) the minimum effect — a declared threshold below which differences are reported as no trend, fixed before comparison; and (iv) the uncertainty rule — how the propagated uncertainties of the compared values enter the judgment (for example, non-overlap of intervals at the stated confidence, or a stated multiple of combined standard uncertainty). A tag missing any component is not a degraded trend statement; it is inadmissible.

The rule is the comparison-layer twin of the banding discipline: just as a band must come from the exact value and not the rendering, a trend must come from a declared comparison rule and not from the visual suggestion of two displayed numbers. The two-decimal display

is again the cautionary instrument: displays of 2.50 and 2.50 can cover exact values 2.4950 and 2.5049 — a difference of 0.0099 in the exact coordinate rendered as perfect stability — while displays of 2.49 and 2.50 can cover exact values differing by one part in 10^4, rendered as a visible step. Display-level trend reading therefore manufactures both false stability and false movement at the rendering quantum, and the admission rule forecloses it by requiring the comparison to be defined on the exact values with an explicit uncertainty rule.

Example 8.16 (An admissible trend tag, fully declared). All values illustrative. An admissible tag reads, in full: "Axis: normalized flow (`I_Q`, Q10) at test point `tp-`⟨digest⟩ under profile ⟨digest⟩. Interval: replay frames $t_1 < t_2$, both inside the profile validity window. Minimum effect: 0.5 Q10 units, declared before comparison. Uncertainty rule: report *rising* only if the lower bound of the later value exceeds the upper bound of the earlier value and the difference of point values exceeds the minimum effect." Under this tag, exact values $6.1 \to 6.9$ with intervals $[5.9, 6.3]$ and $[6.7, 7.1]$ read *rising* (intervals disjoint, difference $0.8 > 0.5$); exact values $6.1 \to 6.5$ with the same half-widths read *no trend* (difference below minimum effect, whatever the intervals); and exact values $6.1 \to 6.9$ with intervals $[5.5, 6.7]$ and $[6.3, 7.5]$ read *no trend* (intervals overlap), even though the displayed values suggest movement. Delete any one of the four declarations and each verdict becomes contestable, which is the rule's point.

No trend tag over live-model data is asserted anywhere in this monograph; the rule is stated for the discipline it imposes on future admitted data, and its components (axis, interval, minimum effect, uncertainty rule) are deliberately the same components a preregistered comparison would freeze in advance. The kinship with dose-response practice is direct: a trend over dose is a model claim, and the modeling tradition insists it be fit and tested on the continuous response with declared error structure rather than read off category transitions [38]; the trend-tag rule is that insistence, restated for the replay-frame axis.

11. What the quantization layer does and does not establish

Claim Boundary 8.17 (Scope of the exactness guarantees). The guarantees of this chapter are guarantees about representation and classification mechanics: exact normalization, total and deterministic banding, display inertness, uncertainty propagation, and comparison admission. They are not guarantees about measurement validity. An exact $Q4$ band over a mismeasured raw flow is an exact classification of a wrong number; a faithfully propagated uncertainty is only as honest as the input uncertainty model; and the full-scale denominators remain method-relative under the preserved full-scale conflict, so cross-profile band comparisons are undefined absent a stated comparison rule. The quantization layer's contribution is negative and structural: of the ways a score pipeline can silently corrupt a classification — rounding across boundaries, clipping away overload, unit slippage, display feedback — none is available here, and the tests that pin each closure are part of the bounded software evidence at the Gate 4 receipt.

CHAPTER 9

Structural Network Operations

1. The capability-gating philosophy

The network layer is where the Ohm analogy is most seductive and least earned. The Quantification Guide presents series addition of resistances, reciprocal aggregation of parallel resistances, and multiplication of stage gains as working rules — arithmetic one may simply perform on constraint values to obtain system-level conclusions. The operationalization addendum, read against it, reclassifies the same arithmetic as *preregistered predictions*: falsifiable hypotheses (the gain-composition and redundancy hypotheses of the registered claim ledger) whose defining rules must be dropped if the corresponding experiments fail. And Appendix I applies the arithmetic to paths while warning, in the same breath, that co-present guardrails are not parallel routes and that flow remains local. This is the validation-status conflict preserved in the source audit, and the candidate's response to it is the organizing principle of this chapter:

Rule 9.1 (Structural-only network operations). Series equivalence, parallel equivalence, and gain composition exist in the candidate only as *unauthenticated structural research calculations*. Each operation is capability-gated: it runs only inside an admitted operation context, under a reviewed and current calibration profile, on typed inputs that satisfy coordinate, unit, topology, region, linearity, and evidence requirements; it emits a result that binds its complete derivation; and it carries an authorization state that is, in every implemented mode, explicitly unauthenticated. No network result asserts that the arithmetic is empirically valid; it asserts only that the arithmetic was performed, on these children, under these gates, with this provenance.

The word "capability" is doing precise work here. The implemented capability enumeration names the network operations (`SERIES_EQUIVALENCE`, `PARALLEL_EQUIVALENCE`, `GAIN_COMPOSITION`) alongside the prediction and diagnosis capabilities, and every operation result embeds an operation binding whose capability field records which gate was exercised. The binding also records a trust mode — `STRUCTURAL_RESEARCH_ONLY` or `INTEGRITY_CONFIGURATION` — and an authorization state that is locked to the trust mode by validation: structural mode must carry `UNAUTHENTICATED_STRUCTURAL`, integrity mode must carry `UNAUTHENTICATED_INTEGRITY`, and any other pairing is a validation error. There is no third mode. The type system does not contain an authenticated authorization state for these operations, so no configuration of the candidate, however keyed or attested, can produce a network result that claims authenticated authority (Section 8).

Why gate so heavily what is, after all, elementary arithmetic? Because the arithmetic is not where the risk lives. Adding two positive decimals is harmless; *calling the sum an equivalent resistance* is a modeling claim with unstated preconditions — that the two components are actually in series in the relevant causal topology, that their local linear regimes are simultaneously applicable, that their fitted parameters share coordinates and units, that their errors combine as declared, and that no interaction term is being silently zeroed. The

electrical formulas are theorems in circuit theory because Kirchhoff's laws supply those preconditions; here, nothing supplies them. The candidate therefore inverts the Guide's presentation: instead of rules that are valid until questioned, it implements calculations that are inert until every stated precondition is demonstrated to the type system, and whose outputs remain, even then, research structures awaiting the preregistered experiments. This is the verification-validation separation [23] applied to network arithmetic: the software verifies its gates exhaustively; validation of the composition laws themselves belongs to the registered hypotheses and has not occurred.

A second, quieter principle governs the layer: *everything replays*. Every network result is factory-only — direct construction is rejected by token check — and every result binds its exact ordered children by digest, so that a verifier holding the result and the evidence registry can re-resolve the children, re-run the factory, and demand digest equality with the original. A network value that cannot be replayed to its exact bytes does not validate. This converts the network layer from a calculator into a closed derivation system: there are no free-floating equivalent resistances anywhere in the candidate, only endpoints of replayable derivation chains.

2. Context admission and immutability

Before any network operation runs, an operation context must be admitted, and the admission mechanics matter enough to the layer's guarantees to be spelled out here rather than assumed.

A context binds five things: the calibration profile; an evidence registry; a capability authority (manifest plus verification machinery); a tuple of capability attestations, possibly empty, with duplicate capabilities rejected; and the caller-supplied replay time, with an optional recorded clock receipt. Admission is not a constructor formality. At admission the context re-validates every object in the evidence registry; resolves the authority's manifest reference and requires byte equality between the registered artifact payload and the manifest's canonical bytes; asserts the profile current at the replay time (reviewed, drift-current, inside the validity window); requires every artifact the profile references — code, dependency lock, corpus, corpus rights, both splits, drift check, directions, full-scale sources, controls — to be resolvable in the registry; and requires the evidence behind every presented attestation. Only then does it take digest snapshots: of the profile, the manifest, each attestation, the replay time, and the clock receipt.

The snapshots are the immutability mechanism. Every subsequent operation — series, parallel, gain, prediction, residual — begins by re-validating the context against its snapshots: the types must be unchanged, the registry must re-validate, and the profile digest, manifest digest, attestation digests, replay time, and clock receipt must equal their admitted values, with any divergence a validation error. A context is therefore not a session that can drift; it is a frozen frame, and every network result records which frame it was computed in. Two results with the same binding digests were computed under identical premises; two results with different profile digests are formally incomparable, and nothing in the layer will combine them — the one-profile requirement of the admission rules is checked against the context's single profile, so cross-profile aggregation is not a forbidden option but an unexpressible one.

One further mechanical point completes the picture: all network arithmetic runs inside a declared exact-decimal arithmetic context, so that sums, reciprocals, products, and square roots are computed at fixed high precision with deterministic behavior across platforms, and

the digest-equality replay obligation of Section 1 is meaningful — bit-identical replay would be unachievable over platform-dependent binary floating point.

3. Series equivalence

3.1. The electrical analogy, bracketed. In circuit theory, resistances in series add because the same current traverses each element and voltages sum along the path. Both premises are physical facts about charge conservation and potential. The candidate's series rule borrows the *arithmetic* — the equivalent value is the sum of component values — while explicitly bracketing the premises: there is no conserved current in a steered activation pathway, no potential whose drops sum, and no theorem that two locally linear response segments compose into one. The bracketing is not rhetorical. It is implemented as the requirement set below, which demands precisely the structural conditions under which the sum is at least *coherent* (same coordinates, same units, same profile, contiguous topology), while the question of whether the sum *predicts* anything is left, deliberately unanswered, to the registered redundancy and composition experiments. Until those close, a series equivalent is a bookkeeping object: the sum of some fitted parameters, with provenance.

3.2. Requirements as implemented.

RULE 9.2 (Series admission). A series equivalence over components $C_1, \ldots, C_n$ ($n \geq 2$) is admitted only when all of the following hold.

(1) *Typed components.* Each C_i is an exact resistance-component object: a component identity, a resistance measurement, a source node, a target node, a declared region, and a declared linearity coefficient. Duplicate component identities, duplicate component digests, and duplicate underlying resistance digests are all rejected — the same fitted resistance cannot be counted twice under two names.

(2) *Calibrated or replayable resistances.* Each component's resistance is of quantity kind `EFFECTIVE_RESISTANCE` with role `CALIBRATED_HELD_OUT` or `NETWORK_EQUIVALENT`; apparent same-sample and design-proposal roles are inadmissible. A network-equivalent child is not taken on faith: it is recursively replayed through its own factory and digest-checked before use.

(3) *One unit.* Every resistance is in the single unit `ALPHA_PER_ACTIVATION`; the equivalent is emitted in the same unit. No unit conversion exists in the layer.

(4) *Compatible coordinates.* Source and target nodes are typed coordinate identities of quantity kind `DRIVE_INTERVENTION` in ALPHA units, and every node in the chain binds the same registered test-point digest; a component whose topology nodes and whose resistance measurement bind different test points is rejected.

(5) *Simple acyclic directed chain.* Adjacent components must link exactly — the target node of C_i equals the source node of C_{i+1} — and the full node path (first source, then each target in order) must contain no repeated node. This excludes cycles, self-loops (already rejected at the component level), branching, and re-visiting: the topology is a simple directed path or nothing.

(6) *Active region.* Every component's declared region is `ACTIVE`; the component type itself rejects any other region at construction. Cutoff, saturated, non-ohmic, unstable, and unassessed segments cannot enter a series computation at all.

(7) *Adequate linearity.* Every component's linearity coefficient (an R^2 in $[0, 1]$) is at or above the profile's declared minimum; the threshold is a profile tolerance, not a constant.

(8) *One profile, with evidence.* Every component resistance is validated against the single admitted calibration profile of the operation context; all components bind one test point; each carries positive-lower-bound uncertainty and a recorded sample size.
(9) *Complete ordered child provenance.* The emitted equivalent's derivation parents are exactly the ordered component digests, in role-stamped positions (`component[0]`, `component[1]`, ...); the result object re-verifies this binding at construction, and any mismatch between claimed children and bound parents is a validation error.

When admitted, the arithmetic is the expected sum: the equivalent value is $\sum_i r_i$; interval bounds sum lower-with-lower and upper-with-upper (with the upper bound floored at the computed value); standard uncertainty combines in quadrature, which silently encodes an independence assumption that the covariance limitation of Section 6 makes loud; the confidence level is the minimum across children; and the sample size is the minimum across children — both deliberately conservative conventions. The result is emitted with resistance role `NETWORK_EQUIVALENT`, which marks it as replayable-derived rather than calibrated, and which downstream consumers (notably the prediction gate) treat as a distinct role with its own replay obligation.

3.3. The arithmetic, explicitly. For admitted components with resistances $r_1, \ldots, r_n$, intervals $[\ell_i, u_i]$, standard uncertainties s_i, confidence levels c_i, and sample sizes m_i, the emitted equivalent is

$$r_{\text{series}} = \sum_{i=1}^{n} r_i, \qquad \left[\, \textstyle\sum_i \ell_i,\ \max\!\left(r_{\text{series}}, \sum_i u_i\right) \,\right], \qquad s_{\text{series}} = \left(\textstyle\sum_{i=1}^{n} s_i^2\right)^{1/2},$$

with confidence $\min_i c_i$ and sample size $\min_i m_i$. The quadrature form for s_{series} is exact for uncorrelated child errors and an approximation otherwise; since cross-input covariance is unavailable in the current evidence model, the zero-covariance approximation is declared on the binding rather than assumed silently (Rule 9.7 applies to series results too). If child errors were positively correlated, the true combined standard uncertainty would exceed the quadrature value — the reported uncertainty would be optimistic — and this direction of possible error is precisely what the declared limitation warns a consumer to expect.

Example 9.3 (Illustrative series computation). All numbers illustrative. Two admitted components on one test point: $r_1 = 2.0$ alpha-per-activation with interval $[1.8, 2.2]$, standard uncertainty 0.1, confidence 0.95, sample size 40; and $r_2 = 3.0$ with $[2.7, 3.3]$, standard uncertainty 0.15, confidence 0.90, sample size 25. Topology: $A \to B$, $B \to C$, distinct nodes, so the chain is simple and acyclic. The equivalent is $r = 5.0$ with interval $[4.5, 5.5]$, standard uncertainty $\sqrt{0.01 + 0.0225} \approx 0.18$, confidence 0.90 (the minimum), sample size 25 (the minimum), role `NETWORK_EQUIVALENT`, derivation parents (`component[0]`, `component[1]`) by digest, and the full operation binding of Definition 9.10. If the same two components were presented in the orders (C_1, C_2) and (C_2, C_1) with correspondingly re-declared node chains, the two results would be distinct objects with distinct digests: child order is part of identity, because the declared topology is directional.

Remark 9.4 (What a series equivalent is not). A series equivalent is not evidence that the two mechanisms are "in series" in any causal sense; the chain topology is asserted by the caller's component construction, not discovered. It is not a claim that end-to-end response equals drive over summed resistance; that is the content of a registered hypothesis, untested. And it is not transferable: it binds one profile, one test point, and one replay frame, like every other quantity in the candidate.

4. Parallel equivalence

4.1. Requirements as implemented. Parallel equivalence shares the entire admission apparatus of Rule 9.2 — typed components, calibrated or replayed resistances, one unit, coordinate compatibility, active region, linearity threshold, one profile, uncertainty and sample-size evidence, ordered child binding, duplicate rejection — and replaces the chain condition with an endpoint condition:

RULE 9.5 (Parallel topology). All components must share *identical directed endpoints*: one common source node and one common target node, as typed coordinate identities on one registered test point. The set of (source, target) pairs across components must have exactly one element. Components that merely coexist, or that connect overlapping-but-distinct coordinates, do not form a parallel topology and are rejected.

The admitted arithmetic is reciprocal aggregation:

$$r_{\text{parallel}} = \Big(\sum_{i=1}^{n} \frac{1}{r_i}\Big)^{-1}, \qquad s_{\text{parallel}} = \left(\sum_{i=1}^{n} \Big(\Big(\frac{r_{\text{parallel}}}{r_i} \Big)^{2} s_i \Big)^{2} \right)^{1/2},$$

with interval bounds computed by the same reciprocal form on the child bounds, and the same minimum-confidence, minimum-sample-size, role-`NETWORK_EQUIVALENT`, factory-only, replay-bound conventions as the series case. The weighting in s_{parallel} is the first-order sensitivity of the reciprocal sum: a child whose resistance is small relative to the equivalent dominates both the value and the uncertainty, matching the electrical intuition that the lowest-resistance branch dominates a parallel combination — an intuition which, once again, is inherited as arithmetic, not as validated mechanism.

EXAMPLE 9.6 (Illustrative parallel computation). All numbers illustrative. The same two components as Example 9.3, now declared with identical directed endpoints $A \to B$ for both. The reciprocal sum is $1/2.0 + 1/3.0 = 5/6$, so $r_{\text{parallel}} = 1.2$ exactly in the declared decimal precision, below both children, with interval bounds $[\,1/(1/1.8 + 1/2.7),\ \max(1.2,\ 1/(1/2.2 + 1/3.3))\,] = [1.08, 1.32]$ and the propagated standard uncertainty computed from the squared-ratio weights $(1.2/2.0)^2 = 0.36$ and $(1.2/3.0)^2 = 0.16$. Reading: under the declared topology, *if* the two components were genuine alternative routes and *if* the reciprocal law held empirically — the registered redundancy hypothesis, untested — the aggregate opposition would be 1.2. Neither conditional is discharged by the computation, and the result object says so through its binding.

4.2. Co-present guardrails are not parallel paths. The electrical intuition behind reciprocal aggregation is that parallel resistors offer *alternative routes* for the same current: adding a path can only decrease equivalent resistance, because flow that cannot pass one way passes another. Imported uncritically into the safety domain, this intuition produces a specific and consequential misreading, flagged in Appendix I and preserved in the source audit: that multiple co-present guardrails — several safety mechanisms bearing on the same behavior — constitute parallel resistances, so that adding a guardrail *lowers* total resistance and thereby *weakens* the system. The arithmetic would even look self-consistent, which is exactly what makes it dangerous. The error is in the premise, not the algebra: co-presence is not alternativeness.

Consider three scenarios, stated at the structural level the candidate can actually represent.

Scenario 1: genuine alternatives. Two mechanisms connect the same source coordinate to the same target coordinate, and the response suppressed along one route demonstrably appears along the other when the first is ablated. This — and only this — is the situation the reciprocal rule's intuition describes. Note that the italicized clause is an experimental outcome, not a topological declaration: it is precisely the registered redundancy hypothesis (ablation of genuine alternative branches produces the specific response predicted by reciprocal aggregation, above matched random-head controls), which is unproven. The candidate will *compute* the parallel equivalent for declared same-endpoint components, but the computation's predictive meaning is suspended pending that experiment.

Scenario 2: co-present guardrails on one pathway. Three safety mechanisms each independently gate the same route from drive to response — serially, as it were: each can block, none provides an alternative route around another. Modeling these as parallel is not a poor approximation; it is a category error with an inverted sign, since each added guardrail here can only maintain or increase opposition to flow, while the reciprocal rule asserts it decreases it. Structurally, the candidate resists the error at the coordinate level: guardrails instantiated at different layers or positions have different test-point or coordinate identities and therefore fail the identical-endpoints gate. Where a caller nonetheless declares them as same-endpoint components, the type system cannot detect the misdeclaration — topology is caller-asserted — and this is stated plainly as a limitation rather than hidden: the parallel gate checks the declared topology's coherence, not its truth.

Scenario 3: correlated mechanisms. Two mechanisms genuinely offer alternative routes, but they share training provenance, circuitry, or failure modes, so that conditions degrading one degrade the other. The reciprocal arithmetic, and its quadrature uncertainty combination, assume away exactly this correlation. The candidate cannot measure the correlation (no cross-input covariance is available in the current evidence model); what it can do is refuse to let the assumption pass silently, which is the function of the mandatory covariance limitation below.

RULE 9.7 (Declared limitations on every parallel result). Interaction, dependence, and covariance are not modeled by the parallel operation and are declared, not waived: every network result's operation binding carries the fixed covariance-limitation marker (`CROSS_INPUT_COVARIANCE_UNAVAI`) and the binding validator rejects any binding in which the marker has been altered or removed. A parallel equivalent thus cannot be excerpted without excerpting its own disclaimer.

5. Gain composition

5.1. Requirements as implemented.

RULE 9.8 (Gain-composition admission). A gain composition over stages $S_1, \ldots, S_n$ ($n \geq 1$) is admitted only when: each stage is an exact typed stage object with a unique stage identity; each stage's gain is a `GAIN_MAGNITUDE` in DIMENSIONLESS units with uncertainty and a recorded sample size, validated against the context's single profile; each stage's source and target coordinates are typed coordinate identities of *identical* quantity kind and unit binding one common test point (a dimensionless gain may not map unlike coordinates); each stage's declared region is `ACTIVE` (enforced at stage construction); each stage's linearity coefficient meets the profile minimum; adjacent stages compose exactly — the target coordinate of S_i equals the source coordinate of S_{i+1} — and the full coordinate path is simple and acyclic (no repeated coordinate); and the emitted composite binds the ordered stage digests as role-stamped derivation parents (`stage[0]`, `stage[1]`, ...), factory-only.

The admitted arithmetic is the product $\prod_i g_i$, with interval bounds as products of child bounds, first-order product-rule propagation of standard uncertainty (each child's contribution weighted by the product of the others), minimum confidence, and minimum sample size. The composite is emitted as a dimensionless gain magnitude named for its end-to-end coordinate pair, with full provenance.

5.2. The Fisher–Rao versus Jacobian distinction, and the missing bridge. The gain gate has a conceptual precondition that no runtime check can discharge, preserved in the source audit as the gain-definition conflict. The PCBA source defines stage gain β by Fisher–Rao displacement ratios on the open probability simplex: the magnitude of induced displacement in an information-geometric metric, at small or large signal, between cause and effect coordinates. The operationalization addendum defines β as a local Jacobian or projection derivative between cause and effect test points: a ratio of infinitesimal changes in two activation projections. These are distinct mathematical objects. One lives on a statistical manifold and measures distributional displacement; the other lives in activation space and measures a directional derivative. They have different invariances (the Fisher–Rao quantity is reparameterization-invariant on the simplex; the Jacobian quantity is basis- and projection-dependent), different units discipline (both can be rendered dimensionless, but by different normalizations), and no theorem in the candidate's corpus identifies them — no chain rule connects a product of projection derivatives to a product of Fisher–Rao displacement ratios except under assumptions (local diffeomorphism between coordinates, metric compatibility) that nobody has stated, let alone verified.

The candidate's handling is deliberately unheroic: the two gains are *separately named profiles*. A gain magnitude enters the composition gate carrying its method identity in provenance, and compositions are computed within a method, under one profile. The missing bridge is recorded as missing. What would supply it is one of two things: a theorem, deriving one quantity from the other under stated conditions on the coordinate maps; or the registered gain-composition experiment, which tests empirically whether local derivative gains compose to end-to-end gain in the active region within stated tolerance. Until one of these exists, a composed gain is a product of numbers that were each, individually, admitted under their own definition — and the composite's meaning is exactly as strong as the weakest link in that chain of definitions, which is to say: structural.

Preserved Conflict 9.9 (No licensed identification of gain profiles). Fisher–Rao displacement gain and local Jacobian/projection gain remain distinct typed profiles in the candidate. No implemented operation converts between them, no composition mixes them, and no result asserts their equality. Any external text that multiplies a Fisher–Rao stage figure by a Jacobian stage figure is performing an operation the candidate refuses.

6. What every aggregate binds

Every network result — series, parallel, or gain — embeds a provenance structure and an operation binding whose fields are enumerated here, because the binding *is* the claim. A network equivalent, stripped of its binding, is a number; with its binding, it is a replayable derivation whose every premise is named.

Definition 9.10 (Aggregate binding contents). An admitted aggregate binds, at minimum:

(1) *Exact child order and identity* — the ordered, role-stamped digests of the components or stages, positionally contiguous from zero, with duplicates rejected;

(2) *Topology* — the declared chain or endpoint structure, re-derivable from the typed children and re-verified on replay;
(3) *Profile identity* — the calibration profile digest, binding model, configuration, tokenizer, weights, code revision, runtime, dtype, corpus, splits, seeds, alpha grid, full-scale method, fit and uncertainty methods, tolerances, controls, drift and expiry, and review state;
(4) *Source and control evidence* — the deduplicated source artifact references of all children (with conflicting metadata for one digest rejected), the profile's control references, and the profile's control bindings, carried on the result's provenance;
(5) *Capability label* — which gate was exercised (`SERIES_EQUIVALENCE`, `PARALLEL_EQUIVALENCE`, `GAIN_COMPOSITION`);
(6) *Authority-manifest identity* — the digest of the trust authority's manifest under which the context was admitted;
(7) *Attestation identities* — the sorted, deduplicated digests of any capability attestations present in the context, *possibly empty*: structural mode carries none, and emptiness is a recorded fact, not an omission;
(8) *Trust mode and authorization state* — locked to each other as described in Section 1, both unauthenticated;
(9) *Historical replay time* — the caller-supplied `replay_as_of`, with the non-freshness semantics of Chapter 7, plus the optional clock-receipt digest when one was recorded;
(10) *Runtime, code, and dependency identities* — the runtime identifier, the code artifact reference, and the dependency-lock artifact reference, each an exact typed reference with digest;
(11) *Covariance limitation* — the fixed marker of Rule 9.7, mandatory on every binding, series and gain included, because quadrature and product-rule propagation embed the same zero-covariance approximation.

RULE 9.11 (Fail-closed topology). Duplicate children (by component identity, component digest, or underlying resistance digest), repeated nodes in a series path, repeated coordinates in a gain path, self-loops, non-contiguous parent positions, non-canonical parent roles, unknown topology methods on replay, and any digest mismatch between a claimed equivalent and its factory replay are validation errors. There is no lenient mode, no warning-and-continue path, and no partial aggregate: an aggregate that cannot bind everything binds nothing.

The replay obligation deserves its own paragraph because it is the layer's strongest structural property. A network-equivalent resistance presented anywhere downstream — as a child of a larger network, or as the resistance input to the prediction gate — is not consumed on the strength of its role field. The consumer resolves its ordered parents from the evidence registry, type-checks each as an exact component, re-runs the declared topology method (series or parallel, and only those; an unknown method identifier fails), and requires the replayed equivalent's digest to equal the presented one. Nesting is therefore sound by induction: a tree of aggregates validates only if every internal node replays exactly over its leaves, and a single altered leaf invalidates every ancestor. The uncertainty carried at each level remains a zero-covariance approximation at that level — the limitation marker propagates with the binding — so what the induction guarantees is integrity of derivation, not validity of the composition law [**37**].

7. Replay, and the prediction gate as consumer

The replay obligation sketched in Section 1 is concrete enough to state as an algorithm, and doing so makes the layer's inductive soundness inspectable.

ALGORITHM 9.12 (Exact factory replay of a network equivalent). Input: an admitted operation context; a measurement value R claimed to be a network equivalent. Output: the replayed result, or a validation error.

(1) Require R to be an exact measurement value with resistance role `NETWORK_EQUIVALENT`; validate R against the context (unit admissibility, profile binding, evidence resolution).
(2) Read R's derivation parents in order. For each position i, require the parent role to be exactly `component[`i`]`; resolve the parent digest through the evidence registry; require the resolved object to be an exact resistance component. Any missing, misrolled, misordered, or mistyped parent fails.
(3) Dispatch on R's provenance method identity: the series topology method replays through the series factory, the parallel topology method through the parallel factory; any other method identity fails as an unknown topology.
(4) Re-run the factory on the resolved children under the current context. This re-executes every admission gate of Rules 9.2 and 9.5 — including recursive replay of any child that is itself a network equivalent — and recomputes the arithmetic in the exact decimal context.
(5) Require digest equality between the replayed equivalent and R. Inequality — from altered children, altered profile, altered tolerances, or any drift whatsoever — fails the replay.

Step 4 is where the induction closes: replay of a parent that is itself an equivalent triggers the same algorithm one level down, so validating the root of an aggregate tree validates every interior node against its leaves. Step 5 is where integrity becomes binary: there is no "approximately replayed" outcome, and a single changed decimal anywhere in the tree surfaces as root-level digest inequality.

The principal consumer of replayed equivalents is the prediction gate. When a network-equivalent resistance is presented as the r_{eff} of a candidate prediction $\hat{y} = a/r_{\text{eff}}$, the gate accepts the role — calibrated-held-out and network-equivalent are the only two roles it admits — but only after running Algorithm 9.12 to completion. The design declines a tempting shortcut: it would have been cheaper to trust the role tag and the stored digest. The gate instead re-derives, because the role tag asserts a derivation history and the candidate's uniform rule is that asserted history is checked history. A prediction computed from a network equivalent therefore carries, transitively, the entire admission record of every component in the tree — and its own status remains what Chapter 7's boundary discipline makes it: a structural research calculation whose empirical adequacy is a registered, open question.

Table 1 summarizes the three operations side by side.

8. The Ed25519 fixture-key mode

The candidate's second trust mode, `INTEGRITY_CONFIGURATION`, admits optional capability attestations: Ed25519 signatures over canonical attestation payloads, verified against a capability authority's manifest, checked for capability match, profile binding, validity window against the replay time, and evidence presence, and replayed like everything else. The keys

	Series	Parallel	Gain composition
Children	≥ 2 components	≥ 2 components	≥ 1 stages
Child quantity	`EFFECTIVE_RESISTANCE`	`EFFECTIVE_RESISTANCE`	`GAIN_MAGNITUDE`
Child unit	`ALPHA_PER_ACTIVATION`	`ALPHA_PER_ACTIVATION`	`DIMENSIONLESS`
Topology	simple directed chain	identical directed endpoints	simple coordinate chain
Region gate	`ACTIVE` only	`ACTIVE` only	`ACTIVE` only
Linearity gate	profile minimum	profile minimum	profile minimum
Value law	$\sum_i r_i$	$(\sum_i r_i^{-1})^{-1}$	$\prod_i g_i$
Result role	`NETWORK_EQUIVALENT`	`NETWORK_EQUIVALENT`	(dimensionless gain)
Capability	`SERIES_EQUIVALENCE`	`PARALLEL_EQUIVALENCE`	`GAIN_COMPOSITION`
Replayable	yes, by digest	yes, by digest	factory-only

TABLE 1. The three structural network operations. All three share the one-profile, one-test-point, uncertainty-and-sample-size, ordered-child-binding, duplicate-rejection, and covariance-limitation requirements.

involved are *fixture keys*: deterministic in-repository test keys, present so that the signature-verification code path is real, exercised, and bounded by tests, rather than a stub awaiting hardware.

What the mode is for: integrity and replay discipline. A signed attestation binds a capability grant to a manifest digest, a profile digest, and a validity window, so that tests can verify the candidate rejects expired attestations, wrong-capability attestations, wrong-profile attestations, tampered payloads, and unknown signers — the whole negative space of a capability system — with real cryptographic verification rather than mocks.

The verification path, when the mode is engaged, checks in sequence: that an attestation for the requested capability is present in the context (absence is a capability error, not a silent downgrade); that the signature verifies over the attestation's canonical bytes against the key identified in the authority manifest; that the attestation binds the context's exact profile digest; that the replay time falls inside the attestation's validity window; and that the attestation's evidence reference resolves in the registry. Each check fails closed, and the attestation digests — sorted, deduplicated — enter the operation binding of every result computed under the context, so that a replayed result re-verifies the same attestations it was computed under. In structural mode, by contrast, the attestation tuple is empty, the integrity requirement is simply not asserted, and the binding records the emptiness. The two modes thus differ observably, in their recorded bindings, while agreeing on the only point that matters for authority: neither is authenticated.

What the mode is not, stated with the bluntness the source material uses:

RULE 9.13 (Fixture keys confer no authority). Verification of an Ed25519 fixture-key attestation establishes only that a payload was signed by a key present in the repository's test configuration and that the structural checks pass. It is never independent reviewer authority, never legal authority, never product authority, and never Evidence-Packet authority. The authorization state of an integrity-mode operation remains `UNAUTHENTICATED_INTEGRITY` — unauthenticated by type — and authenticated Evidence-Packet eligibility remains constant false in this candidate. Deterministic in-repository keys and caller-supplied replay time can never satisfy the eligibility preconditions (independently admitted authority manifest, protected key lifecycle and revocation, trusted current-time receipt, materialized dependency

identities, authenticated instrument evidence, reviewed packet policy). No authenticated capability path is implemented: there is no code in the candidate that, given better keys, would begin asserting authority — the assertion itself is absent from the type system.

The design rationale mirrors the completeness-versus-eligibility split that runs through the whole candidate. It would have been easy, and misleading, to ship the network layer with a configuration flag named something like "trusted mode." The implemented alternative is a pair of modes that differ in *which integrity checks run*, not in *what authority results carry*: both are unauthenticated, and the enumeration contains nothing else. A future authenticated path, if ever built, would be a new admission — new types, new review, new gates — not a configuration change to this one.

9. Chapter boundary

Claim Boundary 9.14 (What the network layer establishes). The network layer establishes that series, parallel, and gain arithmetic can be performed under exhaustive structural gating with complete, replayable provenance, and that every result carries its own limitations — unauthenticated status, zero-covariance approximation, profile relativity, caller-asserted topology — inseparably. It establishes nothing about whether steered activation pathways compose like circuits. The Guide's network rules remain, in this candidate, preregistered predictions with named falsifiers; the redundancy and gain-composition experiments have not been run; and every equivalent resistance and composed gain in existence under this candidate is a structural research calculation, Ring 4, not licensed, held.

CHAPTER 10

Residual Diagnostics and Censoring

1. The residual, defined narrowly

For an admitted active-region prediction, the residual is the signed difference between what was observed and what the calibrated model predicted:

$$\delta_y \;=\; y_{\text{obs}} - y_{\text{pred}},$$

where $y_{\text{pred}} = a/r_{\text{eff}}$ is the registered prediction from a declared alpha dose a and a calibrated effective resistance r_{eff}, and y_{obs} is an instrumented observation of the flow magnitude at the same test point. In the implementation the emitted quantity is named `delta_I` (with the observation and prediction in ACTIVATION units and the residual of quantity kind `FLOW_RESIDUAL`); this chapter writes δ_y for continuity with the response-coordinate notation of the seed paper, and nothing turns on the letter. Alongside the residual, the implementation emits a relative error $\varepsilon = |\delta_y| / \max(y_{\text{pred}}, \phi)$, where ϕ is the profile's declared residual floor — a strictly positive tolerance that prevents division by predictions compatible with zero — and, in the uncensored case only, an apparent resistance a/y_{obs} (explicitly role-tagged `APPARENT_SAME_SAMPLE`, the role the prediction gate refuses as input) and a resistance delta against the calibrated value. Every one of these quantities carries propagated uncertainty; a residual without an uncertainty interval is unrepresentable, since the result type rejects construction when either the residual or the relative error lacks one.

That is the entire positive content of the layer. The residual is a number with an interval, computed by subtraction, bound to its derivation. Everything else in this chapter concerns what the candidate refuses to let that number mean.

2. Admission requirements

The diagnostic entry point is gated more heavily than any other operation in the candidate, because its output is the one most liable to overinterpretation (Section 6). The requirements, as implemented, are these.

RULE 10.1 (Residual admission). A residual is computed only when all of the following hold.

(1) *Instrumented, held-out observation.* The observed value is of quantity kind `FLOW_MAGNITUDE` in ACTIVATION units and of measurement kind `INSTRUMENTED` — which itself requires an instrument identity, a test-point identity, a calibration-profile binding, and raw artifact references — with uncertainty, a recorded sample size, linked controls, and a dataset-split reference equal to the profile's held-out validation split. A score, a behavioral proxy, or a model-derived value cannot occupy the observation slot, and a calibration-split observation cannot masquerade as held-out.

(2) *Matching test points.* Drive, prediction, observation, and calibrated resistance must bind one common test-point identity; any disagreement, or any missing test point, is a validation error.

(3) *Profile identity.* The operating assessment must bind the context's exact calibration-profile digest and a test point registered in that profile; the context itself must have admitted the profile as reviewed, drift-current, and valid at the caller-supplied replay time.
(4) *Registered prediction.* The predicted value is not accepted as a number: the gate recomputes the prediction from the presented drive and calibrated resistance through the full prediction pathway — with all of that pathway's own gates: design-value drive within the calibrated alpha range, calibrated-held-out or replayed network-equivalent resistance bound to the profile's fit method, fit receipt, source and control evidence — and requires digest equality between the recomputation and the presented prediction. A prediction that did not come from the registered pathway cannot have a residual.
(5) *Calibrated resistance.* The resistance carries positive uncertainty bounds, a sample size, and an admissible role; apparent same-sample resistance is rejected at the prediction gate, which forecloses the circular construction $r = a/y$, $\hat{y} = a/r$, $\delta_y = 0$, "validated."
(6) *Active region, assessed outcome.* The operating assessment's region must be `ACTIVE` and its semantic outcome must not be `UNASSESSED`; numeric assessments are replayed against their bound sweep evidence before use. The no-diagnosis conditions of Chapter 7 apply in full.
(7) *Propagated uncertainty.* Prediction and observation must both carry uncertainty; the residual interval is computed by interval subtraction (observed lower minus predicted upper, observed upper minus predicted lower), the standard uncertainty in quadrature, and the confidence level as the minimum of the inputs. The relative error's denominators are floored at ϕ throughout.

The cumulative effect is that a residual exists only inside a fully reconstructed derivation: same profile, same test point, held-out data, registered prediction, calibrated parameter, replayable assessment. There is no API by which two bare numbers can be subtracted into a `FLOW_RESIDUAL`.

Requirement 4 deserves one further paragraph, because it is the admission rule with the longest reach. The gate does not compare the presented prediction to a recomputation *of its value*; it compares canonical digests, which cover value, unit, uncertainty, sample size, provenance, and operation binding jointly. A presented prediction with the right value but the wrong derivation — computed from a different resistance, under a different profile revision, or with its uncertainty altered — fails the digest comparison exactly as a wrong value would. This is the anti-circularity rule of the seed paper made operational at the diagnostic layer: the same-sample construction ($r = a/y$ on a sample, then $\hat{y} = a/r$ on that sample, then $\delta_y = 0$, "validated") cannot reach the residual gate, because the apparent same-sample resistance role is rejected upstream by the prediction pathway the gate replays, and no alternative pathway into the prediction slot exists. Verification of this closure — that the circular route fails at every seam — is part of the bounded software evidence; it is a property of the software, established by tests, in the verification sense rather than the validation sense [**23**].

Example 10.2 (Illustrative admitted residual). All numbers illustrative. Under one admitted context: drive $a = 4.0$ alpha (design value, inside the calibrated grid); calibrated resistance $r_{\text{eff}} = 2.0$ alpha-per-activation, interval $[1.8, 2.2]$, standard uncertainty 0.1, sample size 40, role calibrated-held-out with the profile's fit receipt as sole derivation parent. The registered prediction is $y_{\text{pred}} = 4.0/2.0 = 2.0$ activation units, with interval $[\,4.0/2.2,\ 4.0/1.8\,] \approx$

$[1.82, 2.22]$ and first-order standard uncertainty ≈ 0.10. An instrumented held-out observation at the same test point reads $y_{\text{obs}} = 2.6$ with interval $[2.3, 2.9]$ and standard uncertainty 0.15. The residual is $\delta_y = 0.6$ with interval $[\,2.3 - 2.22,\ 2.9 - 1.82\,] \approx [0.08, 1.08]$ and standard uncertainty $\sqrt{0.15^2 + 0.10^2} \approx 0.18$; the relative error is $0.6/2.0 = 0.30$; the apparent resistance is $4.0/2.6 \approx 1.54$ (role `APPARENT_SAME_SAMPLE`); the resistance delta is ≈ -0.46 with its own propagated interval. The residual interval excludes zero, so the record is uncensored and carries all four quantities. What may be said: the calibrated account underpredicts this observation by 0.6, more than its combined uncertainty. What may not be said: why. Every mechanism in the list of Section 4 remains open, including the boring ones.

Example 10.3 (Illustrative censored case). All numbers illustrative. Same context and prediction, but the observation reads $y_{\text{obs}} = 0.1$ with interval $[-0.05, 0.25]$ (zero-compatible: the instrument cannot distinguish this reading from silence at the stated confidence). The residual $\delta_y = -1.9$ with interval $[\,-0.05 - 2.22,\ 0.25 - 1.82\,] \approx [-2.27, -1.57]$ is emitted — it is informative, and its interval excludes zero — but the apparent resistance $4.0/0.1 = 40$ is *not* emitted, because the denominator's interval includes zero and the value 40 is an artifact of where the point estimate fell: had the point estimate been 0.01, the "apparent resistance" would have been 400 from data the instrument cannot distinguish from the same silence. The result carries the censoring marker `OBSERVED_INTERVAL_INCLUDES_ZERO` and its recorded reason, and the censored result type structurally cannot carry resistance estimates. The flat-response ambiguity of Chapter 7 — cutoff, saturation, wrong direction, insensitivity, measurement failure — applies to this record in full, which is exactly why no ratio is allowed to lend it spurious precision.

3. Censoring at zero-compatible response

Definition 10.4 (Censoring rule). If the observed value's uncertainty interval includes or touches zero (lower bound ≤ 0), the diagnostic emits the residual and relative error — both remain well-defined differences — but *censors* the ratio quantities: no apparent resistance and no resistance delta are constructed, the result carries the censoring marker `OBSERVED_INTERVAL_INCLUDES_ZERO` with a recorded reason, and the censored result type rejects any attempt to attach resistance estimates to it. Conversely an uncensored result must carry both ratio quantities and no reason; the two states are mutually exclusive by construction.

The statistical rationale is elementary but worth stating carefully, because the source material violates it. The apparent resistance is the ratio a/y_{obs}. A ratio whose denominator is compatible with zero has no finite moments and no usable interval: as the denominator's interval approaches zero from above, the ratio's upper bound diverges, and once the interval includes zero the ratio's sampling distribution places mass at arbitrarily large values of both signs. Any finite number reported in that situation is an artifact of where the point estimate happened to fall, not a summary of what the data support. The honest treatments are the standard ones for censored data: report the censoring event itself as the datum (the observation is indistinguishable from zero at the stated confidence), carry the bound rather than a point value, and refuse downstream arithmetic that would consume the undefined ratio. The candidate implements the refusal structurally — the censored result type *cannot* carry resistance estimates — rather than advisorily.

Prediction-side degeneracy is handled by the complementary mechanism: denominators involving the predicted value are floored at the profile's residual floor $\phi > 0$, so a prediction at

or near zero yields a relative error computed against ϕ rather than an unbounded ratio, and the floor value itself is a declared, digested profile tolerance rather than a hidden constant. Between the censoring rule and the floor, there is no code path on which a division by a zero-compatible quantity produces a reported diagnostic.

Note also what censoring does *not* do: it does not suppress the residual. A zero-compatible observation against a nonzero prediction is informative — it is the flat-response situation of Chapter 7, with its five-way causal ambiguity — and the signed difference with its interval is retained. What is suppressed is exactly the derived quantity whose statistical meaning has evaporated.

4. Residuals are non-identifying

4.1. The argument. The central claim of this chapter, and the one the source audit shows the source material itself half-knows, is that a residual — however clean its derivation — cannot identify its own cause. The argument deserves to be made in full rather than gestured at.

A residual is a one-dimensional summary: a signed magnitude (with interval) of disagreement between one observation and one model prediction at one point. Consider the space of generating situations compatible with a given positive residual $\delta_y > 0$ (observed flow exceeding prediction; the negative case is symmetric). At minimum, the following mechanisms each produce it:

(1) *Hidden facilitation or policy.* Some mechanism not in the model assists the driven response — the "hidden bypass" reading: the effective opposition is lower than calibrated.
(2) *Bypass of a modeled constraint.* A constraint that the calibrated resistance implicitly includes is inoperative in this context, for reasons benign or otherwise.
(3) *Opposition elsewhere.* A mechanism opposing a *different* pathway redirects activation such that the measured projection increases; the causal action is not on the measured path at all.
(4) *Measurement error.* The instrument chain — hooks, projections, normalization, full-scale binding — inflates the observation; nothing in the model is wrong.
(5) *Model misspecification.* The true local response is nonlinear, and the linear prediction undershoots at this dose; the discrepancy is structural, not mechanistic [**37**].
(6) *Calibration transfer failure.* The resistance was fit under conditions (corpus, seeds, drift state) that no longer obtain, and the parameter, not the mechanism, changed.
(7) *Sampling fluctuation.* The residual's own interval, honestly computed, may include values consistent with all of the above at once.

These mechanisms are observationally equivalent at the level of the single residual: the datum is one number, the hypothesis space is many-dimensional, and the map from mechanisms to residual sign is many-to-one in both directions (several mechanisms produce either sign, depending on parameters). Sign alone therefore cannot select among them — not weakly, not suggestively, but as a matter of the arithmetic of identification.

The point can be sharpened in the vocabulary of causal inference. Call two generating situations *residual-equivalent* if they induce the same distribution over (a, y_{obs}) at the observed test points. Mechanism attribution from residuals is well-posed only if the attribution target is constant on residual-equivalence classes — that is, only if the observational distribution identifies the mechanism. The mechanism list above exhibits, constructively,

residual-equivalent situations with different mechanisms: a hidden facilitation of the measured path and a measurement-chain gain error can be tuned to produce identical (a, y_{obs}) distributions, as can a bypassed constraint and a miscalibrated r_{eff}. Hence the attribution target is not constant on the equivalence classes, and no amount of statistical sophistication applied to the same observational data can recover it; what changes the situation is intervention. This is the standard identification requirement: causal quantities are defined relative to an explicit causal model and are computable from data only when the model plus the data identify them, typically via interventions or assumptions that break the observational ties [27]. The candidate supplies no causal graph over the mechanism space, asserts no identifying assumptions, and admits no do-operations; its residuals are observational discrepancies, and it says so in their method identity. Distinguishing the mechanisms requires interventions that separate them: manipulating the suspected constraint while holding the dose fixed, ablating the suspected bypass, permuting the labels the instrument depends on, replicating under re-calibration — each an experiment with its own controls, several of them present in the registered closure list. The conclusion is not that residual-based mechanism attribution is difficult; it is that, from the residual alone, it is undefined.

There is also a model-side face of the same coin, familiar from the calibration literature: an observed discrepancy decomposes, in principle, into parameter error, structural model discrepancy, observation error, and numerical error, and the decomposition is itself unidentified without prior structure — fitting the parameter to absorb the discrepancy simply relocates the problem [37]. Mechanism 5 and mechanism 6 in the list are this decomposition's components wearing mechanistic clothes; the candidate's refusal to update r_{eff} from residuals (no feedback path exists from the diagnostic layer to the profile) is the structural form of refusing to let the parameter absorb what may be discrepancy.

Example 10.5 (One residual, three stories). All numbers illustrative. Take the uncensored record of Example 10.2: $\delta_y = +0.6$, interval $[0.08, 1.08]$. Story A (hidden facilitation): a mechanism outside the model contributes response along the measured direction; true opposition is lower than calibrated. Story B (measurement): the full-scale value bound at this test point was set from a corpus percentile that undersamples high-activation contexts, inflating the instrumented reading by a factor compatible with +0.6. Story C (misspecification): the true response is mildly convex in a over the grid, the linear fit undershoots at $a = 4.0$, and the same board would show a *negative* residual at $a = 1.0$. All three stories reproduce the record; the second and third involve no hidden mechanism at all; and the third makes a testable prediction (sign reversal at low dose) that the first two do not — which is a statement about what to probe next, not about which story is true. This is the entire licensed use of the record.

Rule 10.6 (Non-identification). No implemented output of the residual layer asserts, encodes, or ranks a generating mechanism. The emitted quantities are the residual, the relative error, and (uncensored only) the apparent resistance and resistance delta — each a descriptive statistic with propagated uncertainty, none a diagnosis. The method identity stamped on every residual's provenance names the contract: `CAS-OHM-V1-NONIDENTIFYING-RESIDUAL`.

4.2. What residuals are good for. Non-identification does not make residuals useless; it makes them *ordinal evidence for allocation of further measurement*. A large residual (relative to its interval) at one test point, against a small one at another, is a defensible reason to spend the next intervention budget at the first point — not because the residual says what is there, but because it says the calibrated model's account is worse there. Ranking follow-up probes is a decision about experiments, not a claim about mechanisms, and it is robust to the

non-identification argument precisely because every mechanism on the list above, including boring measurement error, is something a follow-up probe can help separate. The resistance delta serves the same triage role in parameter space: a delta whose interval excludes zero flags the calibration, the transfer conditions, or the linearity assumption for re-examination, without saying which.

The ranking criterion itself can be made precise without exceeding the license. For admitted, uncensored residual records $j = 1, \ldots, k$ over one profile, a defensible ordering statistic is the uncertainty-scaled magnitude $|\delta_y^{(j)}|/s^{(j)}$, where $s^{(j)}$ is the record's propagated standard uncertainty — large values mark points where the disagreement is large relative to what the error budget explains. The statistic inherits every caveat of its inputs: it compares only within one profile (cross-profile residuals are in incomparable coordinates), only among uncensored records (censored records enter the queue by their censoring event, not by a ratio), and only as a priority order, since the statistic's distribution under the no-mechanism null is exactly what the shuffled baselines of Section 5 exist to estimate. A priority order is not a significance ranking, and the candidate emits no p-values here.

This is a deliberately modest job description, and the candidate keeps it modest structurally: residual outputs feed no gate, trigger no label, and modify no profile. They are inputs to human research planning, Ring 4, and their provenance chains exist so that the human doing the planning can see exactly what stands behind each number. The asymmetry with the rest of the pipeline is intentional: everywhere else, outputs are consumed by further typed operations; here, the consumer is a person deciding where to measure next, and the type system's last act is to hand that person an honest record rather than a conclusion.

5. The blinded implanted-fault program

The registered residual-localization hypothesis (CLM-CAS-OHM-RESIDUAL-LOCALIZATION) is the candidate's one sanctioned route by which residuals could ever earn more than the triage role — and it is constructed so that failure is meaningful. The program, fixed in the preregistered closure protocol and the synthetic known-fault infrastructure, has the following shape.

Faults with known ground truth are implanted into synthetic boards — hidden constraints, bypasses, cutoffs, saturations, off-axis couplings, non-ohmic responses — by a process separated from the analysis: the analyst receives sweeps and residual diagnostics without fault labels. The diagnostic pipeline ranks candidate fault locations from residual evidence alone. Performance is scored against the implantation record, and — this is the load-bearing clause — against two null references: *matched controls* (boards identical in construction with no implanted fault, so that the false-positive behavior of the ranking is measured on truly clean substrate) and *shuffled baselines* (residual-to-location assignments permuted, so that the information content of the actual assignment is measured against chance under the same marginals). Localization is claimed only if ranking performance exceeds both references at the preregistered margin; negative outcomes, exclusions, and instability are recorded without erasure, per the closure protocol's general terms.

PROTOCOL 10.7 (Withdrawal clause). If the blinded program fails — ranking not above shuffled baseline, or false-positive behavior on matched controls incompatible with the preregistered margin — causal language about residuals is withdrawn, not weakened. The registered hypothesis is closed as refuted; the non-identifying contract of Rule 10.6 remains the permanent ceiling on residual semantics; and no rephrasing ("suggests," "indicates," "is consistent

with a bypass") survives as licensed language, because each of those phrasings is precisely the mechanism attribution the failed test failed to earn.

The synthetic substrate for the program already exists in the bounded software evidence and is worth describing, because it fixes what "implanted fault" means operationally. A synthetic board is a typed generator with a declared base resistance and a declared fault mode drawn from a closed enumeration — none, hidden constraint (base opposition raised by a hidden component), bypass (base opposition reduced), cutoff (zero flow below a drive threshold), saturation (flow capped at a ceiling), off-axis (a declared fraction of flow diverted off the measured axis), and non-ohmic (flow quadratic in drive) — each altering the board's dose-response law in a known, parameterized way. Sweeps over such boards produce observation tuples indistinguishable in type from any other sweep, which is the property the blinding leans on: the analysis pipeline runs the same typed path — classification, prediction, residual — that any admitted data would traverse, and the fault label exists only in the implantation record held outside the analysis. The known-fault suite currently serves as golden-fixture verification (the pipeline detects each implanted signature where detection is defined, and stays silent where silence is required); the blinded *scoring* run, with its matched controls and shuffled baselines and preregistered margins, is the part that has not been executed, and it is the part on which any upgrade of residual semantics depends.

The two null references discipline different failure modes, and both are necessary. Matched controls — fault-free boards drawn from the same construction distribution — estimate the ranking's false-positive behavior: a pipeline that finds "faults" in clean substrate at the same rate as in faulted substrate has a ranking that reflects its own noise, whatever its hit rate. Shuffled baselines — true residuals permuted against locations — estimate how much of the ranking's success is carried by the residual-to-location *assignment* as opposed to the marginal distribution of residual magnitudes: a pipeline that scores no better than its own shuffle is exploiting magnitude statistics that any location would satisfy. Passing one reference and failing the other is a fail; the preregistered claim requires exceeding both at the stated margin, and the margin is fixed before the run, not after.

Two boundaries on the program itself, so that this section cannot be overread. First, the synthetic boards validate *software behavior*: they establish that the pipeline detects what it was built to detect on substrates where ground truth is available by construction. Success on synthetic faults licenses nothing about live models; it is a necessary precondition for the live experiment, which carries its own preregistration, controls, and independent recomputation requirements. Second, even a fully successful program would license localization claims of the preregistered form only — ranking above baselines at stated margins — not the unrestricted "residual proves hidden resistance" reading, which remains on the forbidden-wording list regardless of outcome.

6. The source's residual-overinterpretation tensions, recapped

The preserved residual conflict is the clearest instance in the source corpus of a document warning against a reading and performing it anyway, and the candidate's design is best understood as a mechanical enforcement of the sources' own warnings against the sources' own practice. The tensions, at evidence-record granularity:

The Guide. On the pages where residual analysis is introduced, residual sign is glossed as suggesting hidden opposition (negative) or extra amplification / hidden assistance (positive) — mechanism vocabulary — on the same page as the statement that the residual alone is

insufficient to identify a cause. The warning and the violation share a page; nothing in the Guide's procedure operationalizes the warning.

Appendix I. The quantity is renamed: what the Guide treats as a hidden resistance *component* is called a residual, an explicit step toward the non-identifying posture — and yet the same appendix treats a positive residual on a safety board as a bypass finding, which is mechanism attribution in its most consequential form. The appendix also states that saturated-stage residuals are meaningless, an observation the candidate promotes from remark to gate: the active-region admission requirement of Rule 10.1 makes the meaningless case uncomputable rather than inadvisable.

The addendum. The diagnostic walk places hidden content *last* — after dose, direction, calibration, saturation, and measurement checks — and the falsification row for residuals requires independently identified causes above a shuffled-control baseline before residual-based claims are made. This is the source corpus at its most careful, and it is the position the candidate implements: the blinded program of Section 5 is the addendum's falsification row, given hard edges.

Table 1 aligns each preserved tension with the design element that enforces the source's own warning against the source's own practice.

Source tension	Enforcing design element
Guide: residual sign glossed as hidden opposition or hidden assistance, on the page stating residual alone is insufficient	Non-identifying contract (Rule 10.6); no mechanism field exists in any residual output type
Appendix I: hidden resistance renamed a residual, yet positive residual on a safety board read as a bypass	Apparent resistance quarantined under the `APPARENT_SAME_SAMPLE` role, rejected by the prediction gate; no bypass vocabulary in any emitted quantity
Appendix I: saturated-stage residuals are meaningless	Active-region admission gate; residual computation is impossible, not inadvisable, outside `ACTIVE`
Addendum: hidden content last in the diagnostic walk	Ranking-only license (Section 4.2); residuals allocate probes, they do not name contents
Addendum: residual validity requires independently identified causes above a shuffled-control baseline	Blinded implanted-fault program with matched controls, shuffled baselines, preregistered margins, and the withdrawal clause (Protocol 10.7)
Both: ratios reported at negligible response	Censoring at zero-compatible observation (Definition 10.4) and the residual floor on predicted denominators

TABLE 1. Preserved residual-overinterpretation tensions and their mechanical enforcement in the candidate.

The recapitulation matters because it locates the candidate's contribution precisely. The insight that residuals are non-identifying is *in the sources*; what the sources lack is any mechanism that makes the insight binding on their own procedures. The candidate's residual layer is that mechanism: admission gates that exclude the meaningless cases, censoring that excludes the undefined ones, typed outputs that cannot carry mechanism labels, a method identity that names the contract, and a preregistered program with a withdrawal clause as

the only path to anything more. Where the Guide says "insufficient" and proceeds, the candidate says "insufficient" and stops.

Claim Boundary 10.8 (Chapter boundary). No residual over live-model data has been admitted; every residual computed to date in the candidate's history is a synthetic-board or test-fixture artifact within the bounded software evidence. The residual layer's guarantees are structural: well-defined subtraction under full derivation binding, censoring at zero-compatible denominators, propagated uncertainty everywhere, and the impossibility — at the type level — of a residual that names its cause. Whether residual diagnostics can localize implanted faults above chance is a registered, open, falsifiable question, and this monograph asserts no answer to it.

CHAPTER 11

Structural Completeness versus Authenticated Eligibility

Every evidence architecture eventually confronts the same temptation: a record that has accumulated enough structure—enough hashes, enough cross-references, enough well-typed fields—begins to *look* authoritative, and the system's operators begin to *treat* it as authoritative. The CAS-Ohm 1.0 candidate is built to make that slide impossible by construction. It maintains two entirely separate predicates over a research packet: *structural research-packet completeness*, which asks whether the packet is internally well-formed, fully referenced, and analytically usable; and *authenticated Evidence-Packet eligibility*, which asks whether any independent authority has admitted the packet's claims into a trust regime. The first predicate can be true today, inside this repository, using nothing but the candidate's own validators. The second predicate is *constant false* in the current candidate, and this chapter explains at full resolution why it is false, what six preconditions would have to be independently established before it could ever become true, and why no amount of accumulation along the first axis moves a packet along the second.

The design mirrors a foundational rule of the CAS Core 8 canonical-encoding and admission discipline: *structural consistency is never authorization.* A byte string that canonicalizes deterministically, hashes stably, and round-trips through every validator has demonstrated exactly one thing—that it is a well-formed byte string. It has not demonstrated that its content is true, that its signer had authority, that its clock was honest, or that any reviewer ever read it. CAS-Ohm implements this rule as two independent axes so that no packet can drift from "complete" to "authoritative" by accumulation. The rest of this chapter develops each axis in turn, then examines the boundary between them: the fail-closed rejection classes, the six missing preconditions of authentication, the specific reasons that deterministic fixture keys and caller-supplied replay time can never satisfy those preconditions, the distinction between replay-time-current and authenticated-current, and the general principle—inherited from provenance-modeling practice [22]—that provenance structure alone is not truth.

Throughout, the governing artifact status applies: everything in this chapter describes a candidate whose status is OPEN / RING 4 / NOT LICENSED / HELD. The completeness predicate described here is a tested software behavior at the Gate 4 receipt; the eligibility predicate is a tested *constant*, and the tests confirm precisely that it cannot be made true by any input the candidate accepts.

1. The two-axis design

DEFINITION 11.1 (Structural research-packet completeness). A research packet is *structurally complete* when every one of the following holds: it binds an exact model revision; it carries a valid calibration-profile digest; its raw artifact bytes are present and all artifact references resolve; relevant controls are bound; uncertainty is represented; every field carries lineage and status; the bound calibration profile is reviewed, nonstale, and replay-time-current

at the caller-supplied `replay_as_of`; and at least one field is genuinely instrumented. Completeness is a property of the packet's internal structure and of the artifacts it references; it is decidable by the candidate's own validators without any external party.

Definition 11.2 (Authenticated Evidence-Packet eligibility). A research packet is *eligible* as an authenticated Evidence Packet only when an independently admitted authority manifest, a protected key lifecycle with a revocation policy, a trusted current-time receipt, materialized model, code, and runtime dependency identities, authenticated instrument evidence, and a reviewed packet policy are all simultaneously present and bound to the packet. Eligibility is a property of the packet's relationship to independent external authorities; it is not decidable by the candidate's own validators, and in the current candidate it is a stored constant equal to false.

Invariant 11.3 (No drift by accumulation). No sequence of operations available in the CAS-Ohm 1.0 candidate—adding fields, adding artifacts, adding hashes, adding fixture-key attestations, re-running validators, or re-canonicalizing—changes the value of authenticated Evidence-Packet eligibility. The two predicates are computed from disjoint evidence classes: completeness from the packet's internal structure, eligibility from external authorities that the candidate does not implement. A packet therefore cannot drift from complete to authoritative by accumulation.

The reason this invariant is architectural rather than procedural deserves emphasis. Many evidence systems enforce the same intent by policy: reviewers are instructed not to treat internal records as authoritative. Policy-level enforcement fails in the presence of ordinary institutional pressure, because the artifacts themselves look identical whether or not the policy was followed. The CAS-Ohm design instead makes the eligibility field a distinct typed value with no admitted write path. The tested behavior at the Gate 4 receipt includes the negative capability result: *zero authenticated Evidence-Packet eligibility paths exist in the implementation.* There is nothing to misuse, because there is nothing there.

Remark 11.4 (Mirror of the CAS Core 8 rule). The CAS Core 8 contract, which CAS-Ohm inherits for all canonical encoding, already establishes that a canonical digest proves byte identity and nothing else. The two-axis packet design is the same rule lifted from the encoding layer to the evidence layer. At the encoding layer: a stable hash is not a true statement. At the evidence layer: a complete packet is not an admitted claim. In both cases the dangerous inference runs from syntactic success to semantic authority, and in both cases the candidate blocks the inference by refusing to give the syntactic layer any vocabulary in which the semantic conclusion could even be stated.

Example 11.5 (The accumulation scenario). Consider a hypothetical operator who assembles a packet with an exact model revision, a valid profile digest, complete raw bytes, six bound controls, propagated uncertainty on every numeric field, full per-field lineage, a reviewed and current calibration profile, and three genuinely instrumented fields—twice the minimum. The operator then attaches Ed25519 attestations produced with the repository's deterministic fixture keys, binds a canonical operation record, and replays the whole bundle successfully. Under Definition 11.1 the packet is structurally complete, and every validator confirms it. Under Definition 11.2 the packet is exactly as ineligible as an empty one, because none of the six preconditions concerns anything the operator did: no independent authority admitted a manifest, no protected key ceremony occurred, no trusted timestamp was obtained, no external review of the packet policy exists. The scenario illustrates the design

intent: effort along the completeness axis is real and valuable—it is what makes the packet *analyzable*—but it purchases nothing along the eligibility axis, by construction.

2. Structural completeness requirements

This section discusses each completeness requirement in turn. Each requirement exists because a specific, historically observed failure mode becomes representable when the requirement is dropped; where the source audit of Chapter 4 preserved a conflict on the point, the connection is drawn explicitly.

2.1. Exact model revision. A packet must bind the exact identity of the model under audit: model identifier, revision, and the weights, configuration, and tokenizer digests, as materialized in the candidate's `ModelIdentity` record. The requirement is exact rather than approximate because every CAS-Ohm quantity is profile-relative (Chapter 5): an effective resistance fitted against one checkpoint has no defined meaning against another, and the candidate's comparison rules refuse cross-profile transfer without a stated bridge. An inexact model reference—a family name, a version range, a "latest" pointer—would make the profile digest unstable in exactly the way that silently reintroduces the cross-model comparability that the forbidden-wording list rules out.

The revision binding also serves the replication half of the Gate 5 protocol (Chapter 12): replication on a held-out checkpoint is only a meaningful act if the original checkpoint is pinned beyond dispute. A packet that cannot say precisely which weights produced its observations cannot be replicated, only imitated.

2.2. Valid profile digest. The calibration profile of Chapter 5 binds the full experimental context—model identities, code revision, runtime and dtype, test point, direction vector with norm, sign, and provenance, corpus manifest and rights, seeds, alpha grid, full-scale method, fit and uncertainty methods, tolerances, response-region rules, controls, drift and expiry, and review state—under one canonical digest. Completeness requires that the packet's profile digest be *valid*: well-formed, resolvable to a profile record, and matching the recomputed canonical hash of that record. An invalid or dangling digest is not repaired or defaulted; the packet fails closed as UNKNOWN-PROFILE (Section 3).

The profile digest is the single point through which the eight preserved source conflicts are prevented from re-entering downstream analysis. The three incompatible full-scale methods, for instance, are distinguished precisely by producing different profile digests; a packet whose digest is unverifiable could be quietly mixing rail-and-floor normalization with corpus-percentile normalization, which is the mixed-scale error the audit was built to catch.

2.3. Raw artifact bytes and resolvable references. Every artifact reference in the packet carries an identifier, a URI, a SHA-256 digest, and a media type, and completeness requires both that the referenced raw bytes exist and that every reference resolves to bytes matching its digest. The rule has two halves and both are load-bearing. The digest half prevents substitution: a reference that resolves to different bytes than were hashed is a tampered record. The existence half prevents a subtler failure, the *citation-shaped void*: a packet that references raw evidence which no longer exists—or never existed—retains the typographical appearance of evidential grounding while having none. Requiring the bytes themselves keeps the packet honest about what it can actually show a future auditor.

This requirement is also what makes the Gate 5 retention step (Chapter 12, step seven) enforceable in advance: a protocol that promises to retain full raw evidence can only be audited if raw evidence was a first-class packet obligation from the beginning.

2.4. Relevant controls. A packet must bind the controls relevant to its measurements—not merely reference control artifacts, but bind them through control bindings that name the test point and the relevance method by which the control is claimed to be relevant. The candidate's provenance validator enforces exact agreement between the bound control set and the declared control references; a control that is referenced but not bound, or bound but not referenced, is a structural error.

The insistence on *relevance* rather than mere presence reflects the control taxonomy of the closure protocol: zero-intervention, random same-norm direction, wrong-layer, wrong-token, label-shuffle, and norm-matched controls each rule out a different alternative explanation, and a control of the wrong type provides the appearance of rigor without the discriminating power. A relevance method identifier makes the claimed function of each control explicit and therefore checkable by a reviewer.

2.5. Uncertainty. Numeric fields carry a typed uncertainty record: unit, standard uncertainty, interval bounds, confidence level strictly between zero and one, and the identifier of the method that produced the uncertainty. The measurement value must lie inside its own interval, the interval bounds must themselves satisfy the quantity's unit constraints, and the uncertainty unit must match the measurement unit. This is the packet-level expression of the no-bare-numbers rule of Chapter 3: a displayed number without its uncertainty is not a smaller piece of evidence, it is a different and lesser kind of object, and the completeness predicate refuses to count it.

Uncertainty at the completeness layer is deliberately agnostic about method quality. A packet is complete if its uncertainties are present, typed, and internally consistent; whether the uncertainty method was *appropriate* is a question for review and for the preregistered protocol, not for the structural validator. The two-axis design recurs in miniature here: structural presence of uncertainty is checkable; adequacy of uncertainty is an authority-level judgment.

2.6. Per-field lineage and status. Every field carries provenance: source references, method identifier, and—where applicable—calibration binding, instrument identity, test point, raw artifact references, control bindings, dataset split reference, derivation parents, and operation binding. Derivation parents are ordered, contiguous from position zero, and unique in both role and identity, so that the derivation graph of any value can be reconstructed exactly and in order. Measurement-kind rules bind lineage to type: a model-derived value without derivation parents is invalid; an instrumented value without instrument, test point, calibration binding, and raw artifacts is invalid; behavioral-proxy, human-score, and LLM-score values without raw artifacts are invalid.

Status accompanies lineage: each value is marked as estimate or observation, and missing or censored values carry an explicit reason rather than an empty slot. The censoring reason requirement matters most at the residual boundary (Chapter 10): a zero-compatible predicted response yields no residual quantity, and the packet records *why* the quantity is absent, so that absence-by-censoring can never be confused with absence-by-omission.

2.7. Nonstale reviewed calibration profile at replay time. The bound calibration profile must be in an admissible review state and must be current—within its declared validity interval and drift bounds—*as evaluated at the caller-supplied* `replay_as_of`. Two design decisions are folded into this requirement and both must be stated plainly.

First, staleness is real: profiles expire. A profile with a declared drift model and expiry that has lapsed no longer describes the system it once calibrated, and a packet bound to a lapsed profile is not complete no matter how pristine its other fields are.

Second, and critically, the time against which staleness is evaluated is *supplied by the caller*. The candidate has no trusted clock. The `replay_as_of` value allows historical replay—evaluating whether a packet *was* current at a stated moment—and it makes every currency verdict conditional on an unauthenticated input. The completeness predicate is therefore honest about its own epistemic standing: it asserts "current at the time the caller claims," never "current now." Section 6 develops the consequences of this distinction at length; here it suffices to note that the conditionality is not a defect of the requirement but a designed refusal to overstate it.

2.8. At least one genuinely instrumented field. Finally, a complete packet must contain at least one field whose measurement kind is INSTRUMENTED and which satisfies the full instrumented obligations: instrument identity, test point, calibration-profile binding, and raw artifact references. The kind non-fungibility rule of Chapter 3 governs the word "genuinely": a derived value does not become instrumented because a parent was, and a score does not become telemetry by proximity to a measurement.

PROPOSITION 11.6 (Necessity and insufficiency of instrumentation). *The presence of at least one genuinely instrumented field is necessary for structural research-packet completeness and is not sufficient for it; a fortiori it is not sufficient for authenticated Evidence-Packet eligibility.*

PROOF. Necessity is by definition of the completeness predicate: the zero-instrumentation rejection class (Section 3) fails any packet without such a field, whatever else it contains. Insufficiency for completeness follows from the independence of the remaining requirements: a packet may hold one valid instrumented field while lacking a resolvable profile digest, missing raw bytes for another reference, or binding a stale profile, and each of those conditions independently fails the predicate. Insufficiency for eligibility follows from Invariant 11.3: eligibility is computed from external-authority preconditions disjoint from packet content, so no packet content, instrumented or otherwise, can establish it. □

The rationale for making instrumentation a floor rather than a preference is historical. The addendum source explicitly concedes that its own operationalization contains zero instrumented fields and is therefore not an Evidence Packet; the completeness predicate encodes that concession as a rule. A packet built entirely from design values and scores can be internally flawless and still be, evidentially, a description of intentions. One instrumented field is the minimum contact with the system under audit that distinguishes a research packet about a model from a research packet about a plan.

3. The fail-closed rejection classes

The completeness predicate is enforced negatively by six named rejection classes. Each class fails closed: a packet matching the class is rejected in full, with no partial admission, no repair, and no downgrade to a lesser packet type. The classes are deliberately coarse—each names a whole family of defects—because the purpose of the layer is not diagnosis but refusal; finer diagnosis belongs to tooling that operates on already-rejected packets.

CHECK CLASS 11.7 (Score-only). Every numeric field is a design value, human score, LLM score, or derivation thereof. The packet describes assessments and intentions, not

measurements. Rejected: a score does not become telemetry by aggregation, and a packet of scores does not become a research packet by volume. This class enforces Proposition 11.6 from the negative side and is the packet-level shadow of the CLM-CAS-OHM-PHYSICAL-TELEMETRY prohibition discussed in Chapter 12.

CHECK CLASS 11.8 (Zero-instrumentation). No field satisfies the full instrumented obligations, even if some fields are neither scores nor design values. This class differs from score-only in that it also catches packets whose fields are behavioral proxies or model-derived values with intact lineage: such packets can be analytically interesting, and they are still not complete, because nothing in them was produced by an identified instrument chain against the bound profile.

CHECK CLASS 11.9 (Missing-provenance). Some field lacks required lineage: a model-derived value without derivation parents, an instrumented value without its instrument or test point, a proxy or score without raw artifacts, a control referenced without a binding, or a dangling artifact reference. The class is triggered by any single omission because provenance is only useful when total; a packet that is ninety percent traceable offers an attacker a ten percent unaudited channel.

CHECK CLASS 11.10 (Mixed-scale). A value is presented under a unit inconsistent with its quantity kind, or quantities from incompatible scales are combined—a Q10 score where a Q100 constraint is declared, a fraction compared against a Q10 threshold, an effective resistance without the alpha-per-activation unit. This class is the direct descendant of the audit's first preserved conflict, in which a worked example carried a saturation threshold of 8.5 on a scale where the correct component value was 0.85; the rejection class makes that historical error unrepresentable in an admitted packet.

CHECK CLASS 11.11 (Unknown-profile). The packet's calibration-profile digest is malformed, unresolvable, or fails recomputation against the resolved profile record, or the resolved profile is in an inadmissible review state or stale at the caller-supplied `replay_as_of`. Every quantity in CAS-Ohm is profile-relative; a packet without a verifiable profile is a set of numbers without a coordinate system.

CHECK CLASS 11.12 (Tampered). Any digest mismatch between a reference and its resolved bytes, any canonical-encoding failure, any attestation that fails verification or replay, or any internal inconsistency indicating post-hoc alteration. Tampering is not adjudicated; it is not distinguished from corruption; the packet is rejected identically in both cases, because the completeness layer has no authority to decide intent and does not attempt to.

ADVERSE FINDING 11.13 (Rejection is symmetric across origins). The six classes apply identically to packets generated by the candidate's own synthetic boards, packets assembled by hand, and packets produced by any future harness. The tested Gate 4 behavior includes golden rejection fixtures exercising these classes; the suite verifies not only that bad packets are rejected but that the rejection is closed—no API path admits a partially valid packet under a weaker type.

4. Authenticated eligibility: constant false

RULE 11.14 (Eligibility constant). Authenticated Evidence-Packet eligibility is a separate field from structural completeness and is *always false* in the CAS-Ohm 1.0 candidate. It cannot become true until all six of the following are present: an independently admitted

authority manifest; a protected key lifecycle and revocation policy; a trusted current-time receipt; materialized model, code, and runtime dependency identities; authenticated instrument evidence; and a reviewed packet policy. The candidate implements none of the six, provides no write path to the field, and the Gate 4 receipt records zero authenticated eligibility paths.

The six preconditions are not a checklist that the project has simply not gotten around to; each names a category of external fact that no amount of internal engineering can manufacture. The following subsections analyze them one at a time: what each precondition is, what attack or failure its absence leaves open, and why the candidate's existing machinery—however disciplined—does not and cannot substitute for it.

4.1. Independently admitted authority manifest. An authority manifest declares who may assert what: which parties, identified by which keys, are entitled to sign which classes of statement about which artifacts. The candidate's operation bindings already carry an authority-manifest digest field—the *slot* exists—but the digest currently identifies a document that no independent party has admitted. An authority manifest admitted by its own subject is a description of self-trust, and self-trust is precisely the thing an Evidence Packet exists to exceed.

Threat 11.15 (Self-admitted authority). If the candidate treated its own manifest as admitted, any operator with repository write access could extend the manifest to authorize any key, then sign any statement. The signature chain would verify perfectly, because verification only ever checks consistency with the manifest—and the manifest itself would be the forged object. Independence of admission is therefore not a formality: it is the only property that places the root of the trust chain outside the reach of the parties being trusted.

Independent admission requires an admitting party with its own identity, its own records, and its own accountability—an institutional fact, not a software feature. Until such an admission exists, the manifest digest in every operation binding is honest metadata about an unadmitted document, and the candidate's authorization states say so: both implemented trust modes map to explicitly *unauthenticated* authorization states (`UNAUTHENTICATED_STRUCTURAL` and `UNAUTHENTICATED_INTEGRITY`), and the validator rejects any attempt to pair a trust mode with a stronger authorization claim.

4.2. Protected key lifecycle and revocation. Authenticated evidence requires signing keys whose entire lifecycle is protected: generation in a controlled environment, storage that prevents extraction, use that is logged and rate-limited, rotation on schedule, and—indispensably—revocation that actually propagates. A signature is only as meaningful as the counterfactual it excludes; if a compromised key cannot be revoked, or if verifiers do not consult revocation state, then a valid signature excludes nothing, because the attacker signs as fluently as the owner.

The candidate's Ed25519 machinery is real cryptography with a deliberately unreal key story: the keys are deterministic fixtures committed to the repository. Section 5 treats the consequences in full; the present point is narrower. Even if the fixture keys were replaced tomorrow with well-generated secrets, the lifecycle precondition would remain unmet, because a lifecycle is a set of ongoing institutional processes—custody, ceremony records, rotation calendars, revocation infrastructure—and not a property of any key material as such. The precondition names a practice, not an artifact.

4.3. Trusted current-time receipt. Every currency judgment in the candidate—profile staleness, validity intervals, drift expiry—is evaluated against the caller-supplied `replay_as_of`. This is the correct design for a replay system and a disqualifying design for an authentication system. A trusted current-time receipt is a statement from an accountable time authority, cryptographically bound to the packet, that a specific moment had genuinely arrived when the statement was made. Without it, "current" means "current according to whoever called the function," and a verifier a decade from now cannot distinguish a packet honestly evaluated in 2026 from one assembled later and replayed with a flattering timestamp.

Threat 11.16 (Time travel by parameter). An operator holding a profile that expired in March may simply supply a February `replay_as_of` and obtain a currency verdict of true. Within the candidate's semantics this is not an attack—the verdict genuinely means "current as of the supplied time," and the supplied time is bound into the record—but any consumer who reads the verdict as a statement about the present has been misled by their own reading. The trusted-time precondition exists to close the gap between what the verdict says and what a hurried reader hears.

The operation-binding schema anticipates the missing piece: it carries an optional clock-receipt digest field, present but unpopulated by any trusted source. As with the authority manifest, the slot is engineering; the receipt is an institution.

4.4. Materialized model, code, and runtime dependency identities. The candidate's records *name* their dependencies exactly: model identity with weights, config, and tokenizer digests; code references; dependency-lock references; runtime identifiers. Naming, however, is not materialization. A materialized dependency identity is one whose referent has been captured, archived, and verified end-to-end, so that an independent party can reconstruct the exact execution environment and rerun the computation. The difference is the difference between a bibliography and a library.

The eligibility precondition requires materialization because authenticated evidence must survive the disappearance of its birthplace. A packet whose code reference points into a repository that has since been rewritten, or whose runtime identifier names a container image that no registry still serves, retains perfect structural validity and has lost its reproducibility. Independent recomputation—step five of the Gate 5 protocol—presupposes exactly this materialization, which is why the protocol lists authenticated execution evidence alongside it.

4.5. Authenticated instrument evidence. Structural completeness requires that instrumented fields bind an instrument identity—an identifier, kind, version, and implementation digest. Eligibility requires more: evidence that the named instrument actually ran, on the named test point, at the recorded time, producing the recorded raw bytes. This is attestation of *execution*, not of *identity*. The candidate can verify that a record claims the right instrument shape; it cannot verify that any instrument ever executed, because execution evidence must be produced by the execution environment and vouched for by an authority independent of the packet author.

The gap matters because instrumented status is the load-bearing floor of the completeness predicate (Proposition 11.6). An adversarial packet author who fabricates an instrumented field—plausible raw bytes, a well-formed instrument identity, a consistent test point—passes every structural check. Only authenticated execution evidence distinguishes measurement from the simulation of measurement, and the candidate, having no such evidence channel, correctly refuses to claim the distinction.

4.6. Reviewed packet policy. The final precondition is a packet policy—the rules governing what packets may claim, how they are versioned, who may amend them, and how outcome access is controlled—that has itself been reviewed by parties independent of its authors. The candidate has packet *rules* in abundance; every validator in this chapter enforces some of them. It does not have a *reviewed policy*, because review is an event in the world involving reviewers, and the project's own review states record that internal cross-review and independent external review both remain open.

This precondition is deliberately last in the list because it subsumes the epistemics of the other five: an authority manifest, a key ceremony, a time authority, a materialization archive, and an instrument attestation channel are each only as good as the policy that says how they compose, and that policy is only as good as the scrutiny it has survived. A policy reviewed only by its authors has survived nothing.

5. Why fixture keys and caller-supplied time can never satisfy the preconditions

Two features of the current candidate are sometimes mistaken, by hopeful readers, for down payments on authentication: the Ed25519 fixture-key attestations available in the integrity-configuration trust mode, and the replay-time machinery. Both are genuinely useful. Neither can ever mature into a satisfied precondition, and the reasons are structural, not circumstantial.

The fixture keys are deterministic and committed to the repository. Every party with read access to the repository—which includes every party the attestations might be shown to—can produce a valid signature under them. A signature scheme in which verifiers can sign proves integrity of transport (the bytes were not accidentally corrupted between writer and reader, and replay reproduces them) and proves nothing about origin, because origin is exactly the property the shared key erases. This is not a weakness to be patched by hiding the keys better: the keys are fixtures *by design*, so that the test suite can exercise the verification and replay code paths deterministically on any machine. Their purpose is to test the machinery, and a test instrument that doubled as a production authority would contaminate both roles. The candidate accordingly types the integrity mode's authorization state as `UNAUTHENTICATED_INTEGRITY` and rejects, at validation time, any operation binding that pairs the integrity trust mode with a stronger authorization claim.

Caller-supplied replay time fails the trusted-time precondition for the symmetric reason. The value of a time authority is that it is *not* the caller: its receipts are meaningful because the party asserting the time is distinct from, and unaccountable to, the party benefiting from the assertion. `replay_as_of` is supplied by the most interested party in the transaction. No enrichment of the parameter—finer resolution, signed requests, logged calls—changes who chooses the value. The parameter is the right tool for its actual job, historical replay, precisely because it lets the caller ask counterfactual questions about past moments; that same property makes it permanently unusable as evidence about the present.

Adverse Finding 11.17 (Structural impossibility, not implementation gap). The fixture-key and replay-time mechanisms do not partially satisfy the key-lifecycle and trusted-time preconditions; they satisfy them to degree zero, and further engineering on either mechanism cannot change this. Both preconditions require an independent party—a key custodian with revocation authority, a time authority with accountability—and independence cannot be implemented inside the candidate, only contracted outside it. The Gate 4 suite pins this finding as tested behavior: the eligibility field is false in every fixture, including those with maximal attestation and replay success.

6. Replay-time-current versus authenticated-current

The distinction between the two currency notions deserves its own treatment, because it is the point at which careful readers most often stumble.

Definition 11.18 (Replay-time-current). A calibration profile is *replay-time-current at* t when its declared validity interval and drift bounds contain t, where t is the caller-supplied `replay_as_of`. The judgment is a pure function of the profile record and the parameter; it asserts nothing about when the judgment itself is made.

Definition 11.19 (Authenticated-current). A calibration profile would be *authenticated-current* when it is replay-time-current at a time t attested by a trusted current-time receipt bound to the evaluation. No such receipts exist in the candidate; the notion is defined here only to be distinguished from Definition 11.18.

Proposition 11.20 (Strict one-way entailment). *Authenticated-current would entail replay-time-current at the attested time; replay-time-current entails authenticated-current under no circumstances realizable in the candidate.*

Proof. The first entailment is immediate from the definitions: an attested t inside the validity interval is in particular a supplied t inside the interval. For the second: authenticated currency requires a trusted receipt, the candidate contains no channel by which such a receipt can be produced or verified (Section 4.3), and Invariant 11.3 excludes any accumulation of unauthenticated material from synthesizing one. Hence no evaluation in the candidate, whatever its parameters, yields the authenticated notion. □

The practical reading of the proposition is a discipline for report language. Every currency statement the candidate emits must be read—and, in derived documents, written—with its conditionality intact: "current at the supplied replay time," never "current." The monograph adopts this discipline throughout, and Chapter 14's limitations catalog lists the absence of live calibration as an open boundary rather than a caveat, because a boundary is what it is: the entire authenticated-current half of the vocabulary is unoccupied.

Remark 11.21 (Why keep the weaker notion at all). One might ask why a system that cannot authenticate time bothers evaluating currency at all. The answer is that replay-time currency is exactly what historical audit needs. An auditor reconstructing what the candidate would have concluded on a given date must be able to pose that date; a reviewer checking that a profile was not used past its expiry needs staleness-at-t as a queryable predicate. The weaker notion is not a failed version of the stronger one; it is a different instrument, correctly labeled.

7. Provenance structure is not truth

The chapter closes with the general principle underneath all of its particulars. The candidate's provenance machinery is structurally rich: typed artifact references, ordered derivation parents, control bindings with relevance methods, operation bindings carrying authority, attestation, trust, time, and dependency identities. It is tempting to regard such a structure, once fully populated and internally consistent, as self-certifying—as if a sufficiently detailed account of where a value came from amounted to evidence that the account is true.

Provenance modeling practice itself rejects the temptation. The W3C PROV model [**22**], which the candidate's lineage design consciously parallels, is explicit that a provenance record is an *assertion* about derivation, attributable to whoever asserted it, and that consuming

applications must decide separately how much trust to place in the asserter. PROV gives a vocabulary for saying "entity E was derived from entity F by activity A under agent G"; it deliberately does not give, and cannot give, a procedure for concluding that the saying is so. Structure constrains what can be coherently asserted; it cannot compel the world to have cooperated.

Applied to CAS-Ohm: a research packet's provenance graph, canonically encoded and digest-stable, establishes that a specific, fully articulated *story* about the packet's origin was fixed at a specific byte boundary and has not changed since. Whether the story is true is exactly the question the six eligibility preconditions exist to address—an admitted authority to anchor who may tell stories, protected keys to bind tellers to tellings, trusted time to anchor when, materialized dependencies and authenticated instruments to make the story's physical half checkable, and a reviewed policy to govern the whole. Absent those, the candidate holds the story with perfect fidelity and asserts, about its truth, precisely nothing.

Claim Boundary 11.22 (Chapter claim boundary). This chapter describes tested software behavior of the CAS-Ohm 1.0 candidate at the Gate 4 receipt: the structural completeness predicate, the six fail-closed rejection classes, and the constancy of the authenticated-eligibility field. It establishes no authenticated evidence, no authority, no trusted time, no instrument attestation, no reviewed policy, and no eligibility of any packet, present or future. Statements about what authenticated eligibility *would* require are design analysis, not commitments that the preconditions will be pursued or met. Artifact status OPEN / RING 4 / NOT LICENSED / HELD governs.

CHAPTER 12

Registered Hypotheses and the Preregistered Closure Protocol

The preceding chapters built a type system in which the Ohm analogy can be stated without overstating itself. This chapter presents the part of the program that makes the analogy *rejectable*: five registered, falsifiable hypotheses, held in the shared claim ledger under uniform status, together with the preregistered protocol—the Gate 5 closure path—under which any of them could someday be tested. The chapter is long because each hypothesis deserves a full experimental treatment: not merely its claim text, but the design considerations that shape a fair test, the controls that discriminate its predictions from cheaper explanations, the conditions under which it is refuted, and—most important for claim discipline—the exact boundary of what a success would and would not license.

Three governing facts frame everything below and are repeated deliberately at the outset. First, all five hypotheses carry the status IN REVIEW / RING 4 / NOT LICENSED / HELD; none is established, none is promoted, and none has an admitted Evidence Packet. Second, the ledger additionally records one *product* claim, `CLM-CAS-OHM-PHYSICAL-TELEMETRY`, as NOT LICENSED—and, more strongly, as contradicted by the current evidence; Section 8 treats it separately because a prohibited claim is a different object from an untested one. Third, no experiment is admitted in this monograph. The protocol of Section 9 is a specification of how closure would have to proceed, not a report that it has begun.

1. The claim ledger and the status discipline

Each hypothesis is registered in the shared claim ledger under a stable claim identifier of the form `CLM-CAS-OHM-*`. Registration fixes the claim text, its status vector, and its relationship to the Gate 5 preregistration kernel; the ledger artifact and its digest are bound at the Gate 4 source boundary (Chapter 14). The status vector has four coordinates, and their joint meaning bears spelling out once so that the per-hypothesis sections can use the labels without ceremony.

IN REVIEW states that the claim's own text and framing are still subject to revision by internal cross-review; even the *wording* of the hypotheses is not final. RING 4 places the claims in the outermost admission ring: research material, structurally tracked, with no admission into any inner ring of reviewed or relied-upon results. NOT LICENSED states that no downstream artifact—product, marketing, incident analysis, or derived paper—may present the claim as established or rely on its truth. HELD states that the claims are not disclosed beyond the project boundary. The four coordinates are independent; the discipline of Chapter 11 applies here in miniature, since no accumulation of internal polish moves a claim along any of the four axes.

REMARK 12.1 (Hypotheses are not ordered by confidence). The five hypotheses are presented in dependency order—local response before resistance, resistance before composition, composition before redundancy and localization—because later ones presuppose machinery

whose meaningfulness earlier ones test. The order encodes no confidence ranking. The project records no probability assessments, and the ledger's uniform status makes any such ranking unrepresentable in the artifacts.

REMARK 12.2 (Ledger text governs). The hypothesis statements reproduced in this chapter are faithful restatements for exposition; the registered claim text in the ledger, under its digest, is the governing object. Where this chapter's prose and the ledger's frozen wording could be read differently, the ledger wording controls—the same rule by which the glossary defers to chapters and the monograph defers to its source artifacts. This matters operationally because the Gate 5 freeze (Section 9.1) freezes the *ledger* text, and a future run's permitted claim language will be derived from it, not from any secondary description.

2. CLM-CAS-OHM-LOCAL-RESPONSE: held-out local response to randomized alpha

REGISTERED HYPOTHESIS 12.3 (CLM-CAS-OHM-LOCAL-RESPONSE). Within one calibration profile, the held-out local response of the directional activation projection to randomized steering alpha exceeds the response observed under matched controls.

This is the foundation hypothesis: it asserts only that the dose coordinate does *something* to the response coordinate, in the profile's active region, beyond what null and mismatched interventions do. Every other hypothesis presupposes it, since a resistance, a gain, or a residual over a nonexistent response relation would be arithmetic on noise.

2.1. Design considerations. The operative words are "within one calibration profile," "held-out," and "randomized," and each excludes a familiar failure. Profile confinement excludes the cross-model and cross-test-point aggregation that the sources' vocabulary invites but the audit forbids: a response demonstrated by pooling test points with different directions and full-scale methods would inherit the mixed-scale conflict rather than escape it. The held-out requirement excludes the circularity rejected in Chapter 6: any curve fit on the same samples used to demonstrate it demonstrates only interpolation. Randomization of the alpha schedule excludes confounding between dose level and drift, position in the sweep, or corpus order; a monotone sweep would confound dose with time, and a model drifting over the session would masquerade as dose-responsive.

The design must also fix, in advance, what "local response" is measured as—the fit family, the estimator, and the minimum effect size—because a post-hoc choice among estimators is a multiple-comparisons engine. These are precisely the items the Gate 5 freeze (Section 9.1) locks before any run.

A further consideration is dose-range placement. The hypothesis concerns *local* response, and the classification machinery of Chapter 7 partitions the dose axis into regions; a randomized grid that spends most of its budget in cutoff or saturation measures the flat shoulders of the curve and starves the active region of samples. The frozen alpha grid must therefore be designed against the profile's declared region rules, and—because region boundaries are themselves estimated—the design must state in advance how sessions whose active region turns out narrower than planned are handled: as censored (with reasons, per Chapter 11) or as failures of the profile's region declaration. Either choice is defensible; an after-the-fact choice is a forking path.

2.2. Controls. The applicable controls from the protocol's kit are the zero-intervention control (alpha grid replaced by zeros, establishing the no-dose baseline and its variance); the random same-norm direction control (dose applied along a random unit vector, establishing

how much apparent response any norm-matched perturbation produces); the wrong-layer and wrong-token controls (dose applied at a mismatched site, establishing site specificity); and the norm-matched control family generally, which ensures that no comparison confounds direction with magnitude. The hypothesis text's phrase "matched controls" binds the comparison to these, under the relevance-method bindings of Chapter 11.

2.3. Falsification conditions. The hypothesis is refuted within a profile if the prespecified held-out response statistic fails to exceed the control envelope at the preregistered effect size and confidence—or if it exceeds it only under analysis choices not frozen in advance, which the protocol counts as failure, not as near-success. Refutation is recorded without erasure (Section 9.4); a profile in which the model shows no local response is a result, and under the ledger's rules it is a result that contradicts the foundation of the other four hypotheses in that profile.

2.4. What success would and would not license. Success would license exactly this sentence: in the tested profile, randomized alpha produced held-out-predictable response above matched controls. It would not license: any statement about other profiles, models, layers, or directions; any causal reading beyond the intervention actually randomized; any claim that the response is linear, ohmic, or stable over time; any product language; and no movement of the claim's status vector, which changes only through the ledger's own review process. Success here is a load-bearing floor for the hypotheses below, and nothing more.

3. CLM-CAS-OHM-EFFECTIVE-RESISTANCE: held-out prediction and inverse-slope constraint response

Registered Hypothesis 12.4 (CLM-CAS-OHM-EFFECTIVE-RESISTANCE). A calibrated effective resistance r_{eff} predicts held-out response within the profile's stated tolerance, and an independent manipulation of a non-safety constraint moves the observed response with the inverse slope that the resistance model predicts.

This hypothesis gives the model its first genuinely Ohm-shaped content. It has two conjuncts, and the conjunction is essential. The first conjunct—held-out prediction—establishes that r_{eff}, fit on the calibration split with the declared method and uncertainty, transfers to unseen samples: $\hat{y} = a/r_{\text{eff}}$ within tolerance. The second conjunct—constraint manipulation—establishes that r_{eff} is not merely a fitted slope but tracks something manipulable: increasing the constraint should decrease response as $1/r$, with the specific inverse-slope signature the model predicts, under an intervention chosen independently of the fitting procedure.

3.1. Design considerations. The first conjunct inherits the full split discipline of Chapter 5: registered calibration split, disjoint held-out split, declared fit method, represented parameter uncertainty and model discrepancy. Predicting within tolerance must mean within *propagated* tolerance—a prediction interval, not a point—or the test quietly rewards underclaimed uncertainty.

The second conjunct carries two deliberate restrictions. The manipulated constraint must be *independent*: identified and varied by a procedure that never saw the calibration data, so that the inverse-slope prediction is a genuine out-of-sample commitment rather than a re-description of the fit. And it must be *non-safety*: the design forbids degrading safety-relevant constraints as an experimental convenience. The restriction is both an ethics rule and a claim-discipline rule—a safety constraint manipulated for curve-fitting would entangle the hypothesis with exactly the safety-verdict vocabulary the forbidden-wording list excludes.

The inverse-slope form deserves emphasis because it is the falsifiable heart. Many models predict that more constraint yields less response; monotone decrease is cheap. The resistance model predicts the *shape* of the decrease. A test that checked only the sign would not test this hypothesis.

The dimensional discipline of Chapter 3 also constrains the second conjunct's bookkeeping. The manipulated constraint arrives as a constraint score (Q10 or Q100, per the preserved scale conflict), and a score is not an effective resistance; the design must therefore register, in advance, the bridge by which constraint-level changes are mapped into predicted resistance changes—a fitted `RESISTANCE_DELTA` per constraint step, carried in ALPHA PER ACTIVATION with uncertainty, never a raw score difference. Without the registered bridge, the inverse-slope comparison would silently commit the dimensional-bridge error (Conflict 3) that the type system exists to block, and the candidate's validators would in fact refuse the comparison records.

3.2. Controls. Both conjuncts use the standard kit—zero-intervention, random-direction, wrong-layer, wrong-token, norm-matched—and the second adds constraint-side controls: a sham manipulation (the manipulation procedure executed with null effect) to expose response drift correlated with the act of intervening, and a label-shuffle over constraint levels to establish the no-structure baseline for the slope estimate.

3.3. Falsification conditions. The hypothesis fails if either conjunct fails: held-out prediction outside tolerance refutes the transfer claim; a constraint response that is absent, non-inverse, or inconsistent in slope with the fitted r_{eff} beyond the preregistered comparison rule refutes the manipulability claim. A success on the first conjunct with failure on the second is recorded as overall failure with the partial result retained—the model would then be a predictor but not a resistance, and the ledger language would have to be revised accordingly rather than the result rounded up.

3.4. What success would and would not license. Success would license: in the tested profile, r_{eff} functioned as a calibrated, manipulable inverse-slope parameter. It would not license: the word "resistance" in any physical sense; transfer of r_{eff} across profiles; any statement about *why* the constraint moves the response; any safety claim, since the manipulated constraint is by design non-safety; and no promotion of the network rules of Chapter 9, which depend on the separate composition hypotheses below.

4. CLM-CAS-OHM-GAIN-COMPOSITION: local-to-end-to-end derivative composition

REGISTERED HYPOTHESIS 12.5 (`CLM-CAS-OHM-GAIN-COMPOSITION`). In the active region, locally measured derivative gains along a declared chain compose multiplicatively to the independently measured end-to-end gain within the stated tolerance.

The sources present cascade gain multiplication as a working rule; the audit demoted it to a preregistered prediction (Conflict 6 of Chapter 4), and this hypothesis is that prediction. Its content is a commutation claim: measure each stage's local Jacobian/projection derivative separately, multiply, and compare against the end-to-end derivative measured directly across the whole chain. The chain rule of calculus guarantees this identity for exact derivatives of a smooth composite map evaluated at matched points; the hypothesis is empirical because none of those idealizations holds automatically—the measured gains are finite-difference or fitted estimates, at test points that may not compose, in a system that need not be smooth or stable across the chain.

4.1. Design considerations. Three design constraints do the work. *Coordinate agreement*: adjacent stages must share their interface coordinate exactly—the output projection of stage k must be the input coordinate of stage $k{+}1$, per the capability gate of Chapter 9—or the product is dimensionally well-formed but semantically undefined. *Active-region confinement*: every stage must be classified ACTIVE at the operating point; a saturated intermediate stage clips the composition and the audit's warning that flow is local becomes quantitative. *Matched operating points*: local gains must be measured at the operating point the composite trajectory actually visits, not at each stage's convenient default. The gain-definition conflict (Conflict 4) additionally constrains vocabulary: the gains here are local Jacobian/projection derivatives; Fisher–Rao displacement gain remains a separately named profile, and a success of this hypothesis would *not* bridge the two.

4.2. Controls. The composition claim is tested against a shuffled-chain control (stage gains multiplied in an order or pairing that does not correspond to the declared topology, establishing how often near-agreement arises by accident); wrong-layer stage substitutions (one stage's gain replaced by an off-chain measurement); and the norm-matched random-direction control applied per stage. The end-to-end measurement itself carries the standard intervention controls of Section 2.

4.3. Falsification conditions. The hypothesis fails if the product of local gains disagrees with the end-to-end gain beyond the preregistered tolerance under the frozen comparison rule, in the active region, with coordinates verified. Failure has a specific documented consequence for the candidate: the gain-product capability of Chapter 9 remains structural-only *permanently* in that profile family—the rule "must be dropped if the tests fail" is inherited from the addendum's own falsification rows and is recorded in the ledger, so a failed composition test cannot be quietly survived by the network code that assumed it.

4.4. What success would and would not license. Success would license: within the tested profile and chain, multiplicative composition of local derivative gains within tolerance—a statement about the composability of measurements. It would not license: composition across profiles or untested chains; any identification of Jacobian gain with Fisher–Rao gain; the unguarded use of gain products in downstream scoring; and no causal reading of the chain, since a derivative along a declared path is not evidence that the path is the mechanism.

5. CLM-CAS-OHM-REDUNDANCY: genuine-branch ablation versus matched random-head controls

REGISTERED HYPOTHESIS 12.6 (CLM-CAS-OHM-REDUNDANCY). Ablation of genuine alternative branches produces the specific quantitative response change predicted by reciprocal parallel aggregation, above the change produced by ablating matched random-head controls.

The parallel rule of Chapter 9—conductances add, so resistances aggregate reciprocally—is the sources' second network rule and the audit's second demotion. Its empirical content is a redundancy claim: if two branches are genuinely alternative routes between the same endpoints, then removing one should change the effective resistance by the specific amount reciprocal arithmetic predicts, not merely by *some* amount. The hypothesis therefore has the same shape as the resistance hypothesis: a cheap qualitative prediction (ablation changes something) is sharpened into an expensive quantitative one (ablation changes it by *this much*).

5.1. Design considerations. The design must first establish that the branches are "genuine alternative directed branches with the same endpoints" in the sense of the parallel capability gate—a topological and coordinate condition checked before any ablation, because the audit's sharpest warning here is that co-present guardrails are not automatically parallel routes. Two mechanisms that both suppress a behavior need not be alternative paths for the same flow; they may be sequential, interacting, or entirely disjoint in coordinate. Running the ablation test on non-branches would produce an uninterpretable number, not a refutation.

Second, the ablation itself must be matched: the control ablates randomly selected heads (or units) matched in count, layer distribution, and—where feasible—norm contribution, so that the comparison isolates *branchhood* rather than the generic fragility of the network to any lesion of that size. The audit's covariance limitation applies with full force: reciprocal aggregation assumes independence that real branches need not have, and the declared limitation (cross-input covariance unavailable; zero-covariance approximation) is carried on every aggregate record precisely so that this hypothesis, if tested, tests the approximation too.

5.2. Controls. Matched random-head ablations, as above; sham ablation (the ablation machinery engaged with null mask) to expose harness-induced drift; single-branch versus multi-branch ablation series to check that predicted changes compose across successive removals; and the standard intervention controls on the response measurement itself.

5.3. Falsification conditions. The hypothesis fails if the post-ablation response change is statistically indistinguishable from matched-random-control changes, or distinguishable but quantitatively inconsistent with the reciprocal prediction beyond tolerance. As with gain composition, failure permanently confines the parallel capability to structural-only status in that profile family, and the ledger records the outcome without erasure.

5.4. What success would and would not license. Success would license: in the tested profile and topology, reciprocal aggregation predicted the ablation response of the identified branches. It would not license: the inference that redundancy found here is safety redundancy; any claim about branches not tested; the general validity of parallel arithmetic over co-present mechanisms; and no interpretation of the branches as "the" implementation of the behavior, since ablation identifies contribution under lesion, not unique mechanism.

6. CLM-CAS-OHM-RESIDUAL-LOCALIZATION: blinded implanted faults above shuffled baselines

Registered Hypothesis 12.7 (CLM-CAS-OHM-RESIDUAL-LOCALIZATION). Blinded implanted faults are localized by the residual diagnostics of the candidate at a rate above shuffled and matched-control baselines. On failure, all causal language attached to residual diagnostics is withdrawn.

Chapter 10 established that a residual is a non-identifying diagnostic: it can rank follow-up probes and cannot, by sign alone, identify a hidden policy, bypass, or cause. This hypothesis is the one channel through which residuals could earn a stronger role, and its design is correspondingly the most defensive of the five. The test is an implanted-fault study: known faults—of the kinds the synthetic boards of Chapter 14 model in software—are introduced into a live profile by an implanting party, and a diagnosing party, blind to the implantation schedule, uses residual diagnostics to localize them.

6.1. Design considerations. Blinding is the load-bearing feature. The implanting and diagnosing roles must be separated by an information barrier fixed in the preregistration: fault type, site, magnitude, and schedule are drawn from a frozen distribution by the implanter and revealed to the diagnoser only after localization reports are committed. Without blinding, residual localization is unfalsifiable in practice, because the diagnostician's priors about where faults "should" be leak into threshold and probe choices. The fault distribution must include *no-fault* sessions at a frozen rate, so that the false-positive behavior of the diagnostics is measured, not assumed; a localizer that finds faults everywhere is worse than none, since Chapter 10's whole warning is against residual-driven overinterpretation.

The censoring rules of Chapter 10 apply inside the test: faults implanted at operating points with zero-compatible predicted response are expected to be non-localizable, and the preregistration must decide in advance whether such sessions count against the localizer or are excluded as censored—either is defensible; choosing after unblinding is not.

6.2. Controls. Shuffled baselines (localization reports scored against a permuted fault schedule, establishing chance agreement); matched-control diagnostics (the same reports scored on no-fault sessions); and a strength gradient in the implanted faults, so that the result characterizes a detection curve rather than a single operating point.

6.3. Falsification conditions and the withdrawal clause. The hypothesis fails if blinded localization does not exceed the shuffled and matched baselines at the preregistered rate. Uniquely among the five, this hypothesis carries an explicit *withdrawal clause*, inherited from the audit's residual conflict: on failure, causal language is withdrawn wherever residual diagnostics appear—in the candidate's documentation, in the diagnostic output vocabulary, and in any derived artifact. The clause exists because residuals are the point of maximum drift risk in the entire model: the temptation to read a signed residual as "hidden resistance" or "bypass" is documented in the sources themselves, and a failed localization test converts that temptation from a discipline problem into a refuted claim.

6.4. What success would and would not license. Success would license: in the tested profile, residual diagnostics localized implanted faults of the tested kinds above chance, blind. It would not license: localization claims for fault kinds outside the implanted distribution; any assertion that a residual in the wild indicates a fault (base rates differ); the phrase "proves hidden resistance," which remains on the forbidden-wording list regardless of outcome; and no safety-monitoring role, which would require the separate authority and review machinery of Chapter 11 in addition to statistical success here.

7. Dependency structure and joint outcomes

The five hypotheses are not independent bets, and the program's honesty depends on saying in advance what the possible joint outcomes would mean. The dependency structure is a partial order. `LOCAL-RESPONSE` is the root: if a profile shows no held-out response above controls, then a resistance fitted in that profile is a parameter of nothing, a gain measured there is a derivative of noise, an ablation study has no response variable, and residuals have no registered prediction to deviate from. `EFFECTIVE-RESISTANCE` depends on the root and is depended on, in turn, by the two network hypotheses: `GAIN-COMPOSITION` requires per-stage calibration of the kind the resistance hypothesis tests, and `REDUNDANCY` requires that per-branch resistances be meaningful before their reciprocal aggregate can be. `RESIDUAL-LOCALIZATION` depends on the root and on resistance (a residual is defined against a calibrated prediction) but not on the network pair.

Four joint-outcome regions are worth fixing in advance. *Total failure*: if the root fails in every tested profile, the entire empirical program in those profiles terminates, and the candidate's honest description contracts to what Chapter 14 calls its strongest present achievement—a type system and audit discipline—with the response model marked refuted-as-tested. The ledger records this outcome like any other; the type system's value does not depend on it. *Foundation without law*: the root passes but `EFFECTIVE-RESISTANCE` fails—response exists but is not stably inverse-parameterized. The permitted vocabulary would then include local response and exclude the entire resistance family, and the network capabilities would be permanently confined in the affected profile family. *Law without networks*: resistance passes but one or both network hypotheses fail. The prediction contract survives at single test points; series, parallel, or gain-product arithmetic is dropped per the addendum's own falsification rows; and the audit's warning that flow is local becomes the established result rather than the cautious default. *Asymmetric network outcomes*: composition may pass while redundancy fails, or conversely—the two test different structural commitments (chain smoothness versus branch independence), and neither outcome licenses the other's rule.

The order also disciplines scheduling. A Gate 5 run plan that tested a downstream hypothesis before its prerequisites would spend its frozen design on a question whose meaningfulness was still open; the protocol does not forbid this, but the frozen comparison rules must then state the conditional structure explicitly—a composition result obtained before the resistance result is admissible only under the explicit registered assumption that the prerequisite would pass, and it is voided, not reinterpreted, if the prerequisite later fails.

8. `CLM-CAS-OHM-PHYSICAL-TELEMETRY`: a prohibited product claim

The sixth ledger entry is not a hypothesis and must not be grouped with the five. `CLM-CAS-OHM-PHYSICAL-TELEMETRY` is a *product* claim—the claim that CAS-Ohm quantities constitute physical telemetry of the audited system—registered in the ledger for the specific purpose of being marked NOT LICENSED.

Its evidentiary standing is worse than untested: it is *contradicted by current evidence*. The candidate's own ontology establishes the contradiction internally. Design values are declared, not measured (Chapter 3); behavioral proxies are inferences from external behavior; model-derived values inherit their parents' epistemic type under the kind non-fungibility rule; and the unit ontology contains no physical SI units at all, with volt, ampere, and ohm registered as forbidden aliases. A blanket representation of any of these as physical telemetry would therefore assert a measurement-kind and unit typing that every validator in the candidate rejects. The claim is registered rather than merely omitted because omission is silent and silence drifts: a future artifact could innocently reinvent the claim. A registered NOT LICENSED entry gives reviewers a stable identifier to cite when striking the language, and it gives the forbidden-wording list of the audit—"physical telemetry," "measures attention," "electrical law," "causal by construction," "cross-model comparable," "proves hidden resistance"—an anchor in the claim ledger itself.

Claim Boundary 12.8 (No rehabilitation path inside CAS-Ohm 1.0). No outcome of the five registered hypotheses, singly or jointly, would license `CLM-CAS-OHM-PHYSICAL-TELEMETRY`. The hypotheses concern model-internal response structure under the nonphysical unit ontology; physical telemetry would require the separate, explicitly typed and instrumented physical profile that Chapter 3 names and the candidate does not contain. The claim could only be revisited under a different candidate with different ontology, instrumentation, and review.

9. The Gate-5 preregistered closure protocol

No experiment is admitted in this monograph. The path by which any of the five hypotheses could reach closure is fixed in advance: a future run must use the Gate 5 per-claim preregistration and Evidence-Packet v2 control kernel, and Gate 5's own status is preserved until its separate gates close—the protocol governing experiments is itself an artifact under review, and using it does not promote it.

Protocol 12.9 (Gate-5 closure path). At minimum, a closing run must: (1) freeze the exact claim text, model, checkpoint, tokenizer, and weights, code and environment, test point and direction, corpus, splits, and seeds, alpha grid, fit and uncertainty methods, controls, falsifiers, and permitted claim language; (2) keep calibration and validation splits disjoint; (3) include zero-intervention, random same-norm direction, wrong-layer, wrong-token, label-shuffle, and norm-matched controls as applicable; (4) record cutoff, saturation, instability, drift, missingness, exclusions, and negative outcomes without erasure; (5) require independent numerical recomputation and authenticated execution evidence; (6) replicate on a held-out checkpoint or model under a prespecified comparison rule; (7) retain the full raw evidence and amendment/outcome-access chain; and (8) keep scientific eligibility `NOT_ESTABLISHED` until independent audit, replication, claim review, and external scientific review exist.

The eight steps are now expanded in turn.

9.1. Step one: the freeze. The freeze is the protocol's center of gravity, and its item list is long because every unfrozen item is a degree of freedom that can be spent, after the data arrive, to manufacture success. Freezing the *claim text* prevents outcome-directed rewording—the quiet migration from "predicts within tolerance" to "correlates significantly." Freezing *model, checkpoint, tokenizer, and weights* pins the system under test to exact digests, so that a favorable result cannot be sourced from an unrecorded checkpoint sweep. Freezing *code and environment* closes the analytic garden of forking paths at the implementation level; freezing *test point and direction* closes it at the measurement level, where the choice among layers, positions, and direction-derivation methods is otherwise a silent multiple-comparisons engine. Freezing *corpus, splits, and seeds* makes the randomization real and the split discipline auditable. Freezing the *alpha grid* prevents post-hoc restriction to a flattering dose range; freezing *fit and uncertainty methods* prevents estimator shopping; freezing *controls and falsifiers* converts the control kit from a menu into an obligation; and freezing the *permitted claim language* decides, before the outcome exists, exactly which sentences a success would license—the per-hypothesis "would and would not license" paragraphs above are drafts of precisely this frozen language.

9.2. Step two: disjoint splits. Calibration and validation splits must be disjoint—the anti-circularity rule of Chapter 6 elevated from a type constraint to a protocol obligation. The protocol adds what the type system cannot check: disjointness must hold at the level of *information*, not merely record identity. Near-duplicate prompts, shared templates, or leakage through the direction-derivation corpus can make formally disjoint splits informationally entangled; the frozen corpus manifest and split procedure are required to state the deduplication and leakage rules by which disjointness is claimed.

9.3. Step three: the six control types. The control kit has six members, each excluding a named alternative explanation. *Zero-intervention*: the response exists without dosing; establishes baseline and drift. *Random same-norm direction*: any perturbation of that magnitude does this; establishes direction specificity. *Wrong-layer*: dosing anywhere

does this; establishes site specificity in depth. *Wrong-token*: dosing any position does this; establishes site specificity in position. *Label-shuffle*: the analysis finds structure in noise; establishes the no-structure baseline for fitted statistics. *Norm-matched*: the effect is magnitude, not content; establishes that comparisons never confound the two. The phrase "as applicable" in the protocol is load-bearing and narrow: a control is inapplicable only when its manipulated factor does not exist in the design (a single-layer probe has no wrong-layer variant), never because it is expected to be inconvenient.

Two structural facts about the kit deserve note. First, the controls are typed evidence like everything else: each control run produces records under the same schemas, bound to the packet through the control bindings of Chapter 11 with an explicit relevance method, so that an auditor can verify not merely that controls existed but that the claimed control was the claimed kind. Second, the kit composes: a random same-norm direction at the wrong layer is a meaningful joint control when the design's inferential target requires it, and the freeze must enumerate which compositions are in scope, because an unenumerated control added after unblinding is analytically indistinguishable from a fishing expedition even when innocently intended.

9.4. Step four: non-erasure of negative outcomes. Cutoff, saturation, instability, drift, missingness, exclusions, and negative outcomes are recorded without erasure. This step is where the protocol meets the file-drawer problem: a preregistered design that silently discards its failed sessions is worse than no preregistration, because the registration lends unearned credibility to the surviving subset. The candidate's type system already gives every excluded observation a censoring reason (Chapter 11); the protocol extends the obligation to the run level—an aborted session, a drifted profile, or a refuted conjunct is an artifact with a digest, bound into the packet, visible to the auditor. The per-hypothesis falsification sections above all terminate in this step: refutation is an outcome to be retained, not an absence to be managed.

9.5. Step five: independent recomputation and authenticated execution. Every reported statistic must be independently recomputed—by a party other than the run's operator, from the retained raw evidence, using independently maintained analysis code—and the run must carry authenticated execution evidence in the sense of Chapter 11: attestation that the frozen code actually executed against the frozen model in the declared environment. The two halves close different gaps: recomputation catches analysis error and analysis fraud; execution authentication catches the deeper substitution in which the analysis is honest but its inputs were never produced by the claimed run. Step five is where the protocol's demands exceed the current candidate's capabilities *by design*—authenticated execution evidence requires the instrument-attestation and authority machinery whose absence Chapter 11 documents, which is one of the concrete senses in which no experiment can currently be admitted.

9.6. Step six: held-out replication. A result must replicate on a held-out checkpoint or model under a comparison rule specified before the replication. The prespecification matters as much as the replication: without a frozen comparison rule, any pair of results can be narrated as agreement (both positive) or as expected variation (one null). The rule must state, in advance, which statistic must agree, within what tolerance, and what verdict follows from each cell of the agreement table. Replication on a held-out *checkpoint* tests stability under training progression; replication on a held-out *model* is a strictly stronger and separately stated claim—the protocol permits either, but the frozen language of step

one must say which was committed to, since profile-relativity (Chapter 5) makes cross-model statements expensive. Replication also interacts with the freeze: the replication profile is a *different* calibration profile with its own digest, and the comparison rule is the stated bridge between the two profiles that Chapter 5 requires for any cross-profile statement. A replication reported without its frozen bridge would be exactly the unlicensed cross-profile transfer the type system rejects; the protocol makes the bridge a preregistered object so that the replication claim is well-typed before it is well-evidenced.

9.7. Step seven: raw-evidence retention and amendment chains. The full raw evidence—not summaries, not statistics, the bytes—is retained, together with the amendment and outcome-access chain: a record of every change to the preregistration after freezing (each amendment itself dated, justified, and hashed) and of every access to outcome data (who saw what, when). The amendment chain preserves the distinction between a protocol deviation confessed in advance and one reconstructed afterward; the outcome-access chain makes blinding claims (Section 6) auditable rather than asserted. Retention obligations are packet obligations in the sense of Chapter 11: raw bytes with digests, resolvable indefinitely.

9.8. Step eight: eligibility remains `NOT_ESTABLISHED`. Even a run that executes steps one through seven flawlessly does not, by itself, establish anything. Scientific eligibility remains `NOT_ESTABLISHED` until four independent events occur: independent audit of the run's evidence, independent replication, claim review through the ledger's own process, and external scientific review. Step eight is the protocol-level image of Chapter 11's two-axis design: steps one through seven build a packet that is *complete*; step eight states that completeness is not authority, and that the authority events are performed by parties the protocol cannot conscript. The gate closes from outside or not at all.

10. Preregistration in experimental practice

The protocol's shape is not novel, and its familiarity is a feature. Prospective registration of trials—fixing hypotheses, endpoints, and analyses before data collection—became standard in clinical research precisely because retrospective flexibility had been shown, repeatedly and quantitatively, to inflate apparent effects: outcome switching, selective reporting, and post-hoc subgroup discovery all survive peer review when the reviewer cannot see what was planned. The registered-report format in psychology and neuroscience extends the same logic to publication itself, with journals accepting or rejecting the frozen design before results exist, so that the incentive to narrate whatever occurred as confirmation is removed at the source. Methodological work on researcher degrees of freedom—the garden of forking paths, in the phrase that has become standard—establishes that a analyst making reasonable-seeming choices *after* seeing data can produce significance from noise without any intent to deceive, which is why the Gate 5 freeze enumerates its items so exhaustively: every item on the list is a documented forking point somewhere in the experimental literature.

Three features of Gate 5 are stricter than common preregistration practice, and deliberately so. First, the freeze includes *permitted claim language*, not just analyses—the protocol registers the sentences, because the project's claim discipline treats wording drift as a first-class failure mode. Second, negative-outcome retention is total and typed, with censoring reasons and digests, rather than a reporting norm. Third, step eight decouples even a perfect run from any status change: where a registered report culminates in publication, a Gate 5 run culminates in a packet whose eligibility is still `NOT_ESTABLISHED`—the protocol has no success state that bypasses independent audit, replication, and review. These references to

practice are contextual; the protocol stands on its own text, and no external methodology is incorporated by them.

Claim Boundary 12.10 (Chapter claim boundary). This chapter registers hypotheses and specifies a protocol. It reports no experiment, no data, no partial run, and no pilot. All five hypotheses remain IN REVIEW / RING 4 / NOT LICENSED / HELD with no admitted Evidence Packet; `CLM-CAS-OHM-PHYSICAL-TELEMETRY` remains NOT LICENSED and contradicted by current evidence; the Gate 5 kernel remains under its own open status, and its use in a future run would promote neither the run's claims nor the kernel itself. Artifact status OPEN / RING 4 / NOT LICENSED / HELD governs.

CHAPTER 13

The Memristive Extension Boundary

Every chapter to this point has treated the effective resistance r_{eff} as a fixed parameter of a calibration profile: fitted once on the calibration split, carried with uncertainty, and used—under guard—to predict held-out response. Real systems under repeated intervention rarely oblige such stationarity. A steered model revisited across a long session, a fine-tuned checkpoint drifting under continued training, an activation channel whose sensitivity depends on what was recently driven through it—all of these suggest that the response parameter might itself be a function of history. Classical circuit theory encountered the same structural question half a century ago and produced a precise mathematical answer: the memristor and, more generally, memristive systems [35, 36]. This chapter presents that theory as *motivating mathematics only*; states the provisional, explicitly nonphysical extension that CAS-Ohm 1.0 reserves for history-dependent response; specifies the model-state record that any such extension must carry; erects the typing separations that keep the extension from collapsing into either a memoryless fit or an electrical claim; and fixes the evidentiary bar that the word "memristive" must clear before it is anything more than an analogy. The chapter's conclusion is its title: this is a *boundary* chapter. Nothing stateful, material, or physical is admitted in the current candidate, and the extension described here is a typed placeholder for a hypothesis, not a feature.

1. Classical memristor theory as motivating mathematics

1.1. The missing-element argument. Chua's 1971 argument [35] is an argument from symmetry, and its form matters more to CAS-Ohm than its electrical content. Circuit theory before 1971 recognized four fundamental variables—voltage v, current i, charge q, and flux linkage φ—and three fundamental two-terminal elements, each defined by a constitutive relation between a pair of them: the resistor relating voltage and current, the capacitor relating charge and voltage, and the inductor relating flux and current. Two further pairings are definitional rather than elemental: charge is the time integral of current, and flux linkage is the time integral of voltage. Laying out the six pairings, Chua observed that exactly one relation had no element: the relation between charge and flux,

$$\varphi = f(q) \quad \text{or} \quad q = g(\varphi).$$

He therefore postulated, on grounds of completeness, a fourth element—the *memristor*, for memory resistor—defined by such a constitutive relation. Differentiating the charge-flux relation with respect to time yields

$$v(t) = M(q(t))\, i(t), \qquad M(q) = \frac{d\varphi}{dq},$$

so the memristor behaves at each instant like a resistor whose resistance $M(q)$ depends on the entire history of current through it, since $q(t) = \int_{-\infty}^{t} i(\tau)\, d\tau$. The element is not a resistor

with noise or drift; it is a different element, with a state variable (charge) that integrates its own port history.

The argument's relevance here is methodological. Chua did not begin from an observed device; he began from a *table of relations* and asked which cell was unoccupied. CAS-Ohm's quantity ontology (Chapter 3) is likewise a table—quantity kinds crossed with unit kinds under an admissibility map—and the question this chapter entertains is likewise a table question: the candidate occupies the cell "response parameter, constant within profile," and the memristive literature demonstrates that the adjacent cell, "response parameter, function of integrated history," is mathematically coherent and physically realized in at least one domain. Coherence and realization elsewhere are the full extent of the motivation. A cell being well-defined does not put anything in it.

1.2. Memristive systems: state-dependent constitutive relations. The 1976 generalization by Chua and Kang [**36**] widened the class from the ideal charge-flux element to *memristive systems*: input-output systems of the form

$$\dot{x} = f(x, u, t), \qquad y = g(x, u, t)\, u,$$

where u is the input (current or voltage), y the output (voltage or current), and x an internal state vector evolving under its own dynamics. Two features define the class. First, the output relation is *state-dependent but instantaneously linear in the input*: at frozen state x, the system is an ordinary linear resistance $g(x, u, t)$; all nonlinearity and all memory live in the state evolution. Second, the output is zero whenever the input is zero—$y = g(\cdot)\, u$ vanishes with u—which distinguishes memristive systems from systems with internal sources: they store history, not energy to be released at the port.

This generalization is the mathematically honest ancestor of the CAS-Ohm extension, more so than the ideal memristor, because it makes no completeness claim and requires no conserved integral quantity. It says only: here is a typed decomposition in which a response coefficient is a function of a declared state, and the state has a declared update law. Everything the extension of Section 2 does is transpose that decomposition into the candidate's nonphysical coordinates.

1.3. Pinched hysteresis as signature. The memristive-systems paper also established the class's observable signature: under any bipolar periodic input, the input-output trajectory of a memristive system traces a *pinched hysteresis loop*—a Lissajous figure through the origin. The loop is pinched because output vanishes with input (the zero-crossing property above), and it is a loop because the state, lagging the input, makes the response on the upstroke differ from the response on the downstroke. As input frequency increases, the state has less time to move within a cycle, the loop collapses toward a single-valued curve, and the system approaches an ordinary resistor.

The signature matters to this chapter for one reason: it is a *behavioral* criterion, statable without reference to the device's material or mechanism. A system is exhibiting memristive phenomenology when its dose-response trajectory is path-dependent in the pinched, frequency-collapsing way—whatever the system is made of. That is precisely the kind of criterion a nonphysical extension needs, and Section 6 adopts its structure (though not its electrical variables) as the evidentiary bar: the CAS-Ohm question is whether alpha-response trajectories show held-out path dependence of this character beyond what any memoryless account reproduces. It must be said with equal clarity that the signature has been the site of

the memristor literature's own overclaim disputes—which physical devices are "really" memristors is a contested question that this candidate has no stake in and imports no position on. The mathematics is borrowed; the controversies, and the devices, are not.

2. The provisional nonphysical extension

DEFINITION 13.1 (Provisional memristive-form extension). A *memristive-form response profile* is a calibration profile (Chapter 5) augmented with an explicit state variable x_t and two profile-bound maps

$$x_{t+1} = F_{\text{profile}}(x_t, a_t, y_t, c_t; \theta), \qquad r_{\text{eff},t} = G_{\text{profile}}(x_t; \theta),$$

where a_t is the alpha dose at step t, y_t the observed activation response, c_t the declared context covariates, and θ the profile parameters. The instantaneous prediction remains $\hat{y}_t = a_t / r_{\text{eff},t}$ under all the guards of Chapter 6, with $r_{\text{eff},t}$ carrying the unit ALPHA PER ACTIVATION at every t. The extension is *provisional* (no implementation is admitted) and *nonphysical* (a, y, r, and x carry model units only; no electrical quantity appears anywhere in the typing).

The transposition from Section 1.2 is exact and deliberately narrow. F_{profile} plays the role of the state evolution f; G_{profile} plays the role of the state-dependent coefficient g; and the instantaneous linearity of the memristive class is preserved in the form of the prediction, which at frozen x_t is the ordinary profile prediction of Chapter 6. What is *not* transposed is as important: there is no analog of charge (no claim that any integral of dose is conserved or physically meaningful), no zero-crossing axiom (whether the modeled response vanishes with dose is an empirical property of a profile, not an assumed one), and no port, device, or energy semantics of any kind.

2.1. The model-state record. A state variable that is not itself typed evidence would reintroduce, at the dynamics level, every bare-number pathology that Chapter 3 eliminated at the value level. The extension therefore requires that x_t be carried by a *model-state record* with seven mandatory components, each of which exists to close a specific hole.

Initialization. The record declares x_0 and the procedure that produced it—a fitted initial state, a declared rest state, or a burn-in protocol with its length and acceptance rule. Without a declared initialization, any trajectory can be retrofitted by choosing x_0 after seeing the data; the initial state is a parameter and must be frozen like one.

Update law. The record binds the exact F_{profile}: functional form, parameter values θ, and implementation digest. The update law is the extension's scientific content; an undocumented update law is an unfalsifiable one.

Sampling interval. The record declares the time base on which t advances—per intervention, per token, per session, or per wall-clock unit—and the rule for events between samples. Memristive phenomenology is frequency-dependent (Section 1.3); a state model without a declared sampling interval cannot even state the frequency-collapse prediction, let alone test it.

Reset rule. The record declares what returns the state to a reference value: session boundary, explicit reset intervention, decay to rest, or nothing. The reset rule is where the extension meets the profile-relativity discipline of Chapter 5—if state persists across sessions, then a session is no longer an exchangeable unit, and the split constructions of the closure protocol (Chapter 12) must partition at a coarser grain. Declaring the reset rule is declaring the exchangeability structure of the data.

Drift model. The record separates modeled state dynamics from unmodeled parameter drift: a declared drift model for θ, with bounds, expiry, and the relationship to the profile's

existing drift and staleness machinery (Chapter 11). Without this separation, slow drift is indistinguishable from slow state evolution, and the extension would absorb every calibration failure as "memory."

Uncertainty. The state estimate carries typed uncertainty, propagated through both F and G into $r_{\text{eff},t}$ and thence into every prediction and residual. A point-estimated hidden state feeding a point-estimated resistance would violate the uncertainty obligations of Chapters 3 and 11 twice over, at one remove from visibility—which is exactly where such violations do the most damage.

Provenance. The state record carries full lineage: the observations that informed each update, ordered derivation parents, and the operation bindings of Chapter 11. A history-dependent parameter whose history is not itself evidence would be a rumor with a time constant.

REMARK 13.2 (The record is the admission price). The seven components are not documentation niceties; they are the conditions under which a stateful extension remains inside the candidate's evidence discipline at all. An implementation that produced $r_{\text{eff},t}$ from an untyped internal buffer would fail the completeness predicate of Chapter 11 (missing provenance) regardless of its predictive merit. In this precise sense the extension is already governed before it exists: the type system was built first so that the dynamics, if ever admitted, arrive pre-constrained.

2.2. The refitting alternative. Before any state model is entertained, an implementation must consider the cheaper alternative: if r_{eff} appears to change over time, refit it—new calibration session, new profile, new digest. Refitting stays entirely inside the existing machinery of Chapter 5 and asserts nothing about dynamics; it treats nonstationarity as a sequence of stationary profiles with declared validity intervals, which is exactly what the drift and expiry fields already support. The memristive-form extension is justified only where refitting demonstrably fails: where the parameter's movement is fast relative to any feasible calibration cadence, systematic in its dependence on the dose history, and *predictable* from that history in the held-out sense of the bar. The order of preference is fixed: stationary profile, then refit schedule, then declared drift model, and only last a state model—each step adding machinery only when the previous step's residual misfit is documented. A state model adopted before the cheaper accounts were exhausted would be complexity acquired without evidence, and its apparent success would be uninterpretable against the unexplored alternatives.

3. Typing separation: neither a memoryless fit nor an electrical memristor

The memristive-form profile must be separated by type from the two things it most nearly resembles, and the separations guard against opposite failure modes.

3.1. Not a memoryless fit. A memristive-form profile with a degenerate update law ($F_{\text{profile}}(x, a, y, c; \theta) = x$ for all inputs) is extensionally a constant-resistance profile, and a sufficiently flexible memoryless model—one whose r_{eff} depends on rich instantaneous covariates c_t—can mimic many trajectories that look history-dependent. The typing therefore refuses to let the two share a kind: a profile either declares state dynamics (and carries the full model-state record, and submits to the evidentiary bar of Section 6) or declares none (and is forbidden the vocabulary of memory, hysteresis, and path dependence in every derived artifact). The separation prevents the cheapest failure: a memoryless fit with extra parameters being *described* as memory because the description is evocative. Under the candidate's

discipline, "the model remembers" is a typed claim requiring a state record, not a figure of speech.

3.2. Not an electrical memristor. In the other direction, the separation from physical memristance is total and inherited from the unit ontology. The state x_t is not charge and integrates nothing physical; $r_{\text{eff},t}$ is in ALPHA PER ACTIVATION, a unit whose physical aliases (volt, ampere, ohm) are forbidden (Chapter 3); no material device, physical circuit, or electrical measurement is referenced by any component of the model-state record. The classical theory of Section 1 stands to this extension exactly as Ohm's law stands to the base model: as the *shape* of a mathematical question, borrowed under explicit typing that prevents the shape from smuggling in the substance. A future physical claim of any kind would require the separate, explicitly typed and instrumented physical profile that the ontology names and the candidate does not contain—and it would additionally inherit the authentication preconditions of Chapter 11, since a physical measurement claim without authenticated instrument evidence would be doubly inadmissible.

RULE 13.3 (Vocabulary gate for the word "memristive"). In candidate artifacts, "memristive" may be used only as (i) a citation to the classical literature, (ii) the name of the provisional extension type of Definition 13.1, or (iii) a registered hypothesis about a specific profile, and in case (iii) only after the evidentiary bar of Section 6 has been met for that profile. Unqualified uses—"the system is memristive," "CAS-Ohm implements a memristor"—are forbidden wording in the same class as the physical-telemetry vocabulary of Chapter 12.

4. Interaction with the existing machinery

A stateful extension would touch nearly every structure built in the earlier chapters, and the touch points are worth enumerating now, because each is a place where an implementation could silently weaken an existing guarantee if the interaction were left implicit.

Profiles and digests. F_{profile}, G_{profile}, θ, and all seven components of the model-state record become profile-bound content: they enter the canonical digest, so two profiles differing only in reset rule or sampling interval are different profiles, non-interchangeable without a stated bridge. This is Chapter 5's discipline extended to dynamics, and it forecloses the tempting shortcut of treating state-model variants as "the same profile with options."

Splits and exchangeability. The split constructions of Chapters 6 and 12 assume that the split's units are exchangeable. Under state, observations within a session are coupled through x_t, so the unit of splitting must be at least the maximal state-coupled span—the interval between resets. Held-out evaluation of a stateful profile therefore means held-out *trajectories*, not held-out samples; a sample-level split of state-coupled data would leak history across the boundary and manufacture exactly the spurious "memory" the bar is designed to reject.

Residuals. The residual discipline of Chapter 10 acquires a second uncertainty source: $\hat{y}_t$ inherits variance from the state estimate as well as from θ. Propagation through both F and G is mandatory (Section 2.1), which widens prediction intervals and thereby *raises* the bar for calling any residual anomalous. This is the intended direction: a stateful model that narrowed its intervals by point-estimating its own hidden state would flag more anomalies while knowing less.

Classification. Under state, the dose-response region of Chapter 7 becomes time-indexed: a profile may be ACTIVE at t_1 and SATURATION at t_2 under the same instantaneous dose

because x has moved. The no-diagnosis rules extend pointwise—no diagnostic from a region-excluded *time*, not merely from a region-excluded profile—and the classification record must bind the state estimate it was computed against.

Networks. The structural operations of Chapter 9 assume component stationarity within the aggregation's scope; a state-dependent component resistance makes every series or parallel aggregate a function of the joint state trajectory. Until an admitted stateful profile exists this is moot, but the gate is stated now: network operations over memristive-form components are undefined in the candidate and would require their own capability, their own evidence, and their own registered hypotheses.

The closure protocol. A Gate 5 run on a stateful hypothesis freezes, in addition to everything in Chapter 12's step one, the state model itself: F, G, θ's fit procedure, initialization, reset schedule, and the path-dependence statistic. The six control types acquire stateful analogs—in particular, schedule-permuted controls (the same doses in a different order), which are the sharpest test of path dependence, since a memoryless model is order-invariant by construction and a stateful one is falsified by order-invariance of the data.

5. Failure modes of stateful modeling

The extension's guards are best understood through the failures they anticipate. Four are documented here as standing threats, so that a future implementation review can check each by name.

THREAT 13.4 (State as overfitting sponge). A state vector with enough dimensions can absorb any residual structure, including noise: the fitted trajectory of x_t becomes a redescription of the training data rather than a model of the system. The bar's matched-parameter-budget clause and held-out-trajectory requirement are the countermeasures; the review question is always whether the memoryless baseline was given a genuinely equal budget and equally rich instantaneous covariates.

THREAT 13.5 (Drift absorption). Slow parameter drift—the phenomenon the profile expiry machinery of Chapter 11 exists to catch—can be fitted as slow state evolution, converting a calibration failure into an apparent memory finding. The model-state record's drift-model component forces the separation: a profile must declare which slow changes are modeled state and which are staleness, and a reviewer must be able to ask whether the "memory" survives when the drift model is given first claim on the variance.

THREAT 13.6 (Retrofitted initialization). With x_0 free, a trajectory's early behavior can be fit after the fact by choosing the initial state that makes the model look prescient. The initialization component (declared x_0 or frozen burn-in protocol) removes the freedom; the review question is whether initialization was fixed before the evaluation data were seen.

THREAT 13.7 (Narrative memory). The subtlest failure is linguistic: describing covariate sensitivity, drift, or session-to-session variation as the system "remembering," without any typed state model at all. Rule 13.3 exists for this case. The candidate's discipline is that memory claims are expensive—they cost a state record, a preregistered statistic, and a beaten baseline—and any memory talk that has not paid is forbidden wording.

EXAMPLE 13.8 (A minimal hypothetical state model). For concreteness only: a one-dimensional fatigue state $x_t \in [0, 1]$ with linear relaxation dynamics

$$x_{t+1} = (1 - \lambda)\, x_t + \mu\, |a_t|, \qquad r_{\mathrm{eff},t} = r_0\, (1 + \kappa x_t),$$

$\theta = (\lambda, \mu, r_0, \kappa)$, models a channel whose resistance rises with recent dosing and relaxes at rate λ. The model-state record would declare $x_0 = 0$ at session start (reset rule: session boundary), the per-intervention sampling interval, a drift model on r_0 distinct from the dynamics, and propagated uncertainty on (x_t, θ). The bar of Section 6 would then demand: on held-out trajectories, this model's predictions of order effects (later doses in a burst meeting higher resistance) beat the best memoryless model given (a_t, c_t) at matched budget, under a frozen comparison. The example is illustrative typing, not a proposal; no such profile is registered.

6. The evidentiary bar

What would it take for a memristive-form profile to be more than notation? The bar is fixed now, in advance of any implementation, precisely so that it cannot be lowered to meet whatever an implementation happens to achieve.

Rule 13.9 (Evidentiary bar for history dependence). A memristive-form profile earns its state variable only by predicting *held-out* hysteresis or path dependence beyond a *matched memoryless baseline*: on trajectories not used in fitting, the stateful model's predictions of path-dependent response (upstroke-downstroke asymmetry, sequence effects, frequency-dependent loop collapse, or another prespecified path-dependence statistic) must exceed the best memoryless model given the same instantaneous information (a_t, c_t), the same parameter budget discipline, and the same uncertainty obligations, under a prespecified comparison rule.

Each clause of the bar closes a specific escape route. *Held-out* closes the route by which a flexible state model memorizes its training trajectories—with enough state dimensions, any finite trajectory set can be reproduced, so in-sample loop-fitting is worth nothing. *Matched memoryless baseline* closes the route by which memory is credited for work that covariates could do: the baseline must receive the same instantaneous inputs, so that only genuine dependence on the *past* separates the models. Matching of parameter budgets closes the route by which the comparison rewards capacity rather than structure. And *prespecified comparison* places the whole test inside the Gate 5 discipline of Chapter 12: frozen statistic, frozen tolerance, frozen verdict table, controls as applicable, negative outcomes retained without erasure. A test of this hypothesis would, like the five registered hypotheses, require the full preregistration kernel—and, like them, would leave scientific eligibility `NOT_ESTABLISHED` pending independent audit, replication, and review.

The pinched-hysteresis signature supplies the test's natural geometry but must be adapted honestly. The classical signature is stated for bipolar periodic inputs; an alpha-dose analog would be a prespecified cyclic dose schedule, with the loop statistic computed on the dose-response trajectory and the frequency-collapse prediction tested across declared sampling intervals (this is why the model-state record's sampling-interval component is mandatory). Where bipolar dosing is not meaningful for a profile, the prespecified path-dependence statistic must be defined on the schedules that are meaningful—and the choice, once made, is frozen.

Adverse Finding 13.10 (Status of the bar in the current candidate). No memristive-form profile exists in the CAS-Ohm 1.0 candidate. No state variable, update law, or stateful implementation is admitted; no hysteresis experiment has been designed beyond the level of this chapter, registered, or run; no calibrated history variable exists; and the evidentiary bar of Rule 13.9 is therefore unmet vacuously—there is no candidate profile for it to judge. The bar's presence in this monograph is a precommitment, not a progress report.

7. The boundary restated

Absent the evidence of Rule 13.9, "memristive" is an analogy and a research hypothesis—in that order, and nothing else. The analogy is legitimate: the memristive-systems decomposition is the correct mathematical form for the question "what if the response parameter has memory," and borrowing a correct form is what Chapter 1 argued the whole CAS-Ohm program does. The hypothesis is registrable: a specific profile, a specific state model, and a specific path-dependence statistic could be frozen under Gate 5 tomorrow without any change to the candidate's type system, which is the practical meaning of the extension being *provisional* rather than absent. But between analogy and admitted result lies everything this monograph has been about: typed state records, held-out prediction, matched baselines, preregistered comparison, retained negatives, independent recomputation, and review by parties the project does not control.

Claim Boundary 13.11 (Chapter claim boundary). This chapter admits nothing stateful, material, or physical. No stateful or memristive implementation, no calibrated history variable, no hysteresis or path-dependence experiment, no material device, no physical circuit, and no safety property is admitted in the current evidence. The classical theory cited [**35, 36**] is motivating mathematics only; its electrical content is not imported, and no position is taken on which physical devices realize it. The extension of Definition 13.1 is a typed proposal whose every use is governed by Rules 13.3 and 13.9. Artifact status OPEN / RING 4 / NOT LICENSED / HELD governs.

CHAPTER 14

Evidence, Limitations, and Conclusion

A monograph that spent thirteen chapters teaching its reader to distrust unearned claims owes its final chapter an exact accounting of what it can itself show. The accounting has three parts. First, the evidence: the bound Gate 4 receipt, stated to the digit and the digest, together with a precise description of what the tested behaviors are and—with equal precision—what kind of evidence software tests are and are not. Second, the limitations: the complete catalog of open boundaries, expanded rather than summarized, because a limitations list that compresses is a limitations list that leaks. Third, the conclusion, which restates the monograph's thesis in its final form: CAS-Ohm 1.0 is viable as a profile-specific experimental response and calibration proposal precisely because its type system makes the Ohm analogy testable and rejectable, and everything the analogy might someday assert remains, today, deliberately unasserted.

1. The Gate-4 receipt

Adverse Finding 14.1 (Bound software evidence at the Gate-4 receipt). At the bound Gate 4 source boundary, the CAS-Ohm suite records exactly: 65 passing test methods, with suite test-ID SHA-256 `2E81A685961286D944894A22EB93345B97211B4433B6C71EAF628D84CD8EB1EF`; 58 passing parameterized subtests in the combined focused receipt; eight retained source conflicts or under-specifications; and zero authenticated Evidence-Packet eligibility paths.

Each of the four numbers is a different kind of fact, and the receipt's value lies in keeping the kinds separate. The 65 test methods are the suite's enumerable surface: named test functions, each encoding an obligation of the candidate specification, whose identifiers are canonically hashed so that the receipt binds *which* tests passed, not merely how many—a suite that silently swapped a hard test for an easy one would change the digest. The 58 parameterized subtests are the combinatorial interior: single obligations exercised across systematically varied inputs (boundary values, enum members, malformed variants), counted in the combined focused receipt. The eight retained conflicts are not test results at all but preserved *negative* knowledge—the source audit of Chapter 4 carried forward unresolved, exactly as the audit rule requires, so that the receipt attests the suite passed *without* the sources having been harmonized. And the zero is the receipt's most important number: the tested absence of any code path by which authenticated Evidence-Packet eligibility could become true (Chapter 11), a negative capability result that turns a design promise into a checked property.

Remark 14.2 (What a receipt is). The receipt is historical software evidence: a statement that, at one bound commit, one suite passed under one runtime. It is not a proof of correctness, not a statement about any other commit, and not evidence about any live model. The monograph cites it as what it is—the strongest fact currently available about the candidate, and a fact strictly about software.

The epistemology deserves one further paragraph, because the monograph's claim discipline depends on readers holding three distinctions simultaneously. First, *tested is not proved*: a passing suite establishes that the implementation behaved as specified on the exercised inputs; it bounds neither the unexercised input space nor the specification's own adequacy. Second, *verification is not validation*: in the classical simulation vocabulary adopted in Chapter 6, the suite verifies that the software implements its stated contracts, and validation—whether the contracts describe anything real—is precisely what the unexecuted hypotheses of Chapter 12 would address. Every sentence in this chapter about "tested behavior" is a verification sentence. Third, *historical is not current*: the receipt binds one commit; the monograph is source-bound to the same boundary, and any statement about later states of the repository would require a later receipt. The digest makes this bounding checkable: a reader holding the repository can recompute the suite identity and confirm that the tests this chapter counts are the tests that ran.

2. Threats to the receipt's validity

A receipt this load-bearing should disclose its own weaknesses. Four are structural and admitted here rather than left for reviewers to discover.

Threat 14.3 (Coverage is enumerable, adequacy is not). The suite exercises the behaviors its authors thought to test. The ten areas of Section 3 were derived from the specification and the audit, but no coverage metric—line, branch, or mutation—is bound into the receipt, and a specification obligation with no corresponding test is invisible to it. The receipt attests what passed, not what was omitted.

Threat 14.4 (The specification and the tests share authors). The tests encode the same team's reading of the same sources as the implementation. A shared misreading—of a source conflict, of a boundary convention—would pass symmetrically. Independent recomputation and external review (Chapters 11 and 12) are the designed correctives, and both remain open.

Threat 14.5 (Single runtime, single platform). The receipt binds one runtime environment. Behaviors normalized by the candidate (such as the host integer-conversion case in the canonical layer) are tested on that runtime; other platforms' failure surfaces are untested, and the dependency identities that would make cross-platform replay authoritative are named but not materialized (Chapter 11).

Threat 14.6 (Fixtures resemble their authors' expectations). Golden rejection fixtures and synthetic boards are constructed adversarially, but by the same imagination that built the validators. An adversary with different incentives may find inputs outside the fixture family. The bounded canonical-encoding limits reduce this surface; they do not close it.

None of the four threats is an argument against the receipt; they are the reasons the receipt is bounded evidence rather than assurance, and the reasons steps five through eight of the closure protocol place recomputation, replication, audit, and review outside the project's own hands.

3. What the suite exercises

The suite's coverage groups into ten behavioral areas. For each, the paragraph below states what is tested and what the passing tests establish; the uniform caveat—established *for the software*, at the receipt, and nothing more—applies to all ten and is not repeated.

Quantity and unit rejection. The admissibility map from quantity kinds to unit kinds (sixteen kinds against seven units; Appendix A) is exercised positively and negatively: every allowed pairing constructs, every forbidden pairing raises the candidate's validation error, range constraints hold (Q10 in $[0, 10]$, Q100 in $[0, 100]$, fractions in $[0, 1]$, nonnegativity where declared, strict positivity of effective resistance), and the mixed-scale confusions preserved in Conflict 1 are unrepresentable as admitted values.

Canonical evidence limits. The canonical encoder's declared bounds—depth, node count, container size, text length, output size, and the 2,048-bit integer limit—are tested at and beyond each boundary; exact-type tagging rejects subtype smuggling; cycles are detected; unsafe Unicode is rejected; and a host-runtime integer-conversion failure is normalized to the candidate's validation error rather than leaking a platform exception.

Calibration identity. Profile digests recompute stably; any mutation of a bound field changes the digest; records binding a digest that fails recomputation are rejected; and profile-relative quantities refuse cross-profile use without a stated comparison rule.

Split separation. The anti-circularity contract of Chapter 6 is enforced in type: calibrated resistance requires a held-out dataset split reference, the same-sample role is typed as `APPARENT_SAME_SAMPLE` and refused wherever prediction demands calibration, and the circular procedure—fit and evaluate on one sample—cannot be expressed as an admitted prediction.

Exact Q4 boundaries. The quantization of Chapter 8 is tested at the cut points: exact decimal comparison at 2.5, 5, and 7.5; half-open band assignment including values exactly at a cut; the pre-clipping identity and separate overload ratio (the worked 1.2 full-scale case yielding exact 12, displayed 10, overload 1.2, band Q4); and the boundary-preservation property that display rounding never moves a value across a band.

Structural network constraints. The capability gates of Chapter 9 are exercised: series requires a simple acyclic chain, unit agreement, active-region components, and one profile; parallel requires genuine alternative branches with shared endpoints; gain composition requires adjacent coordinate agreement and dimensionless gains; duplicate and cyclic paths fail closed; and every aggregate binds ordered child provenance, topology, and the declared covariance limitation.

Residual censoring. The rules of Chapter 10: residuals require instrumented held-out observations, matching test points, a registered prediction, and calibrated resistance; at zero or zero-compatible predicted response the ratios are censored with explicit reasons and no residual quantity is emitted.

Uncertainty. Typed uncertainty records validate their internal order (lower bound, value, upper bound), unit agreement with the measurement, confidence strictly inside $(0, 1)$, and unit-constraint satisfaction of the interval endpoints themselves; values without required uncertainty are refused where the schema demands it.

Lineage. Provenance obligations per measurement kind are enforced: instrumented values require instrument, test point, calibration binding, and raw artifacts; model-derived values require ordered, contiguous, duplicate-free derivation parents; proxy and score kinds require raw artifacts; control references and control bindings must agree exactly.

Fail-closed capability behavior. The rejection classes of Chapter 11 (score-only, zero-instrumentation, missing-provenance, mixed-scale, unknown-profile, tampered) reject fully and without partial admission; trust modes pair only with their unauthenticated authorization states; fixture-key attestations verify and replay as integrity checks while the eligibility field remains false in every fixture—the tested constant behind the receipt's fourth number.

4. The seven synthetic known-fault boards

The suite's most instructive fixtures are seven synthetic boards—software-defined response surfaces with declared, known faults—against which the candidate's classification and diagnostic machinery is run end to end. The boards are software-only validation: each is a constructed function, not a model, and what a board can establish is exactly that the candidate's code draws the intended conclusion *when the fault is real by construction.* The boards are, in effect, the unit tests of the diagnostic vocabulary.

The board set is not arbitrary. Its seven members map onto the program's own structure: the first four boards (linear, cutoff, saturation, nonlinear) exercise the dose-response classification axis of Chapter 7 across its principal regions; the fifth (parallel) exercises the network arithmetic of Chapter 9; and the last two (confounded, bypass) are constructed temptations—each realizes, in software, the exact overinterpretation that a preserved source conflict warns against, so that the suite can verify the candidate declines it even when the tempting conclusion happens to be true by construction. Each board declares its fault parameters as design values under the ontology of Chapter 3, runs through the same typed pipeline as any future live data would, and asserts both positive expectations (the intended classification and diagnostics) and negative ones (the absence of every diagnostic the region and censoring rules forbid).

Linear. A clean proportional response within declared bounds. The reference board: classification must find the active region, the fitted resistance must match the constructed slope within the declared tolerance, and residuals must center on zero. A candidate that fails the linear board fails everywhere; a candidate that passes only the linear board has been tested against nothing.

Cutoff. A dead zone below a constructed dose threshold. Tests that the classifier reports CUTOFF rather than fitting a spurious shallow slope through the flat region, and that no diagnostic is emitted from the cutoff region—the flat-response ambiguity of Chapter 7 handled as the sources' own addendum requires.

Saturation. Response flattening above a constructed ceiling. Tests SATURATION classification, the meaninglessness of saturated-stage residuals (Conflict 7), and the separate retention of overload where the normalized value clips.

Nonlinear. A smoothly curved response with no linear region of the declared minimum width. Tests that the candidate refuses the ohmic vocabulary entirely—NON-OHMIC classification, no admitted resistance fit—rather than reporting the least-bad line through a curve.

Parallel. A constructed two-branch topology whose branch resistances and aggregate obey reciprocal arithmetic by construction. Tests the parallel capability gate end to end: branch validation, aggregation, child provenance—while establishing only that the arithmetic is implemented correctly, since the branches are parallel by fiat, which is precisely what no live topology is (Chapter 12).

Confounded. A board whose apparent dose response is constructed to arise from a covariate correlated with dose. Tests that the control machinery detects the confound—the label-shuffle and matched-control comparisons exposing that the "response" survives dose randomization—and that the candidate declines the response claim.

Bypass. A board with a constructed route around a declared constraint, producing the positive-residual signature that the sources' most overinterpreted passage reads as hidden bypass. Tests the discipline of Chapter 10 under maximal temptation: the candidate must rank the anomaly for follow-up *without* emitting a bypass identification, because even here—where the bypass is real by construction—the residual alone does not identify it.

REMARK 14.7 (Why known faults, and why only software). The boards invert the epistemic direction of testing: instead of asking whether the model has a property, they construct the property and ask whether the code sees it. This yields true positives and true negatives for the *software*, with ground truth by fiat. It yields nothing about any live model, because the boards' ground truth is stipulated, not discovered—the boards cannot establish causal or empirical validity, only that the instrument would report correctly if the world were as constructed. The distinction is the verification/validation boundary in classical simulation terms: the boards verify; nothing yet validates.

5. The limitations catalog

The catalog below is exhaustive by intent. Each entry names the boundary, states what is *not* established, and records what closing it would require.

Conflicts are preserved, not resolved. The eight source conflicts of Chapter 4 remain open. The candidate prevents them from contaminating downstream computation by typing—separate quantity kinds, separately named full-scale methods, separately named gain profiles—but the sources still disagree, and no reconciliation, theorem, or experiment has closed any of the eight. Closure would require, per conflict, either a proved bridge or a preregistered experiment; until then, downstream artifacts inherit disciplined disagreement, not resolution.

No live calibration. There is no admitted live calibration of any profile: no held-out resistance estimate on a real model, no drift-threshold validation, no inter-rater reliability result for scored kinds. Every calibrated object in this monograph is schema, fixture, or synthetic. Closure requires Gate 5 runs (Chapter 12) with everything that entails.

A guarded contract, not a validated law. The implemented prediction $\hat{y} = a/r_{\text{eff}}$ is a software contract with guards—positivity, uncertainty, split discipline, role checks—not a validated empirical law. The guards ensure that *if* the relation holds in a profile, the candidate can carry it honestly; whether it holds anywhere is exactly hypothesis `CLM-CAS-OHM-EFFECTIVE-RESISTANCE`, which is untested.

Networks are structural-only. Series, parallel, and gain-product operations exist as unauthenticated structural research calculations behind capability gates. Their empirical content—composition, redundancy—is the untested subject of two registered hypotheses, and their known theoretical weakness (interaction, dependence, covariance; the declared zero-covariance approximation) is carried as a limitation on every aggregate rather than discharged.

Noncausal throughout. Gain, redundancy, and residual diagnostics are noncausal. No causal model is declared, no identification strategy exists, and the one hypothesis that could earn limited causal vocabulary (`CLM-CAS-OHM-RESIDUAL-LOCALIZATION`) carries an explicit withdrawal clause on failure. Causal interpretation requires an explicit causal model and identifiable interventions [27]; the candidate contains neither.

State dynamics are open. Memristive, lacunary, adaptive, and all other history-dependent dynamics remain open (Chapter 13): no state variable, no update law, no hysteresis experiment, and an evidentiary bar that is unmet vacuously.

No physical quantities. No CAS-Ohm quantity is a physical voltage, current, resistance, charge, flux, or device property; physical SI units are absent from the ontology and their names are forbidden aliases. The registered product claim asserting otherwise is NOT LICENSED and contradicted by current evidence (Chapter 12).

No authority. Authenticated Evidence-Packet eligibility is constant false; none of its six preconditions—independently admitted authority manifest, protected key lifecycle, trusted

time, materialized dependencies, authenticated instrument evidence, reviewed packet policy—exists (Chapter 11). No safety, clinical, regulatory, product, canon, license, or disclosure authority exists anywhere in the candidate.

No review. Internal cross-review and independent external review have not occurred. The monograph itself is DRAFT / OPEN / RING 4 / NOT LICENSED / HELD, and drafting it promotes nothing.

CLAIM BOUNDARY 14.8 (Reading rule for the catalog). The entries above are boundaries, not caveats: each marks vocabulary that is entirely unoccupied, not claims that are weakly supported. A derived artifact may cite this monograph only in ways consistent with every entry simultaneously.

6. Related work

The candidate's intellectual debts have been cited where incurred; this section gathers them by role rather than repeating their content.

The *metrological* frame—that a quantity's identity, procedure, and uncertainty are inseparable from its value—is the international vocabulary of metrology and the uncertainty-expression guide [32, 33]. CAS-Ohm borrows their discipline for a nonphysical setting without claiming to be metrology: the no-bare-numbers rule of Chapter 3 is their central lesson transposed into a type system, and the transposition is acknowledged as such—nothing in the borrowed discipline validates the candidate's instruments, which remain unauthenticated (Chapter 11).

The *calibration* frame—that fitting a model to data must represent parameter uncertainty and model discrepancy rather than crowning the best fit as truth—follows the Bayesian calibration tradition [37]. It shapes the anti-circularity rule of Chapter 6 and the uncertainty obligations threaded through every schema; the candidate implements the discipline's bookkeeping while leaving its statistical practice to the unexecuted Gate 5 runs.

The *causal* frame—that causal claims require causal models and identifiable interventions, and cannot be read off associations or residuals—is Pearl's [27]. It governs by prohibition: Chapters 10 and 12 use it to fence what residuals and ablations may say, and the forbidden-wording list is in large part its enforcement arm.

The *provenance* frame—typed, attributable derivation records whose structure is explicitly not truth—parallels the W3C PROV model [22]. Chapter 11's two-axis design is the candidate's rendering of PROV's own caution: provenance is assertion, and trust in the asserter is a separate decision that no record structure can make for its consumer.

The *dynamical* frame—state-dependent constitutive relations, and a behavioral signature for memory—is the memristor and memristive-systems literature [35, 36], borrowed in Chapter 13 as motivating mathematics under an explicit vocabulary gate, with its electrical content and its controversies left where they were found.

What the candidate takes from all five frames is the same single habit: separating what a formalism can carry from what it has earned. What it adds is the enforcement of that habit in software—typed to the point where the historically observed failures of its own sources are unrepresentable, and tested to a receipt.

7. Conclusion

The thesis of this monograph can now be stated in one paragraph and defended in three.

CAS-Ohm 1.0 is viable as a profile-specific experimental response and calibration proposal precisely because its type system makes the Ohm analogy *testable and rejectable*. Not tested:

the five registered hypotheses stand open, their Evidence Packets unadmitted, their protocol frozen but unexecuted. Not accepted: every status label in the artifact chain—OPEN, RING 4, NOT LICENSED, HELD—asserts the absence of promotion. Testable and rejectable: the analogy has been given exact enough types, exact enough splits, exact enough controls, and exact enough falsifiers that a future run must return a verdict one way or the other, and the negative verdict has a prepared, enforceable meaning—capabilities confined, vocabulary withdrawn, outcomes retained.

The first defense is the ontology. Sixteen quantity kinds, seven units, six measurement kinds, and an admissibility map make the historically observed failure modes of the sources—the mixed-scale saturation threshold, the dimensionally unmoored V/R, the score-as-telemetry slide, the residual-as-cause leap—unrepresentable or fail-closed. A proposal that cannot state the old errors cannot silently repeat them.

The second defense is the audit. Eight conflicts preserved at evidence-record granularity, none harmonized, each bound to a disposition rule, so that no downstream artifact can claim an inherited coherence the three sources demonstrably lack. The monograph's willingness to keep its sources' disagreements on the record is the same discipline it demands of future experiments: negative knowledge is retained, not managed.

The third defense is the falsification program. Five hypotheses with frozen claim identifiers; a closure protocol whose eight steps fix the analyses, the controls, the falsifiers, and even the permitted sentences before any data exist; an evidentiary bar for the stateful extension set in advance of any implementation; and an eligibility axis that no internal accumulation can move. Whether the model predicts anything is exactly the question this program leaves to controlled, independently audited, independently reviewed experiments—and nothing in this monograph, from its first chapter to this sentence, preempts their answer.

CLAIM BOUNDARY 14.9 (Final claim boundary). This monograph establishes tested software behavior at the Gate 4 receipt and nothing beyond it: no live execution, no scientific conclusion, no physical measurement, no causal effect, no safety, clinical, or regulatory conclusion, no product capability, no license, no canon admission, and no disclosure. All five registered hypotheses remain IN REVIEW / RING 4 / NOT LICENSED / HELD without admitted Evidence Packets; authenticated Evidence-Packet eligibility remains constant false; internal cross-review and independent external review remain open. Artifact status DRAFT / OPEN / RING 4 / NOT LICENSED / HELD governs this text and every use of it.

APPENDIX A

Enumerations and Schema Tables

This appendix collects, in tabular form, the enumerations and record schemas of the CAS-Ohm 1.0 candidate as implemented at the Gate 4 receipt. The tables are normative transcriptions, not summaries: every enumeration member is listed exactly, under its implementation spelling, and the prose notes attached to each table record the validation rules that give the members their force. Where a chapter develops a table's subject at length, the chapter is named in the table's introduction. All member names are implementation identifiers and are typeset verbatim in typewriter face.

1. Quantity kinds

The sixteen quantity kinds (Chapter 3) partition every numeric value the candidate can carry. Each kind admits only the unit or units shown; the admissibility map is enforced at construction, and a kind-unit pairing outside the map raises the candidate's validation error. Range rules attach to the unit (Q10 in $[0, 10]$, Q100 in $[0, 100]$, `FRACTION` in $[0, 1]$) and to the kind (nonnegativity and strict positivity as noted).

Table 1: The sixteen quantity kinds and their admissible units.

Quantity kind	Admissible unit(s)	Notes
`DRIVE_INTERVENTION`	`ALPHA`	steering dose coordinate
`DRIVE_SCORE`	`Q10`	scored drive level; not a dose
`SIGNED_ACTIVATION`	`ACTIVATION`	signed directional projection
`FLOW_MAGNITUDE`	`ACTIVATION`	nonnegative
`NORMALIZED_FLOW`	`Q10`	full-scale-relative display value
`CONSTRAINT_SCORE`	`Q10`, `Q100`	the sources' two constraint scales, kept distinct
`EFFECTIVE_RESISTANCE`	`ALPHA_PER_ACTIVATION`	strictly positive; requires a resistance role
`RESISTANCE_DELTA`	`ALPHA_PER_ACTIVATION`	paired-difference of resistances
`CAPABILITY_SCORE`	`Q10`	scored capability level
`GAIN_MAGNITUDE`	`DIMENSIONLESS`	nonnegative local gain
`EFFECT_PROJECTION`	`ACTIVATION`	effect-axis projection
`SATURATION_FRACTION`	`FRACTION`	region fraction in $[0, 1]$
`CUTOFF_FRACTION`	`FRACTION`	region fraction in $[0, 1]$
`OVERLOAD_RATIO`	`DIMENSIONLESS`	nonnegative; retained separately from clipped display
`FLOW_RESIDUAL`	`ACTIVATION`	censored at zero-compatible prediction
`RELATIVE_ERROR`	`DIMENSIONLESS`	nonnegative

The kinds encode the audit's first and third conflicts directly: Q10 scores, Q100 constraint scores, fractions, alpha doses, activation projections, and effective resistance are distinct kinds

precisely so that no silent conversion among the sources' mixed scales is expressible, and so that no score kind can occupy the resistance position in a prediction.

2. Measurement kinds

The six measurement kinds (Chapter 3) classify how a value came to be. Kind non-fungibility governs: derivation preserves parent lineage and never upgrades a value's kind.

Table 2. The six measurement kinds.

Measurement kind	Meaning
`DESIGN_VALUE`	declared by a board or profile; not measured
`INSTRUMENTED`	produced by an identified instrument chain with calibration binding
`BEHAVIORAL_PROXY`	inferred from external behavior
`MODEL_DERIVED`	computed from other records
`HUMAN_SCORE`	rated by a person
`LLM_SCORE`	rated by a language model

Per-kind obligations (Chapter 11): `INSTRUMENTED` requires instrument identity, test point, calibration-profile binding, and raw artifact references; `MODEL_DERIVED` requires ordered derivation parents; `BEHAVIORAL_PROXY`, `HUMAN_SCORE`, and `LLM_SCORE` require raw artifact references.

3. Unit kinds

The seven unit kinds (Chapter 3) are the complete unit ontology. Physical SI units are absent by design: `volt`, `ampere`, and `ohm` are forbidden aliases for model quantities, and a physical profile would require separate explicit typing and instrumentation.

Table 3. The unit kinds.

Unit kind	Meaning
`ALPHA`	steering dose
`ACTIVATION`	directional projection magnitude
`Q10`	score scale $[0, 10]$
`Q100`	score scale $[0, 100]$
`ALPHA_PER_ACTIVATION`	effective resistance
`DIMENSIONLESS`	gains, ratios, relative errors
`FRACTION`	normalized region fractions $[0, 1]$

4. Dose-response regions

The dose-response region axis (Chapter 7) classifies where on its response curve a profile is operating. The implemented enumeration has seven members; the source's fifth `BREAKDOWN` region is retained in the implementation alongside the audit's five-region summary, and `UNASSESSED` is the typed absence of a classification.

No diagnostic is emitted from a `CUTOFF`, `SATURATION`, `UNSTABLE`, uncalibrated, replay-time-stale, wrong-model, or semantically unassessed profile.

TABLE 4. Dose-response regions.

Region	Meaning
UNASSESSED	no classification performed
CUTOFF	dose below response threshold; flat
ACTIVE	locally responsive operating region
SATURATION	response flattened at ceiling
BREAKDOWN	source-retained beyond-saturation regime
NON_OHMIC	responsive but without an admissible linear region
UNSTABLE	classification not reproducible under declared rules

5. Semantic outcomes

The semantic outcome axis (Chapter 7) is classified independently of region; no region mechanically determines an outcome.

TABLE 5. Semantic outcomes.

Outcome	Meaning
UNASSESSED	no semantic assessment performed
INTENDED	response on the intended effect axis
OPPOSITE	response opposed to the intended axis
OFF_AXIS	response on a different axis

6. ASOM labels

ASOM is a derived display mnemonic only (Conflict 5 of Chapter 4); the label is computed from the two independent axes and never replaces them. MISALIGNED derives from outcome alone (OPPOSITE or OFF_AXIS); the remaining labels require outcome INTENDED and derive from region; combinations outside the map yield no label.

TABLE 6. ASOM display labels and their derivations.

Label	Derived from region	Derived from outcome
STARVED	CUTOFF	INTENDED
ALIGNED	ACTIVE	INTENDED
OVERSATURATED	SATURATION	INTENDED
MISALIGNED	any	OPPOSITE or OFF_AXIS

7. Full-scale methods

The three full-scale (I_{FS}) methods (Conflict 2 of Chapter 4) are the sources' three incompatible normalization procedures, kept as separately named calibration methods with different profile digests. They are not interchangeable, and no conversion between them is defined.

Table 7. Full-scale methods (`IFSMethod`).

Method	Source procedure
`RAIL_AND_FLOOR_PER_PATH`	measured once per path at drive rail and constraint floor
`CORPUS_PERCENTILE_PER_TEST_POINT`	95th percentile of absolute projection over a fixed hashed corpus
`DECLARED_DESIGN_FULL_SCALE`	board-selected design value

8. Direction provenance

Every test point binds a direction vector with unit norm (`direction_norm` must equal exactly one), an explicit sign (`POSITIVE` or `NEGATIVE`), and a provenance member recording how the direction was derived.

Table 8. Direction provenance.

Provenance	Derivation
`PROBE_DERIVED`	from a trained probe
`JACOBIAN_DERIVED`	from a local Jacobian computation
`DIFF_DERIVED`	from a contrastive activation difference

9. Review states

Review states attach to calibration profiles and reviewable records (Chapters 5 and 11). Completeness requires an admissible review state; the flag state routes a record to a person and is never cleared mechanically.

Table 9. Review states.

State	Meaning
`UNASSESSED`	no review performed
`IN_REVIEW`	review open
`REVIEWED`	review completed at the recorded scope
`FLAG_FOR_HUMAN_REVIEW`	routed to mandatory human attention

10. The ObservedValue schema

The observed-value schema (Chapter 3, Definition of observed value) is the no-bare-numbers rule in record form: every numeric record carries at least the fourteen fields below. The implementation realizes the schema as the measurement-value record with typed provenance; field names below follow the audit's schema listing.

Table 10: The ObservedValue schema.

Field	Content
`quantity_kind`	one of the sixteen quantity kinds
`exact_decimal_value`	exact decimal, never a bare binary float

Field	Content
`unit_id`	one of the seven unit kinds, admissible for the quantity kind
`measurement_kind`	one of the six measurement kinds
`method_id`	identifier of the producing method
`calibration_profile_sha256`	digest of the binding calibration profile
`estimate_or_observation`	epistemic status marker
`uncertainty`	typed uncertainty: unit, standard uncertainty, bounds, method
`confidence_interval`	interval containing the value, with confidence level in $(0, 1)$
`sample_size`	positive integer where applicable
`raw_artifact_refs`	references to raw evidence bytes, digest-bound
`control_refs`	references to bound controls, matching the control bindings
`derivation_parents`	ordered, contiguous, duplicate-free parent references
`missing_or_censored_reason`	explicit reason wherever a value is absent

Additional typed fields attach conditionally in the implementation: `resistance_role` is mandatory for—and valid only for—`EFFECTIVE_RESISTANCE` values, and a held-out dataset split reference is mandatory for the calibrated role.

11. Calibration-profile fields

The calibration profile (Chapter 5) binds the complete experimental context under one canonical digest. All CAS-Ohm quantities are profile-relative; nothing transfers across profiles without a stated comparison rule.

Table 11: Calibration-profile fields.

Field group	Bound content
Model identity	model id, revision, weights/config/tokenizer SHA-256 digests
Code identity	exact code revision
Runtime	runtime identity and dtype
Test point	layer, token positions, hook type, bound model identity
Direction	direction id and digest, norm (exactly one), sign, provenance
Corpus	corpus manifest and rights
Randomization	seeds and randomization procedure
Alpha grid	the declared dose schedule
Full-scale method	one `IFSMethod` member (Table 7)
Fit method	declared resistance-fit procedure
Uncertainty method	declared uncertainty procedure
Tolerances	prediction and comparison tolerances
Region rules	response-region classification rules
Controls	bound control set with relevance methods
Drift and expiry	drift model, validity interval, staleness rule
Review state	one member of Table 9
Canonical digest	SHA-256 over the canonical encoding of all of the above

The digest closes the record: any mutation of any bound field changes the profile identity, which is what makes the audit's three full-scale methods, and every other profile-level disagreement, non-interchangeable by construction.

APPENDIX B

Glossary

This glossary defines the technical vocabulary of the monograph. Each entry gives the term, its definition as used in the CAS-Ohm 1.0 candidate, and a plain-text reference to the chapter or chapters where the term is developed. Enumeration members and implementation identifiers appear in typewriter face. Definitions here are convenience restatements; where a glossary entry and a chapter's normative text could be read differently, the chapter governs.

Activation response (y).: The observed value of the directional activation projection at a test point, carried in the unit `ACTIVATION`. The response coordinate of the CAS-Ohm model. (Chapters 3 and 6.)

Active region.: The dose-response region (`ACTIVE`) in which a profile is locally responsive and an admissible linear fit is possible. Most capabilities—prediction, residuals, network operations—require active-region status. (Chapter 7.)

Alpha dose (a).: The declared steering intervention coordinate, carried in the unit `ALPHA`. The dose coordinate of the CAS-Ohm model; deliberately never called voltage. (Chapters 1, 3, and 6.)

Anti-circularity rule.: The rule rejecting the procedure that computes $r = a/y$ on a sample, predicts $\hat{y} = a/r$ on the same sample, and reports agreement as validation. Calibration and evaluation must use disjoint splits. (Chapter 6.)

Apparent same-sample resistance.: The resistance role (`APPARENT_SAME_SAMPLE`) typing a ratio computed on the sample it would be evaluated on; refused wherever prediction requires calibration. (Chapters 5 and 6.)

Artifact reference.: A typed reference carrying artifact identifier, URI, SHA-256 digest, and media type. Completeness requires every reference to resolve to bytes matching its digest. (Chapters 3 and 11.)

ASOM.: The four-label display mnemonic (`STARVED`, `ALIGNED`, `OVERSATURATED`, `MISALIGNED`) derived from, and never replacing, the two independent classification axes. (Chapters 4 and 7; Appendix A.)

Authenticated-current.: The currency notion that would hold if replay-time currency were evaluated at a time attested by a trusted current-time receipt. Unoccupied in the current candidate. (Chapter 11.)

Authenticated Evidence-Packet eligibility.: The second axis of the packet design: whether independent authorities have admitted a packet into a trust regime. Constant false in the CAS-Ohm 1.0 candidate; requires six missing preconditions. (Chapter 11.)

Authority manifest.: The document declaring which parties, under which keys, may assert which classes of statement. Its digest is carried in operation bindings; no independently admitted manifest exists. (Chapter 11.)

Authorization state.: The typed statement of what an operation's trust mode establishes: `UNAUTHENTICATED_STRUCTURAL` or `UNAUTHENTICATED_INTEGRITY`. Both implemented states are explicitly unauthenticated. (Chapters 9 and 11.)

Breakdown region.: The source-retained dose-response region (`BREAKDOWN`) beyond saturation, kept in the implementation alongside the audit's five-region summary. (Chapter 7; Appendix A.)

Calibration profile.: The record binding the complete experimental context—model, code, runtime, test point, direction, corpus, seeds, alpha grid, full-scale method, fit and uncertainty methods, tolerances, region rules, controls, drift and expiry, review state—under one canonical digest. All quantities are profile-relative. (Chapter 5; Appendix A.)

Calibration split.: The registered data split on which a resistance or response curve is fit. Must be disjoint from the held-out split. (Chapters 5, 6, and 12.)

Canonical evidence encoding.: The exact-type-tagged, cycle-aware, unsafe-Unicode-rejecting, bounded encoding under which all records are hashed. Inherited from the CAS Core 8 contract. (Chapter 3.)

Capability gate.: The set of preconditions (coordinates, units, topology, region, profile, evidence) under which a network operation is permitted to run at all. (Chapter 9.)

Censoring.: The typed refusal to emit a quantity—with an explicit recorded reason—where it would be undefined or meaningless, as with residual ratios at zero-compatible predicted response. (Chapters 10 and 11.)

Claim identifier.: The stable ledger identifier of a registered claim, of the form `CLM-CAS-OHM-*`. (Chapter 12.)

Claim ledger.: The shared registry binding claim texts, status vectors, and their relation to the preregistration kernel. (Chapters 12 and 14.)

Control binding.: The typed binding of a control artifact to a test point together with the relevance method by which the control is claimed relevant. Bound controls and control references must agree exactly. (Chapter 11.)

Covariance limitation.: The declared limitation carried on every network aggregate: cross-input covariance is unavailable and a zero-covariance approximation is in use. (Chapters 9 and 12.)

Derivation parents.: The ordered, contiguous, duplicate-free references from a derived value to the records it was computed from. Required for every model-derived value. (Chapters 3 and 11.)

Direction provenance.: The typed record of how a test point's direction vector was derived: `PROBE_DERIVED`, `JACOBIAN_DERIVED`, or `DIFF_DERIVED`. (Chapter 5; Appendix A.)

Dose-response region.: The first classification axis: where on its response curve a profile is operating (`UNASSESSED`, `CUTOFF`, `ACTIVE`, `SATURATION`, `BREAKDOWN`, `NON_OHMIC`, `UNSTABLE`). Classified independently of semantic outcome. (Chapter 7.)

Effective resistance (r_{eff}).: The fitted proportionality parameter of the CAS-Ohm model, strictly positive, in the unit `ALPHA_PER_ACTIVATION`, carrying uncertainty and a resistance role. A model parameter, never a physical resistance. (Chapters 5 and 6.)

Evidence Packet.: The authenticated packet class that the candidate does not produce: a packet whose claims have been admitted under independent authority, protected keys, trusted time, materialized dependencies, authenticated instruments, and reviewed policy. (Chapter 11.)

Fail-closed.: The design rule that any validation failure rejects the whole record with no partial admission, repair, or downgrade. (Chapters 3, 9, and 11.)

Fixture keys.: The deterministic Ed25519 keys committed to the repository for testing the attestation machinery. They provide integrity and replay checks only and can never satisfy the key-lifecycle precondition. (Chapter 11.)

Forbidden wording.: The list of product phrases excluded from every artifact, including physical telemetry, measures attention, electrical law, causal by construction, cross-model comparable, and proves hidden resistance. (Chapters 12 and 14.)

Full-scale activation (I_{FS}).: The profile-specific normalization denominator for Q10 display values. Its three inherited methods are separately named and not interchangeable. (Chapter 8; Appendix A.)

Gain composition.: The structural network operation multiplying adjacent dimensionless local gains along a declared chain. Its empirical validity is the untested hypothesis `CLM-CAS-OHM-GAIN-COMPOSITION`. (Chapters 9 and 12.)

Gate-4 receipt.: The bound historical software evidence: 65 passing test methods under a canonical suite test-ID digest, 58 parameterized subtests, eight retained conflicts, zero authenticated eligibility paths. (Chapter 14.)

Gate-5 closure protocol.: The eight-step preregistered path—freeze, disjoint splits, six controls, non-erasure, independent recomputation and authenticated execution, held-out replication, raw-evidence retention, eligibility `NOT_ESTABLISHED` pending independent review—under which any registered hypothesis could be tested. (Chapter 12.)

Held-out split.: The registered data split, disjoint from the calibration split, on which predictions are evaluated. (Chapters 5, 6, and 12.)

Instrumented field.: A value of measurement kind `INSTRUMENTED` satisfying the full obligations: instrument identity, test point, calibration binding, raw artifacts. At least one is necessary—and not sufficient—for packet completeness. (Chapters 3 and 11.)

Kind non-fungibility.: The rule that derivation never upgrades epistemic type: a derived value does not become instrumented because a parent was, and a score does not become telemetry by storage adjacency. (Chapter 3.)

Limiting vocabulary of status.: The four-coordinate status vector: OPEN or IN REVIEW (text under revision), RING 4 (outermost admission ring), NOT LICENSED (no downstream reliance), HELD (undisclosed). (Chapters 1 and 12.)

Memristive-form profile.: The provisional nonphysical extension typing a state-dependent resistance $x_{t+1} = F_{\mathrm{profile}}(x_t, a_t, y_t, c_t; \theta)$, $r_{\mathrm{eff},t} = G_{\mathrm{profile}}(x_t; \theta)$, admissible only with a full model-state record and only past the evidentiary bar. Not implemented. (Chapter 13.)

Memristive system.: In the classical literature, an input-output system whose instantaneous response coefficient depends on an internal state with its own dynamics. Cited as motivating mathematics only. (Chapter 13.)

Memristor.: The postulated fourth circuit element relating charge and flux, whose resistance depends on integrated current history. Cited as motivating mathematics only; no electrical content is imported. (Chapter 13.)

Model-state record.: The seven-component record (initialization, update law, sampling interval, reset rule, drift model, uncertainty, provenance) that any stateful extension must carry. (Chapter 13.)

Operation binding.: The typed record attached to every network operation: authority-manifest digest, attestation digests, capability, trust mode, authorization state, replay time, optional clock receipt, runtime, covariance limitation, dependency-lock and code references. (Chapters 9 and 11.)

Overload ratio.: The retained pre-clipping ratio of raw response to full scale, kept separately from the clipped Q10 display value. (Chapter 8.)

Pinched hysteresis.: The classical behavioral signature of memristive systems: a dose-response loop through the origin that collapses with increasing input frequency. Adapted, in prespecified nonphysical form, as part of the evidentiary bar. (Chapter 13.)

Preregistration.: The practice of freezing hypotheses, designs, analyses, and permitted claim language before data collection. The Gate-5 protocol is the candidate's preregistration kernel. (Chapter 12.)

Profile digest.: The canonical SHA-256 identity of a calibration profile. Any mutation of a bound field changes the digest; an unverifiable digest fails the packet closed. (Chapters 5 and 11.)

Q4 quantization.: The exact half-open banding of Q10 display values at cut points 2.5, 5, and 7.5, computed on pre-rounding decimals so that display rounding can never move a value across a band. (Chapter 8.)

Q10 / Q100.: The two score scales of the sources, kept as distinct unit kinds with enforced ranges so that the inherited mixed-scale conflict is unrepresentable. (Chapters 3, 4, and 8.)

Registered hypothesis.: A falsifiable claim held in the claim ledger under a stable identifier and the uniform status IN REVIEW / RING 4 / NOT LICENSED / HELD, with no admitted Evidence Packet. Five exist. (Chapter 12.)

Rejection classes.: The six fail-closed packet classes: score-only, zero-instrumentation, missing-provenance, mixed-scale, unknown-profile, tampered. (Chapter 11.)

Replay-as-of.: The caller-supplied time parameter (`replay_as_of`) against which all currency judgments are evaluated. An unauthenticated input by design. (Chapter 11.)

Replay-time-current.: Currency of a profile evaluated at the caller-supplied replay time. Entailed by, and never entailing, authenticated currency. (Chapter 11.)

Research packet.: The structural evidence bundle the candidate can assess for completeness: typed values, provenance, controls, uncertainty, profile binding, raw bytes. Distinct from an Evidence Packet. (Chapter 11.)

Residual (δ_y).: The difference between observed and predicted response for an admitted active-region prediction, with propagated uncertainty. A non-identifying diagnostic: it may rank follow-up probes and never, by sign alone, identifies a hidden policy, bypass, or cause. (Chapter 10.)

Residual localization.: The registered hypothesis (`CLM-CAS-OHM-RESIDUAL-LOCALIZATION`) that blinded implanted faults are localized above shuffled and matched baselines, with causal language withdrawn on failure. (Chapters 10 and 12.)

Resistance role.: The mandatory typing of every effective resistance: `CALIBRATED_HELD_OUT`, `APPARENT_SAME_SAMPLE`, `NETWORK_EQUIVALENT`, or `DESIGN_PROPOSAL`. Prediction requires the calibrated or separately validated network-equivalent role. (Chapters 5, 6, and 9.)

Review state.: The typed review status of a profile or record: `UNASSESSED`, `IN_REVIEW`, `REVIEWED`, `FLAG_FOR_HUMAN_REVIEW`. (Chapters 5 and 11; Appendix A.)

Semantic outcome.: The second classification axis: whether the response was `INTENDED`, `OPPOSITE`, `OFF_AXIS`, or `UNASSESSED`, classified independently of region. (Chapter 7.)

Series / parallel operations.: The structural network operations of chain addition and reciprocal branch aggregation over effective resistances, capability-gated and unauthenticated; their empirical validity is the subject of registered hypotheses. (Chapters 9 and 12.)

Structural completeness.: The first axis of the packet design: internal well-formedness of a research packet (exact model revision, valid profile digest, raw bytes, controls, uncertainty, lineage, nonstale reviewed profile at replay time, at least one instrumented field). Never authorization. (Chapter 11.)

Synthetic known-fault board.: A software-defined response surface with a constructed fault (linear, cutoff, saturation, nonlinear, parallel, confounded, bypass) used to validate the candidate's code against ground truth by fiat. Software-only evidence. (Chapter 14.)

Test point.: The typed site of measurement: model identity, layer, token positions, hook type, and direction vector with digest, unit norm, sign, and provenance. (Chapter 5.)

Trust mode.: The declared operating mode of a network operation: `STRUCTURAL_RESEARCH_ONLY` (no keys) or `INTEGRITY_CONFIGURATION` (fixture-key attestation, verified and replayed, still unauthenticated). (Chapters 9 and 11.)

Trusted current-time receipt.: The missing precondition: a cryptographically bound statement from an accountable time authority that a moment had genuinely arrived. No channel for it exists in the candidate. (Chapter 11.)

Uncertainty record.: The typed uncertainty attached to numeric values: unit, standard uncertainty, interval bounds containing the value, confidence level strictly in $(0, 1)$, and method identifier. (Chapters 3 and 11.)

Unit separation.: The rule keeping `ALPHA`, `ACTIVATION`, `ALPHA_PER_ACTIVATION`, score scales, fractions, and dimensionless ratios as distinct unit kinds, with physical volt, ampere, and ohm as forbidden aliases. (Chapter 3.)

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